BY ORDER OF THE COMMANDER
HOLLOMAN AIR FORCE BASE (AETC)

HOLLOMAN AFB INSTRUCTION 13-250

30 JULY 2025

Nuclear, Space, Missile, Command and Control

AIRFIELD OPERATIONS

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available on the e-Publishing website at **www.e-publishing.af.mil** for downloading or ordering.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: 54 OSS/OSA

Certified by: 54 OSS/CC
(Lt Col Joseph M. Bisson)

Supersedes: HOLLOMANAFBI 11-250, 25 Jan 2018

Pages: 197

This instruction contains information and directives pertaining to the operation of 49th Wing (WG) airfield, Air Traffic Control (ATC) facilities, and associated airspace. It implements Air Force Policy Directives (AFPD) 13-2, Air Traffic, Airfield, Airspace, and Range Management, with additional guidance derived from Air Force Manual (AFMAN) 13-204v1, Management of Airfield Operations; AFMAN 13-204v2, Airfield Management; AFMAN 13- 204v3, Air Traffic Control; and Department of the Air Force Instruction (DAFI) 13- 213, Airfield Driving. This instruction applies to all assigned, attached, and hosted aircrew members, civilian employees and uniformed members of the Regular Air Force, the Air Force Reserve, and those who are contractually obligated to comply with Department of the Air Force publications involved in Holloman Air Force Base (AFB) flying activities. This publication does not apply to the Air National Guard or United States Space Force. Individuals responsible for complying with this guidance should carefully review this volume. This instruction is the source document for Holloman AFB Airfield and ATC operations. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the DAF Form 847, Recommendation for Change of Publication to the OPR, to 54th Operations Support Squadron, Airfield Operations (54 OSS/OSA), Building 577, 1148 Rescue Road, Holloman AFB NM 88330. Route DAF Form 847 from the field through the appropriate functional's chain of command. Ensure that all records created because of processes prescribed in this instruction are maintained in accordance with Air Force Instruction (AFI) 33-322, Records Management and Information Governance Program, and disposed of in accordance with the Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS).

SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. Major changes include: the publication control number renamed 13-250, major command change, unit name changes, local airspace changes, updated figures, updated diagrams, removed German Air Force, and reference updates. Expanded procedures for aircraft arresting systems, flight planning, standardized departures, formation flight operations, aerial refueling, uncontrolled airfield operations, and aircraft priorities. Incorporated High Speed Test Track, MQ-4 Triton, Small Unmanned Aircraft Systems, Explosive Ordnance Disposal, and Support of Aircraft Carrying Hazardous Materials.

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Chapter 1

GENERAL INFORMATION

1.1. Compliance with this Instruction. HMN based units and those units on temporary duty (TDY) to HMN flying local missions will comply with this instruction. Requests for deviations from this instruction must be submitted to 54 OSS/OSA for coordination.

1.2. Responsibilities.

1.2.1. 54th Fighter Group Commander (54 FG/CC) is the waiver authority for this instruction. 54 FG/CC or 54 FG Brickholder will ensure this instruction is maintained, enforced, and viable.

1.2.1.1. The 54 FG Brickholder is the 54 FG/CC's designated representative.

1.2.2. 54 OSS Commander (54 OSS/CC) will ensure revisions to this directive are fully coordinated with applicable base agencies.

1.2.3. 54 FG Standardization and Evaluation office (54 FG/FGV) and 49th Operations Group Standardization and Evaluation office (49 OG/OGV) will validate and disseminate interim changes to local flying procedures.

1.2.4. 54 OSS/OSA will compile, consolidate, and coordinate changes to this instruction and is responsible for its content.

1.2.5. 49 WG A5, Plans and Programs (49 WG/A5) will ensure TDY units flying local missions from HMN are assigned a host unit. The host unit will coordinate with 54 FG/FGV and 49 OG/OGV to ensure all TDY aircrew are briefed on applicable procedures contained in this instruction, to include a local area briefing, before a TDY unit begins flying local sorties. When coordinated in advance, 54 OSS can support TDY unit briefings.

1.3. Dimensional Units. Visibility distances are in statute miles (SM). All other distances referred to in this instruction are in nautical miles (NM) unless otherwise identified. Altitudes will be identified by feet (ft) in either Above Ground Level (AGL) or Mean Sea Level (MSL). Tactical Air Navigation (TACAN) radial and distance measuring equipment (DME) fixes are listed as HMN R-###/## DME (e.g., CLOUD, HMN R-078/14 DME).

1.4. Office Identification. In addition to the offices listed in paragraph 1.2., the following office identifications will be used throughout for standardization.

1.4.1. White Sands Radar Facility (54 OSS/OSAR) is identified as WSRF and includes Radar Approach Control (RAPCON) functions, Approach Control, Arrival, Departure Control, Clearance Delivery, and United States (US) Army Military Radar Unit, known as CHEROKEE. All WSRF positions can provide full Air Traffic Control services, except for "CHEROKEE".

1.4.2. Tower refers to HMN Tower (54 OSS/OSAT) and includes Local Control, Ground Control, and Flight Data functions.

1.4.3. AM refers to Airfield Management (54 OSS/OSAA).

1.4.4. RAWS refers to Radar, Airfield, Weather Systems, or the maintenance office (54 OSS/OSAM).

1.4.5. Air Traffic Control (ATC) refers to Tower and/or RAPCON.

1.4.6. HAM refers to HMN Airspace Manager (54 OSS/OSP) and Scheduling refers to Wing Scheduling (54 OSS/OSOS).

1.4.7. Ops Sup refers to flying unit Top-3, Operations Supervisor, Duty Officer, etc.

Chapter 2

AIRFIELD FACILITIES AND SERVICES

2.1. Runway and Taxiways.

2.1.1. The airfield is defined as all movement areas including runways, taxiways, arm/de- arm areas, and aircraft parking ramps and service areas, which are used for towing, taxiing, takeoff, and landing of aircraft.

2.1.2. Reference Figure 2.1. and HMN Airport Diagram in Department of Defense (DoD) Flight Information Publication (FLIP) *Low Altitude US, Vol. 6* for - airfield elevation, runway slope gradients, runway magnetic headings, tower blind spots, and other runway information.

2.1.3. Visual Flight Rules (VFR) hold lines exist at the intersections of Runway 07/25 and Runway 16/34. The VFR hold lines referred to in this paragraph are for vehicles and taxiing aircraft only. Aircraft given clearance to land or takeoff must disregard the hold lines at the intersections of Runway 07/25, Runway 16/34, and the intersection of Runways 16/34 and 04/22 during landing and takeoff roll.

2.1.4. Arm/De-Arm areas are Alpha, Bravo, Delta, Echo, Foxtrot, and Hotel End of Runway (EOR) areas. Echo Live Ordnance Loading Area (LOLA) may be use for arming/de-arming when not reserved for live ordnance loading.

2.1.5. Taxiway Alpha becomes Taxilane Alpha starting from the west edge of Hangar 565 entrance to Taxiway Bravo. Wingtip clearances are determined utilizing Unified Facilities Criteria (UFC) 3-260-01, *Airfield and Heliport Planning and Design*.

2.2. Runway Selection Procedures. During airfield operating hours, the Tower Watch Supervisor (WS) will use Table 2.1. to determine the runway(s) in use after coordinating with WSRF WS and the Supervisor of Flying (SOF).

Table 2.1. Runway Selection Criteria.

Wind Direction	Wind Speed	Runway In Use
Any	Less than 5 knots	16
070° clockwise to 250°	5 to 10 knots	16
251° clockwise to 071°	5 to 10 knots	34
Any	11 knots or more	Runway most nearly aligned with the wind

2.2.1. Runway Configuration. The standard runway configuration is single runway operations (e.g., Runway 34 arrivals/departures). Runway 16 is the primary instrument and calm wind runway. All aircraft will depart/recover on the single runway in use except for operational needs for safety (i.e., aircraft using Alpha LOLA for live loads may depart Runway 25 while Runway 16 is primary) or whenever an expeditious arrival and departure traffic flow can be achieved by using multiple runways.

2.2.1.1. Arrivals and departures, to or from runways other than the runway(s) in use, will be handled on a case-by-case basis and are subject to ATC authorization (e.g., T-38 departure Runway 25 for web barrier requirements, QF-16 scheduled use of Runway 22/04 for drone operations, DA- 40 departure Runway 07).

2.2.1.2. 54 FG F-16s may depart Runway 25 while Runways 16, 34, or 22 are in use only when the Runway 25 departure does not conflict with or delay the VFR patterns or Instrument Flight Rules (IFR) arrivals to Runways 16, 34, or 22.

2.2.2. Runway Change. Tower WS will coordinate with WSRF WS and the SOF before changing the runway in use. Tower will notify WSRF, AM, 54 OSS Weather Flight (54 OSS/OSW), and 49th Maintenance Group (49 MXG) - Maintenance Operations Center (MOC) when the runway change is complete. AM will notify 49 WG Command Post (49 WG/CP). The SOF will notify 54 FG/CC, 54 FG Brickholder, 49 OG/CC, and tenant unit Ops Sups as required.

2.2.3. Transient heavy aircraft primary arrival and departure runway is Runway 04/22. If wind or other factors dictate, heavy aircraft may arrive and depart Runway 16/34 with AM coordination and approval. Runway 07/25 is not usable for heavy aircraft arrival or departures but may be used for taxi operations IAW runway bearing strength restrictions (see DoD FLIP *Area Planning - North and South America* (AP/1).

2.3. Runway Surface Condition (RSC) and Runway Condition Reading (RCR). IAW AFMAN 13-204V2, Chapter 7, HMN is not required to maintain a decelerometer or report RCR. RSC inspections are conducted by AM and reported to Tower, WSRF, 54 OSS/OSW, and 49 WG/CP.

2.4. Runway Inspections and Checks. IAW AFMAN 13-204V2 and local checklists, AM will conduct daily airfield inspections and additional airfield checks as required in response to in- flight emergencies, Foreign Object Damage (FOD), Bird Aircraft Strike Hazard (BASH), construction, daytime and nighttime airfield lighting serviceability, to determine RSC, and any other items affecting the status of the airfield (also see paragraph 8.3 Suspension of Runway Operations due to Aircraft Emergencies).

2.5. Opening and Closing a Runway. Procedures for opening and closing the runways will be accomplished IAW AFMAN 13-204V2. Only AM personnel are authorized to open or close runway operations. Refer to AFMAN 13-204V1 for permanent closure and activation of runways.

2.6. Suspending and Resuming Runway Operations. Tower, SOF, or AM will suspend operations to a runway when there is reason to believe a hazard exists on or near the runway or in the immediate approach area. After suspending a runway, AM personnel will contact SOF/Tower WS and describe the type and location of the FOD and an estimated time of correction. See paragraphs 8.3. and 8.4.1.6. for additional runway suspension procedures.

2.6.1. Only AM personnel are authorized to resume runway operations. 54 FG/CC or 54 FG Brickholder can temporarily resume a suspended runway for a declared In-flight Emergency (IFE). Runway will be suspended to facilitate AM checks and procedures once IFE is terminated until AM declares runway operations resumed.

2.6.2. Anytime HMN anticipates or reverts to only one available runway or no runway operations, Tower will broadcast this information on Ultra High Frequency (UHF) 243.0/Very High Frequency (VHF) 121.5 Emergency Frequency with landing runway, alternate airport (if applicable), and estimated time of runway closure. Tower will broadcast on UHF/VHF Emergency Frequency when runways have reopened, and operations are resumed. See paragraph 8.16. for divert procedures.

2.7. Airfield Lighting Systems.

2.7.1. Lighting systems available:

2.7.1.1. Precision Approach Path Indicators (PAPI) - Runways 16, 34, 22, and 25.

2.7.1.2. High Intensity Runway Lights (HIRL) - all runways.

2.7.1.3. Runway Distance Remaining Markers (DRM) - all runways.

2.7.1.4. Approach Lighting System with Sequenced Flashing Lights (SFL) in Instrument Landing System (ILS) Category-I configuration (ALSF-1) - Runways 16 and 22.

2.7.1.5. Runway End Lights / Gated Threshold Lighting – all runways.

2.7.1.6. Rotating beacon.

2.7.1.7. Taxiway lights.

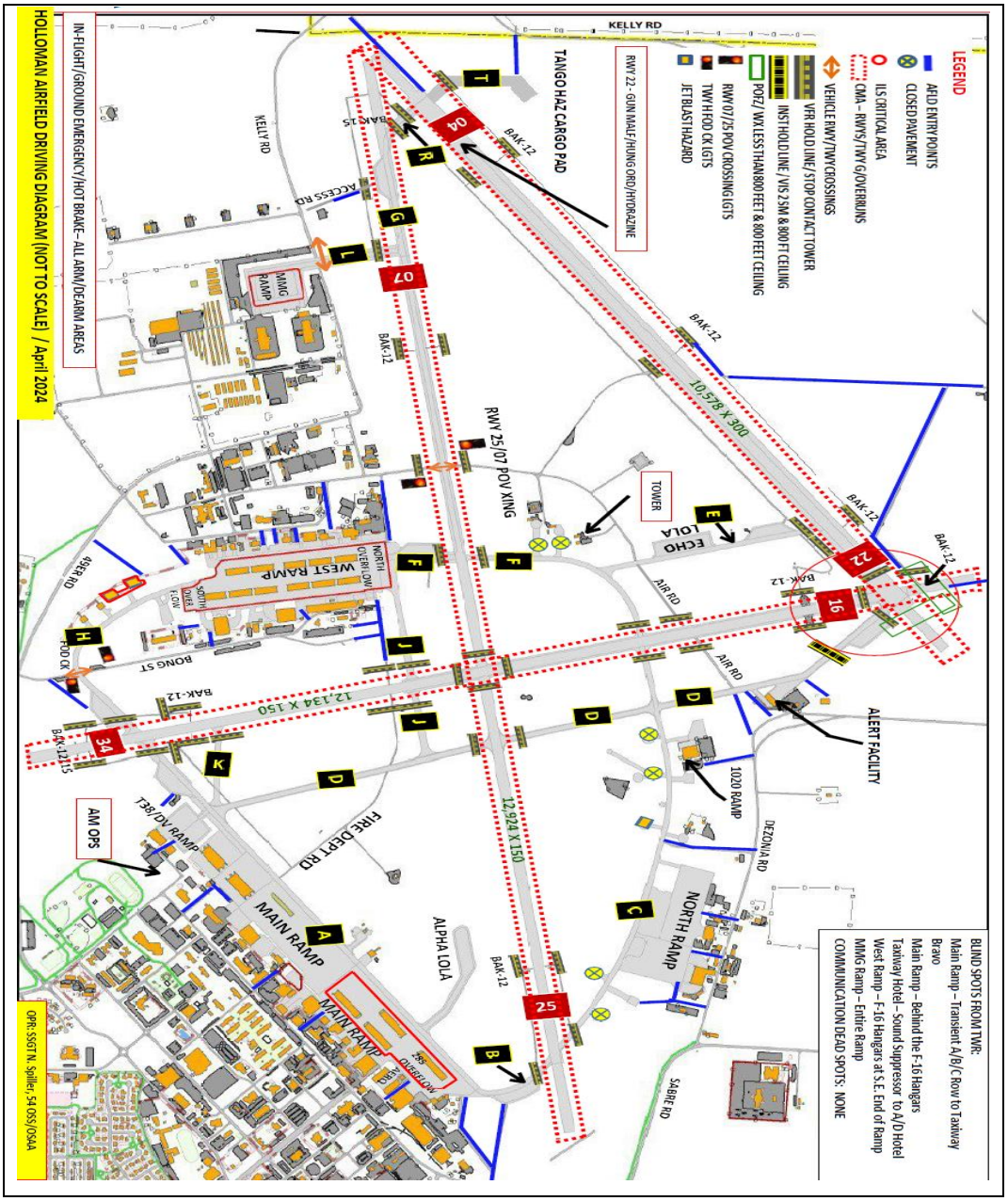
2.7.1.8. Runway Guard Lights.

2.7.2. 49th Civil Engineer Squadron - Exterior Electric (49 CES/CEOFE) performs preventive maintenance inspection(s) (PMI) on the airfield lighting system at weekly, monthly, and 90-day intervals. PMIs should be scheduled around local flying and must be coordinated with Tower WS. Any problems found with the system will be reported to AM immediately.

2.7.3. AM will conduct a daily airfield lighting serviceability check IAW AFMAN 13- 204V2. Outages will be reported to 49 CES/CEOFE daily and documented on the airfield lighting diagram. Outages that adversely affect flying will be reported to Tower and a Notice to Airmen (NOTAM) will be issued. Refer to Air Force Flight Standards Agency (AFFSA) - Airfield Lighting Chart for fixed-wing aircraft restrictions during lighting system outages.

2.7.4. If an emergency requires airfield lighting to be turned on when Tower is unmanned, contact 49 CES Fire and Emergency Services Flight (49 CES/CEF) - Fire Department (FD) personnel in the Emergency Communication Center (ECC) to recall airfield lighting personnel.

Figure 2.1. Airfield Diagram (Drawing not to scale).



2.8. Aircraft Arresting Systems (AAS). HMN has eleven AAS (two Barrier Arresting Kit (BAK)-15s and nine BAK-12s) installed to support flying operations. For standardized radio/telephone coordination between local aircrew, Tower, AM, and 49 CES/CEOFP - Power Production Barrier Maintenance (BM), the AAS are designated in Table 2.2. regardless of the runway in use (also see Figure 2.2.).

Table 2.2. AAS Designations and Locations.

Runway (RY) 16/34 AAS from North to South are located and designated:						
RY 16 →	← 12,197 ft Overrun **"*North BAK-12*"** Underrun -63 ft →	← 10,634 ft **"*RY 16 BAK-12*"** 1,500 ft →	← 1,507 ft **"*RY 34 BAK-12*"** 10,627 ft →	← -59 ft Underrun **"*South BAK-12*"** Overrun 12,193 ft →	← -121 ft Underrun **"*South BAK-15*"** Overrun 12,255 ft →	← RY 34
Runway 07/25 AAS from West to East are located and designated:						
RY 07→	← 15,200 ft Overrun ***"West BAK-15"*** Underrun -2,276 ft →		← 11,424 ft ***"RY 07 BAK-12"*** 1,500 ft →	← 1,630 ft ***"RY 25 BAK-12"*** 11,294 ft →		← RY 25
Runway 04/22 from West to East are designated:						
RY 04→	← 9,128 ft ***"RY 04 BAK-12"*** 1,450 ft →		← 5,287 ft ***"Midfield BAK-12"*** 5,291 ft →	← 1,498 ft ***"RY 22 BAK-12"*** 9,080 ft →		← RY 22

2.8.1. The locally developed AAS designations annotated in Table 2.2. will not be used when issuing AAS information to transient aircrew or crash phone activations. Instead, when communicating the location of an AAS on the active runway, controllers/dispatchers will use the active runway designation and refer to the system in question by approach, midfield, or departure end cable, or approach or departure end barrier. In this case, it is important to differentiate between cables and barriers. BAK-12 systems are cable systems, not barriers. BAK-15 systems are barriers. Standard AAS terminology may also be used between local aircrew, Tower, AM, and BM when deemed necessary for clarification.

2.8.2. The standard AAS configuration and status is managed and monitored by Tower, AM, and BM. Contact Tower to coordinate configuration changes. Any changes to standard configuration will be published in a NOTAM or broadcast in the Automatic Terminal Information Service (ATIS) message. Standard AAS configuration:

2.8.2.1. Runway 16 in use for arrivals/departures:
-North BAK-12 on the underrun.
-RY 16 BAK-12 removed, off runway.
-RY 34 BAK-12 on runway.
-South BAK-12 on overrun and connected to South BAK-15.
-South BAK-15 armed (Tower can raise or lower as needed).

2.8.2.2. Runway 34 in use for arrivals/departures:
-South BAK-15 lowered & rendered safe (Tower cannot raise).
-South BAK-12 disconnected from South BAK-15 but still on the underrun.
-RY 34 BAK-12 removed, off runway.
-RY 16 BAK-12 on runway.
-North BAK-12 on overrun.

2.8.2.3. Runway 07 in use for departures/arrivals:
-West BAK-15 lowered & rendered safe (Tower cannot raise).
-RY 07 BAK-12 removed, off runway.
-RY 25 BAK-12 on runway.

2.8.2.4. Runway 25 in use for arrivals/departures:
-RY 25 BAK-12 removed, off runway.
-RY 07 BAK-12 on runway.
-West BAK-15 armed (Tower can raise or lower as needed).

2.8.2.5. Runway 04 in use for arrivals/departures:
-RY 04 BAK-12 removed, off runway.
-Midfield BAK-12 removed, off runway.
-RY 22 BAK-12 on runway.

2.8.2.6. Runway 22 in use for arrivals/departures:
-RY 22 BAK-12 removed, off runway.
-Midfield BAK-12 removed, off runway.
-RY 04 BAK-12 on runway.

2.8.2.7. The AAS on inactive runways will be configured IAW current and forecast weather conditions to allow for short-notice runway changes and emergency AAS use. During scheduled tailhook-equipped aircraft flying operations, Tower and the SOF will ensure at least one runway that is compatible with aircraft crosswind and tailwind limitations is configured for an approach-end arrestment. This includes configuring the approach end of both the active and multiple inactive runways if weather conditions necessitate.

2.8.3. Special AAS Configuration. Special configurations include AAS outages or non-standard AAS connections/removal (i.e., Midfield BAK-12 connected for unmanned QF-16 drone operations, Runway 25 BAK-12 connected because runway 04/22 closed, manually raised BAK- 15 due to remote control system malfunction, BAK-12 removed for runway rubber removal, etc.). Any non-standard AAS configuration will be issued to aircrew by AM via NOTAM system and by Tower via ATIS.

2.8.4. AAS Coordination Terms:

2.8.4.1. The terms "*in-service*" and "*out-of-service*" will be used when reporting AAS operational status.

2.8.4.2. The terms "*re-configure*" and "*configured*" will be used during runway changes and reporting AAS configuration (see 2.8.2).

2.8.4.3. The terms "*remove*" and "*connect*" will be used when disconnecting AAS or attaching AAS to the runways. Requests to remove or connect AAS must include a reason. Additionally, the past tense of these terms is used to inform local aircrew of a cable's position (e.g., *FLINT 21 runway two five BAK-twelve removed*).

2.8.4.4. The terms "*raise*" and "*lower*" will be used during operation of the BAK-15 systems. Additionally, the term "*safe*" will be used when a BAK-15 net is lowered and cannot be raised remotely by the Tower. The term "*armed*" will be used when Tower has remote control of a BAK- 15 and can raise or lower the net from the Tower cab.

2.8.5. AAS Configuration Coordination. AAS configurations for changes to the runway in use, aircraft incompatibility (e.g., unmanned MQ-9), aircrew request, etc. will be coordinated in the following manner:

2.8.5.1. Tower will initiate the request with BM via Land Mobile Radio (LMR) Tower Net by requesting the runway reconfiguration or by specifying the AAS to remove, connect, safe, or arm.

2.8.5.2. Immediately following BM acknowledgement; Tower will relay the AAS configuration change request to AM via landline.

2.8.5.3. Tower will temporarily suspend runway operations when BM receives approval to enter a runway for AAS configuration. Tower will relay runway suspension information to AM via landline.

2.8.5.4. Immediately following receiving runway suspension information, AM will respond to the AAS being configured in preparation of AAS inspection IAW AFMAN 13-204V2.

2.8.5.5. Upon completion of configuration, AM will inspect the AAS and runway prior to declaring the runway usable for aircraft operations. If the entire runway was used or several AAS were re-configured, the entire runway will be inspected. If the configuration only involves one AAS, AM will check that system and 200 ft either side of the BAK-12.

2.8.6. AAS Configuration Times.

2.8.6.1. The following times are approximate and subject to change based on traffic conditions, weather, manning constraints, and AAS malfunctions. Tower, AM, and BM must ensure all AAS coordination is initiated in a timely manner. Anytime AAS maintenance, re-configuration, or connection is expected to exceed the times noted below, BM will notify Tower and AM the reason of delay and new estimated time of completion.

2.8.6.2. The Midfield BAK-12 and Runway 25 BAK-12 require 30 minutes prior notice to BM for connection.

2.8.6.3. Time intervals between successive AAS engagements (AAS inspection and reset):

2.8.6.3.1. BAK-12 cable: 30 minutes.

2.8.6.3.2. Runway 16 BAK-15 net (departure end only): 4 hours.

2.8.6.3.3. Runway 25 BAK-15 net (departure end only): 8 hours.

2.8.6.4. The following runway AAS re-configurations times are based on the time BM enters the runway until AM accomplishes a runway inspection and declares the runway useable:

2.8.6.4.1. RY 16 reconfigure for RY 34 arrivals (and vice versa) – 30 minutes.

2.8.6.4.2. RY 04 reconfigure for RY 22 arrivals (and vice versa) – 30 minutes.

2.8.6.4.3. RY 07 reconfigure for RY 25 arrivals (and vice versa) – 25 minutes.

2.8.7. AAS Maintenance Coordination.

2.8.7.1. During airfield operating hours, BM will coordinate all AAS maintenance that requires runway access or AAS outage with AM. AM will consider scheduled/proposed flying activities before approving requested AAS maintenance. AM will notify Tower and affected flying units (as necessary) of proposed AAS maintenance.

2.8.7.2. During airfield closed hours, AAS maintenance must be coordinated with AM in advance. AM will ensure the AAS and/or runway is inspected prior to opening the airfield.

2.8.7.3. AAS malfunctions (damaged cable tie-downs, slack cables, inoperative remote barrier control, etc.) information will be exchanged amongst Tower, AM, and BM in a timely manner.

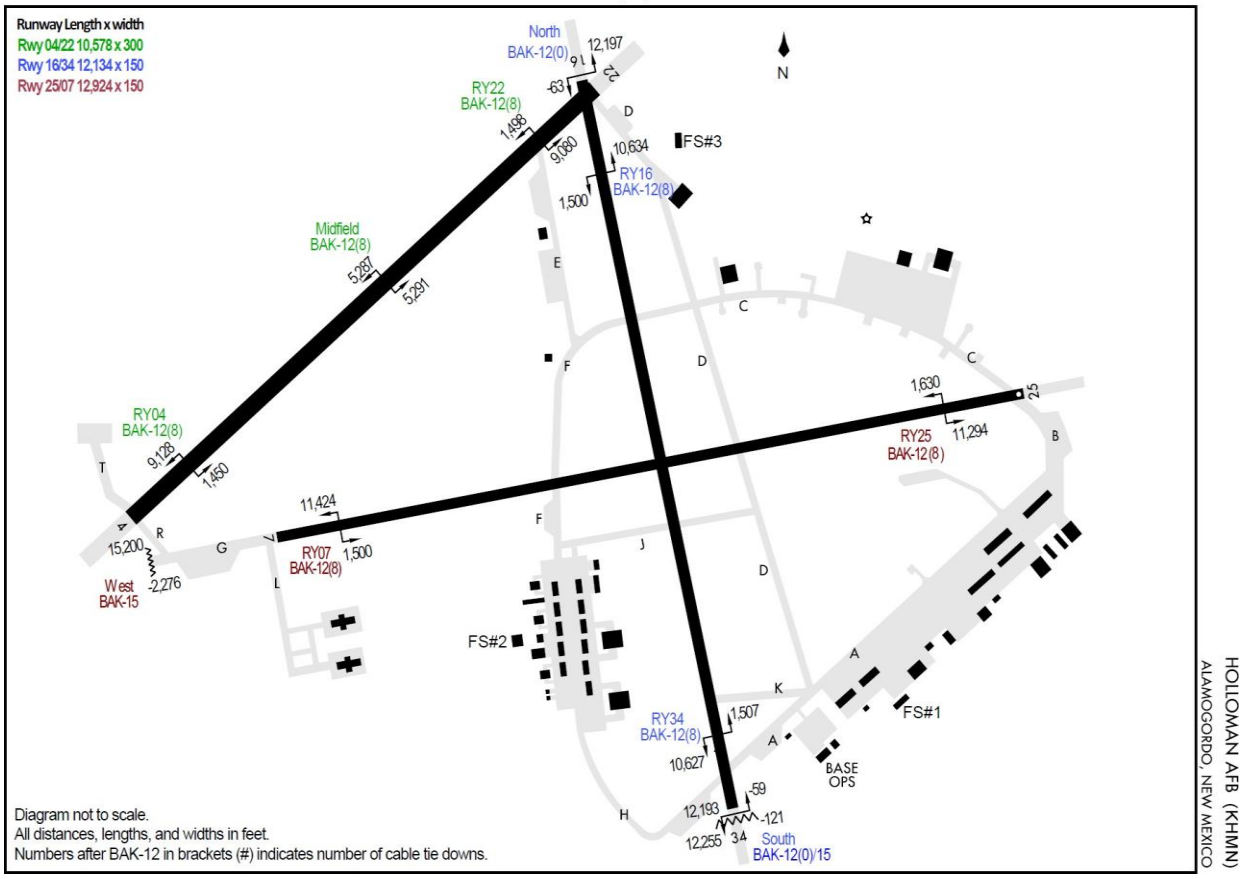
2.8.8. Exceptions to AAS Use. Tower will normally raise the associated runway BAK-15s when aircraft are landing with known or suspected radio failure or conditions (drag chute/hydraulic/electrical failure, etc.) which usually indicate an arresting system may be needed. BAK-15s pose a potential hazard to MQ-9 aircraft and they are exempt from BAK-15 use.

2.8.9. Table 2.2. lists and Figure 2.2. illustrates available AAS for each runway and the distances from runway thresholds.

2.8.10. AAS training program. Familiarization training on HMN AAS locations, capabilities, and procedures must be coordinated through BM.

2.8.11. See paragraphs 5.6.3., 5.6.4., 8.1., and 8.12. for additional AAS procedures.

Figure 2.2. Arresting System Diagram.



2.9. Permanently Closed or Unusable Portions of the Airfield. Permanently closed and unusable portions of the airfield are depicted on the Airfield Diagram, Figure 2.1. Aircraft operations and parking on closed or unusable areas are prohibited.

2.10. Restricted and Classified Areas. The restricted and classified areas are identified in Holloman AFB (HAFB) Integrated Defense Plan (IDP).

2.11. Dangerous/Hazardous Cargo. Support of aircraft carrying dangerous or hazardous cargo will be executed IAW Defense Explosives Safety Regulation (DESR) 6055.09_AFMAN 91-201, *Explosives Safety Standards* and Attachment 13 Aircraft Carrying Hazardous Cargo.

2.12. Non-Standard Airfield Systems and Configurations. Non-standard airfield systems and configurations are as follows:

2.12.1. Airfield Markings: Centerline marking located in Runway 16 underrun F-16 for Barrier Engagement guide and to avoid hooking flushed approach lighting.

2.12.2. Camera box container and aircraft hold marking on Taxiway Foxtrot and Hotel EOR.

2.12.3. Required airfield signage is incorrectly installed and/or missing throughout airfield.

2.12.4. Grounding cables (approximately 600 ft in length) are installed at three locations for munitions related emergency response (see paragraphs 8.8. and 8.9.).

2.12.4.1. Runway 04/22 grounding cable is laid parallel to the runway in the south infield between the Runway 04 aim point markers and the Runway 04 approach end BAK-12.

2.12.4.2. Taxiway Golf grounding cable is laid parallel to Taxiway Golf in the south infield between the Taxiway Lima intersection and the TACAN Checkpoint.

2.12.4.3. Taxiway Hotel EOR grounding cable is laid parallel to the EOR in the south infield.

2.13. Airfield Waivers. Permanent and temporary airfield waivers are maintained by 49 CES Engineering Flight Portfolio Optimization (CENP). Temporary airfield construction waivers are maintained by AM.

2.14. Airfield Maintenance. AM will coordinate with 49 CES to maintain airfield vegetation growth (to include grass mowing and brush trimming). Daily coordination with 49 CES is required to ensure sweeper is available during airfield operating hours, IAW AFMAN 13-204V2 and Airfield Sweeper Operations Letter of Procedure (LOP).

2.15. Airfield Snow Removal Operations. The 49 CES Commander (49 CES/CC) has determined there is no requirement for a HMN snow and ice removal plan IAW Air Force Instruction (AFI) 32-1001, Chapter 13 *Snow and Ice Control Program*.

2.16. Airfield Photography. Airfield photography must comply with HAFB IDP. Aerial photography (photogrammetric flights, aerial surveys, news media, etc.) operations conducted by non-military organizations within HMN Class D airspace must be coordinated through 54 OSS/OSA at least 72 hours in advance.

2.17. Contractors Working on the Airfield. All contractors will notify AM at Defense Switched Network (DSN) 572-5411 or commercial (575) 572-5410 prior to initiating work on the airfield. Additionally, procedures will be followed IAW Department of the Air Force Instruction (DAFI) 13-213, *Airfield Driving*, and the project's Statement of Work. See paragraph 2.10 for restricted and controlled area locations and credentials required.

2.18. Wear of Hats on Airfield. Only seasonally approved cold weather type hats are authorized on the airfield. Seasonally approved hats are not to be worn within danger areas of operating jet engines IAW HAFBI 21-370, *Foreign Object Debris/Damage Prevention Program,* and DAFMAN 91-203, *Air Force Occupational Safety Fire and Health Standards*.

2.19. Smoking on Airfield. Smoking on the airfield is approved in designated areas only.

2.20. Airfield Operating Hours. Airfield hours are established IAW AFMAN 13-204V1 and are subject to change based on seasonal conditions, test mission support, or 49 WG mission needs. Check NOTAMs for actual airfield and ATC operating hours. RAPCON and HMN operating hours are not always coincident. HMN airfield opening outside of published hours must be

approved by the 54 FG/CC or 54 FG Brickholder.

2.20.1. 49 WG/CP is the point of contact for initiating short-notice emergency procedures to open the airfield outside normal operating hours (DSN 572-4900, commercial 575-572-4900). 49 WG/CP will request authorization from 54 FG/CC or 54 FG Brickholder. 49 WG/CP will then notify 54 OSS/OSA and 54 OSS/OSW IAW the airfield standby roster and additional agencies as required. 54 OSS/OSA and 54 OSS/OSW will ensure 49 WG/CP has a current and updated standby roster. The response time for 54 OSS/OSA personnel to open facilities is 1 hour.

2.21. Uncontrolled Airfield Operations (UAO). See Attachment 10, UAO. Attachment 10 outlines flying procedures for 49th Force Support Squadron Aero Club (49 FSS/AC), 704th Test Group (704 TG), Army Air (AA), and authorized TDY units to operate during periods when the airfield is closed.

2.22. Local Frequencies and Channelization. Local frequencies are listed in Attachment 2. Local channelization numbers listed in Table A2.1. may be used in lieu of frequencies.

2.23. Radar, Airfield, and Weather Systems (RAWS).

2.23.1. Navigational Aids (NAVAIDs) available at HMN:

2.23.1.1. TACAN - Runways 16, 22, and 34.

2.23.1.2. ILS - Runways 16 and 22.

2.23.1.3. Airport Rotating Beacon is atop the red-and-white water tower, located at the north-east corner of the airfield.

2.23.2. Tower is designated as the primary NAVAID monitoring facility. When open, Tower continuously monitors the ILS and TACAN for proper operation. Pilots should report any weak or abnormal signals to ATC as soon as practical.

2.23.3. NAVAID PMI schedule is published in the DoD FLIP *IFR Supplement US*.

2.23.4. The following ATC and Landing Systems (ATCALS) have auxiliary backup power generators with auto-start capabilities:

2.23.4.1. Control Tower.

2.23.4.2. WSRF.

2.23.4.3. AM.

2.23.4.4. Runways 16 and 22 ILS.

2.23.4.5. TACAN.

2.23.4.6. Airfield Lighting Systems.

2.23.4.7. HMN Digital Airport Surveillance Radar (DASR).

2.23.5. Civil Use of Military ATCALS. Civilian (non-49 FSS/AC) aircraft may conduct instrument approaches at HMN on a non-interference basis.

2.23.6. TACAN ground receiver checkpoints on airfield:

2.23.6.1. Taxiway Bravo located just south of Runway 25 hold line.

2.23.6.2. Taxiway Charlie located just north of Runway 25 hold line.

2.23.6.3. Alpha and Hotel EOR areas and Taxiway Romeo.

2.24. ATIS Procedures. ATIS must be operational at least 30 minutes prior to the first scheduled departure until the termination of local scheduled flying.

2.25. NOTAM Procedures. Tower is the NOTAM monitor facility for HMN. AM is the NOTAM authority for HMN and disseminates NOTAMs IAW AFI 11-208 Interservice Publication (IP), *DoD Notice to Airmen (NOTAM) System*. AETC/Airspace Ranges, and Airfield Operations (A3AO) Terminal Instrument Procedures (TERPS) or 54 OSS/OSA TERPS will issue "V" series NOTAMs concerning published instrument flight procedures.

2.26. FLIP Accounts. HMN flying units are responsible for creating and maintaining their own FLIP accounts. AM will provide 54 OSS flights a publication disk as they are issued, when required. Request FLIP changes through FLIP Manager or the non-commissioned officer in charge (NCOIC) AM Operations (NAMO). NAMO prepares and coordinates DoD FLIP changes IAW DoD FLIP *General Planning* - Chapter 11. All procedural DoD FLIP changes are coordinated through HAM, TERPS, or AM.

2.27. Weather Dissemination and Coordination Procedures.

2.27.1. ATC will issue variable wind information when there is a wind direction change of 60 degrees or more in the preceding 2 minutes and when the wind speed is greater than 6 knots. When present, ATC will issue wind gusts to all aircraft.

2.27.2. AM will disseminate hazardous, severe weather, and lightning information via the Secondary Crash Net (SCN).

2.27.3. Lightning procedures.

2.27.3.1. Lightning Watch. Informative only and will be issued when lightning is forecast to be within 5 NM radius of the airfield in the next 30 minutes.

2.27.3.2. Lightning Warning. Issued when lightning is observed within 5 NM of HMN

airfield. When a Lightning Warning is issued, the following will be accomplished:

2.27.3.2.1. Cease refueling and all airfield activity. Evacuate personnel from the airfield to indoor cover or within an airfield vehicle as a minimum.

2.27.3.2.2. Aircraft WILL NOT be armed or de-armed. In the chocks, clear the crew chief off and remain running or shut down and clear the airfield. The intent is to expeditiously get the crew chief off the airfield and to safety. If taxiing, contact SOF and expect guidance to either continue taxiing, return to arm/de-arm area (primary), or ramp (secondary).

2.27.3.2.3. Aircraft WILL NOT takeoff, land or fly approaches at HMN during periods of lightning within 5 NM of the airfield area. If airborne, aircraft will hold until reaching divert fuel and will coordinate their next actions with the SOF. If an aircraft is required to land during periods of lightning within 5, the 54 FG/CC or 54 FG Brickholder is the approval authority (49 OG/CC is the approval authority for MQ-9s). This guidance does not preclude the SOF from making time-sensitive safety of flight decisions based on his/her situational awareness and Operational Risk Management (ORM) assessment. See paragraph 8.16. for divert procedures.

2.27.4. See HAFBI 15-101, *Weather Support Procedures*, for additional responsibilities and services.

2.28. Transient Alert. Transient Alert is available beginning at 0530L on Monday through 2300L Friday and from 0800L-1500L on Saturday unless otherwise prearranged. Services and facilities available to support transient aircraft are published in the DoD FLIP *IFR Supplement US* along with the current airport hours of operation.

2.29. Aircraft Fuel Servicing Priorities. Table 2.3. will be adhered to during simultaneous fuel support requests. AM will determine priority for special mission fuel requirements and notify 49th Logistics Readiness Squadron Fuels Service Center (49 LRS/LGRF) for appropriate response.

Table 2.3. Fuel Servicing Priority.

Priority	Aircraft Mission / Activity
1	- Medical Evacuation (MEDEVAC) or Air Evacuation (AIREVAC) - Alert
2	- President of the US - Special Air Mission (SAM) or Special Assignment Airlift Mission (SAAM) - AA
3	- 704 TG
4	- 49 WG
5	- Detachment 1, 82nd Aerial Target Squadron (82 ATRS)
6	- Transient

2.30. Local Aircraft Priorities. ATC will provide highest priority to emergency aircraft then to other missions identified in Federal Aviation Administration (FAA) Order (FAAO) Joint Order

(JO) 7110.65, *Air Traffic Control.* Aircraft established on final inside 5 NM normally are not broken out for higher priorities.

2.30.1. Local operational priority will be as follows:

2.30.1.1. FAA Flight Check aircraft actively engaged in flight inspection.

2.30.1.2. Minimum fuel aircraft.

2.30.1.3. Weather recall aircraft.

2.30.1.4. Active flight test mission support (approaching controlled mission time). Aircrew will use the phrase "*A-T Mission*" with ATC when declaring this priority.

2.30.1.5. Distinguished Visitor (DV) aircraft.

2.30.1.6. Full stop landings requested on initial call-up to Tower.

2.30.1.7. Departures.

2.30.1.8. Locally assigned aircraft requesting touch-and-go/low approach landings IAW paragraph 2.30.2.

2.30.1.9. Transient aircraft conducting practice approaches. During periods of high traffic, transient aircraft should expect one approach to a full stop landing.

2.30.1.10. 49 FSS/AC aircraft.

2.30.1.11. Civilian Aircraft Operations. USAF Auxiliary - Civil Air Patrol (CAP) aircraft may execute practice approaches on a non-interference basis. Civilian (non- 49 FSS/AC and non-CAP) aircraft may execute a low approach only on a non-interference basis.

2.30.2. During periods of conflicting priorities (i.e., departures vs. practice patterns), the SOF will determine aircraft priority, based on knowledge of the flying schedule and unique circumstances of individual unit missions. Ops Sups can assist SOF in priority coordination by relaying timely mission information. Priority will be as follows if multiple aircraft are requesting pattern operations:

2.30.2.1. Check ride or certification aircraft requiring instrument or VFR pattern work. Check ride aircraft will append "91" to their call sign (e.g., BOBA 91).

2.30.2.2. Upgrade training aircraft requiring a certain amount of pattern work (e.g., F-16 Training Ride, MQ-9 Launch and Recovery (LR) training).

2.30.2.3. Aircrew needing patterns for currency.

2.31. Noise Abatement and Quiet Hour Procedures.

2.31.1. The term "quiet hours" is used to denote a period of reduced airfield noise levels at HMN. AM will normally publish a NOTAM for quiet hours 72 hours prior to the event.

2.31.1.1. The 54 FG/CC or 54 FG Brickholder must approve all takeoffs, landings, and multiple approaches (other than scheduled local flying, exercises, and approved Attachment 10 UAO) during quiet hours.

2.31.1.2. See paragraph 5.1. for Engine Test & Run Procedures.

2.31.2. Airfield quiet hours for special events (DV's, ceremonies, etc.) are coordinated through Scheduling and are subject to the approval of the 54 FG/CC or 54 FG Brickholder. The 54 FG/CC or 54 FG Brickholder will specify aircraft taxi, takeoff, landing, and MQ-9 restrictions during special event quiet hours. See HAFBI 11-101 *Squadron Programming/Airspace/Range Scheduling* paragraph 4.5. for special event quiet hours coordination procedures.

2.31.3. See paragraph 3.4.1. Avoidance Areas, paragraph 3.6.9. Supersonic Airspace, and paragraph 6.13.1.4. Nighttime Overhead for additional noise abatement procedures.

Chapter 3

AIRSPACE AND RANGE OPERATIONS

3.1. Airspace Letter of Agreement (LOA), Certificate of Authorization (COA), Scheduling, and Figures.

3.1.1. Contact HAM, 54 OSS/OSA, or Scheduling for copies of any airfield or airspace associated LOAs.

3.1.2. HAM maintains and updates airspace COAs that authorize MQ-9s to operate within the National Airspace System (NAS) outside of restricted airspace. MQ-9 units will inform Scheduling when planning to operate in Wiley East ATC Assigned Airspace (ATCAA). Scheduling will coordinate with the FAA IAW established agreements. Unit schedulers will advise Scheduling no later than (NLT) 2 duty days prior to execution to allow time for proper coordination. Contact the HAM for copies of the COAs.

3.1.3. Airspace and Range Scheduling (see Table 3.2.). Local scheduling operations will be conducted IAW HAFBI 11-101, *Squadron Programming/Airspace/Range Scheduling*. All locally assigned aircrew will review the daily airspace schedule and restrictions on Center Scheduling Enterprise (CSE) (https://cseaf.eglin.af.mil/cse/Home.aspx) prior to conducting local area flight operations.

3.1.4. Depictions of local airspace and ranges are available in the 49 WG In-Flight Guide (IFG). Contact the HAM, 54 OSS/OSA, 54 FG/FGV, or 49 OG/OGV for copies the 49 WG IFG or specific airspace depictions.

3.2. Range Operations. Local range operations will be IAW AFI 13-212, HAFB ADDENDUM-A, *Holloman Primary Training Ranges*, and when inside White Sands Missile Range (WSMR) airspace, WSMR Flight Safety Operational Plan (FSOP) governs the specific flight mission.

3.3. Joint Test and Training Operations Cell (JTTOC) Procedures. The JTTOC is in the WSMR Operations Control Division to facilitate dynamic allocation of test and training airspace to maximize both. The JTTOC will communicate over HMN WAVE radio network using the call sign “BADGER”.

3.3.1. Pilot Procedures.

3.3.1.1. Airspace and range extensions/changes prior to departure are coordinated with JTTOC through your squadron Ops Sup. JTTOC will relay changes to WSRF.

3.3.1.2. Airspace and range extensions/changes while airborne are coordinated directly with BADGER on 315.9. Airborne airspace extensions will not be accepted within 5 minutes of scheduled airspace end time. An airspace extension does not imply an impact range extension. Requestor must ensure they request an impact range extension if one is needed. BADGER will attempt to coordinate extensions and/or, if needed, verify aircrew acceptance of Military Authority Assumes Responsibility for Separation of Aircraft (MARSA)

between flights.

3.3.1.2.1. Climbs above 10,000 ft MSL may be required to establish contact in Lava/Mesa airspace or if redundant radio sites have failed.

3.3.1.3. If two-way communication with BADGER cannot be established or if BADGER is unmanned, make your airborne airspace requests with HMN Approach (UHF 269.225 or 336.2).

3.3.2. Ops Sup Procedures.

3.3.2.1. If JTTOC is unmanned, make day-of airspace requests via Scheduling, DSN 572-3536. The JTTOC will positively notify Ops Sups via TransVerse® or voice when the position status (manned or unmanned) changes.

3.3.2.2. Notify aircrew at step of the JTTOC operating window.

3.3.2.3. Keep the JTTOC informed of significant changes to the flying schedule.

3.3.2.4. For other requests/assistance, contact the JTTOC at DSN 258-6520.

3.3.2.5. Scheduling is the point of contact for next-day and beyond requests, regardless of JTTOC status.

3.3.3. Scheduling Procedures.

3.3.3.1. Scheduling will publish the JTTOC operating window on SharePoint®.

3.3.3.2. When the JTTOC is manned, refer all day-of airspace change requests to the JTTOC.

3.4. Local Flying Area. HMN local flying area is defined as a 200 NM radius from the HMN TACAN, excluding Mexico's airspace (see 49 WG IFG Local Airspace Overview).

3.4.1. Avoidance Areas: (see 49 WG IFG Avoidance Area Map).

3.4.1.1. When HMN High Speed Test Track (HSTT) is in use, comply with CSE restrictions or as directed by WSRF. See paragraph 3.10. for HSTT coordination procedures.

3.4.1.2. Explosive Ordnance Disposal (EOD) Demolition Range. When HMN EOD demolition range (HMN R-318/1.5 DME or N32 52.9676 W106 07.4715) is in use, comply with NOTAM or restrictions issued by Tower. See Attachment 4 for additional EOD procedures.

3.4.1.3. Maintain vigilance near Mesa Verde Ranch Strip (FAA identifier 7NM1, HMN R- 027/5.4 DME), N32 56.117, W106 02.717, or roughly 5 final Runway 22.

3.4.1.4. Veterans Memorial Park (HMN R-049/7 DME) is occasionally used on weekends and holidays for model rocket and high-power rocket launch activities up to 17,000 ft MSL. When in use, avoid the park by 2 NM radius or as instructed by ATC. See Tripoli NM-White Sands-FAA Waiver for additional procedures.

3.4.1.5. White Sands National Park (WSNP) Visitor Center (HMN R-203/6 DME), N32 46.78 W106 10.35. Avoid the WSNP Visitor Center by at least 2,000 ft AGL or 1 NM (see 49 WG IFG, HMN R-203/6 DME, N32 46.78 W106 10.35).

3.4.1.6. Avoid overflight of Apache Point Observatory (HMN R-096/15 DME), N32 46.8133, W105 49.1850, when active during recurring lunar laser research activity. Laser activity is hazardous at an angle of 90 degrees from the surface to 60,000 ft MSL. When active, avoid Sunspot by 5 NM. Check Albuquerque (ABQ) Air Route Traffic Control Center (ARTCC) NOTAMs for Apache Point laser research schedule and restrictions.

3.4.1.7. Avoid overflight of Alamogordo White Sands Regional Airport (KALM) by 2,000 ft AGL or 2 NM (N32 50.4766 W105 59.3316). Use caution as parachute landings can occur at the White Sands Landing Zone (N32 52.9116 W105 58.6750) seven days a week from sunrise to sunset. Maximum altitude of jumps will be 14,000 ft MSL. Alternate drop zone is located at Griggs Field (N32 55.7016 W105 57.0133) in Alamogordo.

3.4.1.8. Remain at or above 3,000 ft AGL within 1.5 NM of the towns of Cloudcroft (N32 57.4 W105 44.4) and High Rolls (N32 56.9 W105 49.9). This restriction is not applicable to helicopters.

3.4.1.9. DO NOT fly within 25 NM of El Paso (ELP) VHF Omni-Directional Radio Range/Tactical Air Navigation (VORTAC) (Channel 99) within Class E airspace without contacting ELP Approach.

3.4.1.10. Aerial Cable. Crews must stay above 9,000 ft MSL within 1 NM of the aerial cable. This 2.5 inch thick, 15,000 ft cable spans between Little Burro Mountains (southwest end of cable - N33 36.4650 W106 24.8883 [HMN R-340/47.85 DME]) and Jim Peak (northeast end of cable - N33 37.3416 W106 22.0650 [HMN R-345/47.9 DME]) and presents a hazard to low flying aircraft. The site is located near the border of Lava East and Lava West at the northwest corner of Oscura (see 49 WG IFG.).

3.4.1.11. Chaff and Flare Restrictions. See 49 WG IFG Chaff Area and Flare Area diagrams. Additional fire danger information is available at US Forest Service – Lincoln National Forest website https://www.fs.usda.gov/lincoln.

3.4.1.11.1. During all fire danger levels except for “Very High” and “Extreme”, in WSMR or R- 5103B/C, flares may be dropped from a minimum altitude of 2,000 ft AGL. Within designated impact areas of Red Rio, Oscura, and Yonder, aircraft are authorized to drop flares from a minimum altitude of 500 ft AGL.

3.4.1.11.2. In other Military Operations Areas (MOA) and ATCAA east of HMN (Wiley East, Beaks, Pecos, and Talon), flares WILL NOT be deployed during periods

of “Very High” or “Extreme” fire danger. During periods of “High” fire danger, aircraft WILL NOT expend flares below 18,000 ft MSL. If fire danger is less than “High” then minimum altitude for flare release is 2,000 ft AGL.

3.4.1.11.3. Aircraft are authorized to drop chaff in all WSMR airspace areas subject to the limitations of the current chaff permit. The permit is available for review at https://cs3.eis.af.mil/sites/OO-OP-AC-30/Holloman/SitePages/Home.aspx. All chaff use must be scheduled through Scheduling on the CSE.

3.4.1.12. Terminal High Altitude Area Defense (THAAD) anti-ballistic missile system. Minimum Operating Altitudes over THAAD when active: Current restrictions require manned aircraft to maintain a minimum altitude of approximately 18,000 ft AGL (23,000 ft MSL) while operating in the vicinity of the THAAD radar sites in Yonder South and 5103B airspace (see 49 WG IFG). This minimum altitude is based upon aircraft carrying electro-explosive devices (EED). Presently, HMN MQ-9s do not carry any type of live weapons or devices activated by an electrical charge. All non-EED aircraft must maintain a minimum altitude 8,000 ft AGL (13,000 ft MSL) while operating in the vicinity of THAAD radar sites.

3.4.1.13. Radar Advanced Measurement System (RAMS). The RAMS test facility located on WSMR is used to conduct testing that is sensitive in nature. Due to the capabilities of some Full- Motion Video (FMV) equipped aircraft, careful attention must be given to ensure imagery of these tests are not inadvertently collected or to be used for illicit purposes.

3.4.1.13.1. An HMN/WSMR “No Look” area has been established for aircraft that have the capability to view/record/transmit video or imagery. This area lies within the boundaries of Yonder airspace and is centered at N33 11.1666 W106 33.85.

3.4.1.13.2. FMV-equipped aircraft including but not limited to MQ-9s, aircraft with targeting pods or FMV sensor suites, or aircraft having the capability to record ground imagery of any kind will not use those sensors while in proximity of the HMN/WSMR "No-Look" area. When active (see CSE restrictions page), crews must stay above 9,600 ft MSL within 2 NM of the HMN/WSMR “No-Look” area.

3.4.1.13.3. Aircraft falling into the categories named above will have their sensors either turned off or trained to areas outside of the HMN/WSMR "No-Look" area. Any aircraft in violation of this directive is subject to a security investigation by the 54 FG. Any aircraft which believes they have inadvertently violated this directive should immediately contact the 54 FG while quarantining the offending sensor or recording device.

3.4.1.14. WSMR Main Post. MQ-9 flights over WSMR Main Post (N32 22.7593 W106 29.1241) are not authorized at any altitude within 1 NM.

3.4.1.15. Global Positioning System (GPS) Jamming. GPS interference testing on WSMR

may result in unreliable or unavailable GPS signal. Check CSE or ABQ ARTCC NOTAMs for GPS testing schedule, affected area, and restrictions.

3.5. ATC Airspace.

3.5.1. HMN Class D airspace extends from the surface (SFC) up to and including 6,600 ft MSL (2,500 ft AGL) within a 4.8 NM radius of HMN and within 1 NM each side of the HMN ILS localizer northwest course extending from the 4.8 NM radius to 5.4 NM northwest of the airport excluding that airspace within a 2 NM radius of the KALM (see 49 WG IFG).

3.5.1.1. Unusual Maneuvers in Class D Airspace. Unusual maneuvers are not authorized in HMN Class D airspace except as specifically authorized by local operating procedures. Unusual maneuvers include any intentional maneuver involving an abrupt change in an aircraft's attitude, an abnormal altitude, or abnormal acceleration not necessary for normal flight. All requests must be coordinated through 54 OSS/OSA and approved by 54 FG/CC or 54 FG Brickholder.

3.5.1.2. Small Unmanned Aircraft System (sUAS) in Class D. All sUAS activity within Class D airspace must be coordinated through 54 OSS/OSA 72 hours in advance. Any recreational, commercial, or governmental sUAS policies, procedures, agreements, etc., which task 54 OSS/OSA services (on or off airfield) must be pre-coordinated and approved through AETC/A3AO. See paragraph 3.6.2.2. and Attachment 3 for additional sUAS procedures.

3.5.2. WSRF and Delegated Airspace. WSRF is located at WSMR. WSRF airspace is within the lateral limits described in Annex 1 of the ABQ ARTCC, 49 WG, and WSMR LOA from SFC to 22,000 ft MSL.

3.5.2.1. VFR Training Areas. VFR training areas in HMN airspace are the 49 FSS/AC training area southeast of HMN and AA/49 FSS/AC training area to the northeast (see 49 WG IFG). Use of either VFR training area is coordinated real-time with WSRF and subject to restriction based on HMN runway configuration and traffic conditions.

3.5.2.2. Civilian Flyway. A flyway for light aircraft extends between WSMR and Beak MOA/McGregor Airspace. It extends from JUPTR (HMN R-357/90 DME) through KALM to ELP (Newman VORTAC (EWM), HMN R-174/71 DME).

3.5.2.3. Minimum Vectoring Altitudes (MVA). HMN DASR MVA charts are maintained by and available from 54 OSS/OSA TERPS. WSRF may vector VFR aircraft at any altitude. Vectors do not imply an IFR clearance. The pilot is responsible for terrain and obstruction clearance. When a pilot cannot accept an ATC instruction because of terrain, weather, or obstructions, they must advise ATC and obtain an acceptable instruction.

3.5.2.4. Diverse Vector Area (DVA). Aircraft may be vectored below the MVA while executing a departure or climb-out instructions. DVA diagrams are available from WSRF.

DVA provides separation from obstacles IAW TERPS diverse departure criteria.

3.5.2.4.1. When a published instrument departure conflicts with active restricted airspace, aircrews have the option to request a DVA departure with Clearance Delivery. E.g., "…*ON DEPARTURE CLIMB AND MAINTAIN (ALTITUDE). CROSS DEPARTURE END AT OR BELOW 5,600 FT, LEAVING 6,100 FT TURN RIGHT HEADING 010, REMAIN WITHIN 8 DME OF HMN UNTIL ESTABLISHED ON THE 010 HEADING"* or as directed by ATC.

3.5.2.4.2. ATC must inform pilots of the climb gradient for the applicable departure runway (see Table 3.1.) and obtain pilot concurrence prior to issuing an ATC departure clearance. Non- standard climb-gradient notification is not required for locally assigned aircrew IAW AETC waiver maintained by 54 FG/FGV.

Table 3.1. DVA Non-Standard Climb Gradients.

Departure Runway	Non-Standard Climb Gradient
16	229 ft per NM
34	257 ft per NM
25	264 ft per NM
22	266 ft per NM
04	278 ft per NM

3.6. Special Use Airspace (SUA). HMN local flying airspace utilizes WSMR, multiple MOAs/ATCAAs, and Fort Bliss. Additionally, HMN Working Areas are established within WSMR and Fort Bliss restricted airspace for scheduling and aircraft deconfliction. Local SUA depictions are available in the 49 WG IFG.

3.6.1. AETC units require "exclusive use" in AETC-managed airspace (see DAFMAN 13-201, *Airspace Management*).

3.6.2. HMN lies within R5107D and can be reserved for research & development and training missions. When portions of R5107D are active, IFR/VFR arrivals/departures to/from HMN may be prohibited or restricted.

3.6.2.1. Delta Shelf is defined as R5107D airspace south of WSMR 50 grid line Papa to Tango. "Delta Shelf" may be used during ATC or ATC/pilot communications. E.g., "*unable max climb to one five thousand, Delta Shelf active above one zero thousand, say intentions*". See paragraph 3.6.7.2. for additional Delta Shelf procedures.

3.6.2.2. sUAS in R5107D. All sUAS activity within R5107D must be coordinated/scheduled with WSMR 14 days in advance.

3.6.3. WSMR. Several restricted areas are combined to form WSMR to include R5107B, R5107C, R5107D, R5107E, R5107F, R5107G, R5107H, R5107J, R5109A, R5109B, R5111A, R5111B, R5111C, and R5111D.

3.6.3.1. AA, 704 TG and 704 TG sponsored units, and TDY units will abide by applicable WSMR airspace directives for test aircraft. Where there is a conflict between those directives and this instruction, the WSMR directives take precedence for test aircraft.

3.6.3.2. WSMR Grid has been established for range activity coordination, restriction, and control. Copies of the WSMR grid are available from the HAM. Latitude and Longitude conversions for the grid are listed in Tables 3.4. and 3.5.

3.6.4. Fort Bliss. Several restricted areas are combined to form Fort Bliss, to include R5103A/B/C and R5107A/K.

3.6.4.1. High Altitude Mountainous Environmental Training Strategy (HAMETS). The HAMETS area is a helicopter training area located in the northern section of R-5103C. The area is designated a free fly area and is continuously active. Davis Dome/Bliss Radio provides advisory service to HAMETS aircraft. WSRF must coordinate with Davis Dome/Bliss Radio prior to allowing an aircraft to transit the area. Helicopters will maintain at or below 9,000 ft MSL. When HAMETS is active aircraft transit the HAMETS area at or above 10,000 ft MSL. Units can schedule the HAMETS area through Scheduling if their mission requires access to the airspace. For real time crossing through the area, contact WSRF who will in turn coordinate with Bliss Radio for availability.

3.6.4.2. Wilde Benton Area. This area is designated a free fly area and is always active. Helicopters will maintain at or below 6,500 ft MSL, and when present all other aircraft must transit the Wilde Benton area at or above 7,500 ft MSL.

3.6.4.3. HAMETS area has two ingress and egress routes from Fort Bliss proper, the White Corridor on the eastern boundary of 5103C and the Red Corridor on the western boundary of 5103C. These corridors are active from the SFC to 800 ft AGL and must be avoided by non-HAMETS aircraft except during the times and for the missions that the Centennial low-level entry has been scheduled.

3.6.4.4. The Northern Buffer (NB) is an area defined in the Fort Bliss/Holloman MOA. WSRF may request the use of the NB at or above 8,500' MSL, or as otherwise coordinated with Fort Bliss Airspace Information Center (Bliss AIC). 49 WG aircraft require this airspace during certain departure methods and occasionally during recovery.

3.6.4.4.1. At the start of each shift, WSRF will coordinate a request with Bliss AIC for release of the NB to WSRF. The JTTOC will query WSRF for status of the NB.

3.6.4.4.2. "NB Cold" indicates the airspace is under WSRF control and "NB Hot" indicates the airspace is under Bliss AIC control. Should the status change during flying operations, WSRF will relay the updated status to the JTTOC. The JTTOC will

advise the SOF, the SOF will update the airfield status, and WSRF will advise departing aircraft that are routed through the NB.

3.6.4.4.3. Pilots stepping to aircraft should be aware of the status of the NB prior to departing. Aircrews will not enter the NB without approval from ATC (WSRF) via IFR/VFR Departure clearance or pilot airborne request to transit.

3.6.5. There are five MOAs/ATCAAs controlled by HMN when released by ABQ ARTCC:

3.6.5.1. Beak Alpha, Bravo, and Charlie MOAs/ATCAAs from 12,500 ft MSL to 23,000 ft MSL and Beak Delta ATCAA from 18,000 ft MSL to 40,000 ft MSL or as requested. These are published active from 0600L through 1800L, Monday through Friday and other times by NOTAM. Beak ATCAAs cannot be used concurrently with Wiley East ATCAA.

3.6.5.2. Wiley East ATCAA from 18,000 ft MSL up to 40,000 ft MSL or as scheduled.

3.6.5.3. Valmont ATCAA. The Valmont ATCAA contains the airspace from 18,000 ft MSL to 60,000 ft MSL or as scheduled and overlays the VFR corridor that extends from HMN to El Paso. This ATCAA can be utilized to bridge the gap between R5107B/K and R5103 or commensurate HMN Working Areas above 18,000 ft MSL.

3.6.6. HMN Working Areas. There are areas within WSMR, and Fort Bliss restricted airspace designated for use by HMN scheduled aircraft conducting training flights also known as HMN Working Areas. These include Wiley West, Wiley East, Wiley West Extension, Lava/Mesa, Red Rio, Oscura, Yonder, Yonder East, Yonder South, Dona East, Dona West, McGregor North, McGregor South, and the Centennial Fly Area (CFA). Additionally, the Centennial, Oscura, and Red Rio ranges are available for air-to-surface weapons operations (see HAFB Addendum to AFI 13-212 for details).

3.6.6.1. Entry Procedures: Contact HMN Approach for entry into Wiley West, Wiley East, Wiley West Extension, Lava/Mesa, Red Rio, Oscura, Yonder, and Yonder East, Yonder South. Contact CHEROKEE for entry into McGregor North, McGregor South, and the CFA.

3.6.6.1.1. WSRF will advise aircraft of other known traffic cleared within their assigned airspace.

3.6.6.2. Exit Procedures: Request approval/clearance from WSRF prior to leaving the assigned area. Aircrews will be responsible to maintain their own separation with other participating aircraft until exiting their assigned SUA/ATCAA. If a pilot receives an IFR clearance leaving their airspace, which includes a clearance limit, altitude to maintain and a route of flight, ATC will assume responsibility for separation of that aircraft. See paragraph 3.3. for airspace and range extension/change procedures.

3.6.6.3. NON WSRF Operations.

3.6.6.3.1. When WSRF is non-operational due to building evacuation or loss of

communication, aircraft may conduct operations inside WSMR restricted airspace as scheduled (no deviations) and IAW CSE airspace schedule and restrictions. Aircrews must obtain specific permission from the WSRF Chief Controller (CCTLR)/ Assistant CCTLR (ACCTLR) to operate within R5107 (excluding Dona West and Dona East) when WSRF is closed. This authority is not delegated to any other 49th WG agencies.

3.6.6.3.2. Flights going to Mesa must enter at GALLAGHER; flights to Lava-Mesa must enter at BUELR; and flights to Yonder must enter at BOOOZ.

3.6.6.3.3. Airspace amendments and extension are NOT authorized during non- WSRF contingency operations. Aircrews must NOT enter restricted airspace prior to their published scheduled range time and MUST exit NLT their scheduled exit time. Flights WILL NOT enter any airspace/range in which other aircraft are present unless all flights are scheduled MARSA on CSE.

3.6.6.3.4. Wiley East ATCAA is NOT authorized for use during non-WSRF contingency operations. ABQ ARTCC is the controlling agency for Beak MOA/ATCAA when WSRF is not available.

3.6.7. Wiley All and HMN Working Area Procedures.

3.6.7.1. Wiley All airspace is composed of three subsections: Wiley East, Wiley West, and Wiley West Extension.

3.6.7.1.1. Wiley East consists of Wiley East ATCAA FL180 – FL400 or as required, Beak A/B/C MOAs (see Table 3.3. for boundary information).

3.6.7.1.2. Wiley West is defined as all of R5107B/C/D/H/J, and portions of R5107F/G, and the Yonder South portion of R5107B above 30,000 ft MSL as scheduled.

3.6.7.1.3. Wiley West Extension is defined as R5111A/B and R5107E as scheduled.

3.6.7.2. Delta Shelf Procedures.

3.6.7.2.1. Delta Shelf, for 54 FG F-16's, is normally scheduled 18,000 to 22,000 ft. MSL, and may be requested as low as 10,000 ft MSL in conjunction with Wiley West airspace. Make requests lower than 18,000 ft MSL on the CSE schedule. Additionally, the requesting Ops Sup should initiate initial coordination with the WSRF WS through JTTOC at least 1 hour prior to the mission start time.

3.6.7.2.2. Restrictions when releasing Delta Shelf below the standard 18,000 ft MSL:

3.6.7.2.2.1. Delta Shelf at or above 14,000 ft MSL:

3.6.7.2.2.1.1. F-16 Simulated Flameout (SFO) pattern altitude restricted below 13,500 ft MSL.

3.6.7.2.2.1.2. Max performance climbs restricted to 13,500 ft MSL.

3.6.7.2.2.1.3. MQ-9 departures and arrivals on Green Route South must cross the WSMR 24 grid line at or below 13,500 ft MSL.

3.6.7.2.2.2. Delta Shelf at or above 10,000 ft MSL:

3.6.7.2.2.2.1. Runway 22 instrument approaches must terminate in a full-stop or VFR.

3.6.7.2.2.2.2. No high-altitude instrument approaches to Runways 16/34.

3.6.7.2.2.2.3. No F-16 High Key SFO pattern altitudes, Low Key only.

3.6.7.2.2.2.4. Max performance climbs restricted to 9,000 ft MSL.

3.6.7.2.2.3. Delta Shelf will not be released at or above 10,000 ft MSL whenever:

3.6.7.2.2.3.1. Runway 22 is in use during IMC conditions.

3.6.7.2.2.3.2. During MQ-9 departures and recoveries on Green Route South.

3.6.7.2.2.4. WSRF WS is the final authority on operations within Delta Shelf and can deny or recall the airspace at any time if conditions require.

3.6.7.3. Fort Bliss Working Areas are composed of multiple areas: Dona East, Dona West, McGregor North, McGregor South, and CFA (see Table 3.3. for boundary information).

3.6.7.3.1. Dona West is the airspace contained in R5107A. The altitude is SFC to 60,000 ft MSL.

3.6.7.3.2. Dona East is the airspace contained in R5107K. The altitude is SFC to 60,000 ft MSL.

3.6.7.3.3. CFA altitudes are SFC to 60,000 ft MSL.

3.6.7.3.4. McGregor North. The airspace contained in R-5103C excluding the portion of the CFA which lies within R-5103C. The altitude is SFC to 60,000 ft MSL.

3.6.7.3.5. McGregor South. The airspace contained in R-5103B excluding the portion of the CFA which lies within R-5103B. The altitude is SFC to 60,000 ft MSL.

3.6.7.4. Fort Bliss Working Areas Procedures.

3.6.7.4.1. CHEROKEE provides airspace radar monitoring in McGregor North, McGregor South, CFA (R5103B/C), and Dona West/Dona East (R5107A/K).

3.6.7.4.2. CFA is designated a protected operating airspace dedicated primarily to high-speed strike aircraft as scheduled with Fort Bliss Range Scheduling. Transit through the CFA by non–participating aircraft will not be authorized during scheduled operations except for emergency aircraft approved by CHEROKEE/Range Operations Center (ROC). When scheduling Centennial range, fast moving aircraft will also request the CFA. Aircraft will be scheduled to fly outside the CFA in McGregor North and/or McGregor South but must observe all Fort Bliss restrictions as depicted in CSE restriction page. Ops Sups and aircrews will be notified that the CFA is active via comments in the "Operating Airspace" section of the daily Airspace Schedule available via Graduate Training Integration Management System (GTIMS).

3.6.7.4.3. Entry Procedures. Enter only within scheduled range time and after receiving clearance.

3.6.7.4.3.1. Radio Contact. Establish two-way radio communications with CHEROKEE prior to entering and departing CFA, McGregor North/South, and Dona East/West airspace. Upon clearance to enter CFA, aircrews may contact ROC for real-time scoring of air-ground munitions (must be pre-coordinated through ROC). If the ROC frequency is not being used for Joint Advanced Weapons Scoring System (JAWSS) scoring, aircrews may use that frequency as a discrete frequency for air-to-air inter-flight communications.

3.6.7.4.4. Exit Procedures. Notify CHEROKEE with your intentions 5 minutes prior to exiting the area. Request approval/clearance from CHEROKEE prior to leaving assigned area. DO NOT exit without a clearance.

3.6.8. Talon MOA/ATCAA. See ABQ ARTCC, 49 WG, and WSMR LOA for Talon MOA/ATCAA procedures.

3.6.9. Supersonic Airspace. Supersonic flight is currently conducted only when necessary to accomplish the mission and is managed IAW DAFMAN 13-201.

3.6.9.1. Supersonic operations are authorized in WSMR at or above 10,000 ft MSL, Wiley East ATCAA at or above 23,000 ft MSL, R5103B/C at or above 10,000 ft MSL, Valmont ATCAA at or above 18,000 ft MSL, and Talon ATCAA at or above 30,000 ft MSL.

3.6.9.2. Minimizing Sonic Boom Impact. The following three areas apply to all aircraft operating within the confines of Wiley East ATCAA.

3.6.9.2.1. Sonic Boom Restriction 1 (Subsonic only): N33 15 W105 45, N33 15 W105 32, N33 38 W105 28, and N33 38 W105 48.

3.6.9.2.2. Sonic Boom Restriction 2 is Non-maneuvering Supersonic: 5NM radius circle with the center point at Carrizozo, N33 39.5 W105 53.

3.6.9.2.3. Sonic Boom Restriction 3 (Non-maneuvering Supersonic): N33 11 W105

52, N33 01 W106 05, N32 50 W105 59, N32 51 W105 40, and N33 05 W105 38.

3.6.9.3. NON-MANEUVERING is defined as no transition from subsonic to supersonic and no turning under G (level or descending) while supersonic. Approaching the confines of non- maneuvering supersonic areas, pilots should not maneuver aggressively but instead decelerate and climb slightly to avoid "throwing" focused booms.

3.6.9.4. Mission planners and all pilots are responsible to mitigate sonic boom exposure in these areas.

3.7. Air Refueling Operations in Conjunction with Wiley Airspace.

3.7.1. Scheduling is the designated military unit responsible for scheduling all military aircraft operations in Wiley airspace, Air Refueling Track (AR)-644, and AR-121. Scheduling will not reserve AR-121 whenever there is a WSMR R&D mission north of the WSMR 50 grid line.

3.7.2. Tanker aircraft established in AR-644 or AR-121 will be MARSA with Wiley airspace participating aircraft. RAPCON is responsible for separation (i.e., altitude) between the enroute tanker aircraft and Wiley participating aircraft until the tanker aircraft is established within their refueling track or other airspace delegated for refueling operations (e.g., Beak MOA/ATCAA or Wiley East).

3.7.2.1. Once the tanker is established inside AR-644, or AR-121 or other delegated airspace, RAPCON will advise all aircraft, and pilots will acknowledge that the refueling airspace is active.

3.7.3. RAPCON will clear tanker aircraft onto AR-644, AR-121, or other delegated airspace and advise tanker aircraft of known traffic established in Wiley airspace (e.g., *TURBO 71 traffic 20 miles west SWAG 1 flight of 4 F-16 established in Wiley airspac*e).

3.7.4. RAPCON will advise Wiley aircraft on their discrete frequency and request acknowledgment once the tanker aircraft is enroute or established inside their refueling track.

3.7.5. Aircraft operating in Wiley will adhere to the AR-644 or AR-121 scheduled times and altitudes in the CSE. The tanker aircraft will transit Wiley airspace enroute to and from AR-644 or AR-121 at the CSE scheduled block altitudes.

3.7.6. Aircrews must coordinate with all other aircrew members involved in the operation and receive approval from the controlling agency prior to entering active SUA/ATCAA. MARSA will be applied for all required IFR operations (i.e., Air Refueling within Wiley airspace).

3.8. Reduced/Covert/Lights-Out Operations in ATCAA. See ABQ ARTCC, 49 WG, and WSMR LOA.

3.9. Military Training Routes (MTR). See ABQ ARTCC, 49 WG, and WSMR LOA for additional information and procedures. All MTRs must be reserved through Scheduling. Aircrews

must receive a local MTR brief from the HAM (DSN 572-2638) or read the MTR brief on the Scheduling SharePoint® prior to operating these MTRs.

3.9.1. Procedures: Visual Route (VR)-176, Instrument Route (IR)-133, IR-134, IR-142, IR-192, IR-194, or IR-195 (managed by HMN) can be filed by local stereos or Department of Defense (DD) Form 1801 *DoD International Flight Plan,* for other entries and exits.

3.9.2. Aircrews using low-level routes are responsible for all military training route procedures in DoD FLIP *Area Planning - Military Training Routes* (AP/1B). Scheduling will deconflict entry times, but MARSA applies at the entry point, alternate entry point, along the route, and at the exit and alternate exit points. Avoid all noise sensitive areas.

3.9.3. Expect clearance from ABQ ARTCC for VR-176. Expect enroute entry clearance from WSRF for IR-133, IR-134, IR-192, IR-194 and IR-195. If unsure of clearance into the route, contact ATC.

3.9.4. Clearance for VR-176 and IR-133 stereo does not mean that WSRF has approved entry into R5107B or R5107H. DO NOT enter WSMR airspace without WSRF approval. For IR-134M, maintain 12,000 ft MSL until point Alpha.

3.9.5. Clearance to fly IR-134, IR-192, IR-194, or IR-195 does not authorize entry into R5103B/C. DO NOT enter R5103 without approval from the controlling agency.

3.9.6. Information on bird hazards and periods of peak activity is available for low-level routes on the following US BASH website: **http://www.usahas.com**.

3.10. HSTT Operations. 846th Test Squadron will use Airspace and Range Scheduling procedures for all HSTT activities requiring range support or airspace that will affect range operations. HSTT will coordinate airspace activation/de-activation with WMSR Range Control.

3.10.1. HSTT Control will notify WSMR Range Control (DSN 258-2222) when:

3.10.1.1. Firings go into a hold status.

3.10.1.2. Firings are complete and airspace is returned or no longer requires protection.

3.11. Use of MARSA. It is the responsibility of all aircraft requesting MARSA to coordinate de-confliction with all other participating aircraft assigned to the same working airspace prior to range entry. Declaration of MARSA ensures that all required coordination between ALL users has been accomplished, and aircraft commanders assume responsibility for separation within assigned working airspace. MARSA will only be used for IFR operations requiring its use and must be scheduled in CSE. ATC will neither invoke nor deny the use of MARSA.

Table 3.2. Local Working Areas.

Area	Altitude (ft)	Contact Agency
Lava E/W (R5107B/C/F/G/J)	500 AGL to 60,000 MSL	54 OSS/OSOS
Mesa E/W (R5107C/H)	500 AGL to 60,000 MSL	54 OSS/OSOS
Oscura (R5107B)	SFC to 15,000 MSL	54 OSS/OSOS
R5107E	SFC to Unlimited	54 OSS/OSOS
R5111A/C	13,000 MSL to Unlimited	54 OSS/OSOS
R5111B/D	SFC to 13,000 MSL	54 OSS/OSOS
Red Rio (R5107B/J) + (R5107C above 9,000 MSL)	SFC to 15,000 MSL	54 OSS/OSOS
Yonder (R5107B/F/G)	SFC to 60,000 MSL	54 OSS/OSOS
Yonder E (R5107B/D)	18,000 MSL to 60,000 MSL	54 OSS/OSOS
Yonder S (R5107B)	3,000 AGL to 60,000 MSL	54 OSS/OSOS
Beak A/B/C ATCAA	18,000 MSL to 23,000 MSL	54 OSS/OSOS
Beak A/B/C MOA	12,500 MSL to 17,999 MSL	54 OSS/OSOS
CFA (R5103B/C)	SFC to 60,000 MSL	54 OSS/OSOS
McGregor North (R5103C)	SFC to 60,000 MSL	54 OSS/OSOS
McGregor South (R5103B)	SFC to 60,000 MSL	54 OSS/OSOS
Dona West (R5107A)	SFC to Unlimited	54 OSS/OSOS
Dona East (R5107K)	SFC to Unlimited	54 OSS/OSOS
MTR (IR133/142)	100 AGL varies to 12,000 MSL	54 OSS/OSOS
MTR (IR134/195)	100 AGL varies to 13,000 MSL	54 OSS/OSOS
MTR (IR192/194)	100 AGL varies to 13,000 MSL	54 OSS/OSOS
MTR (VR176)	100 AGL varies to 5,000 AGL	54 OSS/OSOS
Talon High E/W ATCAA	18,000 MSL to 29,000 MSL	54 OSS/OSOS
Talon High A/B/C MOA	12,500 MSL to 17,999 MSL	54 OSS/OSOS
Talon Low A/B MOA	500 AGL to 12,499 MSL	54 OSS/OSOS
Valentine ATCAA	18,000 MSL to 51,000 MSL	54 OSS/OSOS
Valmont ATCAA	18,000 MSL to 60,000 MSL	54 OSS/OSOS
Wiley East ATCAA	18,000 MSL to 60,000 MSL	54 OSS/OSOS
Capitan Bridge	18,000 MSL to 32,000 MSL	27 SOW/Cannon AFB
Melrose Range, A	SFC to 17,999 MSL	27 SOW/Cannon AFB
Melrose Range, B	18,000 MSL to 23,000 MSL	27 SOW/Cannon AFB
Pecos E/W MOA (LOW)	500 AGL to 11,000 MSL	27 SOW/Cannon AFB
Pecos E/W/S ATCAA	18,000 MSL to assigned	27 SOW/Cannon AFB
Pecos E/W/S MOA	11,000 MSL to 17,999 MSL	27 SOW/Cannon AFB
Taiban MOA	500 AGL to 17,999 MSL	27 SOW/Cannon AFB

Table 3.3. Local Airspace Boundaries.

Area	Coordinate Boundaries
Mesa East	N34° 16.4747’ W106° 24.0015’, N34° 17.0000’ W106° 12.0334’, N34° 17.0000’ W106° 04.0333’, N33° 52.1540’ W106° 04.0333’, N33° 52.0633’ W106° 23.9832’
Mesa West	N34° 15.7500’ W106° 40.5334’, N34° 16.4747’ W106° 24.0015’, N33° 48.7649’ W106° 23.9806’, N33° 48.6668’ W106° 45.5638’
Lava East	N33° 19.0780’ W106° 23.9583’, N33° 19.1673’ W106° 04.0333’, N33° 52.1540’ W106° 04.0333’, N33° 52.0633’ W106° 23.9831’
Lava West	N33° 18.9570’ W106° 50.9559’, N33° 19.0780’ W106° 23.9583’, N33° 48.7648’ W106° 23.9807’, N33° 48.6668’ W106° 45.5638’
Red Rio Airspace	N33° 52.0633’ W106° 23.9832’, N33° 37.2200’ W106° 23.9720’, N33° 37.3100’ W106° 04.0333’, N33° 52.1540’ W106° 04.0333’
Oscura Airspace	N33° 52.0633’ W106° 23.9832’, N33° 37.2200’ W106° 23.9720’, N33° 37.3100’ W106° 04.0333’, N33° 52.1540’ W106° 04.0333’
Yonder	N33° 13.0000’ W106° 52.0333’, N33° 15.6557’ W106° 51.5531’, N33° 15.7969’ W106° 20.0273’, N32° 32.9107’ W106° 20.0270’, N32° 32.8115’ W106° 42.6318’
Yonder East	N32° 36.0000’ W106° 06.0343’, N32° 50.0000’ W106° 04.0333’, N33° 15.8685’ W106° 04.0333’, N33° 15.7968’ W106° 20.0273’, N32° 32.9107’ W106° 20.0271’, N32° 32.9722’ W106° 06.0333’
Yonder South	N32° 32.9722’ W106° 06.0333’, N32° 25.0000’ W106° 06.0333’, N32° 23.3000’ W106° 07.0833’, N32° 24.8000’ W106° 09.0333’, N32° 19.5000’ W106° 20.6333’, N32° 19.5000’ W106° 39.5333’, N32° 32.8114’ W106° 42.6318’
McGregor North	N32° 30.0165’ W105° 30.0191’, N32° 29.9201’ W105° 48.5405’, N32° 29.3790’ W105° 48.5357’, N32° 29.3728’ W105° 49.4935’, N32° 26.1105’ W105° 51.6975’, N32° 14.9997’ W105° 51.5919’, N32° 15.0000’ W106° 10.0333’, then along the Southern Pacific Railroad, N32° 28.0000’ W106° 02.0333’, N32° 27.6667’ W106° 00.0334’, N32° 36.0000’ W106° 00.0000’, N32° 45.0000’ W105° 59.0333’, N32° 45.0000’ W105° 53.0333’, N32° 45.0000’ W105° 52.3667’, N32° 33.3333’ W105° 30.0333’
McGregor South	N32° 15.0000’ W106° 10.0333’, N32° 15.0002’ W105° 51.5933’, N32° 10.4152’ W105° 51.5492’, N32° 10.4460’ W105° 46.7051’, N32° 03.5000’ W105° 53.8333’, N32° 03.9167’ W106° 10.0000’, N32° 05.0333’ W106° 09.3667’, N32° 06.0000’ W106° 15.5333’

Area	Coordinate Boundaries
CFA	N32° 30.0165' W105° 30.0191', N32° 29.9201' W105° 48.5405', N32° 29.3790' W105° 48.5357', N32° 29.3728' W105° 49.4935', N32° 26.1105' W105° 51.6975', N32° 10.4152' W105° 51.5492', N32° 10.4460' W105° 46.7765', N32° 26.2279' W105° 29.9983'
Wiley East ATCAA	N34° 17.0000' W106° 04.0333', N34° 17.0000' W105° 51.0000', N33° 34.0000' W105° 00.0334', N33° 00.0334' W105° 00.0334', N32° 45.0334' W105° 12.0000', N32° 26.3333' W105° 30.0334', N32° 33.3334' W105° 30.0334', N32° 45.0000' W105° 53.0333', N32° 45.0000' W106° 06.0334', N32° 50.0000' W106° 04.0333'
Valmont ATCAA	N32° 45.0000' W106° 04.7666', N32° 36.0000' W106° 06.0334', N32° 25.0000' W106° 06.0334', N32° 23.3000' W106° 07.0834', N32° 06.0000' W106° 17.7500', N32° 06.0000' W106° 15.5333', N32° 15.0000' W106° 10.0333', N32° 24.0667' W106° 03.1667', N32° 27.8833' W106° 01.5334', N32° 27.8333' W106° 00.3167', N32° 36.0000' W106° 00.0000', N32° 45.0000' W105° 59.0333'
Dona East	*Same as R5107K
Dona West	*Same as R5107A
Wiley West	*Same as outer boundaries of R510B/C
Wiley West Extension	*Same as outer boundaries of R5111A/R5107E
R-5103B/C and R5107A/K (Fort Bliss)	Available in FAAO JO 7400.10 SUA or DoD FLIP *Aera Planning – SUA North and South America* (AP/1A)
R5107B/C/D/E/F/G/H/J, R5109A/B (WSMR)	Available in FAAO JO 7400.10 SUA or DoD FLIP AP/1A
Beak A/B/C, Talon Low, Talon High East/West, Pecos North Low/High, Pecos South, Taiban, Cato/Smitty MOAs	Available in FAAO JO 7400.10 SUA or DoD FLIP AP/1A

Table 3.4. WSMR Grid (A – N, 0 – 44) Conversion to Latitude and Longitude.

Vertical lines			Horizontal Lines		
ID	**Starting Point**	**End Point**	**ID**	**Starting Point**	**End Point**
	N34.465918 W107.926751	N31.882502 W107.88066	0	N31.890876 W105.818083	N31.881701 W107.945092
	N34.466765 W107.860399	N31.88327 W107.816226	2	N31.945872 W105.817777	N31.936678 W107.946051
	N34.467576 W107.794045	N31.884006 W107.751789	4	N32.000867 W105.817469	N31.991653 W107.947012
	N34.468351 W107.727687	N31.884708 W107.68735	6	N32.055862 W105.817161	N32.046629 W107.947977
	N34.46909 W107.661328	N31.885379 W107.622908	8	N32.110856 W105.816852	N32.101603 W107.948943

Vertical Lines			Horizontal Lines		
ID	**Starting Point**	**End Point**	**ID**	**Starting Point**	**End Point**
	N34.469793 W107.594966	N31.886016 W107.558465	10	N32.16585 W105.816542	N32.156578 W107.949913
	N34.47046 W107.528601	N31.886621 W107.49402	12	N32.220843 W105.816231	N32.211551 W107.950885
	N34.471091 W107.462235	N31.887193 W107.429572	14	N32.275836 W105.815919	N32.266525 W107.951859
	N34.471686 W107.395866	N31.887733 W107.365123	16	N32.330829 W105.815606	N32.321497 W107.952837
A	N34.472245 W107.329496	N31.888239 W107.300672	18	N32.385821 W105.815293	N32.37647 W107.953817
B	N34.472768 W107.263124	N31.888714 W107.23622	20	N32.440812 W105.814979	N32.431441 W107.954799
C	N34.473255 W107.19675	N31.889155 W107.171766	22	N32.495803 W105.814664	N32.486413 W107.955785
D	N34.473706 W107.130375	N31.889564 W107.107311	24	N32.550794 W105.814348	N32.541383 W107.956773
E	N34.474121 W107.063998	N31.88994 W107.042855	26	N32.605784 W105.814031	N32.596354 W107.957763
F	N34.4745 W106.99762	N31.890283 W106.978397	28	N32.660773 W105.813713	N32.651323 W107.958757
G	N34.474842 W106.931241	N31.890594 W106.913939	30	N32.715762 W105.813394	N32.706292 W107.959753
H	N34.475149 W106.864861	N31.890872 W106.849479	32	N32.770751 W105.813075	N32.761261 W107.960752
I	N34.47542 W106.798479	N31.891118 W106.785019	34	N32.825739 W105.812755	N32.816229 W107.961753
J	N34.475654 W106.732097	N31.89133 W106.720558	36	N32.880727 W105.812433	N32.871197 W107.962758
K	N34.475853 W106.665715	N31.89151 W106.656096	38	N32.935714 W105.812111	N32.926164 W107.963765
L	N34.476015 W106.599331	N31.891658 W106.591634	40	N32.9907 W105.811788	N32.981131 W107.964774
M	N34.476142 W106.532948	N31.891772 W106.527171	42	N33.045686 W105.811464	N33.036097 W107.965787
N	N34.476232 W106.466564	N31.891854 W106.462708	44	N33.100672 W105.81114	N33.091062 W107.966802

Table 3.5. WSMR Grid (O – W, 46 – 94) Conversion Latitude and Longitude.

Vertical Lines			Horizontal Lines		
ID	Starting Point	End Point	ID	Starting Point	End Point
O	N34.476287 W106.400179	N31.891904 W106.398245	46	N33.155657 W105.810814	N33.146028 W107.96782
P	N34.476305 W106.333795	N31.89192 W106.333781	48	N33.210642 W105.810488	N33.200992 W107.968841
Q	N34.476287 W106.26741	N31.891904 W106.269318	50	N33.265626 W105.81016	N33.255956 W107.969865
R	N34.476233 W106.201026	N31.891855 W106.204855	52	N33.32061 W105.809832	N33.31092 W107.970891
S	N34.476143 W106.134642	N31.891774 W106.140392	54	N33.375593 W105.809503	N33.365883 W107.97192
T	N34.476017 W106.068258	N31.89166 W106.075929	56	N33.430576 W105.809173	N33.420845 W107.972952
U	N34.475855 W106.001874	N31.891513 W106.011467	58	N33.485558 W105.808842	N33.475807 W107.973987
V	N34.475657 W105.935492	N31.891333 W105.947005	60	N33.54054 W105.80851	N33.530769 W107.975024
W	N34.475423 W105.86911	N31.891121 W105.882544	62	N33.595521 W105.808177	N33.58573 W107.976065
			64	N33.650502 W105.807844	N33.64069 W107.977108
			66	N33.705482 W105.807509	N33.69565 W107.978154
			68	N33.760462 W105.807174	N33.75061 W107.979203
			70	N33.815441 W105.806837	N33.805568 W107.980255
			72	N33.87042 W105.8065	N33.860527 W107.981309
			74	N33.925398 W105.806162	N33.915485 W107.982367
			76	N33.980376 W105.805822	N33.970442 W107.983427
			78	N34.035353 W105.805482	N34.025399 W107.98449
			80	N34.09033 W105.805141	N34.080355 W107.985556
			82	N34.145306 W105.8048	N34.135311 W107.986625
			84	N34.200282 W105.804457	N34.190266 W107.987697
			86	N34.255257 W105.804113	N34.245221 W107.988772
			88	N34.310232 W105.803768	N34.300175 W107.98985
			90	N34.365206 W105.803423	N34.355129 W107.99093
			92	N34.42018 W105.803076	N34.410082 W107.992014
			94	N34.475153 W105.802729	N34.465035 W107.9931

Chapter 4

FLIGHT PLANNING

4.1. Flight Plan Procedures.

4.1.1. Standardized Routes and Stereo Flight Plans. HMN standardized routes and stereos (See 49 WG IFG) are designed to reduce adverse environmental impact, provide smooth- flow sequencing, and enhance flight safety.

4.1.2. All aircraft departing HMN IFR, VFR, or on a stereo must have a signed flight plan on file in either their designated squadron(s) or at AM.

4.1.2.1. HMN assigned aircrew and tenant assigned units have the options of emailing and calling, or faxing as a back-up procedure, DD Form 1801 to file VFR or stereo departures to AM for sorties originating from HMN.

4.1.2.2. DD Form 1801 must be emailed or called to AM not later than 1 hour prior to departure time, faxed as a back-up. The agency submitting flight plans must confirm via telephone at least 30 minutes prior to proposed departure time to ensure receipt and acceptance of flight plan(s) by AM.

4.1.2.3. HMN assigned and tenant units must maintain original flight plan (DD Form 1801 *Plan* or Stereo), passenger manifest, weight and balance forms, and any other forms associated with the flight plan for 3 months, IAW RDS, Table 13-07, Rule 03.00.

4.1.2.3.1. Units may use an AF Form 4327, *Aviation Resource Management System (ARMS) Fighter Flight Authorization (FA)*; or AF Form 4327a, *Crew Flight Authorization (FA)*, to meet the 3-month requirement. A unit approved remark or remarks code, or legend as required on the AF Form 4327/4327a to reference a HMN 49 WG IFG standardized route or stereo may fulfill the signature requirement as follows:

4.1.2.3.1.1. The pilot in command's (PIC) signature next to the applicable remark or remarks code on the AF Form 4327/4327a constitutes the act of filing a VFR or stereo flight plan.

4.1.2.3.1.2. The Ops Sup (or representative) will convey, verbatim, to AM the applicable blocks from the AF Form 4327/4327a for which the PIC signed.

4.1.2.3.2. HMN assigned and tenant units will comply with paragraph 4.1.2.3. for the TDY units that they host.

4.1.2.4. Transient aircrew filing flight plans with AM in person may submit computer generated flight plans via email.

4.1.2.5. NAMO, or a designated representative will inspect squadrons quarterly to ensure records are maintained IAW paragraph 4.1.2. and 4.1.2.3.

4.1.3. AM will file flight plans through ABQ ARTCC for all aircraft and provide flight tracking for locally assigned and inbound transient aircraft.

4.1.4. Coordinate changes to local flight plans by calling AM at least 15 minutes prior to takeoff with the revision. After departure, a HMN assigned, or tenant unit PIC will coordinate with AM or RAPCON for changes to flight plans.

4.1.5. Formation Flights. Aircraft aborting/separated from the original formation must re- file prior to departure.

4.1.5.1. Formation flight join-up of multiple single aircraft, multiple formations, or single aircraft joining established formations during departure and arrival will avoid using "MARSA" unless participating in an approved MARSA activity (see Chapter 3). Instead, the lead pilot will request "*flight join-up with (joining aircraft call sign)*".

4.1.5.2. Dissimilar formations (e.g., 2/F-16 with 2/F-35) will ensure the dissimilar aircraft are annotated in the remarks section of their flight plan (e.g., "NUM 3 and 4 F35/I).

4.1.5.3. Preplanned flight breakaways (e.g., 4/F-16 departing as a flight and splitting up after air refueling) will ensure a separate flight plan is filed for the aircraft intending to breakaway. Notify ATC when ready to activate the separate flight plan.

4.1.6. Changes to Transient or Outbound flights. The PIC may re-file or amend the flight plan with AM via any means provided. AM personnel must verify an original flight plan was filed if the change does not penetrate an Air Defense Identification Zone. AM may verify original flight plans by contacting the original departure location via telephone or flight plan processing computer (stopover, weather/maintenance divert, etc.). AM will NOT modify/change flight plans for Tanker Airlift Control Center flight managed missions without approval from the flight planning cell/flight managers.

4.1.7. Daily Flying Schedule. HMN assigned units and locally hosted TDY units will provide AM a daily flying schedule NLT 1630L the day before the flight. If able, include mission details (e.g., check rides, extended VFR pattern work, cross country) that will be useful for ATC/AM to plan/coordinate staffing, airfield maintenance, and training. AM will compile the unit schedules into the HMN Daily Flying Schedule. Contact AM for inclusion to the daily flying schedule distribution list.

4.1.8. VFR departures may be made on local stereos if weather and mission requirements dictate. AA flight operations not annotated on the daily flying schedule and changes to AA call signs or departure times will be coordinated by AA Operations with the Tower WS.

4.2. Prior Permission Required (PPR) Procedures. AM administers PPR restrictions IAW AFMAN 13-204V2, DoD FLIP *IFR Supplement US*, and NOTAMs until IFR Supplement changes are in effect.

4.3. Unscheduled and Unauthorized Aircraft Arrivals. See HAFB IDP, Appendix 18, A*rrival of Unidentified or Unannounced Military or Civilian Aircraft.* Immediately report unauthorized aircraft movement via the Primary Crash Alarm System (PCAS) and SCN to 49 WG/CP and Base Defense Operations Center (BDOC). Unscheduled aircraft arrivals will be IAW Airfield Operations Flight Operations Letter.

4.4. DV Notification Procedures. Upon notification, AM will notify WSRF, Tower, and 49 WG/CP of inbound DVs. Workload permitting, WSRF will notify AM of the DV aircraft position when 50-flying miles out and Tower will notify AM at 15-flying miles. Airborne request to ATC for priority will be forwarded to 49 WG/CP for protocol notification or verification.

4.5. Civilian Aircraft Landing Permits. The Airfield Manager (AFM) and/or Assistant AFM (AAFM) are the designated representatives to process and approve Civil Aircraft Landing Permits IAW AFI 10-1001, *Civil Aircraft Landing Permits.* Civil aircraft requesting to operate at HMN must comply with procedures in AFI 10-1001, and AFI 10-1002, *Joint Use Agreements for Military and Civilian Flying Facilities,* as applicable.

Chapter 5

GROUND OPERATIONS

5.1. Engine Test and Run-up Procedures.

5.1.1. Engine Run-up Areas are located on Runway 25 approach end, Runway 34 approach end, on run-up pad west of Taxiway Hotel near Building 801, and Pad 19.

5.1.1.1. Unrestrained engine runs are authorized IAW applicable aircraft technical guidance on all parking aprons for aircraft with a wingspan of 55 ft or less.

5.1.2. Reciprocating or Turboprop Run-up areas for aircraft with a wingspan of 55 ft or less the designated location is 200 ft short of the VFR hold lines for Runway 16/34 at Taxiways Alpha and Delta. Run-up for Runway 07/25 is at Taxiway Bravo favoring the infield edge of the taxiway. For all other runways, run-up on appropriate taxiway or approach end of runway as approved by AM. For aircraft with wingspan greater than 55 ft the locations outlined in paragraph 5.1.1. must be used.

5.1.3. Military Power Engine Runs. Suppressed engine runs required for maintenance may be accomplished during quiet hours. Non-suppressed engine runs will only be accomplished on the first 1,000 ft of the approach end of any runway (to avoid surface damage) or Taxiway Hotel Trim Pad. Non-suppressed engine runs must be approved by 54 FG/CC or 54 FG Brickholder if conducted outside of wing flying hours or during quiet hours.

5.1.3.1. Coordinate all requests for military power engine runs by HMN assigned aircraft with MOC by radio or phone (DSN 572-3342). For transient aircraft, contact AM (DSN 572-5411). AM will determine the location for transient engine runs and AM will advise Tower of time, location, and call sign.

5.1.4. Unless specifically restricted during quiet hours, MQ-9 engine runs, all fighter aircraft engine runs up to 80%, engines requiring scavenge prior to shut down, and all engine runs in a hush house are exempt.

5.2. Aircraft Special Operations Areas and Ramps.

5.2.1. Drag Chute Jettison Areas. Aircrew will inform Tower as soon as practical when anticipating use of a drag chute during landing. Aircrew intending to use a drag chute can expect to be sequenced to last position in the pattern or to an alternate runway. After landing aircrew will report when ready to jettison a deployed chute. When possible, aircrew will attempt to jettison a deployed drag chute off the runway.

5.2.2. Live Ordnance Loading Areas (Alpha and Echo LOLA). Live load operations on the Alpha and/or Echo LOLA must be coordinated with AM 5 days in advance.

5.2.3. Hazardous Cargo (HC) Ramp. The primary parking area for HC loading and unloading is at the north end of Taxiway Tango. See Attachment 13 Aircraft Carrying Hazardous Cargo, paragraphs 8.8., 8.9. and 8.14. for additional HC procedures.

5.2.4. Hot Pit Refueling Area. There are no designated hot pit refueling areas on HMN.

5.3. Aircraft Taxiing Requirements and Routes.

5.3.1. Taxi speed is restricted to a maximum of 30 knots on all taxiways. DO NOT exceed 10 knots ground speed in parking areas, ramps, and in the West Ramp.

5.3.2. Only base-assigned aircraft may park on and taxi through the West Ramp. **Note**: Transient crews may request to park on West Ramp overflow due to security requirements. AM and BDOC will review all requests and maintain approval authority.

5.3.2.1. Taxiway Hotel Warning Lights (north and south of Taxiway Hotel at Bong Street intersection) will be activated by the SOF or Tower whenever Hotel is in use for taxi operations or Arm/De-arm Hotel operations are in progress.

5.3.3. Taxiway Golf Hot Spot (Figure 5.1.). **WARNING:** Taxiway Golf has a history of aircraft runway incursions due to multiple runway/taxiway intersections and short distance between Runway 04/22 and Taxiway Golf. Be aware of the VFR hold lines on Taxiways Tango, Romeo, and Lima. Taxiway Golf is an extension of the Runway 07/25 controlled movement area and requires Tower approval prior to taxiing on the taxiway. Tower will issue specific runway exiting and taxi instructions before aircraft exit Runway 22 or before aircraft enter Taxiways Golf and Romeo.

Figure 5.1. Taxiway Golf Hot Spot.



5.3.4. Taxi routes will be per Tower instructions.

5.3.5. FOD Abatement. Before authorizing multi-engine (i.e., 4 or more wing mounted engines) aircraft (e.g., C-130, C-17, AN-124) to taxi, Tower must request the aircraft taxi with outboard engines at idle. E.g., *"SHADOW 11 IF ABLE, TAXI WITH OUTBOARD ENGINES AT IDLE."* or *"HAL 1950 IF ABLE, TAXI WITH INBOARD ENGINES ONLY"*.

5.3.6. During arrival operations, Tower will not authorize vehicles/aircraft to cross a runway in front of a landing MQ-9 or fighter type aircraft that has verbally committed to exiting the runway prior to the point at which the crossing is intended. See FAAO JO 7110.65 paragraph 3-7- 2g.2.(b)(1).

5.3.7. 180-degree turns on runway asphalt surfaces are not authorized. **Exception:** A-10, F-1, F-15, F-16, F-18, F-22, F-35, T-7, T-38, MQ-9, C-12, and 49 FSS/AC aircraft are exempt from this restriction.

5.3.8. Wingtip, Jet Blast, and Parking Restrictions. 49 WG Master Aircraft Parking Plan is maintained IAW AFMAN 13-204V2 using UFC 3-260-01. This plan ensures wingtip clearances, jet blast safe distance requirements, and parking restrictions are maintained to protect personnel, aircraft, facilities, and equipment. HMN Master Aircraft Parking Plan is maintained by AM. See DoD FLIP AP/1 or DoD FLIP *IFR Supplement US* for wingtip clearance restrictions.

5.3.9. Precision Approach Critical Areas. When weather conditions are reported to be less than an 800 ft ceiling and/or visibility is less than 2 SM, Tower will protect the ILS Critical Areas IAW AFMAN 13-204V3.

5.3.9.1. An Instrument Hold Line is located on Taxiway Delta to protect Runways 16 and 22 Glideslope signals and both Precision Obstacle Free Zones (POFZ). See Figure 5.2. for ILS Glideslope critical areas and POFZs. See Figures 5.3 and 5.4. for ILS Localizer Critical Area locations. When weather conditions dictate, Tower will broadcast "*INSTRUMENT HOLD LINE IN EFFECT*" on Tower frequencies and ATIS.

5.3.9.2. During FAA ILS Flight Inspections, Tower must protect the ILS Localizer Critical Area IAW FAA Order 8240.41, Flight Inspection/Air Traffic On−Site Coordination Requirements.

Figure 5.2. Runways 16 and 22 ILS Glideslope Critical Areas and POFZ.

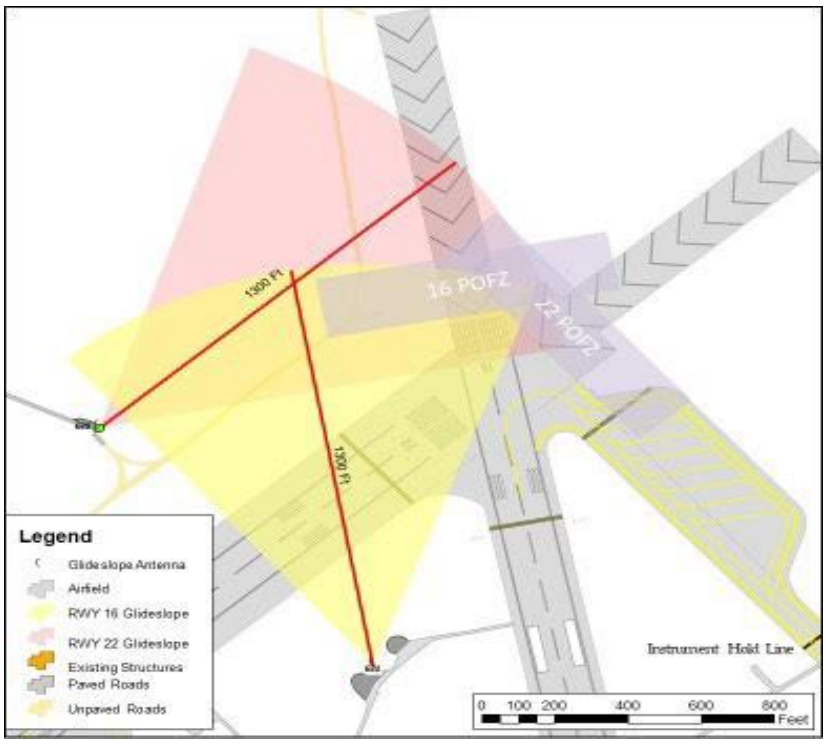


Figure 5.3. Runway 16 ILS Localizer Critical Area.

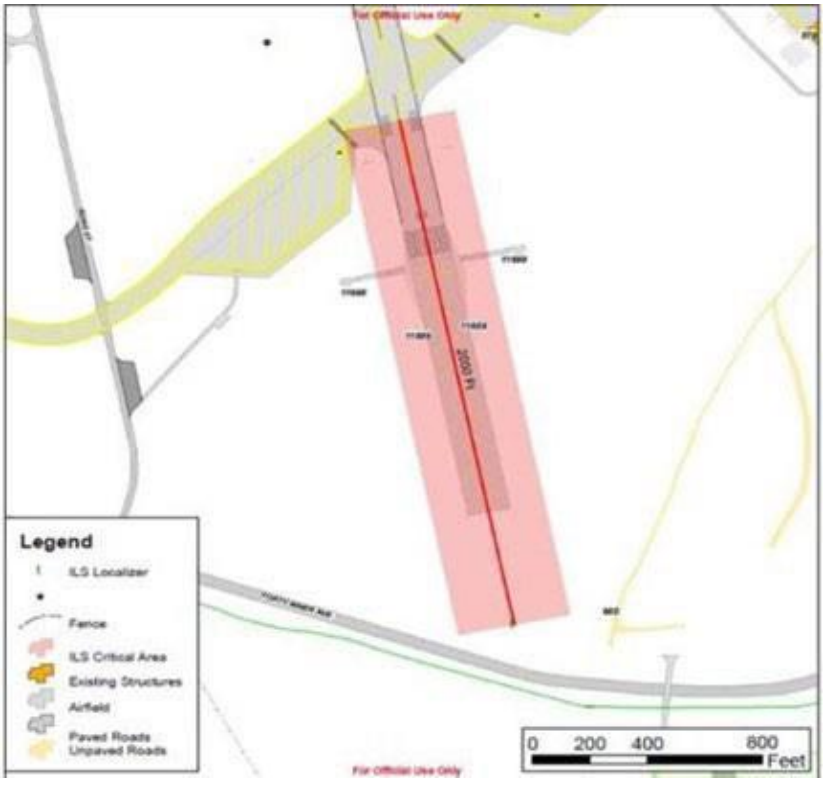
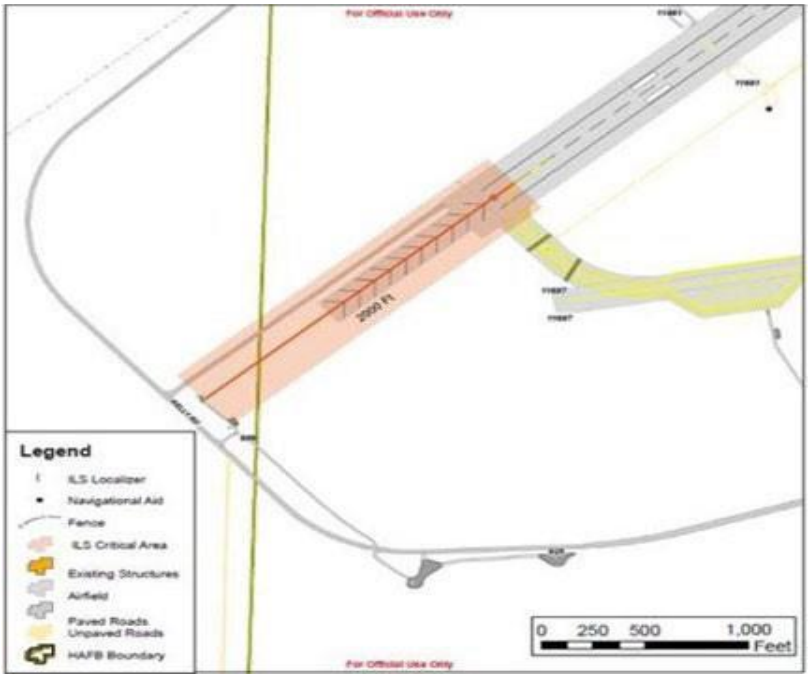


Figure 5.4. Runway 22 ILS Localizer Critical Area.



5.3.10. Fighter Index of Thermal Stress (FITS). Bioenvironmental Engineering (49 OMRS/SGXB) determines the FITS zone IAW DAFI 48-151, *Thermal Stress Program*. When reported, Tower will include "FITS CAUTION" or "FITS DANGER" advisory in the ATIS message.

5.3.11. Taxiway Lighting Outages. During taxiway lighting outages, MQ-9s with operational infrared (IR) camera or sensor may use unlit taxiways at night or low visibility. Movement of all other aircraft on unlit taxiways at night or low visibility must have 54 FG/CC or 54 FG Brickholder approval and the aid of follow-me vehicles or ground support personnel.

5.4. Startup/Taxi Procedures.

5.4.1. Review ATIS prior to calling for IFR/VFR clearance or taxi instructions.

5.4.2. Contact Clearance Delivery no earlier than 30 minutes prior to proposed departure time. If your departure time has changed from your original proposed time, notify Clearance Delivery on initial call-up.

5.4.2.1. Request VFR/IFR departures with Clearance Delivery. Include intentions, state current airspace restriction code, and working frequency/local channel.

5.4.3. Call Ground Control with position for taxi instructions prior to taxiing.

5.4.3.1. Inform Ground Control of any special requirements such as non-standard departures, active test mission priority, pattern delay, pick-up pattern, etc.

5.4.3.2. Include call signs of other flight members if not filed as a single flight. E.g., "*Ground, MANDO 11 with RUTT 13, taxi with Alpha from spot 2 and 5…*"

5.5. General Departure Procedures. Contact Tower when ready for departure and state initial departure intentions (e.g., "*Tower, WOOL 11 ready runway one six at Delta, MEZR1 BOOOZ" or "Tower, STRIKE 91 ready runway three four at Alpha OHDEE One"*. When cleared for takeoff, the pilot will acknowledge clearance and departure runway.

5.6. Line Up Procedures. Line up IAW applicable aircraft flight manual directives except during the following situations.

5.6.1. During the following intersecting runway operations, any aircraft requesting full runway length must advise Tower as soon as possible after completion of EOR checks.

5.6.1.1. Simultaneous Runway 22 departure and Runway 16 or 34 arrival operations, departures will line up at the intersection of Runway 22 and Taxiway Echo.

5.6.1.2. Simultaneous Runway 16 departure and Runway 22 or 04 arrival operations, departures will line up with the last flight element no further north than the BAK-12 located 1,500 ft from Runway 16 threshold.

5.6.1.3 When authorized to line up on intersecting runways (i.e., full length Runway 16 and 22 intersection or Runway 07/25 and Runway 16/34 intersection), absence of holding instructions from Tower authorizes an aircraft to enter the runway intersection (e.g., "*COBRA 21 back-taxi Runway 22 line up and wait").*

5.6.1.4. While heavy or jet aircraft are holding in position for takeoff on Runway 22 approach end, other aircraft conducting approaches to Runway 16 will be limited to restricted low approaches due to the possibility of thrust-induced turbulence in the touchdown zone. Restricted low approaches are not authorized over aircraft in takeoff position.

5.6.2. Intersection Departures. The distance remaining from intersection to end of runway (see Table 5.1.) is not issued to locally assigned aircraft unless requested by the pilot.

5.6.2.1. During Runway 34 departures from Taxiway Alpha or Hotel, advise Tower if full length departure is required. If using full runway length, avoid taxiing onto Runway 34 underrun during line up.

Table 5.1. Intersection Departure Information.

Runway	Departure Intersection (Taxiway or Runway)	Runway Distance Remaining (1,2)	Restrictions
16	TY Charlie or Foxtrot	8,150 ft	None
16	RY 07 or 25	5,750 ft	None
16	TY Juliet	4,750 ft	None
16	TY Kilo	1,800 ft	Helicopters only

Runway	Departure Intersection (Taxiway or Runway)	Runway Distance Remaining (1,2)	Restrictions
16	TY Alpha or Hotel	850 ft	Helicopters only
34	TY Alpha or Hotel	11,250 ft	None
34	TY Kilo	10,250 ft	None
34	TY Juliet	7,350 ft	None
34	RY 07 or 25	6,300 ft	None
34	TY Charlie or Foxtrot	3,950 ft	None
04	TY Echo	1,550 ft	Helicopters only
22	TY Echo	8,950 ft	None
07	TY Foxtrot	8,050 ft	None
07	RY 16 or 34	6,250 ft	None
07	TY Delta	5,100 ft	None
25	TY Delta	7,750 ft	None
25	RY 16 or 34	6,650 ft	None
25	TY Foxtrot	4,800 ft	None
1. Intersection departure distances are rounded down to the nearest 50 ft increment. 2. Intersection departure distances do not include departure end overrun length.			

5.6.3. Takeoff restrictions.

5.6.3.1. For BAK-15 barrier equipped runways, Tower will ensure the departure end BAK-15 barrier is armed and in the lowered position. Aircrew can request the BAK-15 to be raised prior to departure. During BAK-15 malfunctions, the SOF will determine the need to keep the departure end BAK-15 in the raised position. All non-standard barrier/cable configurations will be published in a NOTAM and broadcasted on the ATIS (see paragraph 2.8. AAS)

5.6.3.2. Tower and WSRF approval are required for maximum performance climbs.

5.6.4. Aborting Takeoffs.

5.6.4.1. Aborted takeoffs automatically cancel takeoff clearance for all remaining stationary elements in the flight. If any element of the flight has begun takeoff roll, the decision to continue takeoff or abort rests with the PIC. Takeoff clearance for any remaining elements must be re- issued prior to re-attempting takeoff.

5.6.4.2. If conditions permit, pilots will call "*BARRIER – BARRIER - BARRIER,*" or "*CABLE – CABLE – CABLE - RUNWAY (name*)" on Tower frequency if a barrier or cable engagement is anticipated.

5.7. Airfield Vehicle/Pedestrian Operations. See DAFI 13-213_HAFB Supplement (SUP), *Airfield Driving*.

5.8. Control of Ground Traffic in the Controlled Movement Area (CMA). The CMA at HMN

is defined as all runways, underruns, Taxiway Golf, and all areas within 100 ft of all runways, underruns, and Taxiway Golf. Vehicle operators are required to establish and maintain direct two-way radio contact with the Tower prior to proceeding within the CMA. See DAFI 13-213_HAFBSUP, *Airfield Driving*, for entering, operating within, and exiting instructions.

5.8.1. HMN airfield is closed, vehicle operators will contact 49 WG/CP (LMR Tower Net) prior to proceeding within the CMA. See paragraph A10.5. UAO - Vehicle Operations for additional procedures during scheduled UAO.

5.9. Aircraft Towing Procedures. See DAFI 13-213_HAFBSUP, *Airfield Driving*.

Chapter 6

VISUAL FLIGHT RULES (VFR) PROCEDURES

6.1. VFR Stereo Departures (See 49 WG IFG). MEZR1 departure is the primary Visual Meteorological Conditions (VMC) / VFR procedure intended to allow pilots a flexible VFR departure to an IFR pickup along their intended stereo routing.

6.1.1. Romeo / Yonder Departure. Romeo and Yonder Departures are not authorized for range crossings and may only be used from Runway 04 and Runway 07 with Tower approval. If range crossing is desired, specifically request VFR range crossing.

6.1.1.1. Romeo Departure aircraft will depart VFR on the WSMR Romeo grid line (3 NM west of HSTT) to Salinas Corridor, then to assigned work area at or below 8,500 ft MSL.

6.1.1.2. Yonder Departure aircraft will depart VFR and proceed westbound (into Yonder airspace) at or below 8,500 ft MSL.

6.2. VFR Departures. VFR Departures may be conducted from any runway. Due to restricted airspace, Clearance Delivery will issue initial VFR departure instructions. Use caution for other aircraft in the overhead/VFR patterns on other runways.

6.2.1. VFR Departures to ELP. Fly the appropriate VFR altitudes enroute from HMN to ELP in the VFR corridor. Remain clear of KALM. Follow the railroad tracks south to ELP remaining within 2 NM west of the railroad tracks. The VFR corridor is bounded by R5107K (Dona East) to the west and R5103B/C (McGregor North/South) to the east. DO NOT fly east of the railroad tracks until reaching EWM. This is a civilian flyway with potential live missile firings conducted on either side. If possible, maintain at or above 12,500 ft MSL for better radar identification and deconfliction with light aircraft. Contact ELP Approach abeam Orogrande, NM if radar services are not being provided by WSRF. For return to HMN, follow the VFR corridor northbound and call WSRF abeam Orogrande, NM if ELP Approach is not providing radar services. Helicopters are exempt from altitude restrictions.

6.2.2. Pick-up Pattern Departure is a VFR pattern used by AA, 586 FLTS, and 82 ATRS to achieve a chase formation immediately after departure. Aircrew and Tower will include the phrase "*pick- up pattern*" during line up and takeoff communications. See Attachment 5 for QF-16 pick-up pattern procedures.

6.2.2.1. Prior to departure, aircrew will state pick-up pattern timing, altitude, and direction of turn (e.g., *SHABA 01 ready runway 25 with NIKE 46, two-minute pick-up pattern, 5,600, left turn*).

6.2.2.2. After line up and receiving takeoff clearance, the lead aircraft (chase) will depart while the aircraft being chased remains in line up position.

6.2.2.3. After departure, chase will turn closed traffic and when reaching an appropriate point in the pattern (usually base turn), notify the aircraft being chased when to depart. Subsequent takeoff clearance from Tower for the second aircraft is not required. See Attachment 5 for QF-16 pick- up pattern specifics.

6.3. Position Reporting. When possible, use DME for all position calls. Runway 34 visual straight-in final will always be DME due to the position of the TACAN on the airfield. Aircraft reporting 90-to-initial are not acknowledged by Tower.

6.3.1. Initial Aircraft Contact. Establish initial contact with applicable information, to include call sign, number of aircraft in flight, position, pattern/approach request, and type landing.

6.4. VFR Recoveries. Pilots returning to HMN VFR will contact WSRF NLT than 20 NM from HMN with intentions. Pilots leaving SUA will obtain current ATIS as soon as practical and contact WSRF or ABQ ARTCC prior to leaving the SUA. When possible, maintain 300 Knots Indicated Air Speed (KIAS) when proceeding to initial (350 KIAS tactical initial) or slow to 250 KIAS for straight-in NLT 10 NM from HMN.

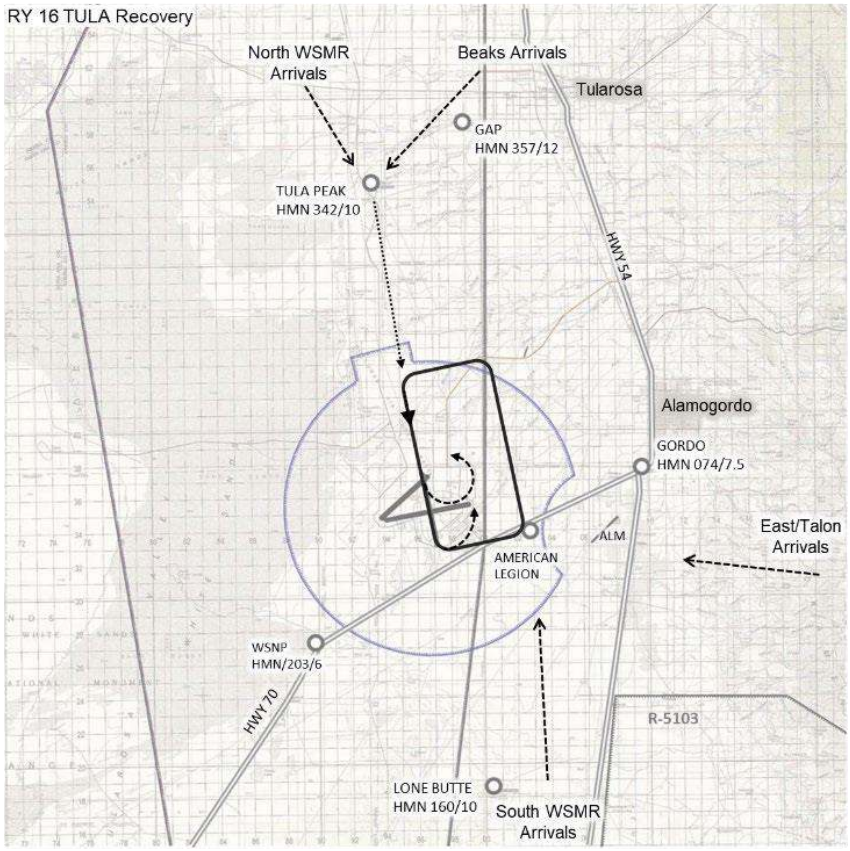
6.5. Radar Vector to Initial/Downwind/VFR Straight-In. Aircraft may request vectors to initial, inside/outside downwind, or VFR straight-in for any runway. **Exception:** See paragraph 6.8. for Runway 25 straight-in restrictions.

6.5.1. IFR aircraft recovering to the VFR traffic pattern, requesting initial or VFR straight-in will advise WSRF of cancellation of IFR or will report the airport/VFR reporting point in sight NLT 15 DME.

6.5.2. A pilot report of the airport or VFR reporting point in sight will constitute automatic cancellation of IFR flight plan. Upon receiving a pilot report of the airport or VFR reporting point in sight and leaving Class A airspace, WSRF will no longer provide standard IFR separation, but will provide basic radar service, sequencing, and advisories until frequency change to Tower.

6.6. VFR Recovery to Runway 16 (Figure 6. 1.). The primary VFR recovery point for Runway 16 is TULA PEAK. When authorized by ATC, aircraft may recover via the outside downwind from the south or east. See paragraph 6.13.8.1. for Runway 16 VFR pattern procedures.

Figure 6.1. Runway 16 VFR Recovery (Drawing not to scale).



6.6.1. Runway 16 Initial transition from the north: report TULA PEAK at 6,100 ft MSL and report initial.

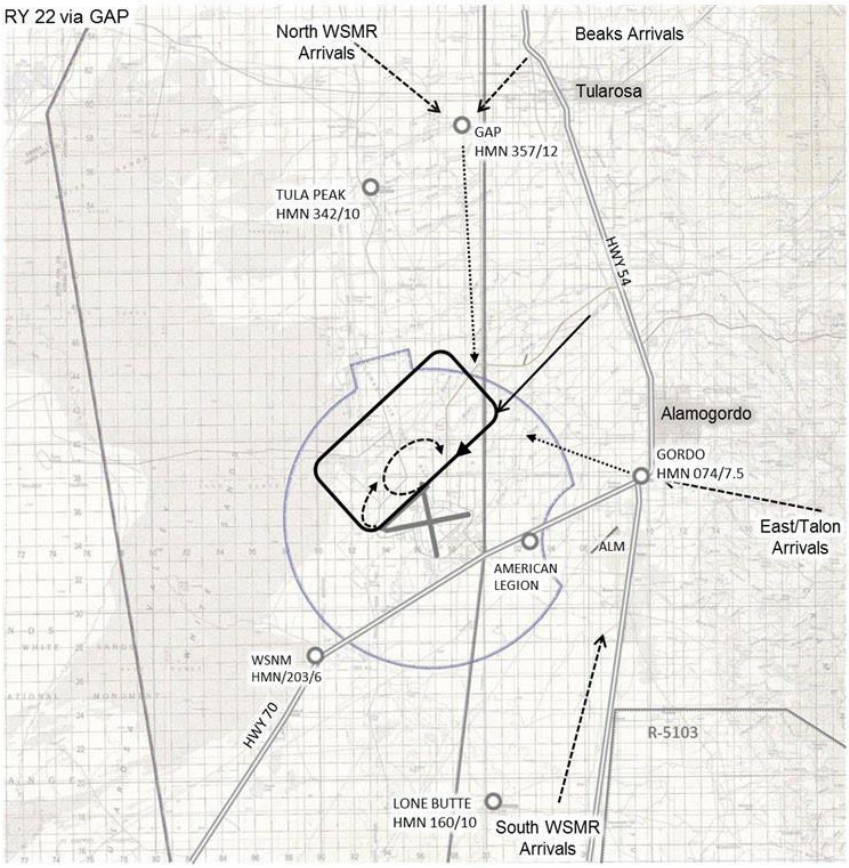
6.6.2. Runway 16 Initial transition from the south or east: request direct to outside downwind from WSRF. Tower will sequence aircraft into outside downwind as conditions permit. When cleared, enter outside downwind at 6,100 ft MSL while visually clearing for traffic. Follow the pattern to initial. If unable to be sequenced into the outside downwind, proceed VFR to Tula Peak.

6.6.3. Runway 16 Straight-in transition from the north: report TULA PEAK at 5,600 ft MSL and report 5 NM final.

6.6.4. Runway 16 Straight-in transition from the south or east: request direct outside downwind from WSRF. Tower will sequence aircraft into outside downwind as conditions permit. When cleared, enter outside downwind at 6,100 ft MSL while visually clearing for traffic. Follow the pattern to straight-in transition. If unable to be sequenced in to the outside downwind, proceed VFR to Tula Peak.

6.7. VFR Recovery to Runway 22 (Figure 6.2). The primary VFR recovery point for Runway 22 is from the north via GAP. When authorized by ATC, aircraft may recover via GORDO from the south or east. See paragraph 6.13.8.3. for Runway 22 VFR pattern procedures.

Figure 6.2. Runway 22 VFR Recovery (Drawing not to scale).



6.7.1. Runway 22 Initial transition from north: report GAP at 6,100 ft MSL. From GAP, turn (approximate heading 160) to intercept 4-5 NM initial Runway 22 and report initial.

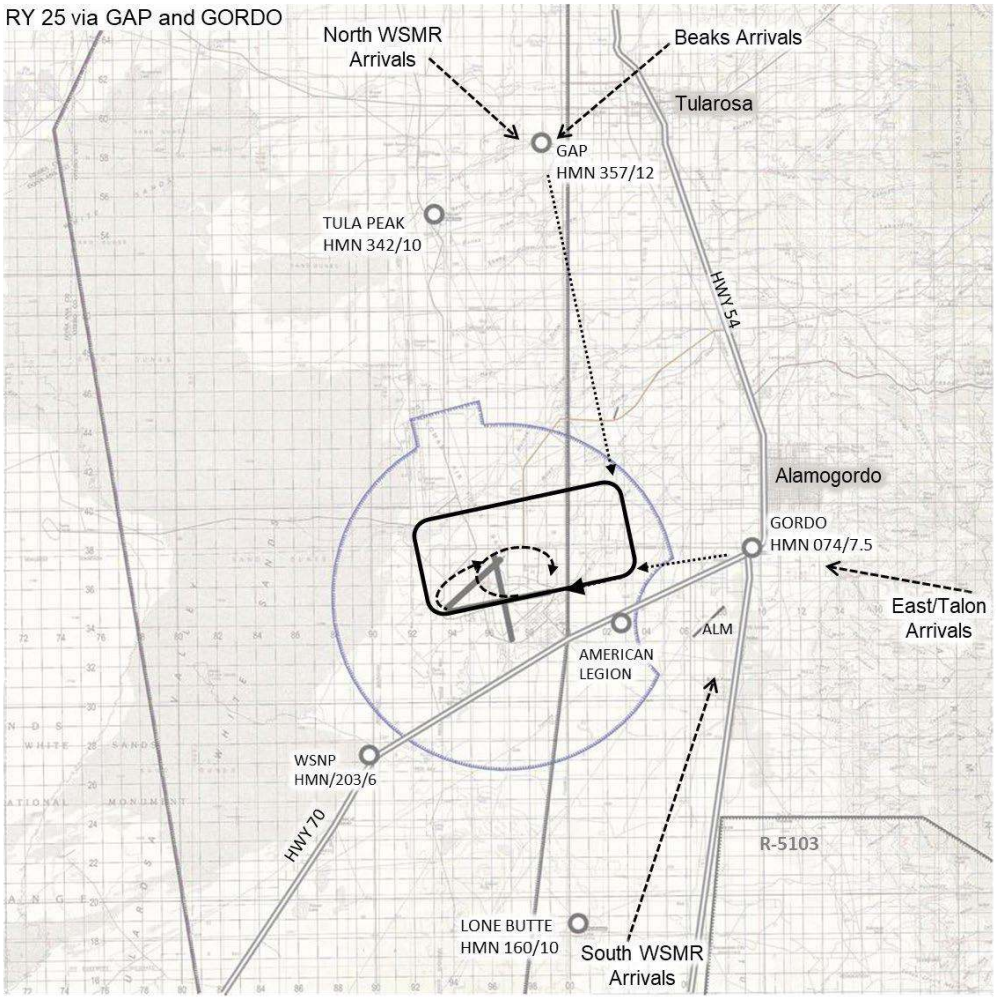
6.7.2. Runway 22 Initial transition from south or east: report GORDO at 6,100 ft MSL. Turn to intercept 4-5 NM initial Runway 22 and report initial. Use caution for KALM traffic.

6.7.3. Runway 22 Straight-In Transition from north: report GAP at 5,600 ft MSL. Turn (approximate heading 145) to intercept 7-DME base turn and report 5 NM final. Runway 22 Straight-In Transition from south or east: report GORDO at 6,100 ft MSL. Descend to 5,600 ft MSL and proceed to a 7-DME base turn and report 5 NM final.

6.7.4. Runway 22 Straight-In Transition from south or east: report GORDO at 6,100 ft MSL. Descend to 5,600 ft MSL and proceed to a 7-DME base turn and report 5 NM final.

6.8. VFR Recovery to Runway 25 (Figure 6.3.). The primary VFR recovery point for runway 25 is from the north via GAP. When authorized by ATC, aircraft may recover via GORDO from the south or east. See paragraph 6.13.8.5. for Runway 25 VFR pattern procedures.

Figure 6.3. Runway 25 VFR Recovery (Drawing not to scale).



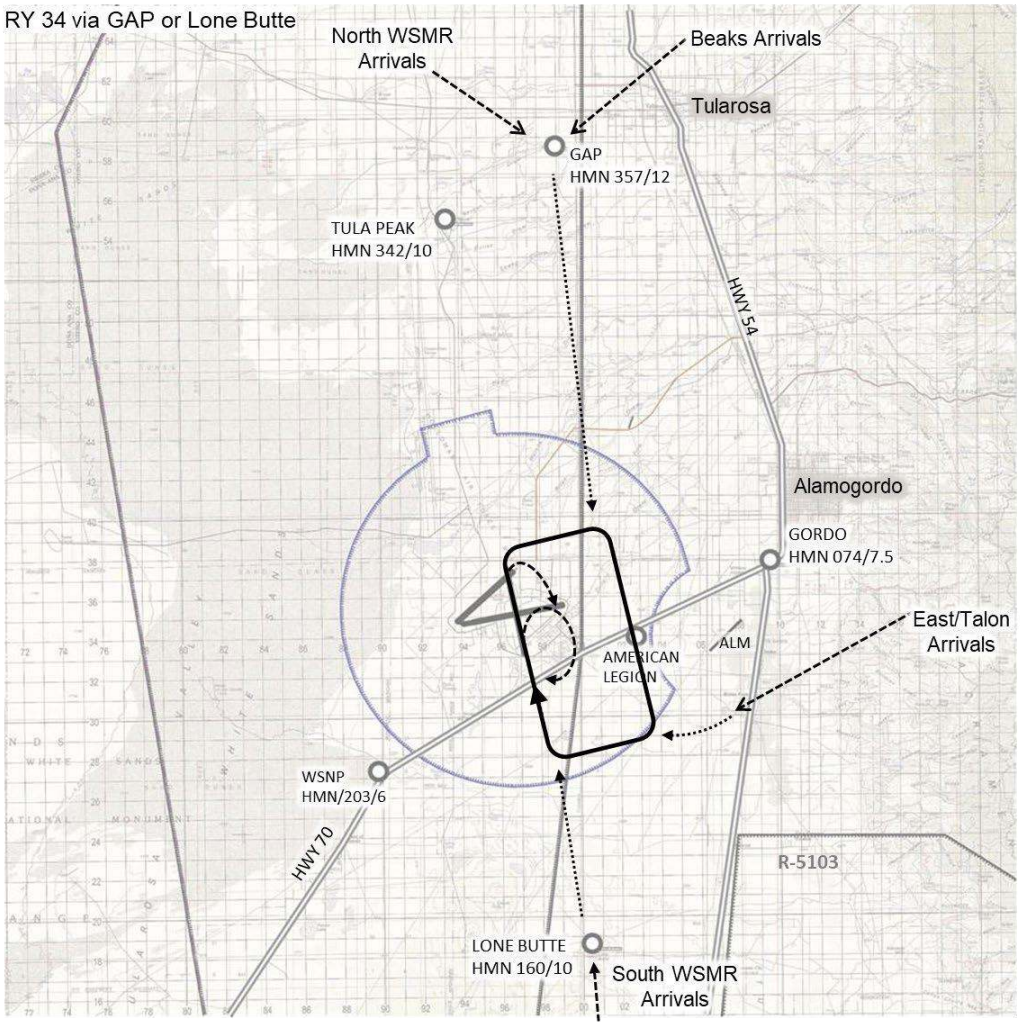
6.8.1. Runway 25 Initial transition from north: report GAP at 6,100 ft MSL. From GAP, turn (approximate heading 155) to intercept 4-5 NM 90-to-initial Runway 25 and report initial.

6.8.2. Runway 25 Initial transition from south or east: report GORDO at 6,100 ft MSL. Turn direct to the airfield and report initial. Use caution for KALM traffic.

6.8.3. Runway 25 Straight-In Transition is not allowed except for emergencies, hung, ordnance, check rides, or when approved by the SOF. Use caution for KALM traffic. Runway 25 Straight- in from the north: report GAP at 5,600 ft MSL, turn (approximate heading 145) to intercept 7 DME base and report 5 NM final. Runway 25 Straight-in from the south or east: report GORDO at 6,100 ft MSL, descend to 5,600 ft MSL and report 5 NM final.

6.9. VFR Recovery to Runway 34 (Figure 6.4.). The primary VFR recovery point for Runway 34 is from the north via GAP. When authorized by ATC, aircraft may recover via LONE BUTTE from the south or direct outside downwind from the east. See paragraph 6.13.8.2. for Runway 34 VFR pattern procedures.

Figure 6.4. Runway 34 VFR Recovery (Drawing not to scale).



6.9.1. Runway 34 Initial transition from the north: report GAP at 6,100 ft MSL. Proceed direct (approximate heading 160 degrees) outside downwind follow standard pattern procedures.

6.9.2. Runway 34 Initial transition from the south: report LONE BUTTE at 6,100 ft MSL and report initial.

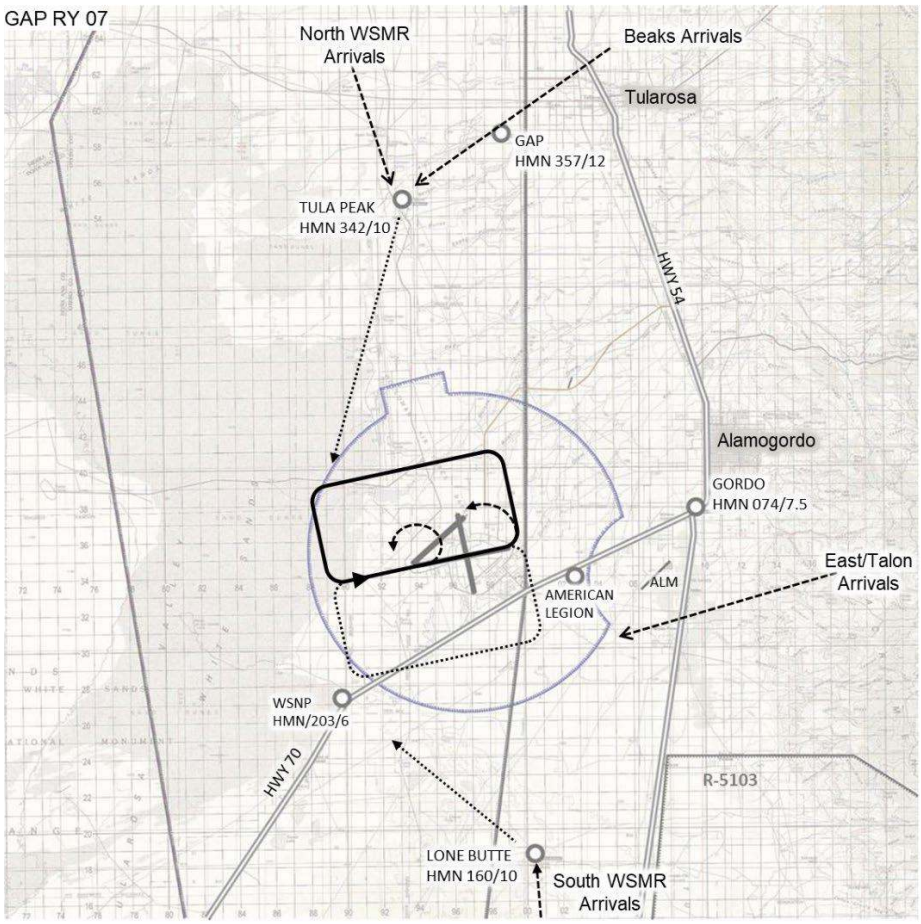
6.9.3. Runway 34 Initial transition from the east: proceed direct 90-to-initial at 6,100 ft MSL and follow standard pattern procedures. Remain south of KALM and outside R5103.

6.9.4. Runway 34 Straight-in Transition from south: report LONE BUTTE at 5,600 ft MSL and report 7 DME final.

6.9.5. Runway 34 Straight-in Transition from north: cross GAP at 6,100 ft MSL then fly heading 160 degrees to outside downwind, report descending straight-in.

6.10. VFR Recovery to Runway 07 (Figure 6.5.). The primary VFR recovery point for Runway 07 is from the north via TULA PEAK. When authorized by ATC, aircraft may recover via LONE BUTTE from the south or direct south outside downwind from the east. See paragraph 6.13.8.6. for Runway 07 VFR pattern procedures.

Figure 6.5. Runway 07 VFR Recovery (Drawing not to scale).



6.10.1. Runway 07 Initial transition from the north: report TULA PEAK at 6,100 ft MSL. Proceed (approximate heading 190 degrees) direct outside downwind and follow standard pattern procedures.

6.10.2. Runway 07 Initial transition from the south: report LONE BUTTE at 6,100 ft MSL. Proceed (approximate heading 300 degrees) direct right 90-to-initial, report 90-to-initial over WSNP Visitor Center.

6.10.3. Runway 07 Initial transition from the east: proceed direct the south outside downwind and follow standard pattern procedures. Remain south of KALM and outside R5103.

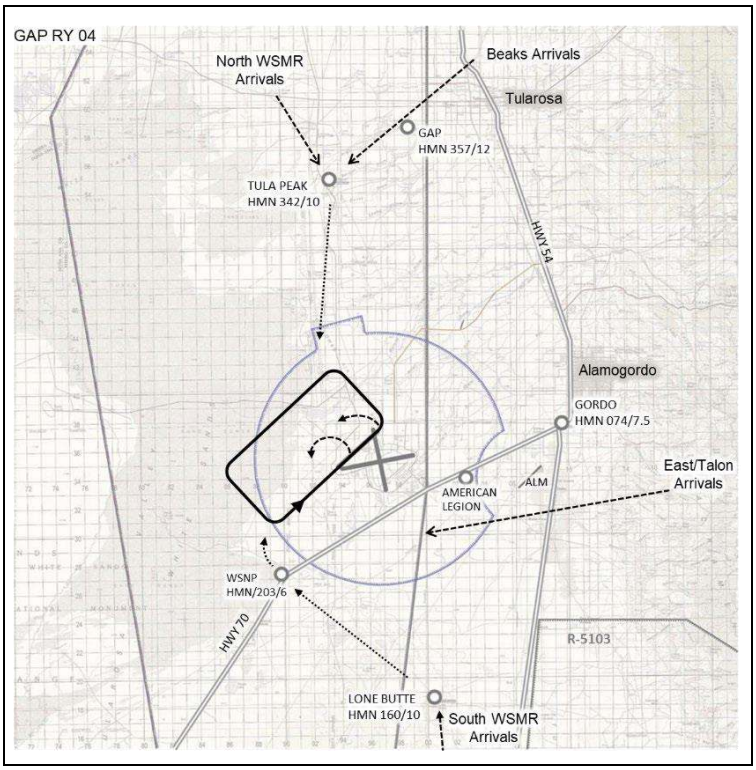
6.10.4. Runway 07 Straight-in transition from the north: report TULA PEAK at 5,600 ft MSL. Proceed (approximate heading 190 degrees) direct outside downwind and follow standard pattern procedures.

6.10.5. Runway 07 Straight-in transition from the south: report LONE BUTTE at 5,600 ft MSL. Proceed (approximate heading 290 degrees) direct an 8 DME base turn runway 07, report 6 DME final.

6.10.6. Runway 07 Straight-in transition from the east: proceed direct the south outside downwind at 5,600 ft MSL, remain south of KALM and outside R5103. Intercept 8 DME base turn runway 07, report 6 DME final.

6.11. VFR Recovery to Runway 04 (Figure 6.6.). The primary VFR recovery point for Runway 04 is from the north via TULA PEAK. When authorized by ATC, aircraft may recover via LONE BUTTE from the south or east. See paragraph 6.13.8.4. for Runway 04 VFR pattern procedures.

Figure 6.6. Runway 04 VFR Recovery (Drawing not to scale).



6.11.1. Runway 04 Initial transition from the north: report TULA PEAK at 6,100 ft MSL. Proceed (approximate heading 190 degrees) direct outside downwind and follow standard pattern procedures.

6.11.2. Runway 04 Initial transition from the south: report LONE BUTTE at 6,100 ft MSL. Proceed (approximate heading 300 degrees) direct right 90-to-initial, report 90-to-initial over WSNP Visitor Center.

6.11.3. Runway 04 Initial transition from the east: at 6,100 ft MSL, proceed direct and report right 90-to-initial over WSNP Visitor Center.

6.11.4. Runway 04 Straight-in Transition from the north: report TULA PEAK at 5,600 ft MSL. Proceed (approximate heading 190 degrees) direct outside downwind and follow standard pattern procedures.

6.11.5. Runway 04 Straight-in Transition from the south: report LONE BUTTE at 5,600 ft MSL. Proceed (approximate heading 290 degrees) direct an 8 DME base turn runway 04, report 6 DME final.

6.11.6. Runway 04 Straight-in Transition from the east: proceed direct WSNP Visitor Center at 6,100 ft MSL, remain south of KALM and outside R5103. After passing WSNP Visitor Center, descend to 5,600 ft MSL and intercept 8 DME base turn runway 04, report 6 DME final.

6.12. VFR Pattern Weather Minimums. VFR weather minimums – 1,500 ft ceiling and 3 SM visibility unless otherwise stated below.

6.12.1. Overhead Pattern – 3,000 ft AGL (7,100 ft MSL) ceiling and 3 SM visibility.

6.12.2. Rectangular Pattern – 3,000 ft AGL (7,100 ft MSL) ceiling and 3 SM visibility

6.12.3. Straight-In (Outside downwind) Pattern – 3,000 ft AGL (7,100 ft MSL) ceiling and 8 SM visibility.

6.12.3.1. Restricted Straight-in Pattern. When weather minimums are at least 2,000 ft AGL (6,100 ft MSL) ceiling and 8 SM visibility, an amended straight-in pattern may be flown at 5,600 ft MSL. The same outside downwind and rectangular pattern ground track will be used. There is no breakout available and no more than four aircraft may be in this pattern at any given time.

6.12.4. Periods of Civil Twilight are considered daytime. Refer to unit-specific instructions for additional guidance regarding use of Civil Twilight.

6.13. VFR Traffic Patterns. The following general pattern information applies to all runways. See paragraph 6.13.8. for runway specific pattern information. See Attachment 5 for QF-16 Drone pattern information. See Attachment 6 for helicopter specific pattern information. See Attachment 9 for MQ-9 specific pattern information. See Attachment 12 for F-16 specific pattern information.

6.13.1. Overhead Pattern. Pattern altitude (initial) is 6,100 ft MSL and airspeed is 300 KIAS.

6.13.1.1. Initial point for all runways is 3 NM from the approach end. At initial, pilots will report: "*(call sign), INITIAL, (type landing request)*".

6.13.1.1.1. Standard break is away from the West Ramp (Runway 16/34 east, Runway 22/04 25/07 north). Tower may change direction of breaks when necessary due to traffic conditions, active airspace, weather, etc. Tower will not change the direction of breaks when aircraft are inside initial unless necessary for safety of flight.

6.13.1.2. Tower will respond with "*(call sign) REPORT BASE*", may be followed with "*BREAK MIDFIELD/DEPARTURE END*", or with sequencing instructions "*BREAK TO FOLLOW*" or "*CARRY INITIAL STRAIGHT THROUGH*". If Tower does not respond with any of the above, carry initial straight through, DO NOT break. To carry initial straight through, remain at 6,100 ft MSL, continue to departure end then turn to enter the outside downwind. When turning base, fly the final turn to roll out on a 1 NM final at 300 ft AGL. Request to return to radar should be made NLT base turn.

6.13.1.3. Tactical Initial. Request tactical initial upon initial contact with WSRF. Maximum flight size will be four aircraft and maximum speed is 350 KIAS. Formation will be within 6,000 ft laterally unless otherwise coordinated. For formations instructed to carry initial straight-through, tactical initial is automatically terminated, and the flight should rejoin as elements as soon as practical.

6.13.1.4. Nighttime Overhead. 49 WG and 586th Flight Test Squadron (586 FLTS), aircraft may conduct night overhead and closed patterns provided they are conducted for training or syllabus requirements. All other aircraft must have 54 FG/CC or 54 FG Brickholder approval to conduct overhead patterns at night. See Attachment 9 for MQ-9 nighttime overhead procedures or see Attachment 12 for 54 FG F-16 nighttime overhead procedures. Tower has the authority to deny night overhead patterns.

6.13.1.4.1. Night overheads will only be flown to Runway 16, 34, 22, or 25. Runway 25 will only be used if the other runways are not available. The minimum operational airport lighting requirements are HIRL, PAPI, and Runway Threshold Lights.

6.13.1.4.2. Maximum of four aircraft in the night overhead/rectangular pattern.

6.13.1.4.3. Outside downwind not available for reentry at night.

6.13.1.4.4. All night breakouts will return to HMN Arrival for re-sequencing. When Tower advises an aircraft to "BREAKOUT", the following procedures apply:

6.13.1.4.4.1. Prior to break point: climb to 6,600 ft MSL, continue to 5 DME outbound, and contact HMN Arrival.

6.13.1.4.4.2. Inside the break point: climb to 6,600 ft MSL, expect suggested heading, and transfer to HMN Arrival for re-sequencing.

6.13.2. Rectangular (Closed Traffic) Pattern. The standard direction of turns is away from the West Ramp (Runways 16/34 east, Runways 22/04 and 25/07 north). Tower may change direction of the turns when necessary. The inside downwind ground track is displaced 1 NM from the runway. Pilots WILL NOT enter the rectangular pattern without Tower approval. The outside downwind ground track is displaced 3 NM from the runway (see paragraphs 6.13.4., 6.13.8.1.- 6.13.6.6.). Rectangular Pattern altitudes are listed in Table 6.1.

Table 6.1. Rectangular Pattern Altitudes and Airspeeds.

Type Aircraft	Altitude MSL	KIAS
Fighter	6,100 ft	300
Reciprocating / Turboprop / MQ-9	5,600 ft	As required
49 FSS/AC	5,100 ft	As required
Helicopters	4,600 ft	As required
Field Elevation 4,100 ft		

6.13.2.1. When turning base, pilots should fly their final turn to roll out on a 1 NM final at 300 ft AGL. Request to return to radar should be made NLT base turn. Initiate closed pattern only after instructed by Tower, normally at departure end of runway, or the extreme west end Taxiway Golf for Runway 25.

6.13.3. Protection of the 360 Degree Overhead Pattern. After completing approach, maintain at or below 5,600 ft MSL until departure end unless otherwise directed by Tower. Advise Tower of intentions (e.g., *request closed traffic, re-enter, re-enter straight-in*). For Runway 25, maintain at or below 5,600 ft MSL until west end of Taxiway Golf.

6.13.4. Outside Downwind. An outside downwind has been established for each runway and is used for pattern sequencing, VFR practice straight-in, pattern re-entry after breakout, return to WSRF for instrument pattern, or initial entry into the traffic pattern. Outside downwind ground track is displaced 3 NM from the runway and intercepts 90-to-initial approximately 45 degrees/5NM from the runway threshold (see paragraphs 6.13.6.1.– 6.13.6.6.).

6.13.4.1. Tower must include reason when directing re-entry (e.g., "*LOBO 11 re-enter due to straight-in traffic", "SUN 12 unable closed, re-enter for departures"*).

6.13.4.2. Aircrew may request re-entry during climb-out from low approach/or touch- and-go. Aircrew must include reason for re-entry (e.g., "*CORONA 13 request renter straight-in", "DANGER 14 request re-enter straight-in SFO").*

6.13.5. Go-Around and Breakout.

6.13.5.1. Standard Go-Around. For Tower issued or pilot-initiated Go-Around, the pilot will maintain at least 500 ft vertical separation from aircraft on the runway (DO NOT overfly the aircraft on the runway). Remain below 5,600 ft MSL until departure end or the west end of Taxiway Golf if using Runway 25. If an aircraft is cleared for a restricted low approach, ATC will issue the appropriate altitude to maintain.

6.13.5.2. Standard Breakout. Pattern breakouts will climb to 6,600 ft MSL in the direction of the pattern until clear of the outside downwind. Aircraft must be wings level prior to 90-to-initial and orbit at 6,600 ft MSL until approved by Tower to re-enter.

6.13.5.3. Tower must include the reason for the Go-Around or breakout when able.

6.13.6. Landing Clearance.

6.13.6.1. Gear reports. Pilots will report position, gear down, intentions, and runway.

6.13.6.2. Formations. Landing clearance for a flight lead is clearance for the entire flight to land. Aircraft after the flight lead will make gear down calls, which will be acknowledged with "Roger". Same type landings (full stop, low approach) are assumed for all members of the flight. If a succeeding flight member wants a clearance different than that of the flight lead, that individual will state intentions with the gear down call. Appropriate runway separation will be set by flights and monitored/enforced by Tower. Unless aircrew state "*with chase*", Tower will consider aircraft no longer in formation on subsequent landings.

6.13.7. MQ-9 Orbit Points. To relieve radio congestion during pattern operations, the MQ-9 orbit points are established for use by HMN-assigned MQ-9s as directed by Tower (see paragraph A9.2.5.).

6.13.7.1. When directed to orbit at a point, the MQ-9 must not fly any closer to any runway centerline than the "No-Closer" point. The orbits are elliptical, 1 NM wide by 1.5 NM long, and are aligned with the expected runway for use. These orbits are positioned to maintain 0.5 NM spacing from the edge of Class D airspace and 1.5 NM spacing from all extended centerlines.

6.13.7.2. Tower controllers will assign each aircraft in orbit a set altitude and include the assigned altitude during all traffic advisories.

6.13.8. Runway Specific Pattern Procedures and Diagrams.

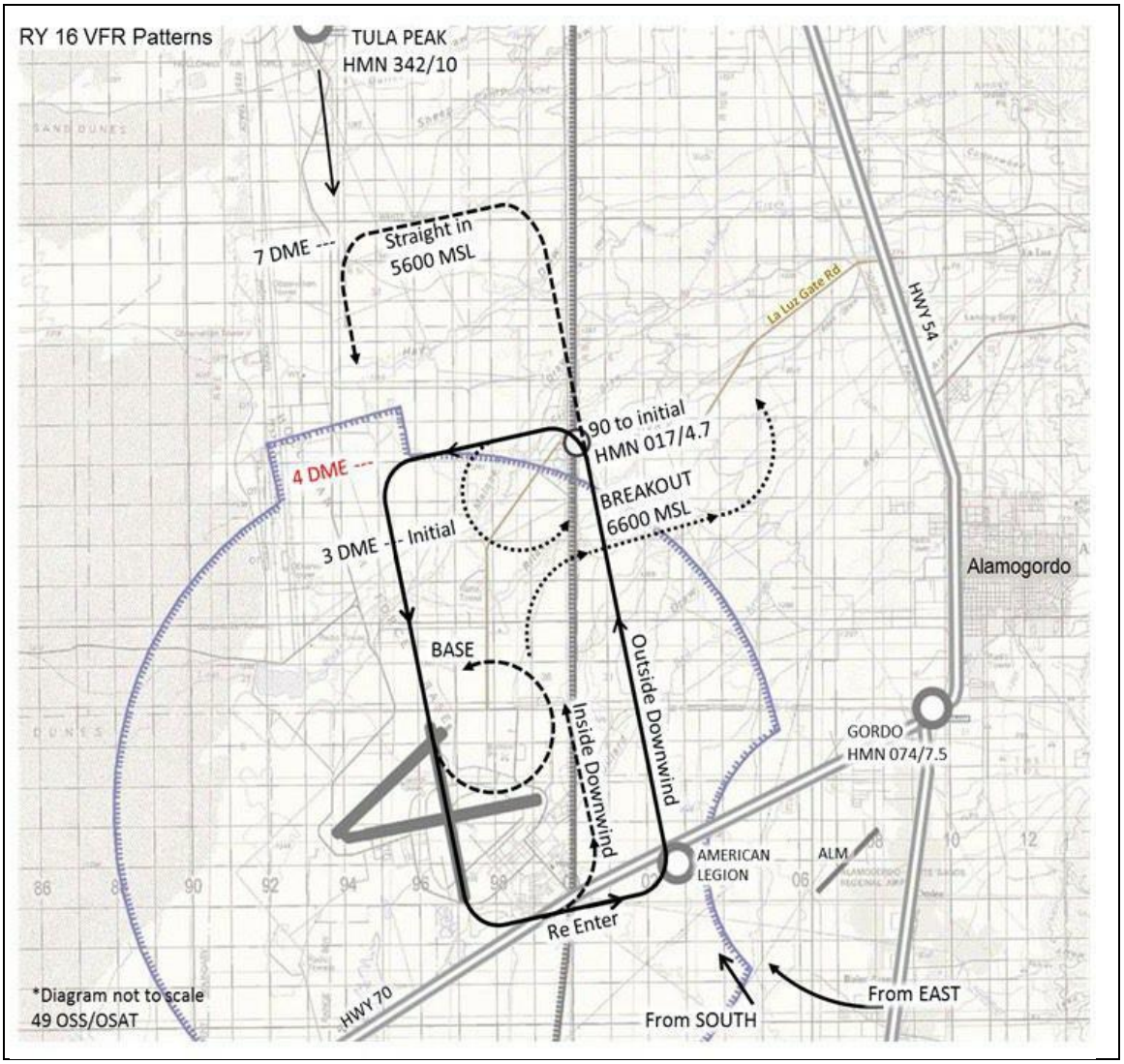
6.13.8.1. Runway 16 (Figure 6.7.) patterns are left traffic/turns/breaks.

6.13.8.1.1. Runway 16 Outside Downwind. The outside downwind ground track is displaced 3 NM east of the runway. Fly the outside downwind pattern ground track at 6,100 ft MSL just inside of the old American Legion heading 340 degrees. Pattern altitude is 6,100 ft MSL and airspeed is 300 KIAS. The 90-to-initial point is the La Luz gate.

6.13.8.1.2. Runway 16 Re-enter Initial. At departure end, initiate a left turn, maintain at or below 5,600 ft MSL until clear of the overhead/closed pattern. At 90- to-initial, pilots will report "*(call sign) 90-TO-INITIAL*".

6.13.8.1.3. Runway 16 Re-enter Straight-In. At 90-to-initial report "*(call sign) DESCENDING STRAIGHT-IN*.", descend to 5,600 ft MSL and slow to 250 KIAS. Extend 2 NM before turning base to intercept a 7 DME final. Report established on final with gear and intentions.

Figure 6.7. VFR Traffic Patterns for Runway 16 (Drawing not to scale).



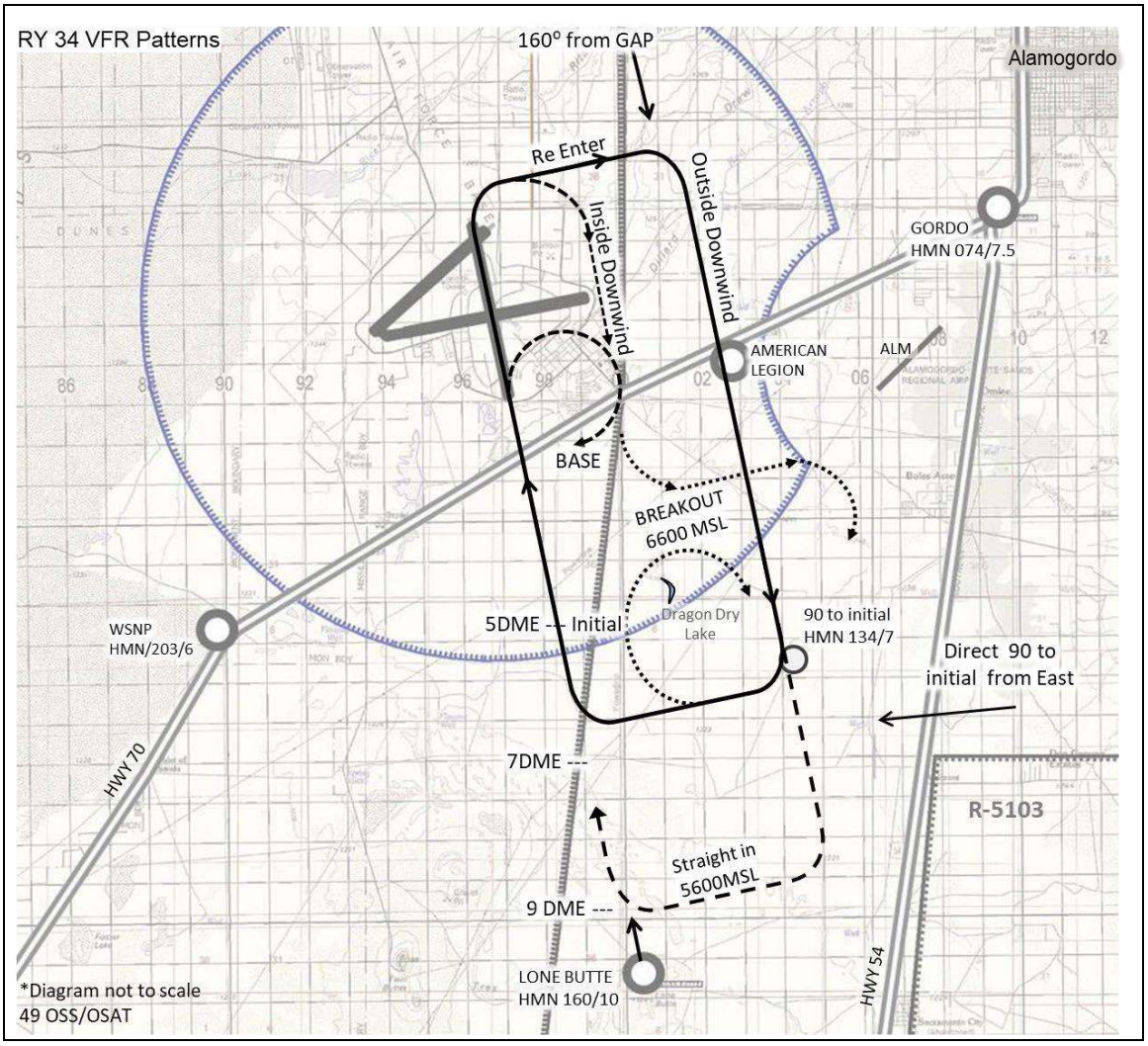
6.13.8.2. Runway 34 (Figure 6.8.) patterns are right turns/traffic/breaks.

6.13.8.2.1. Runway 34 Outside Downwind. The outside downwind ground track is displaced 3 NM east of the runway. Fly the outside downwind pattern ground track at 6,100 ft MSL just inside of the old American Legion heading 160 degrees, airspeed is 300 KIAS. The 90-to-initial point (7 DME) is 2 NM south-east of Dragon Dry Lake.

6.13.8.2.2. Runway 34 Re-enter Initial. At departure end, initiate a right turn, maintain at or below 5,600 ft MSL until clear of the overhead/closed pattern. At 90- to-initial, pilots will report "*(call sign) 90-TO-INITIAL*".

6.13.8.2.3. Runway 34 Re-enter Straight-In. At 90-to-initial report "*(call sign) DESCENDING STRAIGHT-IN*", descend to 5,600 ft MSL and slow to 250 KIAS. Extend 2 NM before turning base to intercept a 9 DME final. Report established on final with gear and intentions.

Figure 6.8. VFR Traffic Patterns for Runway 34 (Drawing not to scale).



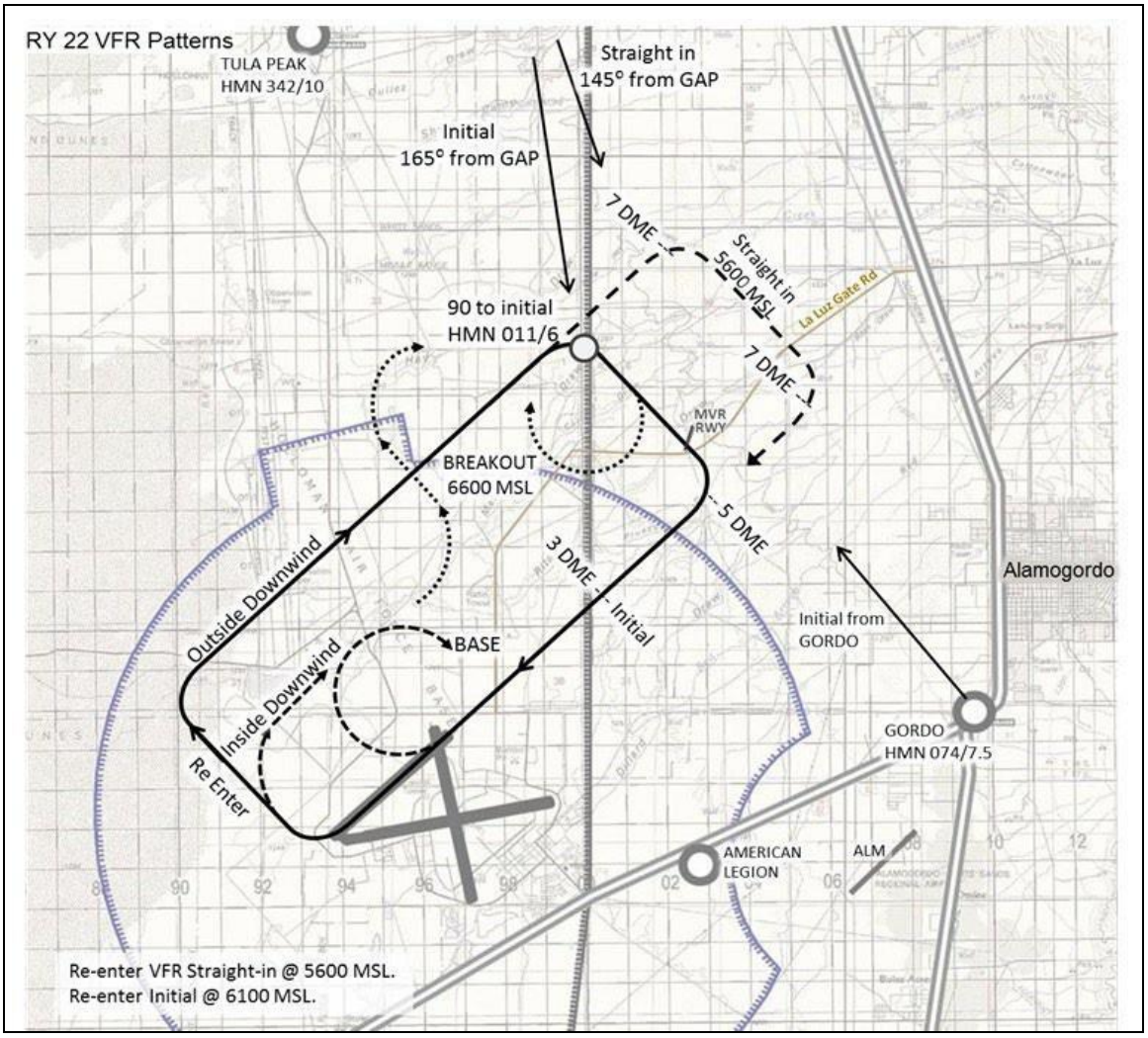
6.13.8.3. Runway 22 (Figure 6.9.) patterns are right turns/traffic/breaks.

6.13.8.3.1. Runway 22 Outside Downwind. The outside downwind ground track is displaced 3 NM north of the runway. Fly the outside downwind pattern ground track at 6,100 ft MSL heading 040 degrees, airspeed is 300 KIAS. 90-to-initial is abeam 7NM1, turn at 5 DME.

6.13.8.3.2. Runway 22 Re-enter Initial. At departure end, initiate a right turn, maintain at or below 5,600 ft MSL until clear of the overhead/closed pattern. At 90- to-initial, pilots will report "*(call sign) 90-TO-INITIAL*".

6.13.8.3.3. Runway 22 Re-enter Straight-In. Fly the outside downwind pattern ground track at 5,600 ft MSL to deconflict the VFR entry from GAP at 6,100 ft MSL. 90-to-initial is abeam 7NM1 at 5 DME. At the 90-to-initial point, report "*(call sign), APPROACHING BASE*" and reduce speed to 250 KIAS. Extend 2 NM before turning base to provide a 7 DME final. Report established on final with gear and intentions.

Figure 6.9. VFR Traffic Patterns for Runway 22 (Drawing not to scale).



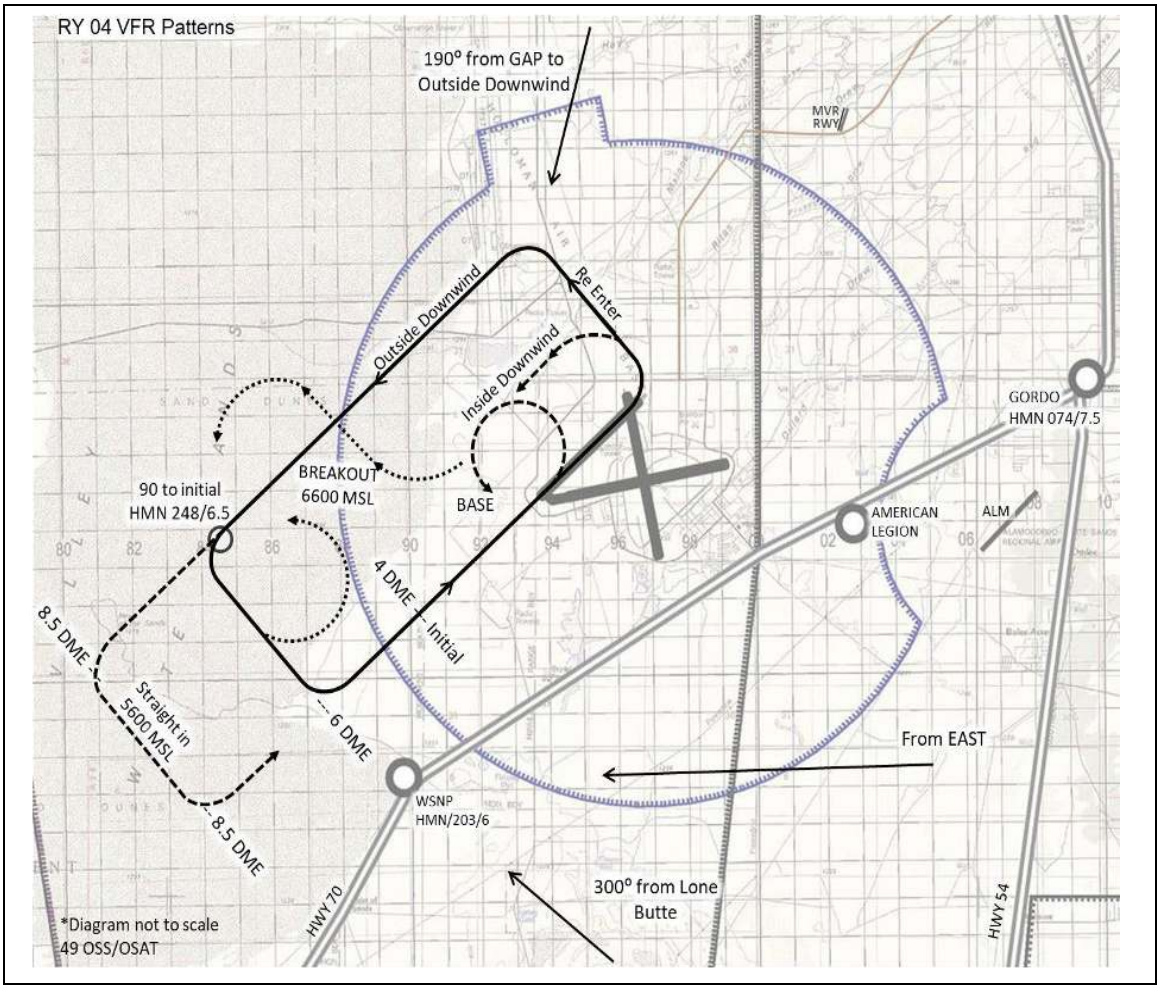
6.13.8.4. Runway 04 (Figure 6.10.) patterns are left turns/traffic/breaks. Use caution, PAPI/Visual Approach Slope Indicator (VASI) not available.

6.13.8.4.1. Runway 04 Outside Downwind. The outside downwind ground track is displaced 3 NM north of the runway. Fly the outside downwind pattern ground track at 6,100 ft MSL heading 220 degrees, airspeed is 300 KIAS. There is no 90-to-initial reference point, turn at 6.5 DME.

6.13.8.4.2. Runway 04 Re-enter Initial. At departure end, initiate a left turn, maintain at or below 5,600 ft MSL until clear of the overhead/closed pattern. At 90- to-initial, pilots will report "*(call sign) 90-TO-INITIAL*".

6.13.8.4.3. Runway 04 Re-enter Straight-In. At 90-to-initial report "*(call sign) DESCENDING STRAIGHT-IN.*", descend to 5,600 ft MSL and slow to 250 KIAS. Extend 2 NM before turning base to intercept an 8.5 DME final. Report established on final with gear and intentions.

Figure 6.10. VFR Traffic Patterns for Runway 04 (Drawing not to scale).



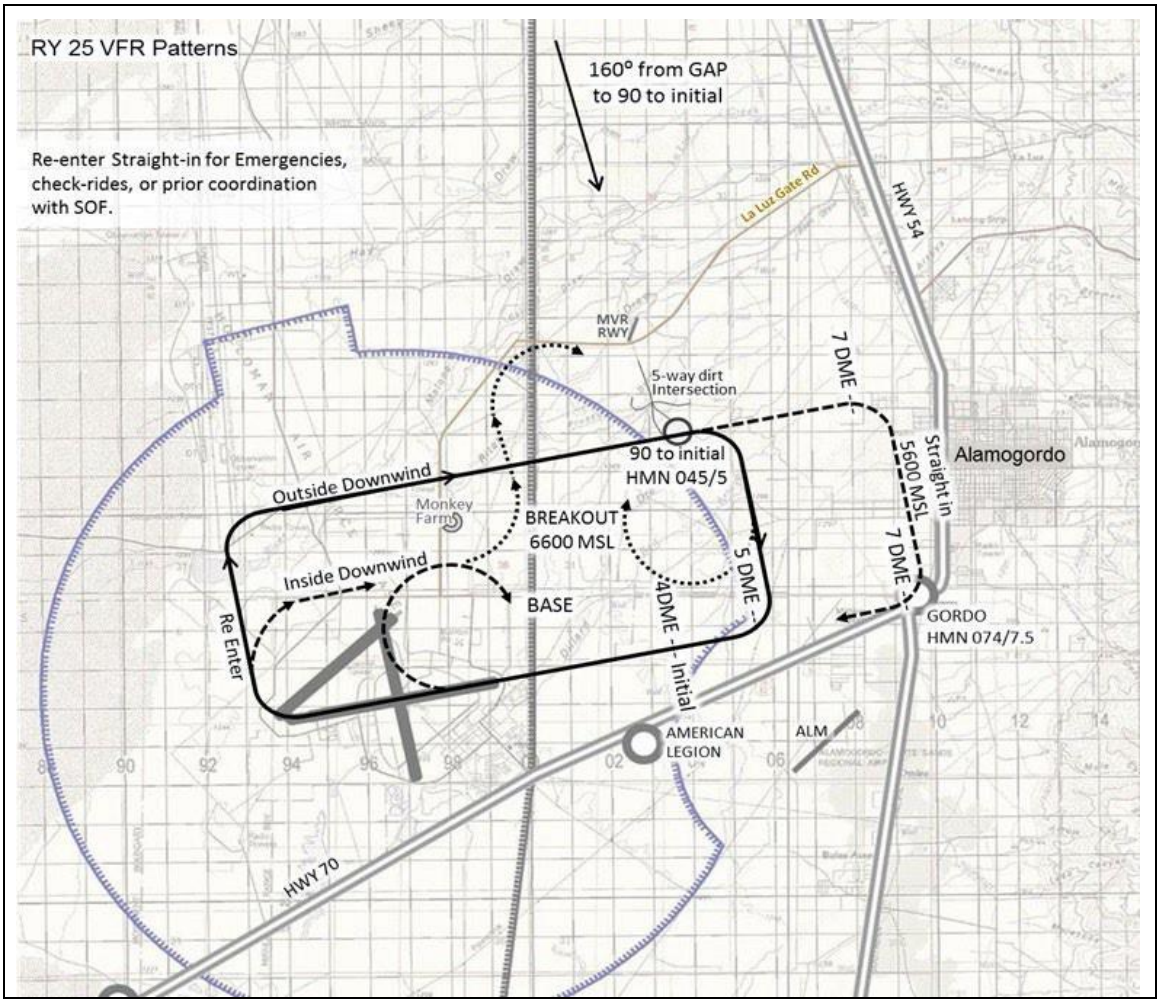
6.13.8.5. Runway 25 (Figure 6.11.) patterns are right turns/traffic/breaks.

6.13.8.5.1. Runway 25 Outside Downwind. The outside downwind ground track is displaced 3 NM north of the runway. Fly the outside downwind pattern ground track at 6,100 ft MSL heading 070 approximately 1 NM north of the Monkey Farm. Base leg point is over a ranch and road intersection at 5 DME. DO NOT extend base turn due to overflight of the city of Alamogordo. Remain vigilant for initial entries.

6.13.8.5.2. Runway 25 Re-enter Initial. At departure end, initiate a right turn, maintain at or below 5,600 ft MSL until west end of Taxiway Golf, clear of the overhead/closed pattern. At 90-to- initial, pilots will report "*(call sign) 90-TO- INITIAL*".

6.13.8.5.3. Runway 25 Re-enter Straight-In. VFR Practice Straight-ins Runway 25 are only allowed for emergencies, check rides, or with prior coordination with the SOF. Fly the outside downwind pattern ground track at 5,600 ft MSL to de- conflict the VFR entry from GAP at 6,100 ft MSL. At the 90-to-initial point, report "*(call sign), APPROACHING BASE*" and reduce speed to 250 KIAS. Extend 2 NM before turning base to provide a 7 DME final. Report established on final with gear and intentions.

Figure 6.11. VFR Traffic Patterns for Runway 25 (Drawing not to scale).



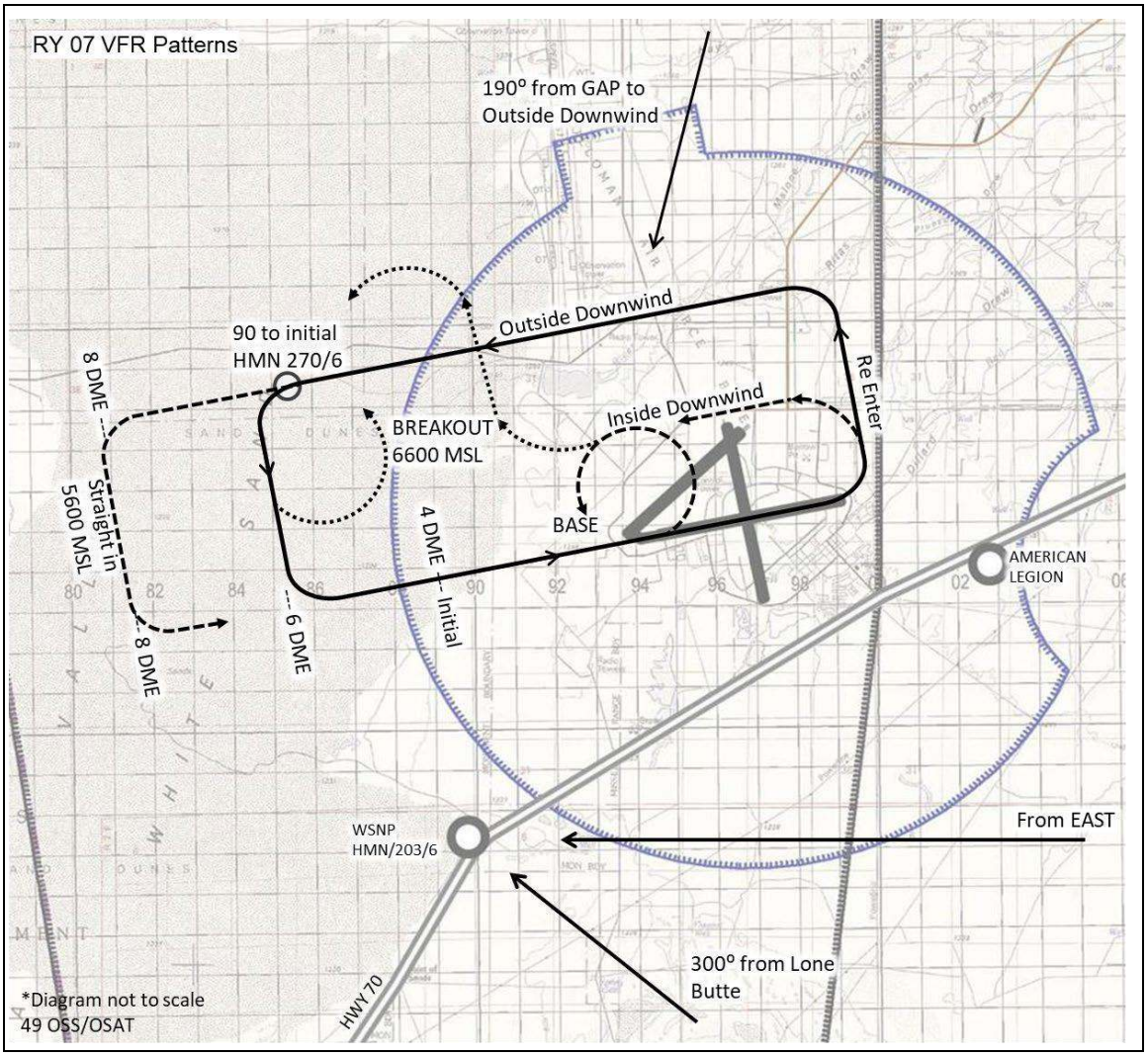
6.13.8.6. Runway 07 (Figure 6.12.): Runway 07 patterns are left turns/traffic/breaks. Avoid landing on Taxiway Golf. Use caution; PAPA/VASI not available.

6.13.8.6.1. Runway 07 Outside Downwind. The outside downwind ground track is displaced 3 NM north of the runway. Fly the outside downwind pattern ground track at 6,100 ft MSL heading 250 degrees, airspeed is 300 KIAS. There is no 90-to-initial reference point, turn at 7 DME.

6.13.8.6.2. Runway 07 Re-enter Initial. At departure end, initiate a left turn, maintain at or below 5,600 ft MSL until clear of the overhead/closed pattern. At 90- to-initial, pilots will report "*(call sign) 90-TO-INITIAL*".

6.13.8.6.3. Runway 07 Re-enter Straight-In. At 90-to-initial report "*(call sign) DESCENDING STRAIGHT-IN.*", descend to 5,600 ft MSL and slow to 250 KIAS. Extend 2 NM before turning base to intercept an 8 DME final. Report established on final with gear and intentions.

Figure 6.12. VFR Traffic Patterns for Runway 07 (Drawing not to scale).



6.14. SFO. Only locally assigned 54 FG F-16, 82 ATRS QF-16, and 49 OG MQ-9s are authorized to conduct SFO approaches at HMN; see Attachment 9 for MQ-9 SFO and Attachment 12 for F-16/QF-16 SFO. Other units requesting SFO operations must establish an LOA with the 54 FG prior to employment.

6.15. Reduced Same Runway Separation (RSRS) Procedures. RSRS criteria are used for AETC assigned, 82 ATRS, 704 TG fighter type, and 49 OG aircraft IAW Tables 6.2. - 6.6. All other non-AETC aircraft officially deployed to HMN will be required to sign a LOA with the 54 FG/CC or 54 FG Brickholder and obtain appropriate training prior to participating in RSRS. Elements within a flight are responsible for their own runway separation.

6.15.1. Pilot-in-command or Tower may refuse RSRS. Additionally, RSRS WILL NOT be applied to the following aircraft or situations.

6.15.1.1. RSRS will not be applied to emergency aircraft.

6.15.1.2. RSRS will not be applied behind any aircraft with a deployed drag chute.

6.15.1.3. RSRS will not be applied to aircraft cleared for the *"option"*.

6.15.1.4. RSRS will not be applied to SFO Touch-and-Go behind a Full Stop.

6.15.1.5. RSRS will not be applied when runway braking action of less than medium is reported or whenever weather conditions are conducive to deteriorating runway conditions.

Table 6.2. Separation Touchdown Minima for Same Fighter Type Aircraft.

Type Landing	Behind	RSRS
Full Stop	Full Stop / Touch-and-Go / Low Approach	3,000 ft
Touch-and-Go	Touch-and-Go / Low Approach	3,000 ft
Low Approach*	Full Stop / Low Approach	3,000 ft
Low Approach	Touch-and-Go	6,000 ft
Touch-and-Go*	Full Stop	6,000 ft
Note: A Touch-and-Go behind a Full Stop is authorized with 6,000 ft RSRS during daylight hours, VFR weather, and dry runway conditions. * Low Approach or Touch-and-Go behind a Full Stop: For all situations involving a Low Approach or Touch-and-Go behind a Full Stop, DO NOT overfly aircraft on the runway.		

Table 6.3. Separation Touchdown Minima for Dissimilar Fighter Type Aircraft.

Type Landing	Behind	RSRS
Full Stop	Full Stop / Touch-and-Go / Low Approach	6,000 ft
Low Approach*	Full Stop / Touch-and-Go / Low Approach	6,000 ft
Touch-and-Go*	Full Stop / Touch-and-Go / Low Approach	6,000 ft
Note: A Touch-and-Go behind a Full Stop is authorized with 6,000 ft RSRS, during daylight hours, VFR weather, and dry runway conditions. * Low Approach or Touch-and-Go behind a Full Stop: For all situations involving a Low Approach or Touch-and-Go behind a Full Stop, DO NOT overfly aircraft on the runway.		

Table 6.4. Separation Touchdown Minima for Fighter Type Aircraft Special Conditions.

Type Landing	Behind	RSRS
Similar Full Stop	Similar **Formation** Full Stop	6,000 ft
Dissimilar Full Stop	Dissimilar **Formation** Full Stop	8,000 ft
Nighttime Similar Full Stop / Low Approach	**Nighttime** Similar Full Stop / Touch-and-Go / Low Approach	6,000 ft
Nighttime Dissimilar Full Stop / Low Approach	**Nighttime** Dissimilar Full Stop / Touch-and-Go / Low Approach	8,000 ft
Wet Runway Full Stop*	Full Stop	6,000 ft
Note: *RSRS must not be applied to all other combinations during wet runway operations. In this situation, ATC will apply FAA same runway separation (SRS) IAW FAAO JO 7110.65.		

Table 6.5. Separation Touchdown Minima for MQ-9.

		Lead MQ-9		
		Full Stop	**Touch-and-Go**	**Low Approach**
Trail MQ-9	**Full Stop**	6,000 ft	FAA SRS	
	Touch-and-Go	6,000 ft**	FAA SRS	
	Low Approach	6,000 ft	FAA SRS	
*****When the lead MQ-9 is using "Auto Land" capability, increase spacing to 8,000 ft. ******Touch-and-Go behind a Full Stop is authorized with 6,000 ft RSRS, during daylight hours, VFR weather, and dry runway conditions.				

Table 6.6. Separation Touchdown Minima for MQ-9 behind Fighter Type Aircraft.

		Lead Fighter Type	
		Full Stop	**Touch-and-Go or Low Approach**
Trail MQ-9	**Full Stop**	6,000 ft	FAA SRS & Wake Turbulence
	Touch-and-Go	8,000 ft*	FAA SRS & Wake Turbulence
	Low Approach	6,000 ft	FAA SRS & Wake Turbulence
*****Touch-and-Go behind a Full Stop is authorized with 8,000 ft RSRS, during daylight hours, VFR weather, and dry runway conditions. **Note:** RSRS NOT authorized when MQ-9 is the lead aircraft (e.g., F-16 Full Stop behind MQ-9).			

6.15.2. Wake Turbulence. Pilots are responsible for wake turbulence separation when maintaining visual separation or operating under VFR. When wake turbulence is expected due to light winds (<5 knots) or light (3-10 knots) tailwind, pilots will increase same type spacing to 8,000 ft (increase brake spacing from 5 seconds to 8 seconds).

6.15.2.1. The SOF and/or Tower WS will determine when increased wake turbulence procedures are in effect. During periods of extended (more than 1 hour) light wind or light tailwind conditions, an advisory "*INCREASED SPACING PROCEDURES ARE IN EFFECT, WIND (*direction and velocity or *CALM)*" will be included in the ATIS broadcast. During variable wind conditions, the advisory will be issued directly to aircrew.

6.15.2.2. Tower will not apply additional separation criteria to aircraft with the potential to experience increased wake turbulence effects. When operating under IFR or under ATC instructions, controllers must ensure FAA or AETC wake turbulence separation.

Chapter 7

INSTRUMENT FLIGHT RULES (IFR) PROCEDURES

7.1. IFR Departures. IFR Departure procedures are listed in Table 7.1.

Table 7.1. IFR Departures.

IFR Procedure	RY 16	RY 34	RY 07	RY 25	RY 22	RY 04
Cloud Four A	✓	NA	NA	✓	✓	NA
Cloud Four B (RNAV)*	✓	NA	NA	✓	✓	NA
DIEZL One	✓	✓	NA	NA	NA	✓
IZZYY One (RNAV)*	✓	NA	NA	✓	✓	NA
OHDEE One	✓	✓	NA	NA	NA	✓
Local Climb-Out	✓	✓	NA	✓	✓	NA
DVA	✓	✓	NA	✓	✓	✓
*RNAV = Area Navigation						

7.1.1. IFR departures are not authorized from Runway 07.

7.1.2. Restrictions: When R5107D is active, IFR departures may not be authorized from Runway 22 and Runway 25 due to separation requirements associated with SUA. If WSRF cannot approve an IFR departure, Clearance Delivery will advise, *"DUE TO RESTRICTED AIRSPACE 5 NM WEST OF THE FIELD, IFR DEPARTURE IS NOT AUTHORIZED SAY INTENTIONS..."*

7.1.2.1. If instructed to depart VFR turn as directed by ATC to avoid the eastern boundary of these areas (approximately 5 DME west of HMN), DO NOT cross the extended centerline of Runways 16 or 34 until above 6,600 ft MSL and proceed VFR.

7.1.2.2. If an IFR pick-up is required, proceed on a northeasterly heading to CLOUD. Expect IFR pickup when clear of the restricted areas and above the applicable MVAs. If weather dictates Instrument Meteorological Conditions (IMC) flight, expect a departure from Runway 16 or Runway 34.

7.1.3. See paragraphs 3.5.2.3. for MVA and 3.5.2.4. for DVA guidelines.

7.1.4. Local Climb-out (see paragraph 7.7.) IAW DVA procedures, may be requested with Clearance Delivery on initial departure for IFR pattern delays, expedited routing, active airspace avoidance, etc.

7.2. Non-Standard Formation Departures. Advise Departure Control of non-standard formation. ATC can expect all F-16 departures to be non-standard. Lead aircraft will squawk the Identification Friend or Foe (IFF) mode 3C code assigned by ATC. Last trailing flight element will squawk 4000 or as assigned by ATC until rejoined to standard formation. Flight leads will advise ATC once standard formation is achieved.

7.2.1. Upon entering any Holloman training area (i.e., Lava, Mesa, Yonder, Red Rio, Oscura, Wiley East, Wiley West, Centennial Fly Area, McGregor North, Beak MOAs/ATCAAs) formation flights must squawk in sequence, when issued a local squawk beginning with 02XX or 03XX, assigned by ATC/Clearance Delivery. Formation flights must continue to squawk while working within the training areas. Upon completion and rejoining as a standard formation flight, trailing elements will squawk standby and lead aircraft will squawk the IFF mode 3C code assigned prior to departure.

7.2.2. Formation flights working Wiley East/West prior to entry into the NAS (e.g., ROW/ABQ/TALONS) assigned a NAS code will be issued additional squawks by ATC as needed, upon range entry clearance into WILEY East/West. At no time will trailing flight elements squawk in sequence when the formation flight is assigned a NAS code.

7.3. IFR Arrivals.

7.3.1. IFR arrivals can expect radar vectors from WSRF for an instrument approach to the active landing runway. Instrument arrival procedures are published for Runways 16, 22, and 34 in DoD FLIP *Low Altitude US, Vol. 6* and DoD FLIP *High-Altitude US, Southwest*.

7.3.2. Pilots will check in with ATIS code, type approach and landing requested, and any special requirements or information. Special requirements or information includes non-standard formation, radar-in-trail spacing, circling approach requests, practice missed approach, multiple practice instrument patterns, SFO request, etc.

7.3.3. Pilots requesting to conduct practice instrument approaches under VFR will execute the following (does not apply to MQ-9s):

7.3.3.1. If approaching HMN from a working area, use the following phrase with HMN Approach: "*Request vectors for VFR practice approach for (desired approach) at (desired altitude)*."

7.3.3.2. If multiple approaches are desired, use the following phrase with HMN Approach: "*Request VFR practice approach for (desired approach) at (desired altitude) followed by local climb-out VFR at (desired altitude)*." Pilots can expect to be switched to HMN Arrival for radar vectors.

7.3.3.3. Pilots may also depart the Tower VFR pattern for a VFR practice approach by making the following request with Tower: "*Request local climb-out VFR at (desired altitude) for a VFR practice approach.*" Pilots can expect to be switched to HMN Arrival for radar vectors.

7.3.3.3. If the local climb-out ground track is not available due to airspace restrictions, expect clearance for a VFR turnout. Pilots will fly toward the base leg of their intended approach by way of their own VFR navigation. Once the airspace restriction is no longer a factor (i.e., in the opposite direction of the current ground track) pilots will request practice approaches IAW the sub-paragraphs above.

7.3.4. Alternate Missed Approach. The published missed approach holding patterns at JOBAM (HMN R-320/12-17 DME) may not be available because of R5107D and/or R5107B activity. During these times, WSRF will advise aircrew and assign Local Climb-out or other alternate missed approach instructions. E.g., "*Published missed approach procedure not available, in the event of missed approach execute local climb-out runway one six east downwind."*

7.3.5. TACAN-Out Procedures.

7.3.5.1. When HMN TACAN DME is not available or unreliable for ILS approaches at HMN (i.e., TACAN out of service or not certified), radar is required for pilots to execute ILS approaches. Aircrew can expect to be provided radar navigational guidance while flying published ILS procedures annotated with the term "RADAR REQUIRED".

7.3.5.2. It is not feasible for ATC to provide descent instructions at every intermediate stepdown fix. The following options are available for pilots to legally and safely recover to HMN.

7.3.5.2.1. Weather conditions allow pilots to get to VMC while being radar vectored at or above the MVA at which point pilots can cancel IFR and proceed to the field in VMC. Pilots will plan accordingly and carry extra fuel during recovery if required.

7.3.5.2.2. Weather conditions in the restricted airspace (WSMR only) allows a descent below the Minimum Safe Altitude (MSA) specified for that airspace at which point pilots can proceed to the field in VMC. Reference 49 WG IFG for local flying area MSAs or specific instrument approach plate MSA information.

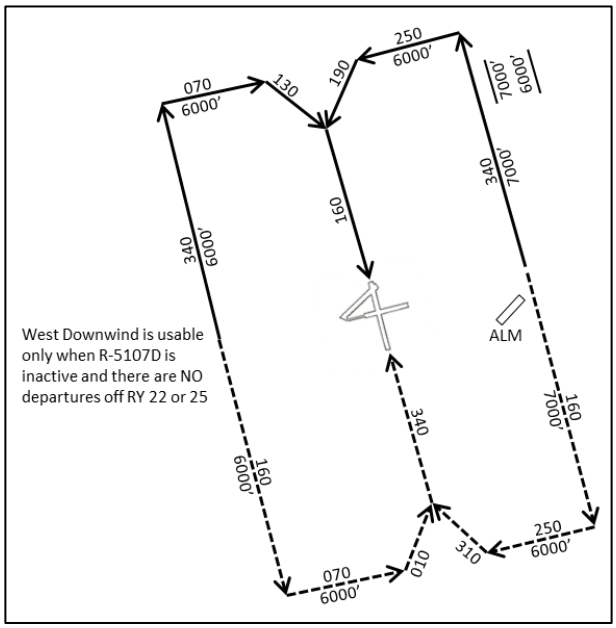
7.3.5.2.3. Weather conditions allow for pilots to get established on an approach and remain VMC until the final approach fix or glide slope intercept altitude at which point pilots can continue the approach until reaching individual Pilot Weather Category (PWC) minimums. At a minimum, for pilot situational awareness, ATC will inform pilots when at 15 DME, 10 DME, and over the final approach fix.

7.3.5.2.4. In all cases, the SOF will work closely with the 54 OSS/OSW and squadron Ops Sups to develop a plan that will be approved by the 54 FG/CC or 54 FG Brickholder. If none of the three options are available, aircraft will be placed on weather hold. If any of the above options are available, but weather is forecast to eliminate all options (+/-1 hour of proposed landing) an appropriate alternate will be declared by the SOF. Pilots arriving at HMN with none of the above options will divert upon reaching divert fuel.

7.4. Radar Traffic Patterns.

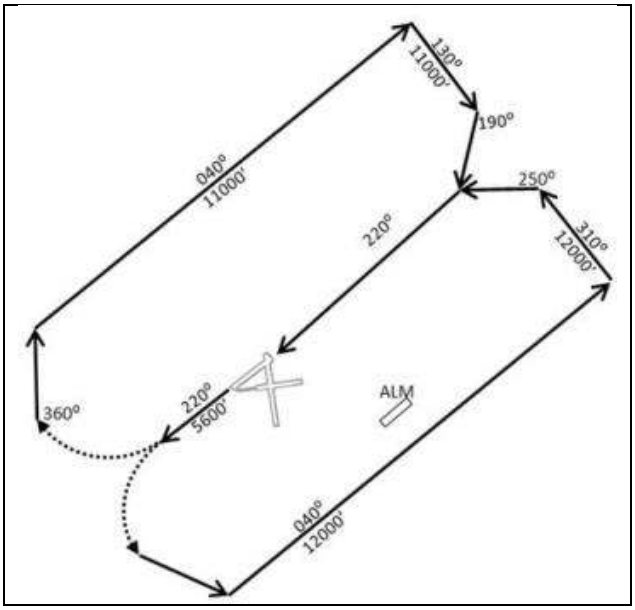
7.4.1. Runway 16/34 radar traffic patterns are normally configured east of HMN (Figure 7.1). A west radar traffic pattern is available only when R5107D is inactive and there are no departures from Runway 22 or Runway 25.

Figure 7.1. Runway 16/34 Radar Traffic Patterns (Drawing not to scale).



7.4.2. Runway 22 radar traffic pattern may be configured to the northeast or southeast of HMN (Figure7.2.). Aircrew can expect vectors to intercept final and be established by 15 DME final; however, aircrews may request vectors to be established inside of 15 DME final.

Figure 7.2. Runway 22 Radar Traffic Patterns (Drawing not to scale).



7.5. Practice Circling Approach. Practice circling approaches in VFR conditions may be flown to any runway with published circling minimums. Once approved, begin circling between 5 and 2 NM final and maintain at or below 5,600 ft MSL during the circling approach maneuver (see Figures 7.3. – 7.4.). Comply with the published circling approach restrictions for the procedure being flown. Transient aircraft may conduct practice circling approaches on a non- interference basis. Tower is the final approval authority for any practice circling approaches.

Figure 7.3. Practice Circling Approach Runway 22 (Drawing not to scale).

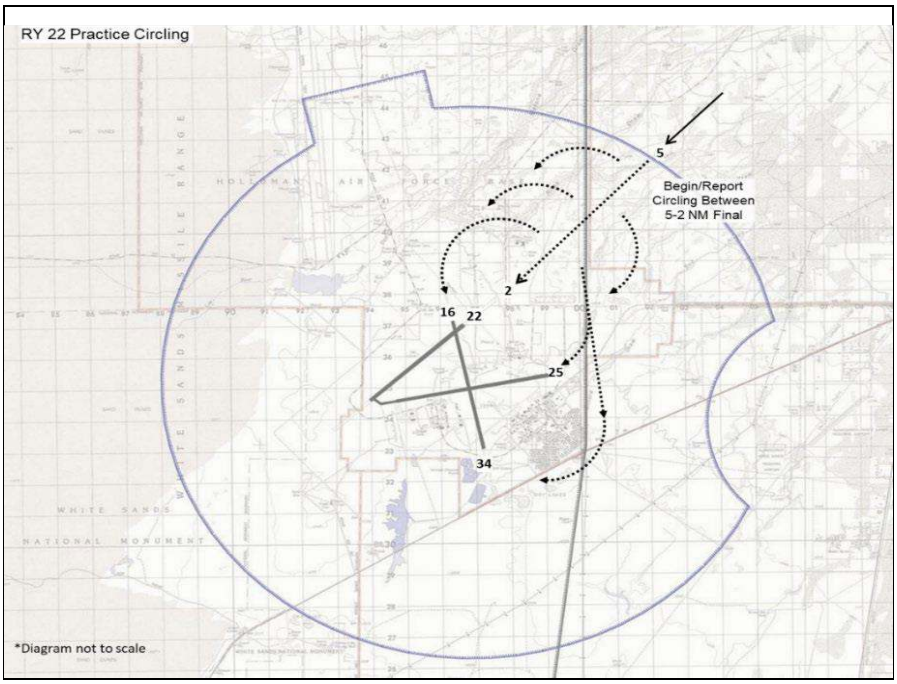


Figure 7.4. Practice Circling Approach Runway 16 (Drawing not to scale).

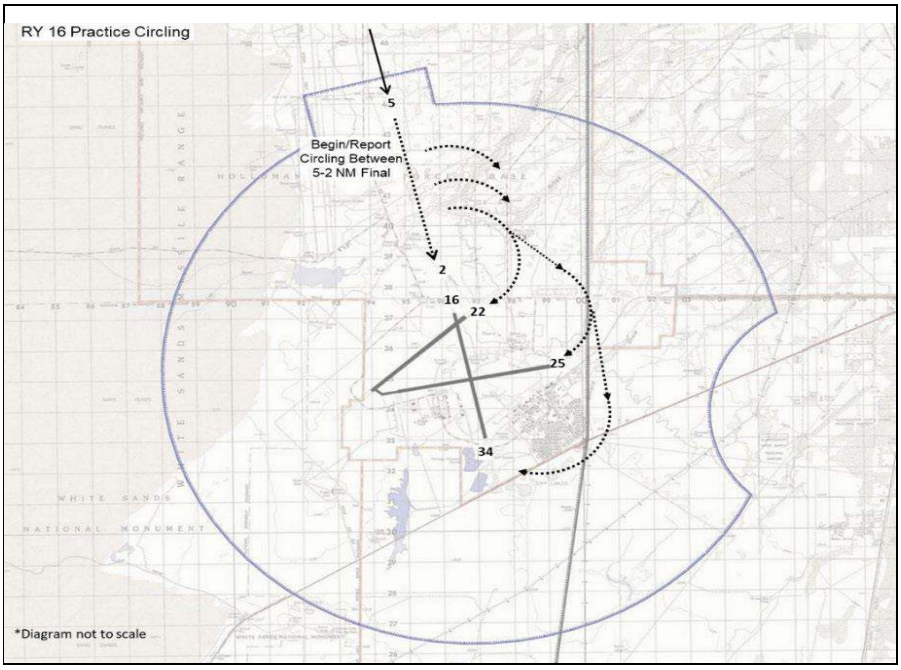
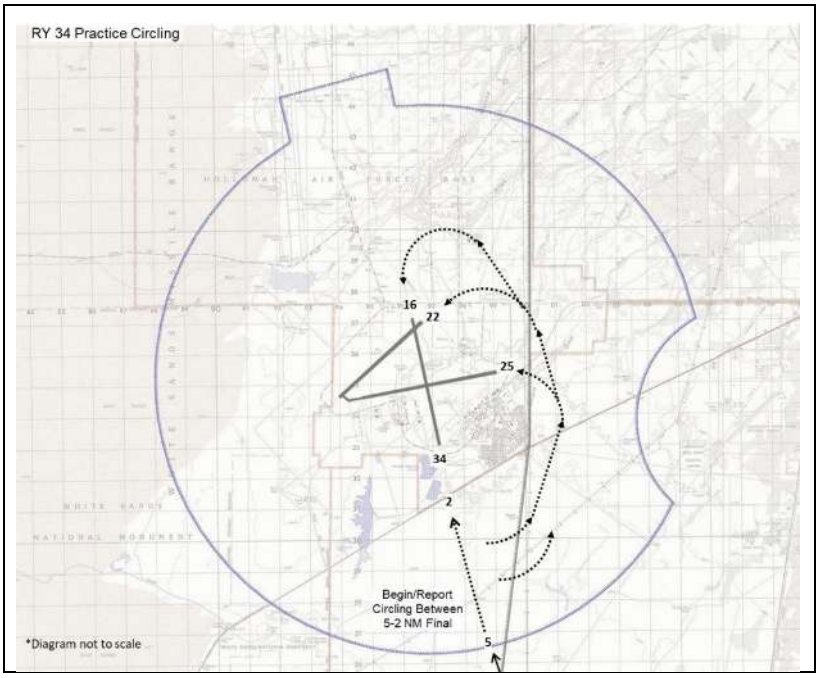


Figure 7.5. Practice Circling Approach Runway 34 (Drawing not to scale).



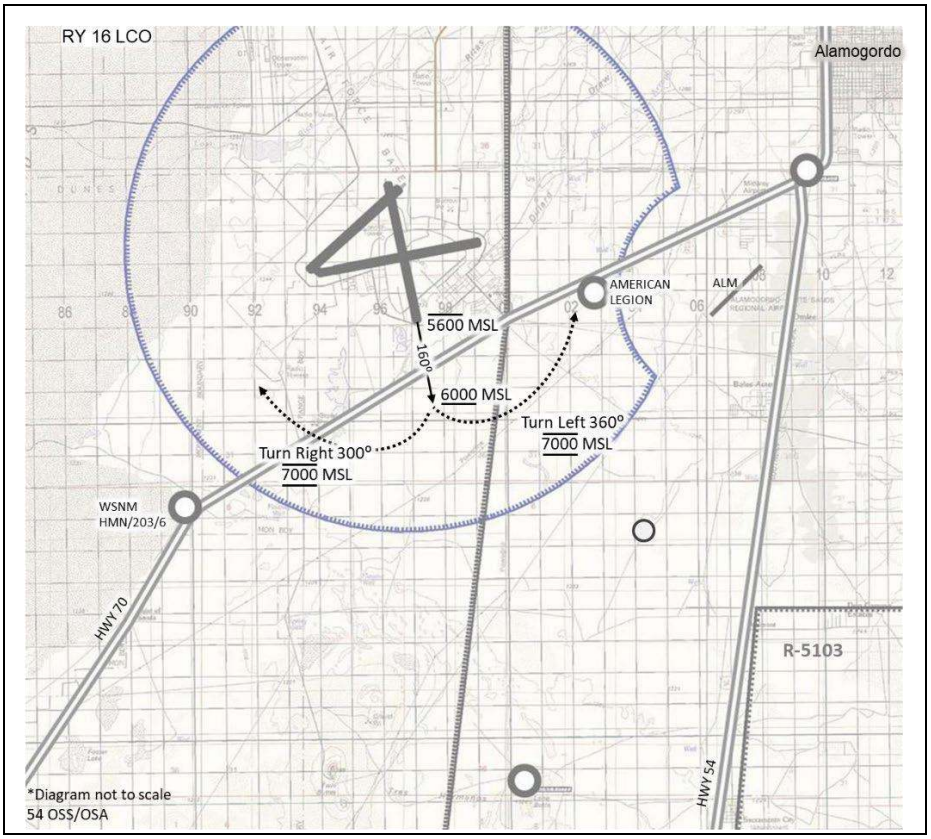
7.6. Availability/Restrictions for Airport Surveillance Radar (ASR) Approaches and Precision Approach Radar (PAR) Approaches/Monitoring. ASR and PAR approaches are not available at HMN.

7.7. Local Climb-out Instructions. Aircrews will use published DVA climb gradients (Table 3.1.) for the specific runway when executing local climb out into the radar patterns or when issued alternate missed approach.

7.7.1. Runway 16 East Downwind (Figure 7.6.). *"EXECUTE LOCAL CLIMB-OUT EAST DOWNWIND*" or "*FLY RUNWAY HEADING, CLIMB AND MAINTAIN 7,000 FT (MSL), CROSS DEPARTURE END AT OR BELOW 5,600 FT (MSL), WHEN LEAVING 6,000 FT (MSL) TURN LEFT HEADING 360 EXPECT RADAR VECTORS*".

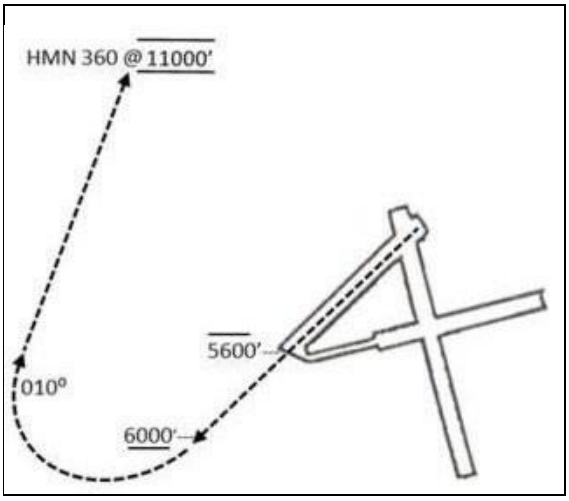
7.7.2. Runway 16 West Downwind (Figure 7.6.). *"EXECUTE LOCAL CLIMB-OUT WEST DOWNWIND*" or "*FLY RUNWAY HEADING, CLIMB AND MAINTAIN 7,000 FT (MSL), CROSS DEPARTURE END AT OR BELOW 5,600 FT (MSL), WHEN LEAVING 6,000 FT (MSL), TURN RIGHT HEADING 300 EXPECT RADAR VECTORS".*

Figure 7.6. Runway 16 Local Climb-out (Drawing not to scale).



7.7.3. Runway 22 (Figure 7.7.). *"EXECUTE LOCAL CLIMB-OUT*" or "*FLY RUNWAY HEADING, CLIMB AND MAINTAIN 11,000 FT (MSL), CROSS DEPARTURE END AT OR BELOW 5,600 FT (MSL), WHEN LEAVING 6,000 FT (MSL). TURN RIGHT HEADING 010, CROSS HMN 360 RADIAL AT 11,000 FT (MSL) EXPECT RADAR VECTORS*".

Figure 7.7. Runway 22 Local Climb-out (Drawing not to scale).



7.7.4. Runway 34. *"EXECUTE LOCAL CLIMB-OUT*" or "*FLY RUNWAY HEADING, CLIMB AND MAINTAIN 7,000 FT (MSL), CROSS DEPARTURE END AT OR BELOW 5,600 FT (MSL), EXPECT RADAR VECTORS*".

7.7.5. Runway 25. *"EXECUTE LOCAL CLIMB-OUT*" or "*FLY RUNWAY HEADING, CLIMB AND MAINTAIN 11,000 FT (MSL), CROSS DEPARTURE END AT OR BELOW 5,600 FT (MSL), WHEN LEAVING 6,000 FT (MSL) TURN RIGHT HEADING 010, CROSS HMN 360 RADIAL AT 11,000 FT (MSL), EXPECT RADAR VECTORS."*

7.7.6. There are no published local climb-out procedures for other Runways; climb-out will be directed by RAPCON.

7.8. HMN Non-Radar Procedures. Use the following procedures whenever WSRF experiences total radar failure.

7.8.1. Cancel IFR if on an IFR flight plan and proceed VFR.

7.8.2. VFR aircraft must adhere to the following procedures.

7.8.2.1. Contact WSRF (Channel 8) NLT 20 DME from HMN TACAN with current ATIS. On initial contact pilots must state position and altitude. Recoveries will be to a full stop or to the VFR traffic pattern.

7.8.2.2. Exit Beak airspace at or below 14,500 ft MSL to deconflict with 16,500 ft MSL departures. McGregor recoveries should exit VFR and contact Tower. Salinas Corridor recoveries will report CAPES.

7.8.2.3. Final approach instrument training may be accomplished by requesting a VFR straight-in, advising Tower of the type of final approach to be flown, and using normal straight-in ground track until established on final. This is a VFR maneuver.

7.8.2.4. All flight briefings will include adherence to flying VFR altitudes, potential traffic conflict areas, and see-and-avoid operations.

7.8.2.5. Aircraft departing VFR will advise Clearance Delivery of intentions. Clearance Delivery will amend flight plans in the Flight Data System as applicable. VFR departures will follow normal ground tracks. All aircraft will report CLOUD and, if going northbound, report BENTT and KARIE. Fly the appropriate VFR hemispheric altitude (16,500 ft MSL or 17,500 ft MSL) until switched to ABQ ARTCC. Departure control will coordinate with ABQ ARTCC for an IFR pick-up if time permits.

7.8.3. Aircraft requesting IFR separation and services can expect up to a 30-minute delay based on air traffic, runway in use, weather, and airspace configuration. If aircraft are unable to depart/arrive VFR, the following procedures are in effect.

7.8.3.1. WSRF will advise ABQ ARTCC of the altitudes assigned. Aircraft held below their final altitude can expect higher with ABQ ARTCC or as higher altitudes become available.

7.8.3.2. Departure Control will request aircraft to provide the estimated time to reach the appropriate transfer of control points: SKEEN (HMN R-078/40 DME), KARIE (HMN R-009/44 DME), SLNGR (HMN R-291/51 DME), or Orogrande, NM (HMN R-176/37 DME). WSRF will coordinate this time with ABQ ARTCC and transfer communications NLT the transfer control point.

7.8.3.3. Arrival aircraft will normally be routed to MONGO (HMN R-034/20 DME) for holding unless an ATC operational advantage can be obtained. Aircraft that are directed to hold will be given an "*Expect Further Clearance*" time. Recoveries will be via a published instrument approach to a full stop or to the VFR traffic pattern.

7.8.3.3.1. Aircraft in the arrival pattern unable to maintain VMC can expect a non- radar routing.

7.8.3.3.2. All aircraft will report commencing the approach at 15 NM and 10 NM final.

Chapter 8

EMERGENCY AND PRECAUTIONARY PROCEDURES

8.1. PCAS and SCN. Tower will activate the PCAS for all known or suspected aircraft mishaps or emergencies. Agencies with two-way voice communication on the PCAS are Tower, FD, AM, and the Ambulance Response Element.

8.1.1. Tower will activate the PCAS for the following reasons (list is not all inclusive).

8.1.1.1. In-flight (IFE) and ground emergency (GE).

8.1.1.2. Aircraft mishaps.

8.1.1.3. Notification of external stores jettison.

8.1.1.4. Planned or unplanned arresting gear/cable engagements excluding barrier certifications.

8.1.1.5. Suspected or confirmed hot brakes.

8.1.1.6. Gun malfunctions and hung forward firing ordnance.

8.1.1.7. Stop Alerts.

8.1.1.8. Hydrazine mishaps.

8.1.1.9. Fluid spills or other emergencies when notified by any first responder base agency.

8.1.1.10. Single ship aircraft experiencing lost communications or no radio (NORDO).

8.1.2. AM will forward all initial and updated information verbatim via the SCN as it becomes available. AM records the SCN.

8.2. Command and Control of IFEs. The primary discrete emergency channel for HMN is the Single Frequency Approach (SFA - Local Channel 18; see Attachment 2). This channel is intended to provide a SFA to emergency aircraft and facilitate communication between the control agency, the SOF, and the aircrew. WSRF will assign the SFA or a UHF/VHF discrete frequency as soon as practical unless otherwise requested by aircrew. Tower will relinquish control of the SFA or UHF/VHF discrete frequency to senior FD officer once the aircraft comes to a complete stop. When an aircraft declares a GE or emergency abort on takeoff and the senior FD officer or designated representative requests use of the SFA or UHF/VHF discrete frequency, Tower will coordinate with WSRF then relinquish the SFA as soon as possible.

8.3. Suspension of Runway Operations due to Aircraft Emergencies. Runway operations will be suspended for all known or suspected FOD conditions and emergencies. 54 FG/CC, 54 FG Brickholder, or SOF may waive requirement to suspend operations for runway inspection following an emergency if in his/her judgment a FOD check is unnecessary. If operations are suspended, AM must conduct a runway check before operations can be resumed.

8.3.1. Emergency aircraft parked on Taxiways Romeo or Golf impact or suspend flying operations to Runways 07, 22, and 25. Discretion should be used when directing emergency aircraft to Taxiway Romeo or Golf.

8.4. Emergency Response Procedures.

8.4.1. IFE.

8.4.1.1. Notify controlling agency of squawk if other than 7700. MQ-9s will set Lost Link squawk 7400.

8.4.1.2. IFEs will be given full runway separation.

8.4.1.3. The preferred runway for cable engagements is Runway 04/22, winds permitting.

8.4.1.4. Except in an emergency, the SOF will not transmit over ATC frequencies without prior coordination with ATC (this does not apply to dedicated SOF frequencies). All radio transmissions to a distressed aircrew, from other than the SOF or ATC, are coordinated through the SOF (this does not apply to the Ops Sup if communicating to the aircrew via a squadron common frequency).

8.4.1.5. Emergency aircraft will normally remain on the emergency frequency until the emergency is terminated.

8.4.1.6. Runway operations will be suspended upon aircraft landing and require AM to conduct a FOD check of the entire runway surface. The SOF may waive the requirement to conduct a FOD check at his/her discretion.

8.4.2. GEs may be worked from the Ground Control frequency.

8.4.3. Tower will direct taxiing aircraft to hold position while emergency response vehicles are responding to the emergency. Emergency vehicles responding to an aircraft emergency will be given priority to cross all active runways over non-emergency airborne aircraft and departures. Ground Control may approve specific aircraft movement if there is no conflict with emergency response vehicles.

8.4.4. Discontinue any operation that would interfere with availability of landing runway and broadcast on the LMR Fire/Crash Net when the emergency aircraft is next to land.

8.4.5. AM will respond to all GEs to conduct a FOD check.

8.4.6. For any IFE or GE (on or off base), the senior FD representative is the incident commander until the emergency is terminated. FD is the only agency with the authority to terminate an emergency. If there is an associated mishap, the incident commander will be designated and assume the responsibilities as directed in 49 WG Mishap Response Plan.

8.4.7. Simulated Emergencies. ATC will exercise the same procedures as in an actual emergency. Unless previously coordinated, simulated emergency aircraft WILL NOT receive priority over other aircraft. The 54 OSS/OSA/CC must be briefed at least 72 hours in advance of any exercise or inspection that involves 54 OSS/OSA personnel, facilities to include RAWS, or airfield (i.e., Major Accident Response Exercise, see AFMAN 13-204V1 paragraph 3.12.).

8.5. External Stores Jettison Procedures.

8.5.1. If able, jettison external and internal stores on Red Rio or Centennial ranges in one of the approved impact areas. Drops in any other areas will be for emergency situations only and must be reported to ATC. Obtain permission from ROC or WSRF to drop stores on the range impact areas. Expect minimum delays if the impact areas are in use or designated hot.

8.5.2. If the areas are not in use or designated cold, WSRF must coordinate with Fort Bliss.

8.5.3. Jettison Area will be at 6,000 ft MSL outbound on the HMN R-321 between 10 and 14 DME. Jettison Area is checked every 5 years for manmade obstacles or hazards. Jettison Area was last verified March 2022 by 54 FG/FGV.

8.5.3.1. Contact RAPCON on the Approach frequency (Channel 8).

8.5.3.2. ATC will provide a vector to intercept the HMN R-321 and advise the pilot when at 10 DME and again at 14 DME. If WSRF radar is out, RAPCON will issue clearance outbound on the HMN R-321 prior to 10 DME at 6,000 ft MSL.

8.5.4. Tower will activate the PCAS when alerted that an aircraft will use the Jettison Area (PCAS WILL NOT be activated for areas over the range) and again upon termination of jettison. AM will alert agencies involved in cleanup/recovery operations.

8.6. Fuel Dumping. There is no defined fuel dumping area. Location of fuel dumping will occur as directed by ATC. Aircrew should attempt to dump above 5,000 ft AGL and avoid populated areas, safety constraints permitting.

8.7. Abandonment of Aircraft. The designated bailout area is northwest of HMN between HMN R-270 and R-300 between 3 and 5 DME. Bailout at or above 10,000 ft MSL on a northwesterly heading.

8.7.1. Whenever aircrew abandons an aircraft, the pilot should make every effort to advise ATC of the aircraft position prior to abandonment. When alerted that an aircrew abandons an aircraft, Tower will plot the position on base crash grid maps or on a sectional chart for crash recovery efforts and activate the PCAS to pass along all applicable information.

8.8. Hung Ordnance Procedures. Primary runway for hung ordnance is Runway 22 and alternate is Runway 25. Prior to recovery, notify the SOF and relay type of munitions and nature of problem. Recover via straight-in with chase (if possible) and avoid overflying populated areas.

8.8.1. Aircraft with hung-secure INERT ordnance (to include training or practice ordnance), will normally not declare an IFE and after landing, may taxi to EOR for de-arming. When possible, use EOR spot closest to the runway (e.g., EOR Foxtrot spot 10, EOR Alpha spot 5) to allow for outlet of other aircraft and standardized emergency response if de-arming escalates to a GE.

8.8.2. Aircraft with hung-secure LIVE or hung-unsecure LIVE or INERT ordnance (to include training or practice ordnance) will declare an IFE. The following areas have prepositioned grounding cables in the south infields for Weapons/EOD use during hung ordnance evaluation and de-arming.

8.8.2.1. After landing Runway 22, exit on Taxiway Tango and stop on HC Ramp. When exiting is not possible or HC Ramp is occupied by equipment or parked aircraft, stop straight ahead on runway between the departure end BAK-12 and the Runway 04 aim point markings (also known as "captain's bars").

8.8.2.2. After landing Runway 25, stop on Taxiway Golf between Taxiway Lima intersection and the TACAN checkpoint.

8.8.2.3. If ordnance can be safely pinned and secured, taxi to LOLA or parking (as required) when released by the incident commander. If not, shut down and have the aircraft towed to LOLA or parking (as required).

8.8.3. MQ-9 with hung-secure LIVE or hung-unsecure LIVE or INERT ordnance will loiter for at least one hour prior to landing. During holding, Tower will coordinate to remove the Runway 22 or 25 departure end BAK-12 as required. After the appropriate cable is removed, MQ-9 will execute the procedures listed in paragraph 8.8.2.

8.8.4. TDY units, 704 TG, or 82 ATRS projects involving ordnance requiring actions different from above must coordinate with 54 FG/CC or 54 FG Brickholder, the SOF, AM, and 49 WG Weapons Safety (49 WG/SEW) prior to launch.

8.9. Gun Malfunctions/Hung Forward Firing Ordnance. Safe switches and declare an IFE. Avoid pointing aircraft at populated areas. As required, recover with chase ship straight-in Runway 22 (primary) or Runway 25 (secondary) or Runway 16 (tertiary).

8.9.1. After landing Runway 22, exit on Taxiway Tango and stop on HC Ramp with the aircraft's nose pointed to heading 040 degrees.

8.9.1.1. When unable to exit runway after landing, stop aircraft just after the departure end BAK- 12 and point the aircraft's nose to heading 270-290 degrees. Prepositioned grounding cable is in the south infield.

8.9.1.2. When Taxiway Tango/HC Ramp is occupied by equipment or parked aircraft, stop straight ahead at the departure end of runway and point aircraft's nose to runway heading. If able, stop on the right/north side of the runway to allow FD/Emergency vehicles access via Taxiway Romeo.

8.9.2. After landing Runway 25, stop straight ahead on Taxiway Golf between Taxiway Lima intersection and the TACAN checkpoint. Prepositioned grounding cable is in the south infield.

8.9.3. Use Runway 16 only as a last resort. This option requires vehicle roadblocks on 49er Avenue and Bong Street. After landing, taxi to A/D Hotel and use the two spots furthest from Runway 16/34 and park with nose of aircraft pointed south. Prepositioned grounding cable is in the south infield.

8.9.4. When gun or ordnance is declared safe, aircraft may taxi to chocks/authorized LOLA area respectively. Expect F-16 aircraft to shut down on the runway with a gun malfunction. Upon notification via the SCN, 49th Security Forces Squadron (49 SFS) will secure the appropriate area(s) IAW 49 SFS Quick Reaction Checklist 3, *In-Flight Emergency.*

8.9.5. De-arming of simultaneous gun malfunctions/hung forward firing ordnance emergencies will be directed by the incident commander on a case-by-case basis.

8.10. Lost Communications Instructions (NORDO). If able, squawk 76XX (16, 34, 07, 25, 04, or 22) for expected landing runway. If a greater emergency exists, squawk 7700. See Attachments 6, 8, and 9 for additional NORDO procedures.

8.10.1. If able to return VFR, use the following procedures:

8.10.1.1. Have a wingman lead the NORDO aircraft to a straight-in if possible. NORDO aircraft will be passed the lead on final with clearance to land.

8.10.1.2. Maintain appropriate VFR altitudes while returning to base.

8.10.1.3. If single ship, proceed to initial, rock wings at initial, and break at departure end.

8.10.1.4. Look for a steady green light gun signal from Tower for clearance to land.

8.10.2. If weather conditions require an IFR return to base (RTB), use the following procedures:

8.10.2.1. Maintain 16,000 ft MSL or the lowest practical altitude where VMC is obtainable at or above 16,000 ft MSL.

8.10.2.2. If recovering from the north or west, proceed direct LAYKZ (HMN R-002/12 DME) and descend to 13,000 ft MSL in holding. If able, execute one complete turn in holding followed by the published approach to the expected landing runway.

8.10.2.3. If recovering from the south or east, proceed direct FITZZ (HMN R-093/20 DME) at 16,000 ft MSL. Cross FITZZ at or above 13,000 ft MSL and execute the published procedure to the expected landing runway.

8.11. Air Evacuation Notification and Response Procedures.

8.11.1. AM is the primary point of contact for AIREVAC or MEDEVAC aircraft. AM will relay AIREVAC or MEDEVAC information to base agencies via the SCN, radio, or direct landline as required. Tower will notify AM and FD of arriving AIREVAC or MEDEVAC aircraft by 15- flying miles.

8.11.2. In addition to routine flight plan information, AIREVAC or MEDEVAC information should include number of patients uploading or off-loading, litter/ambulatory status, special equipment requests, and physician requirements.

8.11.3. Tower will notify AM and FD of MEDEVAC helicopter operations within the Class D airspace but outside the aerodrome. E.g., MEDEVAC helicopter responding to vehicle accident on Highway 70. Notification is not required for VFR itinerant MEDEVAC aircraft.

8.12. Emergency AAS Procedures.

8.12.1. HMN AAS locations are displayed in Figure 2.1. or 2.2. Landing zone for a BAK-15 barrier is displaced. Aircraft may only land over a raised BAK-15 arresting barrier in an emergency. Tower will issue the distance of the barrier from the approach end of the runway.

8.12.2. All BAK-15s are remotely controlled by Tower. When the approach end of the primary landing runway is equipped with a BAK-15 barrier, the barrier will be rendered safe (taking remote capability away from the Tower) and placed in the "*down*" position. The BAK-12 cable will be disconnected from the BAK-15. If weather conditions demand an immediate runway change, it is permissible to land over an armed BAK-15 in the "*down*" position. However, Tower will immediately notify BM or FD to render the barrier safe if continued landings are expected. BM will advise Tower of barrier configuration and whether the BAK-15 barrier is armed or safe.

8.12.3. When Runway 22 approach end BAK-12 engagement is anticipated, and aircraft are holding in Echo EOR or on Taxiway Echo, Tower will either clear the aircraft for takeoff (emergency outside 10 NM) or direct the aircraft to taxi well clear of Runway 22.

8.12.4. Crash Recovery, FD, BM, and AM will respond immediately to all barrier engagements. Barrier certifications are pre-coordinated with all required responding agencies.

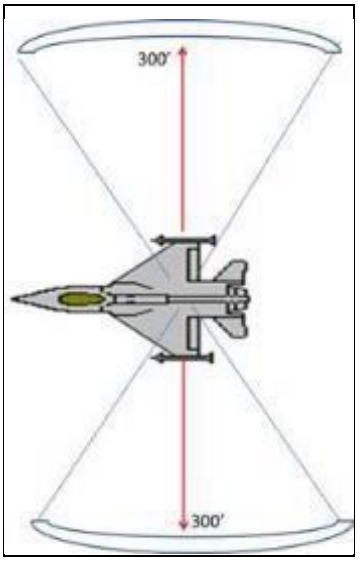
8.13. Hot Brake Area and Procedures.

8.13.1. If able, aircrew will taxi to the nearest hot brake area (nearest EOR, farthest parking spot from the runway). Crash Recovery personnel will determine if hot brakes exist and if the aircraft is cleared to continue taxiing.

8.13.2. Incident commander will ensure a 300 ft safety zone (see Figure 8.1.) that extends outward from the aircraft's tires. Other aircraft WILL NOT taxi within the 300 ft safety zone.

8.13.3. F-16s parked on the Main Ramp that suspect hot brakes while in their hangar/shelter may reposition to Alpha LOLA in lieu of an EOR.

Figure 8.1. Hot Brake Danger Zone.



8.14. Hydrazine Procedures.

8.14.1. Airborne: If conditions permit, the aircraft will land Runway 22 and exit on Taxiway Tango and stop on HC Ramp. When exiting is not possible, stop straight ahead on the runway. Once the aircraft is stopped, evacuate an area at least 150 ft upwind and 300 ft downwind of all non-essential personnel and remain isolated at the scene until FD and Medical personnel can decontaminate and evaluate each potentially exposed patient. If landing on Runway 22 is not possible, aircraft will clear at the first available taxiway.

8.14.2. Ground: Parked aircraft will remain in their spot, all responding personnel will remain outside of the response area until the aircraft and area are cleared by the Hydrazine Response Team (HRT). If conveniently aligned with an intersecting taxiway, affected aircraft will exit the flow of traffic and stop, else taxiing aircraft will stop immediately. In all cases, PIC will declare a GE.

8.15. Personnel/Crash Locator Beacon Signal/Emergency Locator Transmitter (ELT) Response Procedures.

8.15.1. Scheduled operational testing of beacon/ELT will be conducted IAW FAAO JO 7110.65 during the first 5 minutes of the hour for no more than three audible sweeps.

8.15.2. 49 WG/CP is designated as the HMN single point of contact for ATC notification. The PCAS IS NOT activated in response to unscheduled personnel, crash locator beacon/ELT. 49 WG/CP will attempt to determine the source of the signal or personnel and will direct Tower to activate the PCAS as necessary.

8.15.3. Aircrew Emergency Beacon Risk Mitigation Measures. Aircrew will monitor Emergency Frequency (UHF 243.0) while airborne. Aircrew will diligently monitor other flight members and maintain situational awareness on them throughout the sortie. In the event of an ejection, the on-scene Rescue Combat Air Patrol (RESCAP) commander will follow procedures in the HMN 49 WG IFG. The on-scene commander will attempt to contact the downed aircrew on 243.0 and 282.8 while marking the position of the downed aircrew via a GPS mark point. The mark point coordinates will be passed as soon as possible to the SOF and/or Search and Rescue (SAR) forces.

8.16. Weather Recall/Divert Procedures.

8.16.1. 54 OSS/OSW will immediately notify the SOF and Tower with pertinent adverse conditions and location. Any pilot or controller observing such conditions should immediately pass that information to their Ops Sup, SOF, Tower WS or WSRF WS, and 54 OSS/OSW.

8.16.2. The SOF will determine whether to recall, divert, or delay aircraft. Time and conditions permitting, the SOF will coordinate with affected Ops Sups prior to diverting, holding, or recalling aircraft.

8.16.2.1. Alternate airfield advisories will be relayed to aircrew and included in the ATIS broadcast in the following format: "*VFR or IFR ALTERNATE (airport)*". E.g., IFR alternate Roswell.

8.16.3. WSRF will pass weather and route recall notification to ABQ ARTCC and will coordinate range and restricted area crossings for aircraft regarding weather avoidance areas. Route recalls must include the specific call signs of the aircraft affected.

8.16.4. Pilots will acknowledge weather and route recall on their current controlling frequency and arrange to RTB. Range crossings cannot be guaranteed. Pilots will contact the SOF and relay their available hold time prior to contacting WSRF.

8.16.5. IFR Divert to ELP. Contact WSRF for emergency service. Expect hand off to ELP Approach for sequencing into ELP.

8.16.6. Unmanned Aircraft Divert Base Designation. In addition to locally assigned MQ- 9s, US Navy MQ-4C Triton may designate KHMN for emergency divert. See Depart of the Navy – Holloman AFB Unmanned Patrol Squadron 19 and Test and Evaluation Squadron 20 Memorandum of Understanding dated 26 July 2022. Any units requesting to use HMN as a divert location for unmanned aircraft must coordinate with the 54 OSS/OSA and 49 WG/XP.

8.17. Alternate Airfields (Table 8.1.). HMN has two primary alternate airfields for practice patterns and approaches. Biggs Army Airfield (KBIF) located 62 NM to the south-southwest, and Roswell Air Center (KROW), located 84 NM to the northeast. In addition, ELP International Airport (KELP) is available for practice instrument approaches.

Table 8.1. Primary Divert Airfields.

Aerodrome - Identifier		TACAN - Channel	Bearing	Distance
Alamogordo-White Sands	KALM	n/a	090 ^{o}	006 NM
Albuquerque International	KABQ	ABQ 079X*	341 ^{o}	134 NM
Biggs AAF	KBIF	EWM 071X*	185 ^{o}	062 NM
Cannon AFB	KCVS	CVS 053X	048 ^{o}	167 NM
Davis-Monthan AFB	KDMA	DMA 123X	253 ^{o}	246 NM
El Paso International	KELP	ELP 099X*	184 ^{o}	064 NM
Midland International	KMAF	MAF 095X*	096 ^{o}	206 NM
Roswell Air Center	KROW	CME 108X*	063 ^{o}	084 NM
Sierra Blanca Regional	KSRR	n/a	028 ^{o}	046 NM
n/a = not applicable		* = TACAN not co-located with airfield.		

8.18. Anti-Hijacking/Unlawful Seizure Procedures. Refer to the 49 WG Integrated Defense Plan 101, Tab 1 to Appendix 2 to Annex C; *Unauthorized Movement/Hijacking of Aircraft*. Immediately report such incursions via the PCAS/SCN to 49 WG/CP and BDOC.

8.19. Wind Limitations on Control Tower (Building 864). 49 CES has determined the wind velocity Tower can safely withstand to be 70 knots. Tower personnel will evacuate when the winds reach 55 knots sustained or gusts to 65 knots.

8.20. Evacuation/Loss Communications of Airfield Operations Facilities. The 54 FG/CC has determined that there is no requirement for alternate ATC facilities.

8.20.1. Tower Evacuation Procedures.

8.20.1.1. IAW AFMAN 13-204V1, flight operations are not authorized when Tower is closed, except for approved UAO (see Attachment 10) or alternate ATC Tower operations are in use. Divert airfields are listed in Table 8.1.

8.20.1.2. Tower will transmit on all assigned frequencies, to include UHF/VHF Emergency Frequency, that Tower is evacuating and that airfield operations are suspended. Tower will advise all airborne aircraft in the pattern to contact WSRF for further instruction.

8.20.1.3. Aircraft taxiing for departure will pull into the closest arm/de-arm area and monitor Ground Control frequency (Attachment 2) for further instructions.

8.20.2. Mobile Tower - Transition Procedures (Activation).

8.20.2.1. Upon completion of evacuation checklist and securing the SOF truck, Tower will attempt to set up at a predetermined alternate location. If the SOF truck is unavailable or disabled, the WS will relocate using a POV. A general POV pass has been provided by AM and is kept in the Tower evacuation kit. Once established, mobile Tower will notify the following agencies and advise that limited operations are ready to resume from the mobile Tower and additional information as necessary.

8.20.2.1.1. WSRF – Relay cell phone / radio contact information. Confirm runway, approach in use, and missed approach/return to radar climb-out instructions. Request control of Class D and request a briefing on current airborne traffic.

8.20.2.1.2. AM – If evacuation precluded crash phone activation, ensure secondary is activated to alert facilities of mobile Tower activation. Inform AM of mobile Tower location on airfield, runway in use, and if a FOD sweep of the active runway will be necessary to resume operations.

8.20.2.1.3. FD – Advise FD of mobile Tower location and runway in use.

8.20.2.1.4. 49 WG/CP – Advise 49 WG/CP of cell phone / radio contact information.

8.20.2.1.5. 54 OSS/OSW – Request current observation and winds for runway in use.

8.20.2.1.6. Tower CCTLR, 54 OSS/OSAA/CC, or next in chain-of-command.

8.20.2.1.7. The 54 FG/CC or 54 FG Brickholder is the approval authority for all mobile tower operations. Tower WS will confirm with the SOF prior to resuming operations, that the FG/CC or designated representative has approved operations to resume utilizing the mobile tower.

8.20.2.2. Manning permitting, non-mobile Tower personnel will staff the control panel in the Airfield Lighting Vault (ALV, Building 863).

8.20.3. Mobile Tower - Transition Procedures (De-activation):

8.20.3.1. Operation from the primary Control Tower may be resumed at the discretion of the Tower WS/SC once informed by first responders that it is safe to return.

8.20.3.2. The Tower WS will suspend mobile Tower operations and notify all agencies identified in paragraphs 8.20.2.1., that ATC is relocating back to the primary Control Tower with an estimated time of resuming operations. Tower will release the Class D to WSRF, and runway operations will be suspended at this time.

8.20.3.3. The Tower WS/SC will complete appropriate facility opening checklists and ensure AM resumes runway operations before resuming operations in the primary Control Tower.

8.20.4. Mobile Tower - Aircraft Restrictions and Limitations.

8.20.4.1. Due to the airfield complexity, length of runway's, geography and landscape, mobile tower has limited capability in scanning the movement area. Landing clearances are predicated on known traffic and known physical airport conditions.

8.20.4.2. VFR Pattern will be closed. No departures or helicopter operations will be permitted. Transitions for MEDEVAC aircraft through Class D or to HMN are authorized.

8.20.4.3. Active runway prior to tower evacuation will be the preferred choice of operations. Runway 25 may be used only when the ALV is manned.

8.20.4.4. VFR / IFR practice approaches are prohibited.

8.20.4.5. Use of SRS or RSRS is not authorized. The runway must be clear to allow landing between separate flights of aircraft with normal separation authorized between flight members in accordance with MDS guidance. WSRF will provide a minimum of 8 miles separation between arriving flights/single aircraft to facilitate runway separation.

8.20.4.6. Aircraft operations will be limited to a single aircraft, or a single formation made up of no more than four aircraft. Aircraft must recover via straight-in or instrument approach.

8.20.4.7. Airfield information (NOTAM, airfield advisories, etc.) are available from AM on PTD UHF 372.2. Weather information is available from PMSV UHF 344.6.

8.20.4.8. NAVAID (ILS / TACAN) monitoring is not available. Aircrew will report any NAVAID abnormalities to mobile Tower or WSRF immediately.

8.20.4.9. When the ALV is manned, aircrew will expect delays when requesting changes to airfield lighting settings. Additionally, the South and West BAK-15s may be controlled from the ALV. When manned, aircrew can expect a delay when requesting a BAK-15.

8.20.4.10. Mobile Tower may be equipped with a battery-operated ATC light gun. When equipped, mobile Tower will ensure aircrew are aware of the mobile Tower location prior to arrival or departure.

8.20.4.11. Mobile Tower multi-channel radio range and quality may vary based on airfield parking position and weather conditions.

8.20.4.12. Limited additional services will be provided by Tower. Tower applied visual separation will be limited.

8.20.4.13. Mobile Tower will not monitor 243.0/121.5 emergency frequencies.

8.20.4.14. Aircraft must report 5-mile final/final approach fix and 2-mile final to tower unless radar monitoring procedures are in effect.

8.20.5. Mobile Tower – Airfield Vehicle Restrictions.

8.20.5.1. Vehicle operations on, across, or in the CMA will be restricted to essential operations only.

8.20.6. Mobile Tower – Operations.

8.20.6.1. Mobile Tower will use standard ATC radio communications, e.g., *"RUNWAY ONE SIX, WIND ESTIMATED ONE EIGHT ZERO AT FIVE, CLEARED TO LAND"*.

8.20.6.2. Aircraft operations will be limited to Tower WS discretion. All aircraft can expect one approach to a full stop.

8.20.7. WSRF loss of communication. Aircraft under control of WSRF that can maintain VFR will recover with Tower. All other aircraft will contact ABQ ARTCC for an instrument approach. See paragraphs 3.6.6.3. Non WSRF Operations and 7.8. HMN Non-Radar Procedures for additional information.

8.20.8. AM Evacuation Procedures. AM will evacuate IAW AFMAN 13-204V3, and HMN IEMP 10-2, *Installation Emergency Management Plan*.

Chapter 9

MISCELLANEOUS PROCEDURES

9.1. Opposite Direction Departures and Landings. Tower will have final approval authority on opposite direction arrivals. WSRF will have final approval authority on opposite direction IFR departures. Cutoffs prescribed below also apply to helicopter operations except when helicopters are established on arrival/departure routes described in paragraph A6.3.

9.1.1. Arrival versus Arrival (applies to IFR and VFR aircraft).

9.1.1.1. Opposite direction arrivals will be restricted to straight-in full stops.

9.1.1.2. An opposite direction arrival will proceed no closer than 10-flying miles from the runway until the preceding arrival to the active or opposing runway has landed or has completed a practice approach and has turned at least 45 degrees from the reciprocal course of the opposite direction arrival.

9.1.1.3. Aircraft in or entering the VFR traffic pattern will be restricted from base and final until after the opposite direction arrival lands.

9.1.2. Arrival versus Departure (applies to IFR and VFR aircraft).

9.1.2.1. When opposite direction arrivals reach a point 10-flying miles from the runway, IFR departures will be suspended.

9.1.2.2. Aircraft departing VFR, remaining in the VFR pattern, must have departed, or turned to avoid the arrival before the arrival reaches a point 10 miles from the runway.

9.1.2.3. Arrivals to either Runway 04 or Runway 07 will be considered opposite direction to both Runway 22 and Runway 25 departures.

9.1.3. Departure versus Arrival.

9.1.3.1. IFR opposite direction departures will be established on a course diverging by at least 45 degrees from the reciprocal of the final approach course to avoid any conflict prior to an arrival reaching a point 10-flying miles from the active/opposing runway.

9.1.3.2. VFR opposite direction traffic will be airborne and turning to avoid any conflict before an arrival reaches a point 10-flying miles from the runway.

9.2. Special Procedures. Land and Hold Short Operations are not authorized at HMN.

9.3. Bird Watch Conditions/Wildlife Control. See 49 WG Bird Aircraft Strike Hazard (BASH) Plan for local BASH program guidelines.

9.4. SOF Operating in the Tower. The SOF's responsibilities while operating in the Tower are outlined in AFI 11-418_HAFBSUP, *Operations Supervision*.

9.5. Night Vision Device (NVD) Operations. The use of NVDs on HMN airfield for air operations are authorized for AA only (see Attachment 6). Other units requesting NVD operations must establish an LOA with the 54 FG prior to employment.

9.6. Paradrop/Parachute Operations. Paradrop/Parachute operations are not authorized in HMN Class D airspace.

9.7. Functional Check Flights (FCF).

9.7.1. Pilots on FCF profile that require maximum performance climb on takeoff will make their initial request with Clearance Delivery and expect approval/disapproval with Ground Control or Tower. Approval for max climbs will be contingent upon traffic conditions and/or airspace limitations.

9.7.2. Taxi check only FCF must be coordinated with AM.

9.7.3. FCF profiles that require non-standard operations (e.g., extended line up, pattern delay after initial departure, multiple VFR or IFR patterns) should be coordinated with ATC prior to departure. Use of standard FCF call signs (e.g., BREAK 71, HITEST 1) or annotation on daily flying schedule will not afford the FCF priority over other operations.

9.8. Installation Security Plan. See HAFB IDP.

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Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

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None

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Abbreviations and Acronyms

7NM1—Mesa Verde Ranch Strip

72NM—Otero Mill Airport

AAFM—Assistant Airfield Manager

AAS—Aircraft Arresting System

ABQ—Albuquerque, New Mexico

ACCTLR—Assistant Chief Controller

ADPM—Airfield Driving Program Manager

AETC—Air Education and Training Command

AFB—Air Force Base

AFI—Air Force Instruction

AFM—Airfield Manager

AFMAN—Air Force Manual

AFRIMS—AF Records Information Management System

AGL—Above Ground Level

AIC—Airspace Information Center

AICUZ—Air Installation Compatible Use Zone

AIM—Aeronautical Information Manual

AIREVAC—Air Evacuation

ALSF-1—Approach Lighting System with Sequence Flashing Lights

AM—Airfield Management

AOB—Airfield Operations Board

AP—Area Planning

AR—Air Refueling Track

ARMS—Aviation Resource Management System

ARTCC—Air Route Traffic Control Center

ASR—Airport Surveillance Radar

ATC—Air Traffic Control

ATCAA—Air Traffic Control Assigned Airspace

ATCALS—Air Traffic Control and Landing Systems

ATIS—Automatic Terminal Information Service

ATLC—Automatic Takeoff and Land Capability

AUX—Auxiliary

AWOS—Automated Weather Observing System

BAK—Barrier Arresting Kit

BASH—Bird Aircraft Strike Hazard

BDOC—Base Defense Operations Center

BM—Barrier Maintenance

CAP—Civil Air Patrol

CBT—Computer Based Test

CCTLR—Chief Controller

CD—Current Day

CES—Civil Engineer Squadron

CFA—Centennial Fly Area

CFR—Code of Federal Regulations

CMA—Controlled Movement Area

COA—Certificate of Authorization

CSE—Center Scheduling Enterprise

CTAF—Common Traffic Advisory Frequency

DAFI—Department of the Air Force Instruction

DAFM—Deputy Airfield Manager

DAFMAN—Department of the Air Force Manual

DASR—Digital Airport Surveillance Radar

DD or **DoD**—Department of Defense

DESR— Defense Explosives Safety Regulation

DME—Distance Measuring Equipment

DRM—Distance Remaining Marker

DSN—Defense Switched Network

DV—Distinguished Visitor

DVA—Diverse Vector Area

ECC—Emergency Communications Center

EED—Electro-Explosive Devices

ELP—El Paso, Texas

ELT—Emergency Locator Transmitter

EOD—Explosive Ordnance Disposal

EOR—End of Runway

EWM—Newman VORTAC

FA—Flight Authorization

FAA—Federal Aviation Administration

FAAO—FAA Order

FAR—Federal Aviation Regulation

FCF—Functional Check Flight

FCIF—Flight Crew Information File

FD—Fire Department

FITS—Fighter Index of Thermal Stress

FLIP—Flight Information Publication

FM—Frequency Modulation

FMV—Full-Motion Video

FOD—Foreign Object Damage

FSOP—Flight Safety Operational Plan

FSS—Flight Service Station

ft—Feet or Foot

GBU—Guided Bomb Unit

GDT—Ground Data Terminal

GE—Ground Emergency

GPS—Global Positioning System

GTIMS—Graduate Training Integration Management System

HAFB—Holloman Air Force Base

HAM—Holloman Airspace Manager

HAMETS—High Altitude Mountain Environmental Training Strategy

HC—Hazardous Cargo

HC/D—Hazard Class/Division

HELSTF—Hight Energy Laser System Test Facility

HIRL—High Intensity Runway Lights

HMN—Holloman Air Force Base

HRT—Hydrazine Response Team

HSTT—High Speed Test Track

IAW—In Accordance With

IDP—Integrated Defense Plan

IFE—In-Flight Emergency

IFF—Identification Friend or Foe

IFG—In-Flight Guide

IFR—Instrument Flight Rules

ILS—Instrument Landing System

IMC—Instrument Meteorological Conditions

IP—Interservice Publication

IR—Instrument Route

JAWSS—Joint Advanced Weapon Scoring System

JO—Joint Order

JTTOC—Joint Test and Training Operations Cell

KABQ—Albuquerque International Sunport Airport

KALM—Alamogordo-White Sands Regional Airport

KBIF—Biggs Army Airfield

KCVS—Cannon AFB

KDMA—Davis-Monthan AFB

KELP—El Paso International Airport

KIAS—Knots Indicated Air Speed

KMAF—Midland International Air and Space Port

KROW—Roswell Air Center Airport

KSRR—Sierra Blanca Regional Airport

LMR—Land Mobile Radio

LOLA—Live Ordnance Load Area

LOP—Letter of Procedure

LOS—Line of Sight

LR—Launch and Recovery

MARSA—Military Authority Assumes Responsibility for Separation of Aircraft

MCE—Maximum Credible Event

MDS—Mission Design Series

MEDEVAC—Medical Evacuation

MEQ—Mission Essential Quantities

MHE—Material Handling Equipment

MMG—Material Maintenance Group

MOA—Military Operating Area

MOC—Maintenance Operations Center

MSA—Minimum Safe Altitude

MSL—Mean Sea Level

MTR—Military Training Route

MTS—Multi-spectral Targeting System

MVA—Minimum Vectoring Altitude

NAMO—NCOIC Airfield Management Operations

NAVAID—Navigational Aid

NAS—National Airspace System

NB—Northern Buffer

NCOIC—Non-Commissioned Officer In-Charge

ND—Next Day

NLT—No Later Than

NM—Nautical Miles

NORDO—No Radio

NOTAM—Notice to Airmen

NVD—Night Vision Devices

ODO—Operations Duty Officer

OI—Operating Instruction

ORM—Operational Risk Management

PAPI—Precision Approach Path Indicators

PAR—Precision Approach Radar

PCAS—Primary Crash Alarm System

PHETS—Permanent High Explosive Test Site

PIC—Pilot in Command

PM—Program Manager

PMI—Preventative Maintenance Inspection

POFZ—Precision Obstacle Free Zone

POV—Privately Owned Vehicle

PPR—Prior Permission Required

PWC—Pilot Weather Category

RAMS—Radar Advanced Measurement System

RAPCON—Radar Approach Control

RAWS—Radar, Airfield, and Weather Systems

RCO—Range Control Officer

RCR—Runway Condition Reading

RDS—AF Records Disposition Schedule

RESCAP—Rescue Combat Air Patrol

ROC—Range Operations Center

RSC—Runway Surface Condition

RSO—Radiation Safety Officer

RSRS—Reduced Same Runway Separation

RTB—Return to Base

RWR—Radar Warning Receiver

RY—Runway

SAAM—Special Assignment Airlift Mission

SAM—Special Air Mission

SAR—Search and Rescue

SATCOM—Satellite Communications

SCN—Secondary Crash Net

SFA—Single Frequency Approach

SFC—Surface

SFL—Sequenced Flashing Lights

SFO—Simulated Flameout

SI-SFO—Straight-In Simulated Flameout

SM—Statute Miles

SOF—Supervisor of Flying

SRS—Same Runway Separation

SUA—Special Use Airspace

SUAS—Small Unmanned Aircraft System

SUP—Supplement

SVFR—Special Visual Flight Rules

TACAN—Tactical Air Navigation

TDW—Tower Display Workstation

TDY—Temporary Duty

TERPS—Terminal Instrument Procedures

THAAD—Terminal High Altitude Area Defense

UAO—Uncontrolled Airfield Operations

UFC—Unified Facilities Criteria

UHF—Ultra High Frequency

US—United States

VASI—Visual Approach Slope Indicator

VFR—Visual Flight Rules

VHF—Very High Frequency

VMC—Visual Meteorological Conditions

VORTAC—Very High Frequency Omni-Directional Radio Range Tactical Air Navigation

VR—Visual Route

WS—Watch Supervisor

WSMR—White Sands Missile Range

WSNP—White Sands National Park

WSRF—White Sands Radar Facility

Office Symbols

6 ATKS—6th Attack Squadron

8 FS—8th Fighter Squadron

8 AMU—8th Aircraft Maintenance Unit

9 ATKS—9th Attack Squadron

29 ATKS—29th Attack Squadron

49 CES—49th Civil Engineer Squadron

49 CES/CC—49th Civil Engineer Squadron - Commander

49 CES/CEF—49th Civil Engineer Squadron - Fire and Emergency Services Flight

49 CES/CEO—49th Civil Engineer Squadron - Operations Flight

49 CES/CEOFE—49th Civil Engineer Squadron - Exterior Electric

49 CES/CEOFP—49th Civil Engineer Squadron - Power Production Barrier Maintenance

49 CES/CEN—49th Civil Engineer Squadron - Engineering Flight

49 CES/CENP—49th Civil Engineer Squadron - Engineering Flight Portfolio Optimization

49 CS—49th Communications Squadron

49 CS/SCO—49th Communications Squadron - Operations Flight

49 CS/SCOT—49th Communications Squadron - Frequency Manager

49 EMS—49th Equipment Maintenance Squadron

49 FSS/AC—49th Force Support Squadron - Aero Club

49 LRS/LGRD—49th Logistics Readiness Squadron - Deployment and Distribution Flight

49 LRS/LGRDAS—49th Logistics Readiness Squadron - Air Transportation Function

49 LRS/LGRF—49th Logistics Readiness Squadron - Fuels Management Flight

49 MSG/CC—49th Mission Support Group - Commander

49 MXG—49th Maintenance Group

49 MXG/MOC—49th Maintenance Group - Maintenance Operations Center

49 OG—49th Operations Group

49 OG/CC—49th Operations Group - Commander

49 OG/OGV—49th Operations Group - Standardization and Evaluation Office

49 OMRS/SGXB—49th Operational Medical Readiness Squadron - Bioenvironmental Engineering

49 OMRS/SGXF—49th Operational Medical Readiness Squadron - Flight Medicine

49 MDG—49th Medical Group

49 SFS—49th Security Forces Squadron

49 WG—49th Wing

49 WG/A5—49th Wing - A5, Plans and Programs

49 WG/CP—49th Wing - Command Post

49 WG/SEF—49th Wing - Flying Safety

49 WG/SEW—49th Wing - Weapons Safety

54 FG—54th Fighter Group

54 FG/CC—54th Fighter Group - Commander

54 FG/FGV—54th Fighter Group - Standardization and Evaluation

54 OSS—54th Operations Support Squadron

54 OSS/CC—54th Operations Support Squadron - Commander

54 OSS/OSA—54th Operations Support Squadron - Airfield Operations Flight

54 OSS/OSA/CC—54th Operations Support Squadron -Airfield Operations Flight Commander

54 OSS/OSAA—54th Operations Support Squadron - Airfield Management

54 OSS/OSAM—54th Operations Support Squadron - Radar, Airfield, Weather Systems Maintenance

54 OSS/OSAR—54th Operations Support Squadron - White Sands Radar Facility

54 OSS/OSAT—54th Operations Support Squadron - Tower

54 OSS/OSK—54th Operations Support Squadron - Wing Weapons

54 OSS/OSOA—54th Operations Support Squadron - Airspace Manager

54 OSS/OSOS—54th Operations Support Squadron - Wing Scheduling

54 OSS/OSW—54th Operations Support Squadron - Weather Flight

82 ATRS—Detachment 1, 82nd Aerial Target Squadron

311 FS—311th Fighter Squadron

311 AMU—311th Aircraft Maintenance Unit

314 FS—314th Fighter Squadron

314 AMU—314th Aircraft Maintenance Unit

586 FLTS—586th Flight Test Squadron

704 TG—704th Test Group

846 TS—846th Test Squadron

849 AMXS—849th Aircraft Maintenance Squadron

AA—Army Air

AETC/A3AO—AETC/Airspace, Ranges, and Airfield Operations

AFFSA—Air Force Flight Standards Agency

AFRL/RVEP—Air Force Research Laboratory - Balloon Operations

Attachment 2

FREQUENCY MANAGEMENT

A2.1. Frequency Management. HMN Frequency Management Program is administered by the Installation Spectrum Manager, 49 CS/SCOT, IAW DAFI 17-220 *Spectrum Management* paragraph 2.1.9.

A2.1.1. Changes to HMN frequencies or new requests must be made through unit frequency managers and will be coordinated through 54 OSS/OSA (DSN 572-3421) for de-confliction and update requirements for this document. Frequency changes with the 49 CS/SCOT will be directed by the 54 OSA/OSA/CC.

A2.1.2. For radio outages on an ATC frequency, contact RAWS Maintenance Control Center at DSN 572-3395 or commercial (575) 430-8484 for after-hours maintenance.

Table A2.1. HMN Radio Channelization.

Local Channel	UHF	VHF	Agency
1			Squadron Ops
2	275.8	127.05	HMN Ground
3	254.4	119.3	HMN Tower
4	284.0	128.1	HMN Departure
5	305.5	126.95	CHEROKEE Primary
6	336.2		HMN Approach (MOA)
7	257.6	132.65	ABQ Center
8	269.225	120.6	HMN Approach
9	335.625	125.7	HMN Arrival
10	289.4	126.7	Clearance Delivery
11	342.4		Yonder
12	279.7		Lava East / Red Rio
13	311.5		Lava West
14	314.2		Mesa
15	342.2		Oscura Range
16	348.8		Centennial
17	315.9		BADGER
18	277.4		SFA
19	395.15	139.6	SOF
20	273.5		HMN ATIS

Table A2.2. Manual Frequencies.

Freq.	HMN Agency	Freq.	SUA / Range
229.3	AA	315.9	BADGER (JTTOC)
139.025	AA	353.6	Beak A
343.2	311 FS Ops	256.7	Beak B
378.8	314 FS Ops	259.2	Beak C
378.475	8 FS Ops	348.8	Centennial Range
277.1	Reaper Common	305.5	CHEROKEE Primary
149.4	6 ATKS Discrete	346.7	CHEROKEE Alternate
149.4	9 ATKS Discrete	252.5	CHEROKEE Discrete
149.4	29 ATKS Discrete	245.4	CHEROKEE Discrete
253.4	82 ATRS	336.2	HMN Approach (MOA)
143.1	82 ATRS	303.0	IR-133 Discrete
231.55	586 FLTS Common	363.9	IR-134 Discrete
251.25	49 WG/CP (RAYMOND 14)	279.7	Lava East / Red Rio
346.55	HMN Metro	311.5	Lava West
372.2	HMN Pilot-to-Dispatch (AM)	314.2	Mesa West
Freq.	**ATC Facility**	376.15	Melrose Range
353.6	ABQ ARTCC Alternate	356.8	McGregor Discreet
255.4	ABQ Flight Service Station	245.4	McGregor / Dona Ana Control
122.8	ALM Unicom	397.7	McGregor Range Control
342.25	BIF Tower	342.2	Oscura Range
358.3	CVS Approach	348.8	ROC / CFA
298.85	ELP Approach	299.25	ROC Alternate
239.275	ELP Tower	340.8	Talon B/C
239.0	ROW Approach East	297.5	Talon A Low
229.4	ROW Approach West	257.3	Talon A
233.7	ROW Tower	282.7	TOD / Mickey
282.8	SAR Alternate	342.4	Yonder
Freq.	**Air Refueling**	234.8	Yonder South Discrete
249.525	Air Refueling		
255.775	Air Refueling Alternate		
324.4	AR-644 Primary		
319.5	AR-644 Secondary		
229.5	AR-121 Primary		
258.2	AR-121 Secondary		

A2.2. HMN Assigned Frequencies. Ground to Air frequencies listed in this table are serviced by 49 WG at local/remote locations spanning local flying area. Use of these frequencies will be IAW DAFI 17-220 paragraph 1.2 and approved frequency authorizations. Area discrete is reserved for WSRF and aircrew for ATC and tactical use.

Table A2.3. HMN Assigned Frequencies.

Freq.	Purpose	Local Use	DAFI 17-220 Type
119.3	HMN Tower	ATC	ATC
120.6	HMN Approach	ATC	ATC
121.5	Emergency	ATC	ATC
125.7	HMN Arrival	ATC	ATC
126.7	Clearance Delivery	ATC	ATC
127.05	HMN Ground	ATC	ATC
128.1	HMN Departure	ATC	ATC
139.6	SOF	C2	Tactical
143.1	82 ATRS	C2	Tactical
225.875	Not Assigned - Currently Not in Use	Any	Tactical
234.8	HMN Approach (MOA Yonder S Disc.)	Area Discrete	Tactical
243.0	Emergency	ATC	ATC
251.25	HMN Command Post	C2	Tactical
253.4	82 ATRS	C2	Tactical
254.4	HMN Tower	ATC	ATC
256.7	Beak B	Area Discrete	ATC
259.2	Beak C	Area Discrete	Tactical
269.225	HMN Approach	ATC	ATC
273.5	HMN ATIS	ATC	ATC
275.8	HMN Ground	ATC	ATC
277.1	Reaper Common	C2	Tactical
277.4	SFA Discrete	ATC	ATC
279.7	Lava East/Red Rio	Range/Area Disc.	Tactical
284.0	HMN Departure	ATC	ATC
289.4	Clearance Delivery	ATC	ATC
311.5	Lava West	Area Discrete	Tactical
314.2	Mesa West	Area Discrete	Tactical
315.9	JTTOC	Area Discrete	Tactical
335.625	HMN Arrival	ATC	ATC
336.2	HMN Approach (MOA)	ATC	ATC
342.4	Yonder	Area Discrete	Tactical
343.2	311 FS Ops	C2	Tactical
348.8	ROC	Range	Tactical
353.6	Beak A	Area Discrete	Tactical
376.15	Melrose Range		
378.8	314 FS Ops	C2	Tactical
378.475	8 FS Ops	C2	Tactical
395.15	SOF	C2	Tactical
397.875	HMN Discrete	ATC	ATC

Attachment 3

SMALL UNMANNED AIRCRAFT SYSTEMS (sUAS) OPERATIONS

A3.1. Introduction. This attachment establishes local procedures for 49 SFS sUAS operations on or near HMN airfield. It supplements procedures listed in AFMAN 11-502 *Small Unmanned Aircraft Systems*. Manned aircraft procedures will take priority over sUAS operations.

A3.2. sUAS Procedures.

A3.2.1. Scheduling. 49 SFS will coordinate sUAS activity with 54 OSS/OSA NLT five days in advance with location, route, and duration of the activity. AM will publish a NOTAM for all scheduled sUAS activity. sUAS NOTAMs will include coordinates for activation and termination coordinates, HMN TACAN radial/DME, route of flight, and time of activity. When active, Tower will include sUAS activity in ATIS broadcast. The Program Manager (PM) will ensure a NOTAM is created prior to launch and that all information in the NOTAM correctly depicts the intentions for the operation. Any corrections that need to be made will be coordinated through AM.

A3.2.1.1. Any flights operating outside of HMN Class D surface area or outside of published airfield hours will require additional coordination with White Sands Missile Range scheduling. If R5107D is active coordination/approval is required with CHEROKEE on frequency 126.95 or 305.5 or by commercial phone (575) 678-8000.

A3.2.2. Flight Approval Procedures. sUAS operators will obtain approval from the airspace control agency prior to launch. In order of priority, obtain approval from: Tower, AM, and Uncontrolled Airfield Operations (UAO) Operations Duty Officer (ODO). As a last resort contact 49 WG/CP.

A3.2.3. Activation/Termination.

A3.2.3.1. Operations while airfield is open. The lead sUAS operator will maintain two-way radio communication (LMR Tower Net) with the Tower while sUAS operations are in progress. The SUAS team will provide Tower with a 30-minute call prior to launch and will coordinate real-time with Tower for activity extensions or route deviations. The sUAS team will contact Tower when operations are complete, and the unmanned vehicle has landed.

A3.2.3.2. Operations while airfield is closed. The lead sUAS operator will monitor the LMR Tower Net to maintain situational awareness of any activity on the airfield. The sUAS team will provide a 30-minute call prior to launch. The sUAS team will make a notification over the LMR Tower Net that operations are complete once the unmanned vehicle has landed.

A3.2.3.3. Operations while UAO is in effect. If sUAS operation times overlap with UAO, the lead sUAS operator will maintain two-way radio communication via the LMR Tower Net with the UAO ODO. The sUAS team will provide the ODO with a 30-minute call

prior to launch and will coordinate real-time for activity extensions or route deviations. The sUAS team will contact the ODO when operations are complete. The sUAS team is encouraged to coordinate with the UAO ODO 24-48 hours prior to overlapping operations to discuss proposed flights.

A3.2.3.4. Knock It Off. The Tower Watch Supervisor, AM, and the UAO ODO maintains approval/denial authority to all sUAS operations based on impact to flying operations and safety concerns and can terminate sUAS operations at any time.

A3.2.4. Flying Operations. sUAS will only be flown within the designated approved areas at a maximum operating altitude of 400 ft AGL. Lead sUAS operators are responsible for avoiding no fly zones depicted in Figure A3.1 unless prior approval has been granted by the Tower. All sUAS operations will only occur within the de-conflicted sUAS areas up to the maximum altitude allowed in each restricted operating zone. Visual observers will accompany sUAS operators and will always maintain visual contact with the sUAS and surrounding airspace during flight operations. Operators will be responsible for reading all applicable published NOTAMs prior to conduction activities. sUAS will always operate with full lighting (including transponder if equipped).

A3.2.4.1. The patrol sectors (see Figure A3.1.) listed in Table A3.1. will not be used for sUAS operations for traffic pattern deconfliction when each runway is the active landing runway:

Table A3.1. sUAS No-fly zones.

Landing Runway	No-Fly Patrol Sector
16	1
34	4
22	3
25	3
07	4
04	3 and 4

A3.2.4.2. Pilots should note where sUAS will be operating based on NOTAMs and avoid, if possible, in the event of abnormal operations. Despite sUAS operations being approved, pilots will still report any sUAS sightings to appropriate control agencies, even when a NOTAM is published.

A3.2.5. Contingencies. During flight abnormalities or unresponsive sUAS, the operator must attempt to land. If the sUAS experiences uncontrolled flight near or leaving the approved area boundary, immediately notify Tower, AM, or UAO ODO (in that order), with last known location, direction of flight, altitude, and lost link landing location. The SUAS operator will utilize lost link points in accordance with the area they are operating in. Lost link points are indicated in Figure A3.1. The sUAS operator will confirm that the observer has visual

contact with the sUAS. A mishap report and notifications will be completed IAW AFMAN 11-502. The Tower will cease aircraft launches/recoveries until status of affected sUAS is determined and will issue instructions/advisories to ensure safe operations for all aircraft.

A3.2.6. Separation/Sequencing Procedures. Tower will provide ATC services IAW AFMAN 13-204v3, FAAO JO 7110.65, and Memorandum of Agreement, DoD UAS in the National Airspace System.

A3.3. Emergency sUAS Operations. Any sUAS activity requiring airfield or airspace restrictions (e.g., real world force protection requiring a safety cordon or traffic pattern restrictions) will be coordinated with Tower real-time by the incident commander via frequency or program manager via landline.

Figure A3.1. sUAS No Fly Zones and Patrol Sectors.

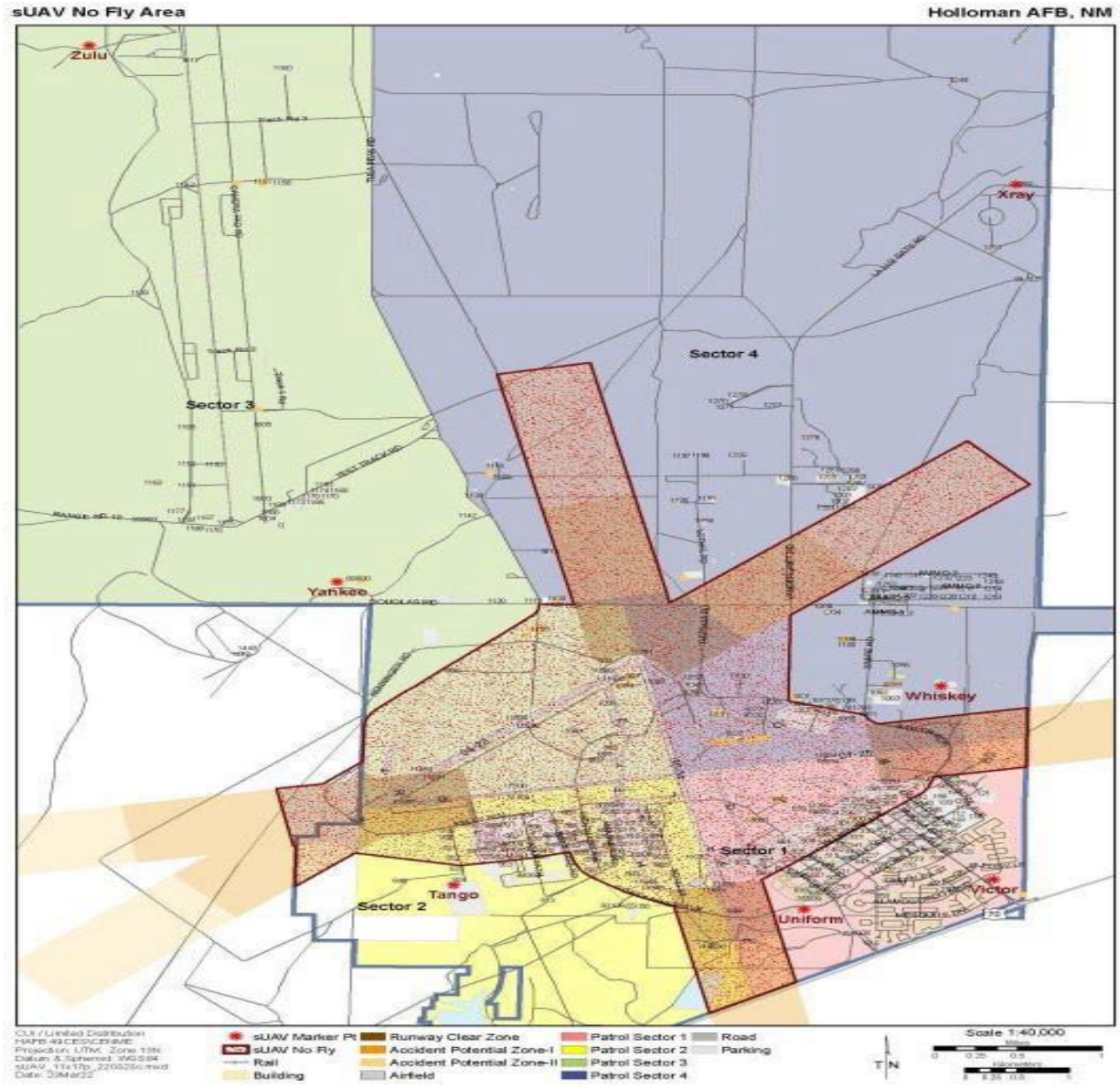


Figure A3.2. Traffic Pattern Hazards Outside KHMN Perimeter.



Attachment 4

EXPLOSIVE ORDNANCE DISPOSAL (EOD) OPERATIONS

A4.1. EOD Operations. This attachment establishes local coordination procedures for EOD operations on or near HMN airfield. It supplements procedures listed in paragraphs 3.4., 8.8., and AFMAN 32-3001 *Explosive Ordnance Disposal (EOD) Program*.

A4.2. Emergency EOD Operations. The on-scene commander will coordinate and obtain approval from Tower for any EOD activity requiring airfield or airspace restrictions (e.g., safety cordon, traffic patter restrictions, supersonic activity restrictions). Any request outside of airfield hours will be coordinated through 49 WG/CP.

A4.3. HMN EOD Demolition Range (see Figure A4.1.) operations may pose a potential hazard to aircraft on the west downwind Runway 16/34 and the north downwind Runway 04/22.

A4.3.1. Scheduling. EOD flight (49 CES/CED) will coordinate Demolition Range activity and restrictions with AM NLT 24 hours prior to use.

A4.3.2. Advisory. AM will publish a NOTAM for all scheduled EOD activity. EOD NOTAMs will include latitude, longitude, HMN TACAN radial/DME, and safety cordon information. When active, Tower will include EOD Demolition Range activity in ATIS broadcast.

A4.3.3. Detonation Activation/Termination. EOD will maintain two-way radio communication (LMR Tower Net) with the Tower while on the Demolition Range. EOD will provide Tower with a 2-minute warning prior to detonation and report when each detonation is complete. Real-time extensions of EOD Demolition Range activity will be coordinated with the Tower WS and are subject to approval/denial based on the impact to flying operations. **Note**: AGL for all detonations will be 1,250 ft.

Figure A4.1. EOD Demolition Range.



Attachment 5

QF-16 OPERATIONS

A5.1. Drone Operations. See 82 ATRS, 49 WG, and WSMR Safety Memorandum of Agreement for additional procedures.

A5.1.1. 82 ATRS Responsibilities.

A5.1.1.1. Brief Tower WS at least 1 hour prior to takeoff for all drone missions. Information will include but is not limited to:

A5.1.1.1.1. Estimated Time of Departure and Arrival.

A5.1.1.1.2. Mission frequency.

A5.1.1.1.3. Manned or unmanned.

A5.1.1.1.4. Mission summary (i.e., type of departure, length of time on range, number of approaches).

A5.1.1.2. Contact CHEROKEE for clearance into scheduled airspace.

A5.1.1.3. Release Runway 22 at Taxiway Echo to Tower for departures until 5 minutes prior to drone takeoff time (manned drone only).

A5.1.1.4. Advise Tower when manned drone is 18 or 10-flying miles (normally at the start of the recovery leg) and at 5 miles from touchdown Runway 04.

A5.1.1.5. Provide two-way radio equipped vehicle to assist in approach/departure area evacuation (unmanned drone operations only).

A5.1.1.6. Advise Tower when all personnel are clear of the runway and that normal operations may resume.

A5.2. Tower Responsibilities.

A5.2.1. Manned QF-16 Drone Procedures.

A5.2.1.1. Issue taxi instructions to aircraft. 82 ATRS support vehicles/personnel (i.e., Attila Launch) will normally lead manned QF-16 during taxi and Runway 22 line-up for preflight checks/alignments. Prior to departure, support vehicles/personnel will reposition to a safe distance behind and left/right of QF-16 at the edge of the Runway 22. Support vehicles/personnel will remain on the runway edge and vacate the runway after successful departure.

A5.2.1.2. Limit Runway 22 to drone operations only. Tower will verbally coordinate with X-Ray Control to confirm authorization of exceptions.

A5.2.2. Chase Aircraft.

A5.2.2.1. Once takeoff clearance has been issued, if pickup is immediate, the chase aircraft is automatically cleared to takeoff from any runway.

A5.2.2.2. If pickup is not immediate (i.e., chase aircraft is already airborne), obtain clearance from Tower before conducting drone takeoff and pickup operations.

A5.2.2.3. Suspend all departures from Runway 25 when the drone is 15-flying miles to Runway 04. Limit straight-in approaches to Runway 16/34 as necessary to allow full runway or missed approach separation for the drone. Arriving aircraft can expect to be delayed when drone operations are in progress.

A5.2.3. Unmanned Drone Procedures.

A5.2.3.1. Clear BM or FD personnel onto the runway for arresting gear configuration.

A5.2.3.2. The area surrounding Runway 22/04 will be evacuated 2,000 ft from the runway during armed drone launch/recovery to include the Delta, Echo EOR Areas, and Taxiway Golf. 82 ATRS personnel will visually ensure the areas are clear prior to permitting armed drone launch and prior to recovery.

A5.2.3.3. Departures. Tower will assist 82 ATRS personnel with evacuating all non-participating vehicles and aircraft within a 2,000 ft radius of the unmanned drone before authorizing the drone destruct system to be armed (see Figure A5.2.). Once notified the destruct system is being armed, Tower will ensure a 2,000 ft cordon from Runway 04/22 is in place for all non-participating vehicles and aircraft until the drone departs. This cordon includes portions of Taxiways Delta, Echo, Romeo, and Golf as well as their corresponding EOR areas (see Figure A5.3.).

A5.2.3.4. Arrivals. Prior to issuing landing clearance, Tower will ensure a 2,000 ft cordon from Runway 04/22 is in place for all non-participating vehicles and aircraft. The cordon will remain in effect until the drone is stopped on the runway, at which time the cordon will be adjusted to a 2,000 ft radius around the aircraft and will be canceled once the drone's destruction system is de-armed.

A5.3. Pick Up Pattern for Chase Aircraft.

A5.3.1. Normally, QF-16 chase aircraft will fly a pattern altitude of 5,100 ft MSL north of Runway 04/22.

A5.3.2. During manned QF-16 operations, the chase aircraft will normally depart from the same runway. During unmanned operations, the chase aircraft will normally depart from Runway 25 or 16.

Figure A5.1. Drone Traffic Pattern/Departure/Recovery Routes (Drawing not to scale).

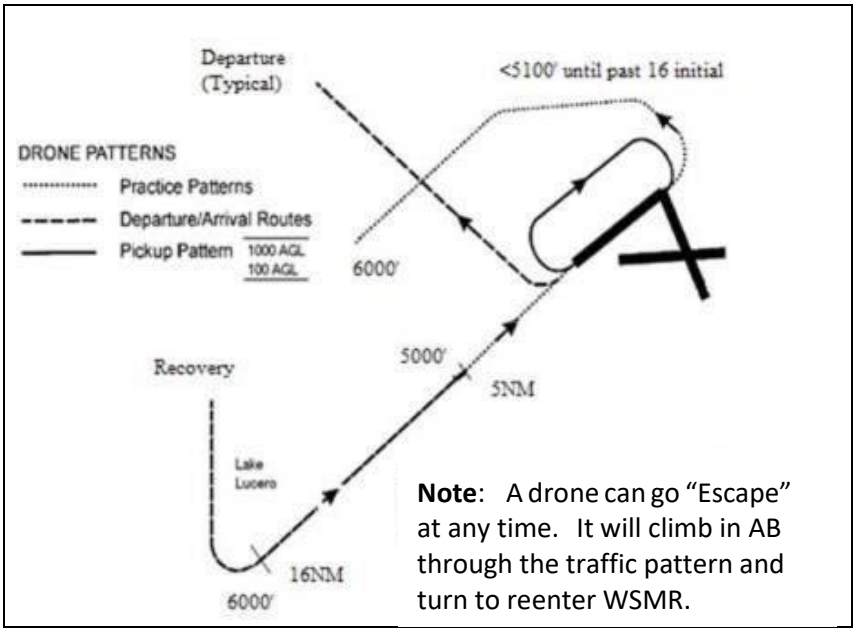


Figure A5.2. Unmanned QF-16 Drone 2,000 ft Line-up Cordon (Drawing not to scale).



Figure A5.3. Unmanned QF-16 Drone 2,000 ft Runway Cordon (Drawing not to scale).



Attachment 6

HELICOPTER OPERATIONS

A6.1. HMN Class D Helicopter Operations.

A6.1.1. Within 5 NM of HMN, VFR arrivals and departures will be at or below 4,600 ft MSL.

A6.1.2. Remain clear of all runways until approval to cross is obtained from Tower or IAW a specific arrival or departure route, which includes clearance to cross. Remain clear of the extended runway centerlines by at least 1,500 ft (lateral) within the airfield boundary. DO NOT overfly aircraft in run-up areas, taxiing, holding in takeoff position or in parking ramps (Transient Parking Area, North Ramp, Drone Parking, or West Ramp).

A6.1.3. Normal airfield traffic arrives and departs Runway 16. Any change of the recovery runway will be relayed by Tower to approaching aircraft. Locally assigned helicopters are authorized to depart and arrive on the northwest section of the North Ramp. Transient helicopters must use the runways for arrival and departure. Arrival or departure operations from non-CMA areas must be authorized by the AFM.

A6.1.4. Locally assigned arriving helicopters will be informed, "*(name of specific arrival) ARRIVAL APPROVED*". When reporting for landing, Tower will state "*LANDING AT NORTH RAMP WILL BE AT YOUR OWN RISK*" and if necessary, "*USE CAUTION (reason)*". Locally assigned departing helicopters will be informed "*DEPARTURE FROM NORTH RAMP WILL BE AT YOUR OWN RISK*" and if necessary, "*USE CAUTION (reason)*".

A6.2. Helicopter Operations Outside HMN Class D. After departure from HMN Class D, helicopter operations will be IAW AA Operations standard operating procedures.

A6.3. Standard VFR Arrival and Departure Routes (Figure A6.1.). Request specific departure/arrival stereo routings by name. Tower will specify an alternate routing if requested route cannot be provided.

A6.3.1. Monument. (Water tower HMN R-195/5.6 DME). When departing the North Side, proceed directly across Runway 16 and Runway 25 intersection and remain west of the West Ramp enroute to the water tower abeam WSNP Visitor Center (departure routing approval from Tower constitutes clearance to cross the Runway 16 and Runway 25 intersection). Aircraft departing from the Transient Ramp will proceed directly across Runway 16. On arrival call "*MONUMENT*" then report southwest of the West Ramp. Tower will sequence the helicopter to cross Runways 16 and Runway 25. Specific clearance from Tower is required. Tower may hold aircraft over the West Ramp or issue instructions to adjust route for traffic separation.

A6.3.2. Track. (HSTT Headquarters HMN R-295/2.4 DME; King 1 HMN R-316/0.9 DME). Departing North Side, cross the approach ends of Runway 16 and Runway 22 enroute to King 1 then to Track. Departure routing approval from Tower constitutes clearance to cross the Runway 16 and Runway 22 intersection. Departures from the Transient Ramp will proceed directly to Runway 16 and Runway 22 intersection enroute to King 1, then to Track. Departure routing approval from Tower constitutes clearance to cross Runway 07/25. On arrival, call Track then report over King 1. Tower will hold aircraft at King 1 or issue instructions to adjust route for traffic separation.

A6.3.3. Alamogordo. Depart North Side to the northeast of Runway 07/25 and east of Runway 04/22. Departure routing approval for aircraft departing the Transient Ramp constitutes clearance to cross Runway 07/25 directly to the North Side. On return call Alamogordo, then report 1-mile from the North Side. When Runway 07/25 is the primary landing runway, exercise caution, and make approach from the north-northeast quadrant. DO NOT overfly the HMN small arms range.

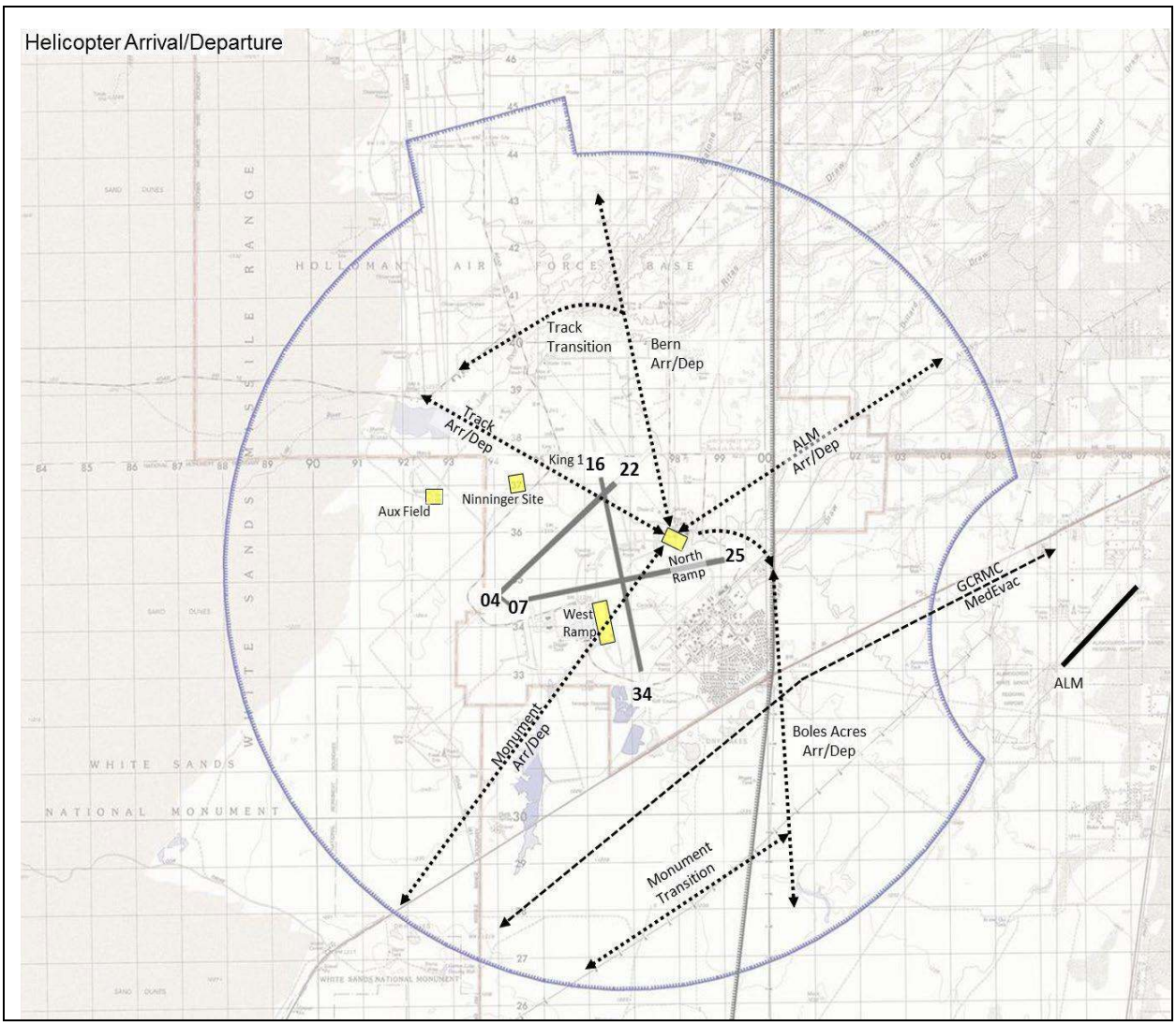
A6.3.4. Bern (HMN R-005/3.0 DME). Depart the North Side to the north, parallel to Runway 16 extended centerline and fly over perimeter road that leads to La Luz Gate. Departure routing approval for aircraft departing the Transient Ramp constitutes clearance to cross Runway 07/25 directly to the North Side. DO NOT overfly the Monkey Farm or small arms range enroute to Malone Draw (Bern), and then proceed to Tula Peak. If transitioning to the Track or Auxiliary (AUX) Field, advise Tower. Turn left at Malone Draw (3 DME), descend to 50 ft AGL, and cross the extended centerline of Runway 16 to the Track. On return, state your location and call for the Bern; report 2 NM north of the North Side. From Track, cross 3 DME north of Runway 16 extended centerline at 50 ft AGL.

A6.3.5. Boles Acres (South of KALM to Twin Butte). Depart the North Side to the east to remain east of the tank farm, then south along power lines staying east of the lines and housing area. Aircraft departing the Transient Ramp will proceed directly to the tank farm without overflying the Main Ramp and remain south of Runway 07/25. Aircraft departing the North Side will remain north of Runway 07/25. At power line intersection proceed to Boles Acres or toward WSMR. If going west toward WSMR from power line intersection, descend to 50 ft AGL to be low when crossing the extended centerline of Runway 34. On arrival (opposite departure routing), call "*BOLES ACRES*" then report the tank farm. When Runway 07/25 is the primary landing Runway, exercise extreme caution. Report the tank farm inbound and DO NOT proceed until clearance is obtained from Tower.

A6.3.6. If instructed to orbit to hold for runway crossing, ensure that orbit pattern does not overfly any ramp areas. In West Ramp Orbit, remain west/southwest of the West Ramp hangars and ramp. In North Side Orbit, remain east of Runway 16/34 and north of Runway 07/25, remain north/northeast of the North Side parking area. In King Orbit, remain northwest of Runway 22. In Transient Orbit remain northwest of Taxiway Alpha and east of Runway 16/34 and south of Runway 07/25.

A6.3.7. AUX Field Transition. Helicopters may modify any of the above departures by requesting an "*AUX FIELD TRANSITION*" in addition to the specific type of departure. All clearances/restrictions of the applicable departure still apply. AUX Field transition simply clarifies the destination of the helicopter. See Bern departure above for applicable restrictions on AUX Field Transitions.

Figure A6.1. Helicopter Arrival and Departure Routes (Drawing not to scale).



A6.4. Non-standard VFR Departures and Arrivals. Traffic permitting, non-standard routes may be approved by Tower. In non-emergency situations, helicopters will cross the fewest possible active Runways at designated points. Helicopters may obtain special VFR clearances from Tower or WSRF to operate within the Class D. Helicopters with radio failure will plan their approach to avoid crossing any extended runway centerline within 5 SM of the airfield.

A6.5. Helicopter NORDO Procedures. During radio failure in the HMN area the following procedures apply.

A6.5.1. If IFR, follow two-way radio failure procedures IAW the DoD FLIP *Flight Information Handbook*.

A6.5.2. If VFR, squawk 76 and the last two number of your call sign.

A6.5.2.1. If communication is lost after initial contact with Tower, continue to the established reporting point for that approach and look for a light gun signal from Tower.
A6.5.2.2. If not established on an approach and have not talked to Tower, circumnavigate the Class D, and proceed inbound on the Alamogordo Arrival. Proceed inbound to a quarter mile from the north side and look for a light gun signal from the Tower.

A6.6. Autorotation/Running Landing Areas. The primary autorotation/running landing area for AA or transient wheeled helicopters is Runway 22 between the approach end and departure end BAK-12 cables. AA may use Runway 25/07 or 16/34 between the approach end and departure end BAK-12 cables when Runway 22/04 is not available. Transient helicopters with skid-type landing gear (e.g., UH-1, OH-6, OH-58, AS-350) must have AM approval to perform autorotation landings on any runway.

A6.7. Helicopter NVD Operations. Only locally assigned AA helicopters are authorized to conduct NVD operations at HMN.

A6.7.1. NVD Scheduling. AA NVD operations will normally be scheduled through Scheduling and coordinated with AM NLT 24 hours in advance. Schedulers will include NVD in the AA aircraft's remarks section of the flying schedule. Changes within 24 hours of NVD start time will be coordinated with AM. Short notice NVD operations, if the pattern is available, may be coordinated directly with Tower real time.

A6.7.2. Notification/Coordination. Prior to commencing and at the conclusion of NVD operations:

A6.7.2.1. Tower must advise AM and WSRF that NVD operations are being conducted or terminated. Tower will include a NVD operations advisory in the ATIS broadcast.

A6.7.2.2. AM will confirm (scheduled) or publish (non-scheduled) a safety NOTAM. An airfield check of the runway and lighting configuration will be accomplished. If NVD ops are deemed unsafe or hazardous condition is found, AM will notify AA, Tower, CP, AFM, NAMO, 54 OSS/OSA/CC & SOF. A check of the affected areas must be complete before resuming normal operations.

A6.7.3. Airfield Lighting. When closed traffic or extended operations within the Class D airspace is requested:

A6.7.3.1. Runway 22/04. Prior to starting NVD operations, Tower will turn off all Runway 22/04 lights (HIRL, PAPI, DRM, ALSF-1, SFL, and Runway Threshold Lights).

A6.7.3.2. Runway 16/34 or 25/07. Lighting associated with Runways 16/34 or 25/07 may be operated as required for non-NVD aircraft arrival/departure/taxi operations.

A6.7.3.3. Taxiway lights on Taxiways Romeo, Echo, and Delta may remain illuminated as required for non-NVD aircraft taxi operations.

A6.7.4. Internal Tower Cab Lighting. Tower cab lights will be set as required for ATC operations. Tower will not wear or use NVDs.

A6.7.5. Aircraft Lighting (Position, Anti-collision, Strobe, etc.). AA will operate helicopter lighting IAW Army Regulation (AR) 95-1.

A6.7.6. Vehicle Lighting Requirements. Vehicle operations in the Runway 22/04 CMA are not authorized during NVD operations. Vehicle operations in the Runway 16/34 or 25/07 CMA are authorized on a non-interference basis.

A6.7.7. Weather Requirements. IAW AR 95-1, HAFBI 15-101, and Memorandum of Agreement between WSMR and HAFB.

A6.7.8. Communication.

A6.7.8.1. Communications with ATC will be normal for the approach being flown. Departure and landing clearances for unlit runway will be as follows: "*DEPARTURE FROM (runway or location) WILL BE AT YOUR OWN RISK" (additional instructions, as necessary). "USE CAUTION" (if applicable)* or "*LANDING AT (runway or location) WILL BE AT YOUR OWN RISK" (additional instructions, as necessary). "USE CAUTION (if applicable)".*

A6.7.8.2. Aircraft performing NVD operations will add the word "*aided*" to the call sign *(*e.g., *NIKE 43 aided).*

A6.7.9. Traffic Pattern Procedures. NVD Closed traffic operations will only be flown to Runway 22/04 IAW VFR pattern procedures listed in Chapter 6.

A6.7.9.1. Non-NVD aircraft will not mix with AA NVD helicopters in the Runway 22/04 traffic patterns or conduct pattern operations on other runways during AA NVD operations on Runway 22.

A6.7.9.2. Non-NVD departures from Runway 16 or 07 during AA NVD operations to Runway 22/04 are authorized. Non-NVD departures from Runway 34 or 25 or arrivals to Runways 16/34 or 25/07 during AA NVD operations to Runway 22/04 are not authorized. Non-NVD AA helicopter arrivals/departures from the North Ramp are authorized on a non-interference basis (do not use the Track Arrival/Departure).

A6.7.9.3. NVD Operational missions departing from or arriving to the Class D airspace may land and takeoff directly from the North Ramp. Tower will issue landing or takeoff "*at pilots' own risk*" instructions.

A6.7.9.4. NVD operations at the AUX field are authorized with proper coordination with Tower.

A6.7.10. Suspension/Restart Procedures.

A6.7.10.1. NVD operations will be suspended when a non-NVD aircraft reaches 15- flying NM from Runway 22/04. Should a non-NVD aircraft need to use Runway 22/04, Tower will inform the NVD aircraft that *"NORMAL LIGHTS COMING UP,* (reason)" and direct them where to go in the pattern. NVD helicopters can hold or continue with overt lighting while waiting to continue with NVD operations.

A6.7.10.2. Resume NVD operations after non-NVD aircraft have exited the landing runway or departed the Class D and the Runway 22/04 lights have been reconfigured for NVD operations.

A6.7.10.3. Emergency/KNOCK-IT-OFF. Aircrew or ATC may initiate a "*KNOCK-IT-OFF*". In such cases, the Tower will turn airfield lighting back on. Tower will state "*NORMAL LIGHTS COMING UP,*" wait 10 seconds, and then turn the lights on. This will allow the aircrew to take off NVD equipment prior to the lights coming up. Tower will issue any additional instructions.

A6.7.11. Taxi Routes. NVD air taxi between North Ramp and Runway 22/04 is authorized. NVD ground and hover taxi is authorized provided aircraft lighting is configured for unaided flight (position, anti-collision, and taxi lights on).

A6.8. AA Training Area (see paragraph 3.5.2.1.). Use of AA training area (also used as maintenance test flight area) is coordinated real-time with WSRF, ABQ ARTCC, or KROW and subject to restriction based on HMN runway configuration, Beak MOA activity, and traffic conditions.

Attachment 7

BALLOON OPERATIONS

A7.1. Balloon Operations.

A7.1.1. Air Force Research Laboratory/Balloon Operations (AFRL/RVEP) responsibilities.

A7.1.1.1. All balloon flights in the US are governed by Code of Federal Regulations (CFR) Title 14-Chapter I-Subchapter F-Part 101, *Moored Balloons, Kites, Amateur Rockets, and Unmanned Free Balloons,* and the FAA-H-8083-11A, *Balloon Flying Handbook*.

A7.1.1.2. Notify AM of estimated time of balloon launch, location, altitude climbing to, and whether it is tethered or un-tethered, 2 weeks before, then a follow-up telephone call 2 days before the scheduled balloon launch.

A7.1.1.3. Notify the FAA and ensure a NOTAM is published. All balloon launches must be approved by the 54 FG/CC or 54 FG Brickholder.

A7.1.1.4. Contact the Tower to request clearance to launch a balloon, stating the location of balloon launch, whether it is tethered or un-tethered, and altitude climbing to, or passing through all altitudes.

A7.1.1.5. After obtaining clearance from Tower, advise Tower when the balloon is passing through 2,000 ft AGL or if the launch is aborted or canceled.

A7.1.2. Tower Responsibilities.

A7.1.2.1. Approve or disapprove a balloon launch based on traffic in the VFR pattern. Ensure traffic is deconflicted with the balloon launch prior to clearance for the balloon launch.

A7.1.2.2. Coordinate with WSRF on any balloon launches affecting their airspace.

Attachment 8

AERO CLUB OPERATIONS

A8.1. 49 FSS/AC Standard VFR Arrival Procedures. 49 FSS/AC aircraft will descend and maintain 5,100 ft MSL and establish two-way radio contact with Tower before entering HMN Class D. Ingress call points (see Figure A8.2.) are the Otero Mill Airport (FAA identifier 72NM) 2 NM southwest of La Luz, GORDO (Highway 54/70 intersection, see Chapter 6), or old American Legion. Call when departing KALM. If Tower is unable to allow the aircraft to proceed inbound, comply with controller holding instructions.

A8.2. 49 FSS/AC Runway Specific Procedures.

A8.2.1. Runway 16.

A8.2.1.1. Runway 16 Departure. Climb to 5,100 ft MSL, cross Highway 70, climb to 5,600 ft MSL, and then proceed east until clearing Class D airspace.

A8.2.1.2. Runway 16 Arrival. Descend and maintain 5,100 ft MSL before entering Class D airspace. Maintain a westerly heading, turn southwest over 72NM. Maintain heading until abeam the extended Runway 16 centerline and turn left to align with the centerline. Watch for helicopters operating at the same altitude.

A8.2.2. Runway 34.

A8.2.2.1. Runway 34 Departure. Reverse Runway 16 arrival procedures. Maintain 5,100 ft MSL until clear of the Class D airspace.

A8.2.2.2. Runway 34 Arrival. Reverse Runway 16 departure procedures.

A8.2.3. Runway 22.

A8.2.3.1. Runway 22 Departure. Maintain 5,100 ft MSL after clearing departure end, and fly south crossing Highway 70, then proceed the same as the Runway 16 departure.

A8.2.3.2. Runway 22 Arrival. Same procedure for Runway 16 arrival, except turn south at the curve and follow the road until abeam extended Runway 22 centerline. Turn to align with the centerline.

A8.2.4. Runway 04.

A8.2.4.1. Runway 04 Departure. Reverse Runway 22 arrival procedures.

A8.2.4.2. Runway 04 Arrival. Reverse Runway 22 departure procedures.

A8.2.5. Runway 25.

A8.2.5.1. Runway 25 Departure. Maintain 5,100 ft MSL after clearing beyond MMG Ramp. Turn left and fly south, remaining east of WSNP Visitor Center, across Highway 70, then proceed as a Runway 16 departure.

A8.2.5.2. Runway 25 Arrival. Enter Class D at 5,100 ft MSL. If exiting KALM Runway 21 right downwind, expect extended left base. If inbound from the north or east, report GORDO and make straight-in approach. If inbound from the south, remain clear of KALM traffic patterns and report old American Legion.

A8.2.6. Runway 07.

A8.2.6.1. Runway 07 Departure. Climb to 5,100 ft MSL on runway heading until clear of the Class D airspace. Turn north or south when approved or when directed.

A8.2.6.1. Runway 07 Arrival. Enter Class D airspace at 5,100 ft MSL. If inbound from the south or exiting KALM pattern, expect right downwind to base. If inbound from the north or east, expect left downwind to base.

A8.3. Pattern Work/Departure. Coordinate with Tower for pattern work. Tower's approval is subject to pattern traffic. If no pattern work is desired or if Tower determines the pattern is not usable for 49 FSS/AC aircraft, 49 FSS/AC aircraft will depart the area.

A8.4. 49 FSS/AC Practice Areas. Procedures for the 49 FSS/AC VFR Training Areas are listed in paragraph 3.5.3.2. The boundaries are depicted in 49 WG IFG and include the altitudes from 500 ft AGL up to but not including 18,000 ft MSL. 49 FSS/AC aircraft will monitor WSRF Approach frequency when using the practice area during 49 WG flying window. The boundaries of the areas are as follows.

A8.4.1. North Practice Area. North: Tularosa; South: La Luz; East: Mountains; West: Highway 70/54.

A8.4.2. South Practice Area. North: Boles Acres; South: 2 NM north of Dog Canyon Road; East: Mountains; West: Highway. Use caution for helicopter periodic operations between KALM and R5103C.

A8.5. 49 FSS/AC NORDO Procedures.

A8.5.1. If outside HMN Class D airspace, land at KALM.

A8.5.2. If inside Class D airspace or landing at HMN is necessary, follow procedures in paragraph 8.10.

A8.5.2.1. 49 FSS/AC NORDO Orbit Area may be used for better visibility of Tower light gun signals or to give way to other aircraft and continue circling.

A8.5.3. After landing, exit the runway at the first available taxiway and look for a light gun signal before taxiing to parking or crossing any runways.

Figure A8.1. Runway 25 NORDO Orbit Area (Drawing not to scale).

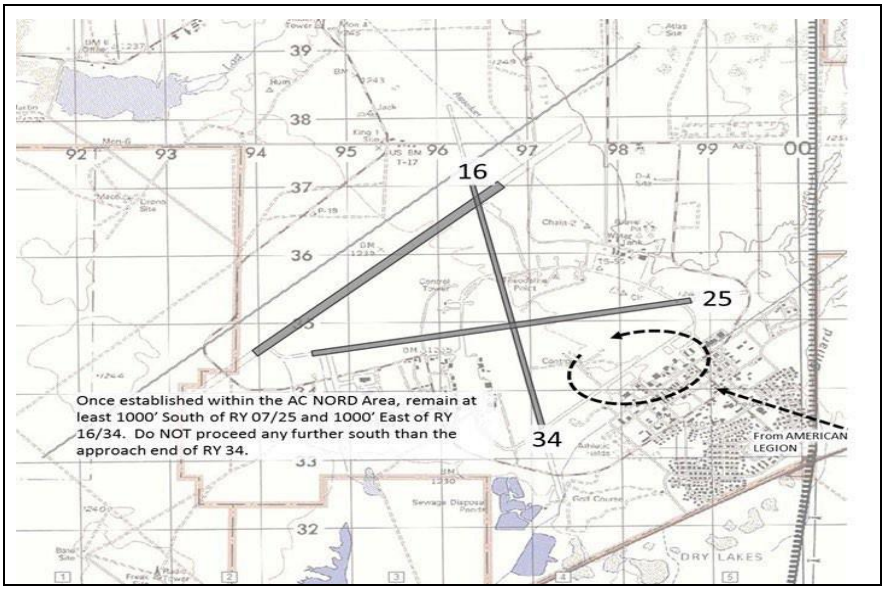
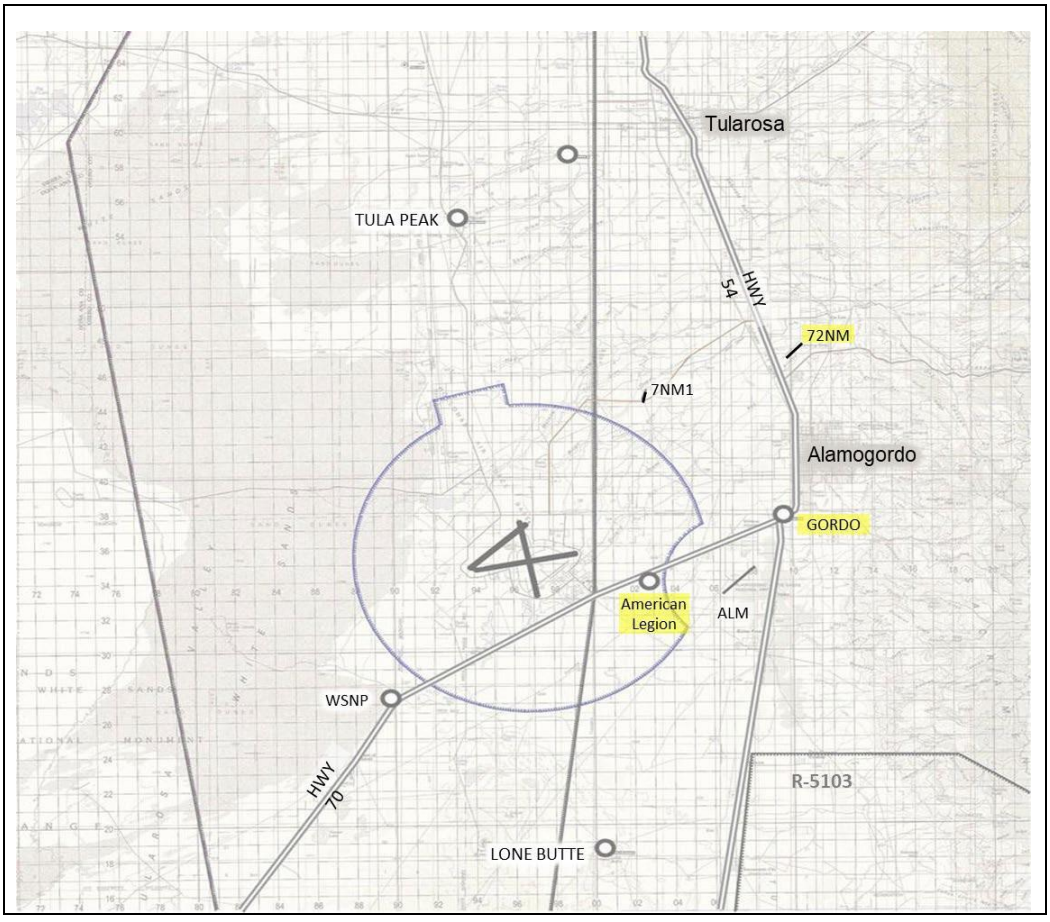


Figure A8.2. 49 FSS/AC Ingress Points (Drawing not to scale).



Attachment 9

MQ-9 OPERATIONS

A9.1. Introduction. This attachment complements AFMAN 11-2MQ-9V3 chapter 8 and AFMAN 13-204 Volumes 1-3.

A9.2. General Policy.

A9.2.1. Primary point of contact for command and control of MQ-9 missions is the Squadron Ops Sup. All communication with the crew not directly associated with the mission will be routed through the Ops Sup.

A9.2.2. MQ-9 Separation. MQ-9s are unable to accept see-and-avoid clearances; therefore, ATC must be more vigilant of separation when potential conflicts arise and must notify all aircraft involved immediately. ATC cannot ask aircrews to maintain visual separation. However, aircrews can report the runway in sight for continuation of VFR maneuvers. ATC will handle MQ-9s as follows.

A9.2.2.1. During Tower control.

A9.2.2.1.1. FAA SRS Category I has been deemed insufficient for MQ-9 separation. Instead, ATC may apply SRS Category III minima and use FAA Wake Turbulence Category “small” for all MQ-9 separation.

A9.2.2.1.2. Issue cautionary wake turbulence advisories, the position, altitude, and direction of flight to landing MQ-9 pilot, if in your opinion, wake turbulence may have an adverse effect on it. Wake turbulence rules cannot be waived by the MQ-9 pilot.

A9.2.2.2. During WSRF control.

A9.2.2.2.1. WSRF will provide an authorized route of flight for MQ-9s arriving and departing SUA. MQ-9 colored routes are not considered an ATC IFR Clearance. An IFR ATC clearance, when requested and issued, will consist of a clearance limit, SID or route of flight, altitude, frequency, and transponder code.

A9.2.2.2.2. MQ-9s versus Manned Aircraft. Standard Radar Separation IAW FAAO JO 7110.65 will be applied until the MQ-9 enters the HMN Class D airspace or upon entering their assigned HMN Working Areas.

A9.2.2.3. Issue cautionary wake turbulence advisories, the position, altitude, and direction of flight to landing MQ-9 pilot, if in your opinion, wake turbulence may have an adverse effect on it. Wake turbulence rules cannot be waived by the MQ-9 pilot.

A9.2.3. MQ-9 Orbit Locations. Due to lack of instrument navigation equipment, aircraft are unable to hold IAW AFMAN 11-202V3, *Flight Operations.* MQ-9s will “*orbit*” or “*delay*” as directed by ATC for aircraft deconfliction.

A9.2.3.1. To relieve radio congestion during pattern operations, the following orbit points (Table A9.1. and Figure A9.1.) are established for use by HMN-assigned MQ-9s as directed by Tower.

Table A9.1. MQ-9 Orbit Point Information.

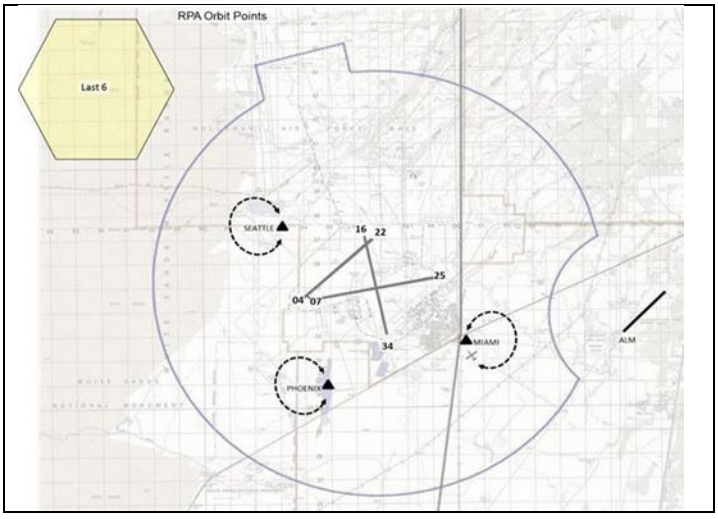
Name	Center Point	Orientation	No-Closer Point	Primary Use	Orbit Offset
Miami	N 32 49 12 W 106 02 57	250 M	N 32 49 40 W 106 03 50	RY 25	Remain South and East of point
Phoenix	N 32 47 49 W 106 08 00	340 M	N 32 48 40 W 106 07 25	RY 34	Remain South and West of point
Seattle	N 32 53 13 W 106 09 04	340 M	N 32 52 30 W 106 08 30	RY 16/22	Remain North and West of point

A9.2.3.2. When directed to orbit at a point, the MQ-9 must not fly any closer to any runway centerline than the "No-Closer" point. The orbits are elliptical, l NM wide by 1.5 NM long, and are aligned with the expected runway for use. These orbits are positioned to maintain 0.5 NM spacing from the edge of Class D airspace and 1.5 NM spacing from all extended centerlines. Direction of turns will be as requested or directed by ATC.

A9.2.3.3. Tower will assign each aircraft in orbit a set altitude and include the assigned altitude during all traffic advisories. Tower will continue to deconflict aircraft in the priority order listed in paragraph 2.30.

A9.2.3.4. Points Phoenix and Seattle are the primary orbit locations; point Miami requires extra vigilance to remain inside the lateral confines of the Class D airspace. Aircraft unable to maintain orbit parameters will inform Tower immediately. MQ-9 crews will not conduct pattern operations without graphical representation of the orbit points available in the cockpit.

Figure A9.1. MQ-9 Class D Orbit Points (Drawing not to scale).



A9.2.4. MQ-9s will normally operate in VMC and are not authorized to operate under Special Visual Flight Rules (SVFR). MQ-9s must avoid IMC to the maximum extent possible due to airframe limitations. MQ-9s may operate in IMC conditions when inside Restricted Airspace while expediting any transitions through precipitation and avoiding icing. Due to a lack of onboard instrumentation, colored routes (A9.3.4.) listed in this addendum will be used to the maximum extent.

A9.2.5. ATIS Procedures. During MQ-9 flight operations, Tower will include in the ATIS broadcast an advisory that "*UNMANNED AIRCRAFT OPERATIONS IN PROGRESS*".

A9.2.6. Filing of Stereo Routings. To comply with flight plan procedures of this instruction, the launching PIC will annotate, and sign for, a stereo routing in the remarks section of the AF Form 4327a which has a RDS of 1 year. To activate the stereo, the launching PIC or Squadron Ops Sup must call AM with the proposed call sign, departure times, arrival times, DSN phone number of the respective cockpit, and working airspace. AM will then file the appropriate stereo. LR crews must contact Clearance Delivery to prior to departure.

A9.2.7. Position Reporting. All MQ-9s assigned to HMN are not TACAN equipped and will use distance from their respective Ground Data Terminal (GDT) for position reporting. ATC must consider that this distance will be different for each MQ-9 as each uses a different GDT. Fixed GDTs are located on the infield approximately 200 ft east of Taxiway Delta midway between Runway 25 and Taxiway Charlie.

A9.2.8. Vehicles/Equipment/Personnel Near Runways. For safety concerns, Tower will ensure all vehicles, equipment, or personnel are outside the CMA of the runway in use before MQ-9 takeoff or landing.

A9.2.9. Automatic Takeoff and Landing Capability (ATLC). When using ATLC, aircrew will include the phrase "Auto" with takeoff and landing requests. Tower will acknowledge ATLC or include "Auto" in the takeoff or landing clearance.

A9.3. Mission Planning/Local Airspace Procedures.

A9.3.1. FAA COA. HAM maintains airspace COA that directly impact HMN local flying area. MQ-9 units will inform Scheduling when planning on operating in National Airspace System and adjacent ranges. Scheduling will do appropriate coordination with FAA and ranges IAW established agreements. Unit schedulers will advise Scheduling NLT 2 duty days prior to execution to allow time for proper coordination. The COAs are located on the 54 OSS SharePoint® and are updated annually or as listed within the COA.

A9.3.2. MQ-9 Airspace Deconfliction.

A9.3.2.1. The primary means of airspace deconfliction must be through the scheduling process with a backup of real time deconfliction utilizing CHEROKEE. This deconfliction does not alleviate the requirement for airspace clearance from WSRF. To facilitate departures and recoveries, WSRF may direct MQ-9s to maintain an altitude and descend over the field, within the confines of HMN Class D airspace.

A9.3.2.2. Yonder East. The use and scheduling of the airspace underlying the confines of Yonder East Areas (WSMR grid lines P – T, 28 – 50) are as follows.

A9.3.2.2.1. The "Yellow Container" (WSMR grid lines Romeo to Tango, 24 to 28, 10,000 ft MSL to assigned altitude) will be scheduled 15 minutes prior to the first scheduled MQ-9 launch until 45 minutes after the last scheduled MQ-9 launch. The confines of the Yellow Container may be extended as agreed upon at the weekly airspace scheduling meeting. When scheduled, WSRF may still use this airspace for departures or recoveries of other aircraft.

A9.3.2.2.2. When scheduled, the Yellow Container will be added to the HMN Airspace Restrictions page by Scheduling. MQ-9 Squadrons that desire to use this airspace in conjunction with Wiley Airspace must coordinate with Wiley Ops Sups. If launches are delayed, MQ-9 Ops Sups must notify Wiley Airspace owning Ops Sups and Scheduling of the extended launch window.

A9.3.2.2.3. Yonder East airspace east of the WSMR Romeo grid line within R5107D 10,000 ft MSL to 22,000 ft MSL, may be scheduled in conjunction with Wiley Airspace. When requested, Scheduling will place this request on the CSE Airspace Schedule. In addition, requesting Ops Sup must gain final approval from the WSRF WS (DSN 258-8000 or 8001) for use of the altitudes 10,000 ft MSL to 18,000 ft MSL NLT 1 hour prior to mission start time.

A9.3.2.2.4. Yonder East airspace west of the WSMR Romeo grid line within R5107D at 10,000 ft MSL and above may be scheduled real-time by requesting Ops Sup through the WSRF WS.

A9.3.2.2.5. WSRF WS is the final authority on operations within Yonder East and can deny or recall the airspace at any time if conditions require. CHEROKEE must contact aircraft on the appropriate working frequency passed by the aircraft, via an Airborne Warning and Control System controller, or on 243.0.

A9.3.3. MQ-9 Colored Routes. MQ-9 aircrew must fly the colored routes (Figure A9.2. and Table A9.2.) to transition restricted airspace unless otherwise directed by CHEROKEE. MQ-9 colored routes are scheduled and used as follows.

A9.3.3.1. Green Route (South) SFC to Assigned Altitude (Green Route points S1 through S9) may be used by MQ-9s to facilitate departures and recoveries between working airspace and HMN, regardless of airspace ownership by 49 WG units. However, MQ-9s are still required to abide by all airspace restrictions affecting the Green Route. The Green Route corridor is ½ NM inside WSMR Restricted Airspace perimeter.

A9.3.3.2. Departures and recoveries along any other colored route require MQ-9 Ops Sup to coordinate with affected airspace owners. MQ-9 Ops Sups will notify WSRF WS of the coordinated route and altitudes.

A9.3.3.3. WSRF will advise airspace users if a colored route is active, i.e., "*Green Route (South) is active at Fight Level 180*." WSRF will include amplifying information if MQ-9s are climbing or descending, by providing the associated block altitude information. Aircraft transiting MQ-9- colored routes will fly altitudes as assigned by CHEROKEE.

Figure A9.2. MQ-9 Colored Routes (Drawing not to scale).

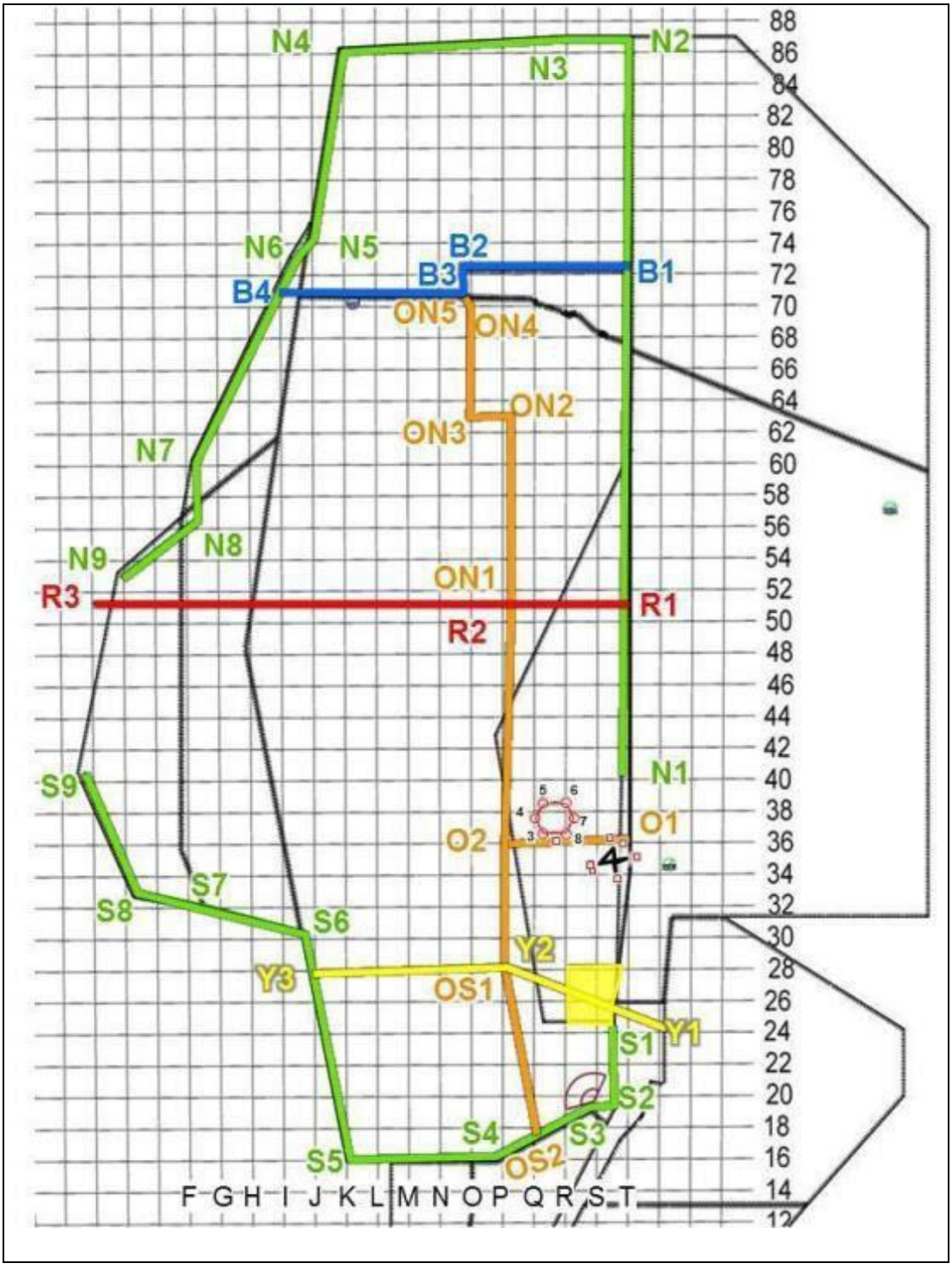


Table A9.2. Departure/Arrival Routing Points, Latitude/Longitude

Green Routes	
N1: N 33° 00’ 00” / W 106° 05’ 00”	S1: N 32° 36’ 00” / W 106° 06’ 30”
N2: N 34° 16’ 30” / W106° 04’ 30”	S2: N 32° 25’ 30” / W 106° 06’ 30”
N3: N 34° 16’ 30” / W 106° 12’ 10”	S3: N 32° 25’ 20” / W 106° 09’ 03”
N4: N 34° 15’ 10” / W 106° 40’ 00”	S4: N 32° 20’ 15” / W 106° 20’ 50”
N5: N 33° 56’ 20” / W 106° 43’ 25”	S5: N 32° 19’ 55” / W 106° 39’ 07”
N6: N 33° 53’ 30” / W 106° 45’ 54”	S6: N 32° 43’ 26” / W 106° 44’ 34”
N7: N 33° 32’ 30” / W 106° 58’ 10”	S7: N 32° 46’ 14” / W 106° 56’ 47”
N8: N 33° 26’ 30” / W 106° 58’ 10”	S8: N 32° 47’ 38” / W 107° 05’ 29”
N9: N 33° 20’ 30” / W 107° 07’ 30”	S9: N 33° 00’ 06” / W 107° 12’ 03”
Blue Route	
B1: N 33° 53’ 00” / W 106° 04’ 45”	B3: N 33° 50’ 15” / W 106° 25’ 10”
B2: N 33° 53’ 00” / W 106° 25’ 10”	B4: N 33° 50’ 15” / W 106° 48’ 00”
Red Route	
R1: N 33° 17’ 45” / W 106° 04’ 30”	R3: N 33° 17’ 45” / W 107° 11’ 00”
R2: N 33° 17’ 45” / W 106° 19’ 00”	
Yellow Route	
Y1: N 32° 33’ 40” / W 106° 00’ 00”	Y3: N 32° 39’ 20” / W 106° 44’ 15”
Y2: N 32° 40’ 00” / W 106° 19’ 45”	
Yellow Container	
NE: N 32° 40’ 00” / W 106° 05’ 20”	SW: N 32° 34’ 00” / W 106° 11’ 50”
E: N 32° 36’ 00” / W 106° 06’ 30”	NW: N 32° 40’ 00” / W 106° 11’ 50”
SE: N32° 34’ 00” / W 106° 06’ 30”	
Orange Routes	
ON1: N 33° 17’ 45” / W 106° 19’ 00”	O1: N 32° 53’ 15” / W 106° 04’ 30”
ON2: N 33° 37’ 20” / W 106° 19’ 20”	O2: N 32° 52’ 45” / W 106° 19’ 55”
ON3: N 33° 37’ 20” / W 106° 24’ 25”	
ON4: N 33° 48’ 45” / W 106° 24’ 25”	OS1: N 32° 40’ 00” / W 106° 19’ 45”
ON5: N 33° 50’ 15” / W 106° 25’ 10”	OS2: N 32° 22’ 10” / W 106° 15’ 45”

A9.4. Ground Operations.

A9.4.1. MQ-9 Designated Start Areas: Main Ramp, Sunshades, Hangar 564, Hangar 565, Alpha LOLA, Echo LOLA, North Ramp, and all EORs.

A9.4.2. MQ-9 Towing Procedures. Towing procedures are IAW DAFI 13-213_HAFBSUP.

A9.4.3. MQ-9 Taxi Restrictions.

A9.4.3.1. MQ-9s using Line-of-Sight (LOS) taxi capability must not be taxied to or through the West Ramp, MMG Ramp, or on Taxilane Alpha abeam the T-38/Alpha engine run hush houses. MQ-9s using SATCOM capability are exempt from this restriction.

A9.4.3.2. Taxiing MQ-9s are extremely susceptible to heat extremes above 95°F. During these times, aircraft should be given the most direct taxi routing and expedited takeoff clearance to reduce heat exposure on the ground.

A9.4.3.3. Sunshades. Aircrews may taxi out of the sunshades from south to north. A marshaller will be used until clear of the sunshade. When recovering to the sunshades, aircrew will use a marshaller to taxi into the parking area and then shut down. Crews will not taxi into the sunshades.

A9.4.4. MQ-9 AAS Configurations.

A9.4.4.1. MQ-9s are unable to taxi over BAK-12 cables therefore Tower will ensure cables are configured appropriately as required utilizing BM and coordinating with AM.

A9.4.4.2. 15 minutes prior to takeoff and landing, Tower must ensure all applicable cables are de- strung. During overlapping fighter/MQ-9 flying, the SOF determines the cable configuration for the runway(s) in use.

A9.4.4.3. Runway 16/34 is the primary MQ-9 departure and arrival runway.

A9.5. Departure Procedures.

A9.5.1. MQ-9s must depart via stereo route listed in 49 WG IFG or Colored Route. PIC must contact Clearance Delivery to request a specific-colored routing. Clearance Delivery must issue the clearance with stereo or colored route, departure frequency, and transponder code.

A9.5.2. If active airspace restrictions dictate, Clearance Delivery must issue crossing restrictions along the route such as "*CROSS THE 44 LINE AT OR ABOVE 13,000*". Outside HMN Class D airspace, MQ-9s must resume the departure route requested via the aircraft's flight plan.

A9.5.3. Example Flight Plan Clearances.

A9.5.3.1. "*BONES 25, MOUNTAIN 2 DEPARTURE AUTHORIZED AT 0000Z, DEPARTURE FREQENCY CHANNEL 4, SQUAWK 0271*".

A9.5.3.2. "*CLAW 20, ORANGE NORTH TO RED WEST DEPARTURE AUTHORIZED, DEPARTURE FREQENCY CHANNEL 4, SQUAWK 0271*".

A9.5.3.3. "*KNIGHT 09, ORANGE NORTH DEPARTURE AUTHORIZED, CROSS THE 44 LINE AT OR ABOVE 13,000, DEPARTURE FREQUENCY CHANNEL 4, SQUAWK 0271*".

A9.5.3.4. *"CLAW 24, SANDS1 DEPARTURE AUTHORIZED, DEPARTURE FREQUENCY CHANNEL 4, SQUAWK 0261."*

A9.6. Range Operations/Weapons Employment.

A9.6.1. Local range operations will be IAW AFI 13-212_HAFB ADDENDA-A, and when inside WSMR airspace, WSMR FSOP governing the specific flight mission.

A9.6.2. Laser boresight runs on Oscura range must be south to north only and IAW WSMR FSOP governing the specific flight mission.

A9.6.3. In case of unexploded ordnance, the PIC will pass aircraft release parameters and intended target to RCO. If impact point is observed, pass coordinates as well.

A9.6.4. HMN and TDY MQ-9s must comply with SUA avoidance area restrictions listed in paragraph 3.4.1.

A9.6.5. Under current agreements, WSRF controls all aircraft above the coordination altitude in R5103, to include aircraft dropping ordnance from above the coordination altitude, along with all fast movers below the coordination altitude. Bliss Radio controls all aircraft below the coordination altitude except for fast movers.

A9.6.5.1. Aircraft that will operate in McGregor North/South and are scheduled to use the CFA/Centennial Range later will contact CHEROKEE 5 minutes prior to using the CFA/Centennial Range (e.g., MQ-9s dropping GBU-12s on Centennial). Once CHEROKEE has coordinated with Bliss Radio, they will clear the aircraft to use the CFA/Centennial Range. Aircrew WILL NOT utilize the CFA/Centennial Range until this clearance has been granted. **Note**: This does not prevent aircraft from overflying the CFA/Centennial Range, above the coordination altitude, as scheduled. However, it will improve safety by ensuring that CHEROKEE provides Bliss Radio notification prior to aircraft either flying below the coordination altitude or releasing ordnance that will pass below the coordination altitude.

A9.6.5.2. Ops Sups and Aircrew will be notified that the CFA is active via comments in the “Operating Airspace” section of the Daily Airspace Schedule available via CSE.

A9.6.6. MQ-9 discrete frequency in assigned airspace. MQ-9 crews will notify CHEROKEE once established in assigned airspace and advise of switching to CHEROKEE discrete (see Attachment 2). If the frequency is not available, crews will continue to monitor CHEROKEE primary.

A9.6.6.1. Crews will continuously monitor CHEROKEE primary or discrete for radio calls and minimize communication to ensure a sterile radio environment. Any airspace coordination calls from the crews will be accomplished on CHEROKEE primary.

A9.7. Arrival Procedures.

A9.7.1. All MQ-9s must recover from restricted airspace via a colored route (see paragraph A9.3.4) or as directed by ATC. If required, appropriate altitude restrictions along recovery route must be issued by CHEROKEE. Generally, fighter aircraft will have traffic priority during their recovery and MQ-9s may be requested to "*ORBIT PRESENT POSITION*", "*PROCEED TO LAST SIX*", or orbit as directed by ATC.

A9.7.2. MQ-9s must enter HMN Class D airspace via the Orange or Green Routes (see paragraph A9.3.4) or as directed by ATC. If necessary for clarification, the controller will specify route points and the order in which to fly them.

A9.7.3. Example recovery phraseology.

A9.7.3.1. ATC: "*CLAW 24, CLEARED OFF RANGE VIA GREEN, SOUTH*".

A9.7.3.2. ATC: "*BONES 38, CLEARED OFF RANGE VIA RED, EAST, ORANGE, SOUTH*".

A9.7.3.3. ATC: "*KNIGHT 30, CLEARED OFF RANGE VIA ORANGE, SOUTH, CROSS THE 44 LINE AT OR ABOVE 13,000*".

A9.8. VFR Traffic Patterns.

A9.8.1. MQ-9 operations in Class D outside of R5107D. Aircrew and ATC will adhere to the following procedures when conducting MQ-9 operations within Class D airspace outside of R5107D.

A9.8.1.1. During nighttime operations, mixing of civil manned and MQ-9 traffic within Class D airspace is prohibited.

A9.8.1.2. Tower will monitor the Tower Display Workstation (TDW), if available, as a supplement to ensure no traffic is approaching the controlled airspace without making the required radio contact. Additionally, information from the TDW will be used to help reduce possible nighttime optical illusions. If the TDW is not operational, night operations will not be authorized.

A9.8.1.3. A ground-based observer will monitor the MQ-9 as it makes its approach to Runway 25. The SOF, Tower WS, or any other controller in the tower cab can act as the Ground Observer, in that order of preference, and in accordance with the following:

A9.8.1.3.1. The local controller will notify the ground observer when the MQ-9 exits R5107D for approach to Runway 25.

A9.8.1.3.2. The Tower WS or any Tower controller will not perform observer duties while also acting in an air traffic capacity. The SOF will not perform observer duties while assisting with a separate emergency or other abnormal situation.

A9.8.1.4. While outside of R5107D airspace, the observer must maintain sufficient visual contact with the MQ-9 to ensure avoidance of other air traffic with the MQ-9. Should a traffic conflict arise, the ground observer is expected to alert ATC to the situation. **Note**: Only ATC personnel may authorize direct communication over ATC frequencies; however, this does not preclude the ground observer from using whatever communication is necessary to prevent an unsafe situation that could immediately jeopardize safety.

A9.8.1.5. Incident/Accident/Mishap reporting procedures. Immediately after an incident or accident, and before additional flights, MQ-9 Ops Sup, or designated representative will inform the HAM.

A9.8.2. MQ-9 VFR Pattern Weather Minimums. Overhead/Rectangular: Ceiling must be 500 ft above the requested Overhead/Rectangular pattern altitude or orbit area and 3 SM visibility. SFO: Ceiling must be 1,000 ft above pattern altitude and visibility 5 SM.

A9.8.3. Overhead Pattern. Pattern altitude (initial point) is 5,600 ft MSL or as directed by ATC. MQ-9 overhead patterns at night are permitted with operational infrared (IR) camera or sensor. Standard MQ-9 overhead break is away from the main base (Runway 16/34 west break, Runway 22/04/25/07 north break).

A9.8.4. Rectangular Pattern. MQ-9 pattern altitude is 5,600 ft MSL.

A9.8.4.1. Runway 16 is normally west patterns. Expect orbits at Phoenix, Seattle, on downwind, or as directed by Tower for deconfliction from departing and arriving traffic.

A9.8.4.2. Runway 34 is normally west patterns. Expect orbits at Phoenix, Seattle, on downwind, or as directed by ATC for deconfliction from departing and arriving traffic.

A9.8.4.3. Runway 07/25 is normally north patterns. Expect orbits at Miami, Phoenix, on downwind, or as directed by Tower for deconfliction from departing and arriving traffic. Avoid left traffic to runway 25 to the max extent possible.

A9.8.4.4. Runway 04/22 is normally north patterns. Expect orbits at Phoenix, Miami, on downwind, or as directed by Tower for deconfliction from departing and arriving traffic.

A9.8.4.5. Unless directed by Tower, MQ-9 crews will not start closed pattern turns from Runways 16/34 and 25 in the direction of the Control Tower until at least 400 ft AGL or at the departure end of the runway.

A9.8.5. Protection of the 360 Degree Overhead Pattern. After completing approach, maintain at or below 5,600 ft MSL until departure end unless otherwise directed by Tower. For Runway 25, maintain at or below 5,600 ft MSL until west end of Taxiway Golf. Fighter overhead pattern breakouts may climb to 6,600 ft MSL in the direction of the pattern until clear of the outside downwind. When an orbit area conflicts with overhead pattern breakout, Tower will direct MQ-9 orbits at or above 7,100 ft MSL.

A9.8.6. MQ-9 Straight-in. MQ-9s will be at or below 5,600 ft MSL prior to departing 3 NM final.

A9.8.7. MQ-9 SFO Restrictions/Cut-offs.

A9.8.7.1. Only locally assigned MQ-9s may conduct SFO approaches at HMN.

A9.8.7.2. Prior coordination with Tower WS is required to conduct an SFO. Prior coordination does not constitute blanket approval. Approval or disapproval must take place real time after considering traffic volume and safety concerns. Straight-in or Alternate-Entry SFOs are not authorized.

A9.8.7.3. No more than two MQ-9s will be allowed to execute SFO procedures simultaneously. Maximum number of aircraft operations while an SFO approach is being conducted are Fighters: two; Heavy aircraft: one; Turboprop: two; MQ-9: three. Aircraft operations include aircraft requesting departure, inbound aircraft within 15- flying miles (regardless of frequency) and over flights within the Class D airspace (regardless of frequency). Tower WS will determine the number of aircraft that can safely operate in the VFR pattern. Tower must consider other aircraft characteristics, climb-out, missed approach, availability of the Tower Radar Display, active restricted airspace, and FAA mandated intersecting/same runway separation before authorizing other operations during SFO patterns.

A9.8.7.4. MQ-9 SFO versus IFR/VFR Straight-in Arrivals. Other aircraft inbound to the same or intersecting runway must be more than 15 NM away for fighters and 10 NM away for all other aircraft before the MQ-9 leaves high key. Aircraft inside the 15/10- NM cutoff may be sequenced behind the MQ-9 or held outside the Class D airspace.

A9.8.7.5. MQ-9 SFO versus VFR Rectangular/Overhead Pattern. Tower may allow MQ-9 SFOs to leave high key and follow traffic that has passed base turn.

A9.8.7.6. MQ-9 SFO versus IFR/VFR Departures. Departures from the same or intersecting runway must be rolling before an SFO reports low key.

A9.8.7.7. SFOs are only authorized to terminate in a low approach.

A9.8.7.8. SFOs will be approved only between sunrise and sunset. Simultaneous non-MQ-9 pattern ops are not authorized.

A9.8.8. SFO Pattern Specifications.

A9.8.8.1. Once Tower approves the SFO recovery, climb or descend to 6,600 ft MSL (MQ-9) along the inside or outside downwind ground track. Report high key and perform engine-out procedures while remaining within the inside downwind ground track.

A9.8.8.2. All SFO patterns begin with the MQ-9 aligned with the runway centerline, at the approach end of the landing runway. SFO patterns will be flown in the same direction as MQ-9 overhead patterns (Figures A9.3. through Figure A9.8.).

A9.8.8.3. All SFO patterns will remain clear of HMN housing when able.

A9.8.8.4. MQ-9 SFO high key is over the runway approach end at 6,600 ft MSL and airspeed approximately 100 knots.

A9.8.8.5. MQ-9 SFO low key is at a point abeam the intended touchdown zone, between 5,200 ft and 5,500 ft MSL.

A9.8.8.6. SFO Breakouts are listed in Table A9.3.

Table A9.3. SFO Breakouts.

Runways 16/34: Westerly climbing turn back to high key altitude.
Runways 04/22/07/25: Northerly climbing turn back to high key altitude.
Different breakout procedures may be directed by Tower for deconfliction purposes or approved upon pilot request. "*BREAKOUT*" will normally not be used once an MQ-9 has departed Low Key, instead Tower will use "*GO-AROUND (additional instructions as necessary)*".

A9.8.9. SFO Phraseology.

A9.8.9.1. Request SFO approach with Tower by stating "*(call sign), REQUEST HIGH KEY*".

A9.8.9.2. Traffic permitting, Tower will approve the SFO by stating "(c*all sign), REPORT HIGH KEY*".

A9.8.9.3. Report High Key, "*(call sign), HIGH KEY*".

A9.8.9.4. Tower will request the MQ-9 to report Low Key and include surface wind information "*(call sign), REPORT LOW KEY, WIND ONE SIX ZERO AT EIGHT*".

A9.8.9.5. Report Low Key, "*(call sign), LOW KEY, GEAR DOWN, LOW APPROACH*".

A9.8.9.6. At Low Key, Tower will issue low approach clearance or alternate instructions.

A9.8.10. West Outside Downwind/Re-Entry (Runway 16/34 only) may be used for training or deconfliction from departing and arriving traffic. West Outside Downwind ground track is displaced 3 NM west of Runway 16 or 34.

A9.8.10.1. At departure end initiate a right turn (Runway 16) or a left turn (Runway 34), maintain at or below 5,600 ft MSL. Higher than standard pattern altitudes may be requested or directed. Coordinate intentions (overhead, straight-in, or SFO) with Tower when established on the West Outside Downwind. Turn 90-to-initial or base to intercept 3 NM. At 90-to-initial, pilots will report: "*(call-sign) 90-TO-INITIAL, or (call-sign) DESCENDING STRAIGH-IN*". Tower will acknowledge the position report and may issue additional reporting instructions or non-standard deconfliction orbits when required.

Figure A9.3. Runway 16 SFO Pattern (Drawing not to scale).

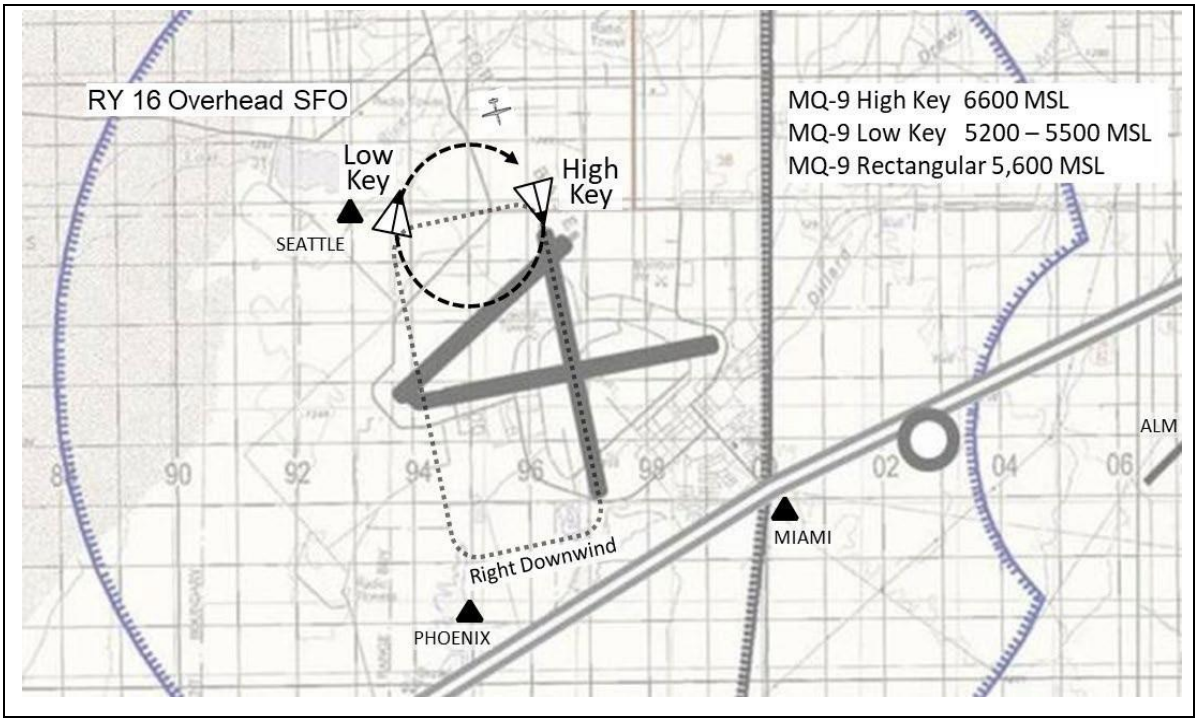


Figure A9.4. Runway 34 SFO Pattern (Drawing not to scale).

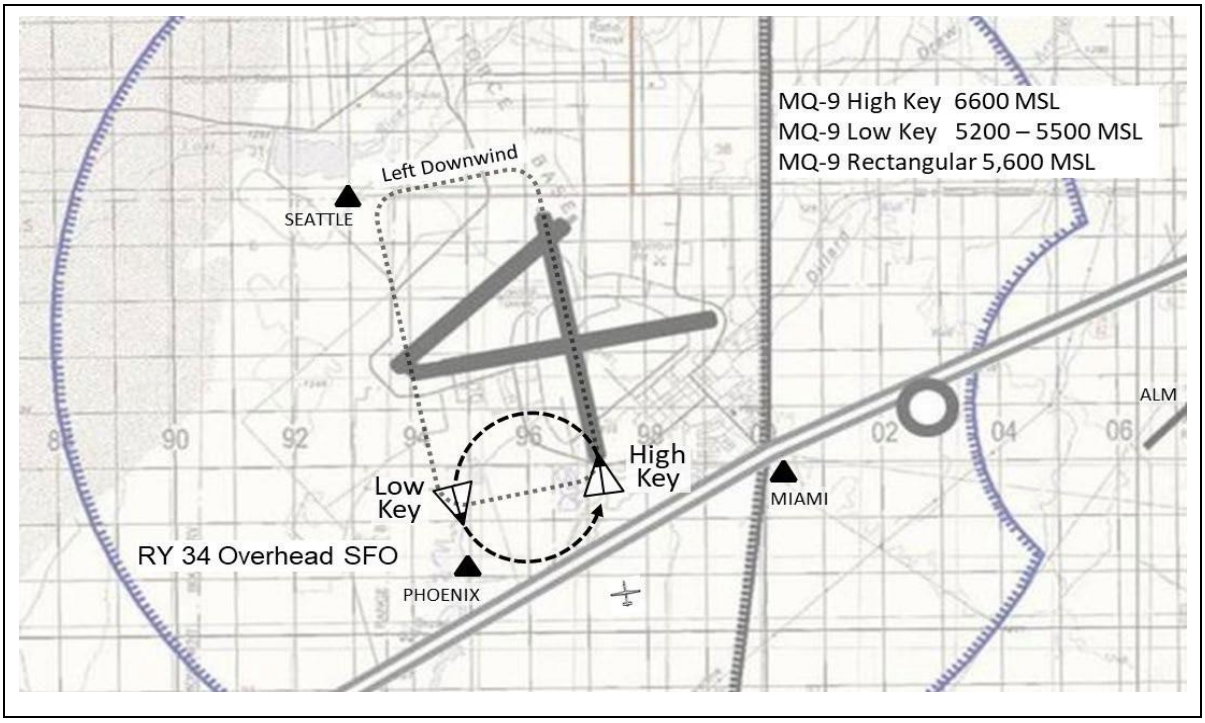


Figure A9.5. Runway 04 SFO Pattern (Drawing not to scale).

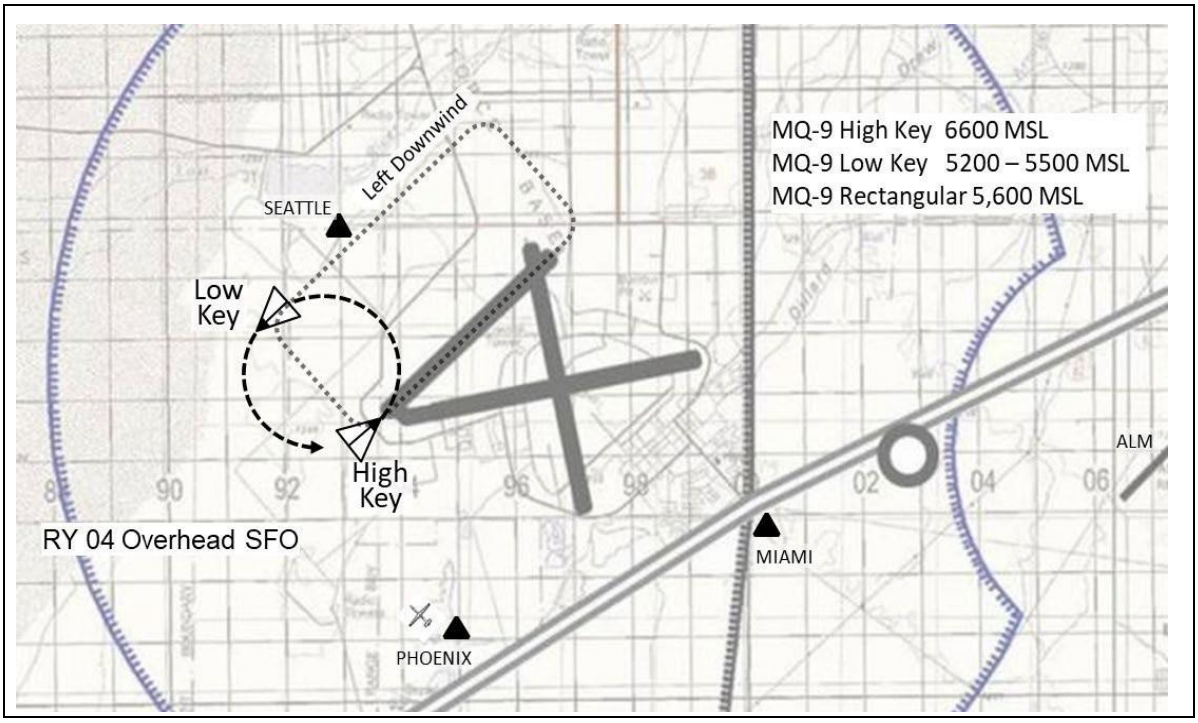


Figure A9.6. Runway 22 SFO Pattern (Drawing not to scale).

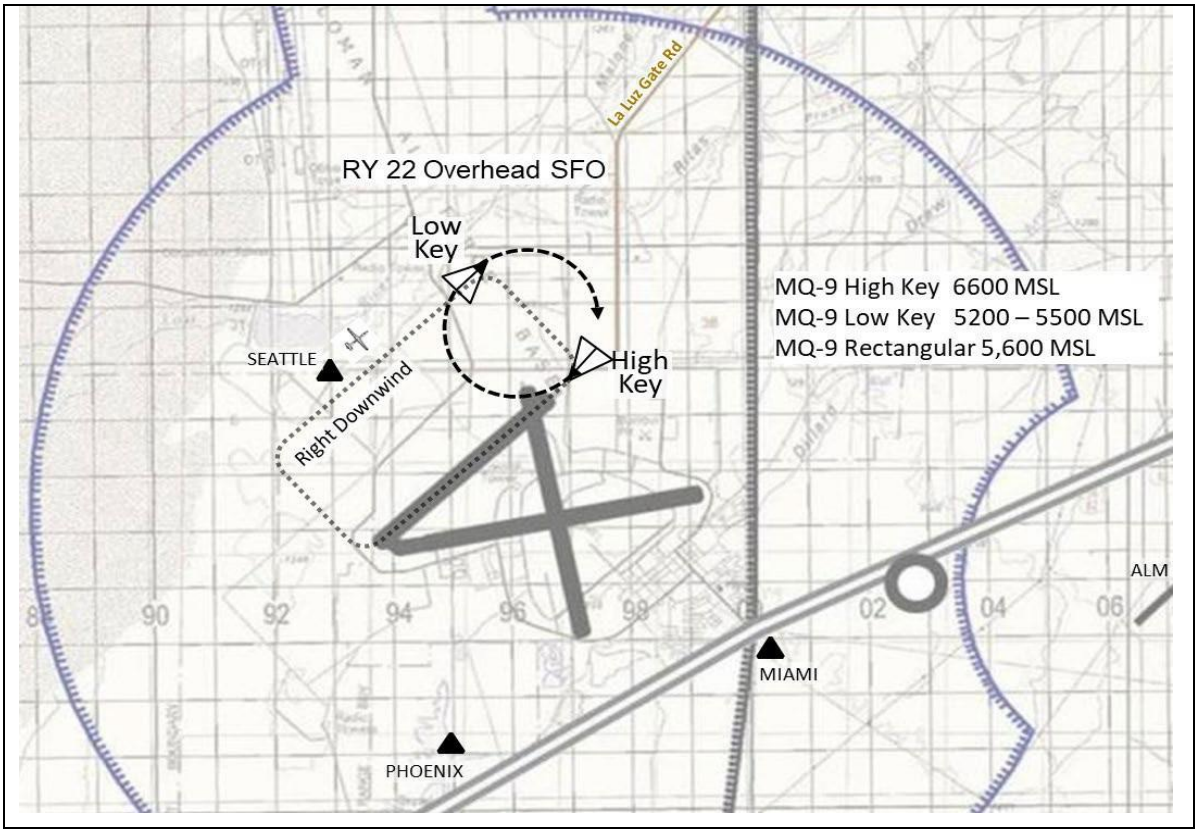


Figure A9.7. Runway 07 SFO Pattern (Drawing not to scale).

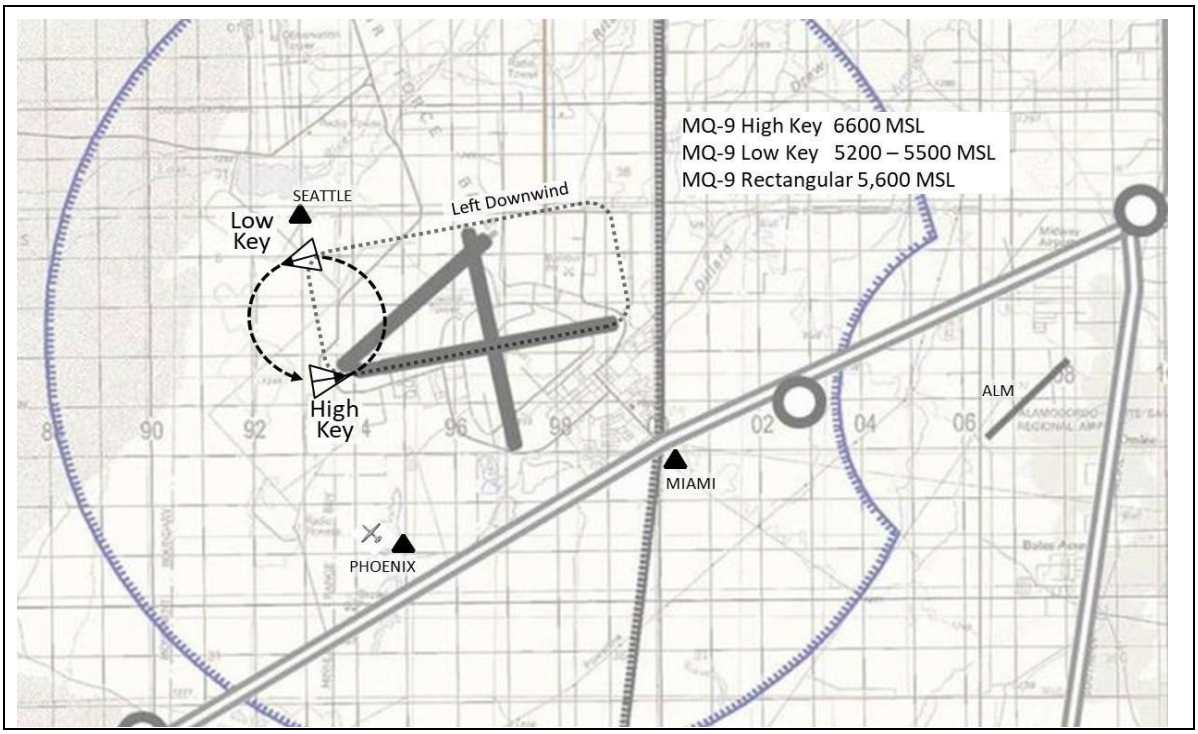
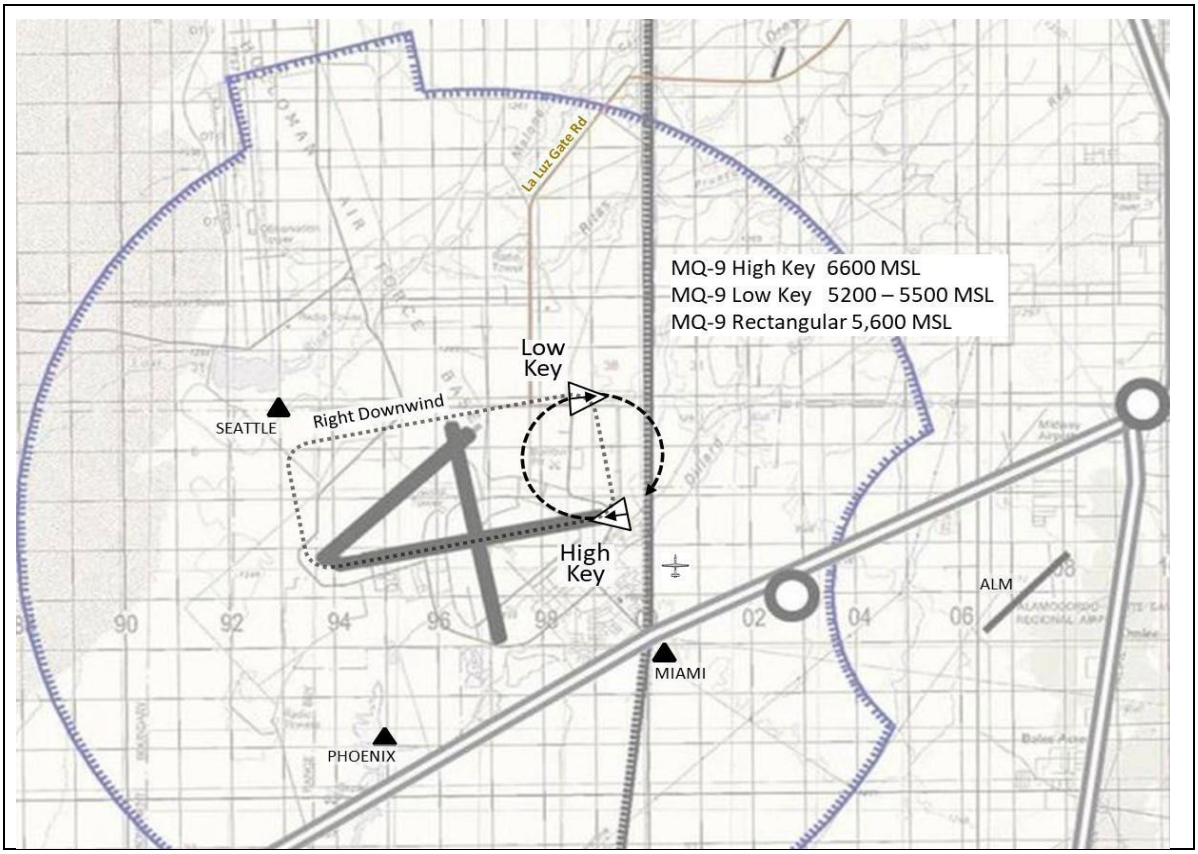


Figure A9.8. Runway 25 SFO Pattern (Drawing not to scale).



A9.9. Abnormal/Emergency Procedures.

A9.9.1. Loss of Thrust or Engine Failure East of the Valmont. An emergency such as a loss of thrust or engine failure that prevents the aircraft from crossing the Valmont corridor at or above 18,000 ft MSL does not preclude the crew from attempting to glide back to HMN. The crew will declare the emergency with the appropriate ATC agency as soon as possible to receive priority as well as assistance with coordination and deconfliction.

A9.9.2. MQ-9 Emergency Divert Base Designation. HMN assigned MQ-9s do not have an emergency divert base.

A9.9.3. Lightning.

A9.9.3.1. MQ-9 cockpits do not have integral lightning protection. The following steps will be followed to mitigate the hazard and risk to aircrew.

A9.9.3.1.1. Mission Control Element operations may continue whenever a lightning within 5 NM warning has been issued; however, personnel will not transit to or from the cockpit until the lightning within 5 NM warning has been rescinded unless authorized by either the 49 OG/CC or 49 MXG/CC.

A9.9.3.1.2. MQ-9 aircrews will not takeoff if severe turbulence is forecast or reported along the route of flight.

A9.9.4. MQ-9 Weather Recall Procedures.

A9.9.4.1. 54 OSS/OSW will immediately notify the SOF and Tower with pertinent adverse conditions and location. Any pilot or controller observing such conditions should immediately pass that information to their Ops Sup, SOF, Tower WS or WSRF WS, and 54 OSS/OSW.

A9.9.4.2. The SOF will coordinate with MQ-9 Ops Sup when determining whether to recall, discontinue departures, or direct full-stops. MQ-9s will recover IAW AFMAN 11-2MQ-1&9V3 specific guidance.

A9.9.5. MQ-9 Forced Landing Procedures. If unable to glide back to HMN, avoid populated areas and glide to the nearest uninhabited, undeveloped site. MQ-9s with live ordnance will make every effort to remain within the impact area of assigned range. Avoid the following areas:

A9.9.5.1. Red Butte (N33.8262 W106.6645).

A9.9.5.2. Stallion Range Center (N33.8157 W106.6679).

A9.9.5.3. Trinity Site (N33.6783 W106.4785).

A9.9.5.4. Permanent High Explosive Test Site (PHETS) (N33.6518 W106.5382).

A9.9.5.5. Aerial Cable Facility (N33.6219 W106.4064).

A9.9.5.6. Oscura (N33.5190 W106.2125).

A9.9.5.7. Salinas Peak (33.2987 W106.5323).

A9.9.5.8. RAMS (N33.1725 W106.5692).

A9.9.5.9. Rhodes Canyon (N33.1612 W106.4871).

A9.9.5.10. Andre Site (N32.8676 W106.4902).

A9.9.5.11. Main Site (N32.8817 W106.3472).

A9.9.5.12. HSTT (N32.9320 W106.1478).

A9.9.5.13. WSNP (N32 46.78 W106.1035).

A9.9.5.14. High Energy Laser System Test Facility (HELSTF) (N32.6321 W106.3323).

A9.9.5.15. Small Missile Range (N32.4803 W106.4190).

A9.9.5.16. Orogrande Range Camp (N32.4053 W106.1520).

A9.9.5.17. 500 ft Meteorological Tower (LC-236) (N32.4119 W106.3267).

A9.9.5.18. Main Post Area + Launch Complexes on Nike Blvd (N32.3784 W106.4814).

A9.9.5.19. Survivability, Vulnerability and Assessment Directorate (N32 20.8645 W106 26.7545).

A9.9.5.20. NASA White Sands Test Facility (N32 30.98 W106 35.895).

A9.9.6. MQ-9 Lost Link Procedures.

A9.9.6.1. If an MQ-9 goes lost link, it will fly its pre-programmed emergency mission which will terminate in the last six waypoints (Figure A9.9.). The PIC must immediately notify the controlling agency (Tower, WSRF, or ABQ ARTCC) of the lost link condition, time of lost link, last known position, altitude, direction of flight, and confirm the aircraft is executing its emergency mission. The PIC will advise the controlling agency of the RTB timetable and routing. ATC will advise any aircraft or agencies that might be affected by the lost link aircraft. Upon declaration of lost link, radar service/IFR clearance is automatically terminated/cancelled. A lost link aircraft that reaches the last six area and is within one hour of fuel exhaustion will be considered an emergency aircraft and the PIC must declare an IFE with the controlling agency.

A9.9.6.2. MQ-9 aircrew will deconflict their emergency missions over the last six in 500-ft increments. When weather and/or airspace preclude use of scheduled orbits, the Ops Sup(s) will coordinate 500-ft altitude stacks or direct MQ-9s to RTB as required.

A9.9.6.2.1. Mission Control Element emergency mission altitudes below 11,100 ft MSL require the PIC to coordinate with the Ops Sup for deconfliction from LR Element mission altitudes.

A9.9.6.3. On departure, the pre-programmed emergency mission will be set to follow the cleared departure routing and delay in the assigned airspace for no less than 30 minutes before recovering.

A9.9.6.4. While in assigned airspace, the pre-programmed emergency mission will be set to begin with no less than a 30-minute delay at the assigned altitude (squawk normal), then recover as described above (squawk 7400). If altitude change is required for de-confliction, aircraft will climb/descend upon departing loiter enroute to WP 2 and remain at new altitude for RTB.

A9.9.6.4.1. If below 18,000 ft MSL in McGregor North/South, crews will set the emergency mission waypoint 2 to a 30-minute loiter climbing to a deconflicted altitude above 18,000 ft MSL; remaining waypoints will stay at the new altitude for RTB. Waypoint 2 will be Ops Sup- assigned/deconflicted (Points A or B from current 49 WG IFG).

A9.9.6.5. During recovery, the pre-programmed emergency mission will be set fly a predictable ground track along the cleared recovery routing. If approaching the field from the southwest to northwest quadrant, the MQ-9 must fly to O2 of the Orange Route and then proceed direct to the last six. If approaching the field from any other quadrant, the MQ-9 will fly to a point overhead the field and then proceed direct to the last six. The pre-programmed emergency mission will not descend the MQ-9 below 7,100 ft MSL until after the LR crew has established two-way communication with the Tower.

A9.9.6.6. Whenever lost link occurs in the Class D airspace the pre-programmed emergency mission will be set to fly the MQ-9 direct to a point one NM past the departure end of the active runway. The MQ-9 will then turn to its second waypoint, 0.3 NM south of the last six, and climb no higher than 1,500 ft AGL. After reaching the second waypoint, the lost link aircraft will then climb to its assigned altitude as it flies the last six waypoints.

A9.9.6.7. If control of the MQ-9 cannot be regained, it will orbit at the last six until fuel exhaustion.

A9.9.6.8. If the datalink cannot be re-established within one hour of fuel exhaustion, the SOF (or Ops Sup, if no SOF on duty) must coordinate with the 49 OG/CC for the probable activation of the HMN Installation Emergency Management Plan 10-2, On/Off Base Aircraft Accident Checklist(s).

A9.9.6.9. The last six orbit area (Figure A9.9. and Table A9.4.) is the primary LOS recovery orbit for aircraft that have lost satellite communications (SATCOM) command and control links but may also be used as a delay orbit during periods of traffic pattern saturation.

A9.9.6.10. Departures may be restricted when a lost link MQ-9 is orbiting at the last six.

A9.9.6.11. Lost link while on ground/taxiing. If an MQ-9 loses link while on the ground (less than approximately 50 KIAS), the throttle will be set to ground idle, brakes will be set, and the engine will be killed within approximately 7 seconds. Approximately 5 minutes later the brakes will be released, and the aircraft may roll. Crash Recovery will have to respond and tow the aircraft off the runway or taxiway.

Figure A9.9. MQ-9 Lost Link Waypoints (Drawing not to scale).

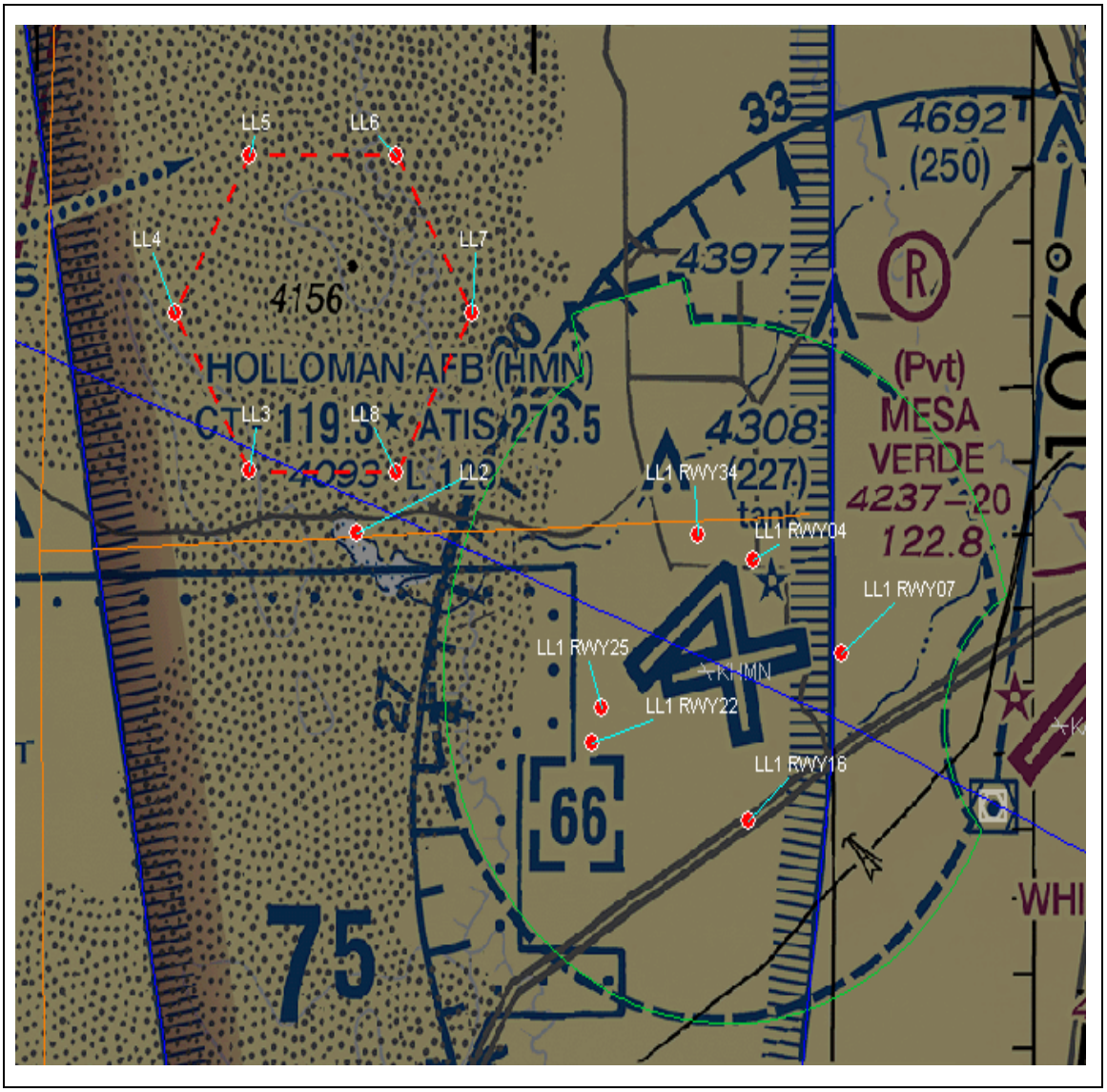


Table A9.4. MQ-9 Lost Link Waypoints.

LL1 RWY04	N 32° 52' 37" W 106° 05' 36"
LL1 RWY07	N 32° 51' 21" W 106° 03' 50"
LL1 RWY16	N 32° 49' 03" W 106° 05' 42"
LL1 RWY22	N 32° 50' 08" W 106° 08' 50"
LL1 RWY25	N 32° 50' 36" W 106° 08' 39"
LL1 RWY34	N 32° 52' 58" W 106° 06' 42"
LL2	N 32° 53' 00" W 106° 13' 32"
LL3	N 32° 53' 50" W 106° 15' 43"
LL4	N 32° 56' 00" W 106° 17' 12"
LL5	N 32° 58' 09" W 106° 15' 43"
LL6	N 32° 58' 09" W 106° 12' 45"
LL7	N 32° 56' 00" W 106° 11' 15"
LL8	N 32° 53' 50" W 106° 12' 45"

A9.9.7. MQ-9 Lost Communications/No Radio (NORDO).

A9.9.7.1. Squawk 76XX (16, 34, 07, 25, 04, 22) for intended landing runway. If a greater emergency exists, squawk 7700.

A9.9.7.2. Direct telephone communications between ATC and a NORDO MQ-9 are the primary means of back-up communication. NORDO MQ-9 WILL NOT conduct multiple VFR practice patterns. NORDO MQ-9 will use the telephone to coordinate recovery and landing clearance. ATC will advise all effected aircraft of a NORDO MQ- 9's intentions.

A9.9.7.3. Using standard light gun signals with a NORDO MQ-9 will likely be ineffective and is not recommended, as MQ-9s configure their MTS to position mode (straight ahead) while operating below 500 ft AGL.

A9.10. GPS Jamming and MQ-9 Operations.

A9.10.1. This section identifies MQ-9 procedures for scheduling and flying during GPS jamming activities. The OPR for guidance in this section is 54 OSS Wing Weapons (54 OSS/OSK).

A9.10.2. Scheduling will coordinate and track GPS jamming in HMN Local Airspace. The following timeline is stated in WSMR OI 70-3 for scheduling GPS jamming. However, in execution, the timeline may happen significantly closer to the execution date.

A9.10.2.1. T-60 - WSMR Range Scheduling enters event into schedule.

A9.10.2.2. T-28 - Scheduling is provided information package at weekly scheduling meeting. This includes program description, jammer locations, GIANT L1 and L2 Jam to Signal (J/S) ratio plots, schedule, etc. GIANT plots will include the scheduled flying window, ±1 hour, for:

A9.10.2.2.1. HMN Ramp at 15 ft AGL, for initial L1 acquisition.

A9.10.2.2.2. HMN Airspace at 10,000, 15,000, 20,000, and 25,000 ft MSL.

A9.10.2.3. T-14 - GPS Jamming Mission submitted for Conduct Approval.

A9.10.2.4. T-7 - WSMR mission is coded and committed to schedule, schedule locked.

A9.10.2.5. T-0 - Test Execution Day.

A9.10.3. Scheduling will forward applicable information to the effected squadrons as soon as possible to enhance squadron awareness. To aid in real-time execution by MQ-9 Ops Sups, Next Day (ND) and Current Day (CD) GIANT plots, and/or lines of deconfliction, for each specific CSE Airspace Restriction Mission ID/Name will be posted or linked in/from the CSE home page under "*Scheduling Documents*".

A9.10.4. Aircrew will also add GPS jamming events into their ORM, by defining GPS jamming as "*Airspace Restrictions*". While not driving overall Mission ORM to any specific level, this should be considered in weighing overall mission risk and mitigation strategies.

A9.10.5. See 49 WG IFG for additional GPS Jamming guidance.

Attachment 10

UNCONTROLLED AIRFIELD OPERATIONS (UAO)

A10.1. Introduction. This attachment implements AFMAN 13-204V1, Attachment 4, *ORM Considerations for Operations When Essential Services Are Not Available.*

A10.1.1. It outlines flying procedures for 49 FSS/AC, 704 TG, AA, and authorized TDY units to operate during periods when HMN ATC Tower and AM are closed.

A10.1.2. This attachment also outlines airfield operating procedures for the units that are authorized access to the runways, Taxiway Golf, and North Ramp when the airfield is closed (see paragraph A10.5.). In addition to the units above, airfield drivers include FD, 49 SFS, Ambulance Response, Medical Field Response Team, Crash Recovery, Operations Duty Officers (ODO), BM, Airfield Maintenance, 49 CES/CEOFE, RAWS, 49 EMS, 49 LRS/LGRF, 49 MXG, 82 ATRS, and its contractors.

A10.1.3. Additionally, the 54 OSS/OSA and 49 WG/CP have duties in this publication. All units listed in A10.1.2. are responsible for understanding the UAO process and their responsibilities within this publication. The intent is to ensure the safe arrival/departure of aircraft during UAO.

A10.2. UAO Limitations. Operations conducted under the scope of this attachment have specific limitations. **Note:** 49 FSS/AC refers to all locally assigned aero club aircraft. 704 TG refers solely to aircraft operating under or hosted by the 704 TG. AA refers to all Army Air rotary wing and fixed wing aircraft assigned to HMN. TDY units must have 49 WG/CC approval and obtain appropriate training prior to participating in UAO. At no time will any organization operate a jet aircraft. Unit will be cognizant of weight restrictions for taxiways and parking areas.

A10.2.1. HMN airfield hours of operation are as published in paragraph 2.20.1. UAO will take place during periods when the airfield is closed. If a unit would like to operate during hours of darkness, the ODO will be trained on airfield lighting system operation.

A10.2.2. Aircraft must not depart if weather is forecasted to go below VFR minimums within 1 hour plus or minus the scheduled departure/arrival time.

A10.2.2.1. Minimum ceiling is 1,500 ft AGL with minimum visibility of 3 SM. **Exception**: Rotary wing aircraft may adhere to service minimums.

A10.2.3. UAO are normally limited to a single departure/arrival for each aircraft however, mission may require multiple departures/arrivals. Pattern work is prohibited. Pilots must depart the aerodrome following initial takeoff unless experiencing a situation that requires an immediate landing. Pilots returning HMN to land must make one pattern to a full stop landing.

A10.2.4. 49 FSS/AC, 704 TG, AA, and authorized TDY units must ensure operations at the airfield do not conflict with scheduled PMI of RAWS or airfield equipment. PMI schedules are published in the DoD FLIP *IFR Supplement US*. If a situation arises where UAO will

impact airfield, RAWS, barrier, or lighting maintenance, the ODO and maintenance unit will coordinate and attempt to resolve. If both the ODO and maintenance unit deem safe, UAO and maintenance may occur simultaneously. If no resolution is possible, the maintenance operation will have priority.

A10.2.4.1. The unit manager for each participating unit will provide AM with up-to- date ODO contact lists, to include assigned ODOs and ODO's cell phone number. AM will provide unit ODOs, 49 WG/CP, 49 CES, 49 SFS, Tower, RAWS, WSRF, FD, and Ambulance Response with the updated ODO contact list.

A10.3. UAO - Procedures.

A10.3.1. AM will ensure critical information/procedures for HMN UAO are published in the appropriate DoD FLIP.

A10.3.2. The ODO is the focal point for operations conducted under the scope of this instruction. In supervising UAO, the ODO operates in lieu of the ATC Tower and AM to ensure operations are conducted safely, airfield security is not compromised, and that emergency response is available as required. If multiple ODOs are present, paragraph A10.3.14.2. identifies who will have the lead until termination of their unit's ops. The ODO will coordinate with other ODOs prior to terminating to hand off the lead to another ODO. 49 WG/CP will be advised by the incoming lead ODO of the "ODO lead" transfer.

A10.3.2.1. The ODO MUST NOT perform ATC functions and may not issue ATC instructions. Each unit conducting operations in the local area during UAO will have an ODO on duty. The ODO will be on duty until their last aircraft shuts down. For aircraft arriving from outside of the local flying area (i.e., cross-country arrivals), the unit will have an ODO on duty 1 hour prior to scheduled arrival and will remain until the aircraft is parked and the engines are shut down. For aircraft departing the local area, the ODO will be on duty from 1 hour prior to departure to 30 min after takeoff. The ODO must provide updated arrival/departure times to the 49 WG/CP.

A10.3.3. ODO Schedule. The unit manager must manage the ODO schedule. Pilots planning to fly when HMN is closed must ensure an ODO has been scheduled.

A10.3.4. Preparation. The ODO accomplishes the following steps prior to commencing operations.

A10.3.4.1. Scheduling UAO. The ODO for the requesting unit must complete the Figure A10.1., UAO Setup Checklist, NLT Wednesday of each week, or the duty day prior to requested operations, if a short notice requirement exists such as range support, or MEDEVAC, etc. If UAO will occur on multiple days, coordination must be accomplished for each day. E.g., if the requested ops are on Saturday or Sunday, the checklist must be accomplished NLT Wednesday by 1600L (assuming it is not a Holiday or "Down" day). If the requested ops are, for example, Wednesday night after the airfield closes or Thursday morning before the airfield opens, the checklist must be completed on Tuesday by 1600L.

A10.3.4.1.1. The ODO must obtain an in-person briefing from the AFM or their designated representative prior to commencing operations. This briefing must be the last checklist item completed. The briefing must address airfield facilities status, construction avoidance areas, and planned airfield maintenance activities during the period of UAO. Once the checklist is complete, a copy will be given to AM for final approval. When ODO calls CE and RAWS, the ODO will ensure that there are no maintenance or PMIs scheduled outside of what is published in the DoD FLIP *IFR Supplement US*. If an irreconcilable conflict exists, maintenance operations will have priority IAW paragraph A10.2.4. **Note**: If the full setup checklist, including AM briefing is not accomplished, the respective unit must not participate in UAO.

A10.3.4.1.2. AM will retain (see Figure A10.1.) UAO Setup Checklist for record keeping. AM will notify 54 FG/CC or 54 FG Brickholder, 49 WG/CP, 49 MXG/MOC, Unit Airfield Driving Program Managers (ADPM) and other units as deemed necessary via email to include a copy of the upcoming UAO schedule and checklist once received.

A10.3.4.1.3. WSRF/CE/RAWS/54 OSS/OSA reserve the right to cancel/modify scheduled UAO at any time due to unforeseen conflicts. These agencies will call 49 WG/CP to verify whether UAO will be occurring. If so, 49 WG/CP will refer the agency to the lead ODO. The agency will contact the lead ODO and attempt to coordinate deconfliction. If deconfliction is not possible, then the agency must inform the ODO that ops will terminate due to mission essential activities. The agency will inform 49 WG/CP of the termination. The ODOs will then notify the 49 WG/CP once ops have terminated.

A10.3.4.2. The ODO must obtain a weather briefing for HMN and the surrounding area. Any official weather source is acceptable including a self-brief from Direct User Access Terminal Service or the National Oceanic and Atmospheric Administration internet aviation weather services.

A10.3.4.3. The ODO must contact 49 WG/CP to coordinate commencement of UAO prior to their unit's first aircraft engine start. 49 WG/CP must in turn notify the 49 MDG Ambulance Response Element, FD, and 49 SFS.

A10.3.4.4. Prior to aircraft movement, the ODO must perform an airfield check on all taxi routes and runway(s) to be utilized by their aircraft to ensure movement areas are clear of FOD, wildlife, obstructions, etc. AM must provide initial and recurring training to all unit ODOs to ensure quality education on appropriate airfield checks (see Figure A10.2., *ODO Training Checklist)*.

A10.3.5. Flight Plans. IAW AFMAN 13-204V2, Airfield Management, all aircraft departing USAF installations must have a flight plan on file with AM (or locally assigned unit with approved flight plan processing procedures in place) prior to takeoff. Use DD Form 1801 or other authorized forms IAW AFMAN 11-202V3, Flight Operations. In the remarks section of the flight plan(s) the pilot will annotate "UAO". Flight plans will be retained by originating unit, pilot must file, activate, and close flight plans with ABQ Flight Service Station (FSS) or other applicable facility.

A10.3.5.1. Units will maintain the original flight plan IAW AFRIMS RDS available from the Air Force Portal link. Suspension of UAO privileges may occur if unit records are not maintained properly.

A10.3.6. Step briefing. The ODO must accomplish an in-person briefing with each aircrew prior to launch. The ODO must ensure knowledge of each aircrew's intentions. Additionally, the ODO must brief aircrews on any airfield taxi, departure, and landing restrictions.

A10.3.7. Taxi Routes. Pilots must taxi in the most direct route to and from the runway consistent with safety and taxi restrictions. Pilots must not taxi into restricted areas (to include the West Ramp) and all taxi operations must be overt. Pilots must give right of way to all emergency vehicles.

A10.3.8. Runways. Operations are unrestricted on Runway 04/22 and Runway 16/34. Runway 07 at or east of Taxiway Foxtrot may be used for departures only. Runway 07 WILL NOT be used for landings under any circumstances. Runway 25 WILL NOT be used for landings and takeoffs under any circumstances. The ODO must determine runway in use based on existing and forecasted wind direction. ODOs must coordinate this information with other participating unit ODOs to alleviate conflicts.

A10.3.8.1. AA Helicopters may depart and land at the North Ramp in lieu of using the runways.

A10.3.9. Patterns. Pilots will adhere to recommended pattern procedures for UAO as described in the FAA, *Aeronautical Information Manual* (AIM) and other related FAA regulations. Observe traffic pattern and VFR procedures identified in Chapter 6, Attachment 6, and Attachment 8.

A10.3.9.1. To ensure deconfliction with KALM, which is within 6 NM of HMN, the southeast sector of HMN Class D airspace must not be used. The southeast sector consists of an imaginary line extending from the departure ends of Runways 16 and 07 to extend to 5 NM. **Exception**: Standard downwind to Runway 16/34 is allowed.

A10.3.10. Altimeter Setting. Pilots must obtain an altimeter setting from the KALM Automated Weather Observing System (AWOS) @ 127.825 or the HMN Weather Observer, if on duty.

A10.3.11. Airfield Lighting Training. All unit ODOs planning flight operations during hours of darkness will schedule training with 49 CES/CEOFE for lighting procedures at the ALV (Building 863) prior to scheduling night ops.

A10.3.12. If aircraft are departing/arriving at night, the ODO must turn on airfield lighting at the ALV no more than 30 minutes prior to taxi and no less than 30 minutes prior to landing. Airfield lighting must be turned off by the ODO no earlier than 30 minutes after departure and prior to terminating operations. **Note**: If airfield lighting is not turned on prior to landing, aircraft must divert to KALM or other airports as required until lighting is turned on.

A10.3.13. Commencement/Termination of Operations. The ODO must report commencement/termination of operations to 49 WG/CP. 49 WG/CP will ensure that a new lead ODO has been coordinated if applicable or will notify the 49 MDG Ambulance Response Element, FD, and SFS that UAO has been terminated.

A10.3.14. Communication procedures.

A10.3.14.1. Radio Communications. Common Traffic Advisory Frequency (CTAF) is HMN Tower frequency, 119.3 (119.3 is continuously recorded as an ATC frequency). **Note**: Use of UHF for the purpose of HMN CTAF is not authorized. Pilots must adhere to the recommended radio procedures for UAO as described in the AIM. Departing pilots must establish contact with the ODO before taxiing to the runway for departure. Arriving pilots must establish contact with the ODO at least 10 NM prior to landing. The ODO must provide airfield advisories as appropriate. All communication between the ODO and the aircraft will be via the CTAF on a VHF radio. The ODO must use unit specific call signs IAW Table A10.1.

Table A10.1. Unit Specific Call Signs.

Unit	Call Sign
AA	ODO One
49 FSS/AC	ODO Two
704 TG	ODO Three
TDY unit	ODO Four
Note: Usually, the above order of ODOs designates a priority to who will be the lead ODO. Example: AA is the first and only ODO on the scene. If 49 FSS/AC arrived to commence ops, the lead ODO would be transferred to 49 FSS/AC. **Exception:** As needed, priority may be delegated differently when agreed upon by overlapping ODOs.	

A10.3.14.2. The primary method for the ODOs to communicate with vehicular traffic will be via LMR "Tower Net". Each participating unit will ensure the ODO has an operable LMR Tower Net. All vehicle operators that will be accessing the runways, Taxiway Golf, or North Ramp will have an operable LMR Tower Net to ensure two-way communication between the ODO and the vehicle operators. ODO should communicate via cell phone as a last resort. The ODO must monitor all ops on VHF and LMR radios.

A10.3.14.3. At a minimum the ODO will transmit a blanket broadcast on the LMR Tower Net at the following times. Each unit ODO will broadcast the arrival and departure advisories of their unit's aircraft.

A10.3.14.3.1. Commencing operations (after calling 49 WG/CP) - "*Attention on the Tower Net, Uncontrolled Airfield Operations is commencing*".

A10.3.14.3.2. When aircraft calls to taxi for departure - "*Attention on the Tower Net, aircraft departing Runway XX or North Ramp*".

A10.3.14.3.3. When aircraft call inbound- "*Attention on the Tower Net, aircraft XX minutes from Holloman, will be landing Runway XX or North Ramp*".

A10.3.14.3.4. Terminating operations (prior to calling 49 WG/CP) - "*Attention on the Tower Net, Uncontrolled Airfield Operations is terminated*". **Note**: Ensure that no other ODOs are operating prior to making this broadcast.

A10.3.14.4. AM will have a limited number of LMRs with Tower Net capability that will be available for use, if needed, by the ODOs. LMRs will be hand receipted out on a first come, first served basis, when the ODO comes to AM to complete the (see Figure A10.1. *UAO setup checklist*). On the next duty day, the ODO will return the LMR to AM. Failure to return the LMR could result in the loss of the unit's UAO privilege.

A10.3.14.5. The 49 WG/CP will provide vehicle operators the cell phone number of the lead ODO if the vehicle operator cannot establish communication with the ODO via LMR Tower Net.

A10.4. UAO - Aircraft Mishap/Emergency.

A10.4.1. The PIC and the ODO have the authority to declare an emergency at any time. In case of an aircraft mishap or declared emergency, the ODO must immediately notify FD with pertinent information (i.e., type aircraft and location) to assist rescue efforts to the maximum extent possible. The ODO must then notify 49 WG/CP, who then must, at a minimum, again notify FD, Ambulance Response Element, Crash Recovery, and SFS for immediate mishap/emergency response. 49 WG/CP will also notify 54 OSS/OSA/CC or OSA Director of Operations (54 OSS/OSA/DO) and 49 WG/SE immediately, to ensure they are aware of potential impact to airfield operations. If neither 54 OSS/OSA/CC nor DO can be contacted, 49 WG/CP will call 54 OSS/DO or 54 OSS/CC. If unavailable, 49 WG/CP will call the 54 FG/CC or 54 FG Brickholder.

A10.4.2. ODO must restrict vehicle access to emergency response vehicles only until the emergency has been terminated by the FD. After an emergency, the ODOs are responsible for ensuring that the runway or North Ramp is clear of all debris, damage, and fluid spills. Following an aircraft mishap, aircraft debris WILL NOT be moved without Emergency Operations Control Commander approval. ODO will notify AM on the next duty day of any declared emergencies or any pertinent info to include damage to airfield or fluid spills that occurred.

A10.5. UAO - Vehicle Operations.

A10.5.1. To minimize the potential for incidents, only a select few agencies must come within 100 ft of any runway, Taxiway Golf, and or North Ramp when the airfield is closed. These agencies are FD, SFS, Ambulance Response, Medical Field Response Team, Crash Recovery, ODOs, Tower, AM, BM, Airfield Maintenance, 49 WG/SE, 49 CES/CEOFE, RAWS, 49 EMS, 49 LRS/LGRF, 49 MXG, 82 ATRS, and 586 FLTS. **Note**: The 54 FG/CC or 54 FG Brickholder may authorize additional base agencies to come within 100 ft of any runway, Taxiway Golf, and or the North Ramp when the airfield is closed as the mission dictates.

A10.5.1.1. If any other unit needs access to a runway, Taxiway Golf, and/or North Ramp for mission essential reasons when the airfield is closed, prior coordination and training must be accomplished with the AFM. The AFM will be the approval authority. Under no circumstances will any non-approved units drive on any portion of a runway, Taxiway Golf, and/or North Ramp when the airfield is closed.

A10.5.1.2. If live load operations and UAO have a conflict, live loads will take priority. AM must monitor and approve UAO when live loads (that require runway crossings) are scheduled to be in effect. The ODO will be advised of non-availability of UAO when ODO comes to AM to complete the setup checklist. Supplemental training is not required for vehicle operators who are involved in live load operations because AM will deconflict and disapprove UAO during that time if live load vehicle operators require runway access.

A10.5.2. Airfield Driving Training. See DAFI 13-213_HAFBSUP for UAO training. Units not listed in paragraph A10.5.1. ARE NOT allowed access to the runway, Taxiway Golf, and/or North Ramp when the airfield is closed unless prior coordination is accomplished with AFM or DAFM.

A10.5.3. Vehicles operating on the airfield when the airfield is closed must exercise extreme caution when operating in the proximity of taxiways/runways and/or North Ramp.

A10.5.4. Vehicle operations on the runways, Taxiway Golf, and North Ramp when the airfield is closed will be kept to the absolute minimum. Vehicles WILL NOT be left unattended on the runway at any time. Whenever a vehicle requires access to a runway, Taxiway Golf, and/or North Ramp when the airfield is closed, they must contact 49 WG/CP (LMR Tower Net) prior to entering the runway (defined as the runway and the area within 100 ft of any runway surface, Taxiway Golf, and North Ramp). If 49 WG/CP verifies that UAO are not being conducted, the vehicle may proceed to enter the runway environment.

A10.5.4.1. If UAO are or will be in effect, the 49 WG/CP will provide a schedule of UAO and ODO contact information and instruct them to contact the ODO for access. If UAO is scheduled during the time the vehicle requires access to a runway, Taxiway Golf, and/or North Ramp, the vehicle operator must contact the lead ODO for permission via LMR Tower Net to cross or enter any runway, Taxiway Golf, and/or North Ramp. The driver will advise the ODO of number of vehicles entering the runway, Taxiway Golf, and/or North Ramp and notify the ODO once crossing/access is complete. Vehicle operators must visually clear the runway, Taxiway Golf, and/or North Ramp and adjoining airspace in both directions prior to proceeding across or onto a runway, Taxiway Golf, and/or North Ramp. If the 49 WG/CP or vehicle attempting to access the runway, Taxiway Golf, and/or North Ramp cannot contact the scheduled ODO via LMR Tower Net (primary) or commercial phone (575) 572-4900 (secondary), the vehicle must not enter or cross the runway, Taxiway Golf, and/or North Ramp. **Note**: Scheduled ODOs must be accessible via LMR and the provided contact numbers or risk suspension from UAO at the discretion of the 54 FG/CC or 54 FG Brickholder.

A10.5.4.2. The unit ODO must transmit a blanket broadcast via LMR Tower Net prior to commencing UAO, upon receiving notification of a landing or departing unit aircraft, and

prior to terminating UAO. Refer to paragraph A10.3.14.3.

A10.5.4.3. Vehicles responding to an emergency must contact the ODOs prior to entering the runways, Taxiway Golf, and/or North Ramp. Ensure the emergency aircraft is at a full stop before entering the runway in use. All participating emergency response vehicles must use extreme caution when entering, exiting, and operating within the runways, Taxiway Golf, and or North Ramp environment during UAO. FD will advise lead ODO and 49 WG/CP when emergency is terminated, and all emergency vehicles are off the airfield.

A10.5.4.4. When the airfield lights are turned on and the airfield is closed, all vehicle traffic will immediately clear the runways and use increased caution while operating on the airfield until the airfield lights are turned off. Operators will call 49 WG/CP to verify UAO status. If active, the driver will contact the ODO for approval prior to crossing or entering any runway, Taxiway Golf, and or North Ramp and will state their intentions.

A10.6. UAO - ODO Qualifications and Selection.

A10.6.1. ODOs must be knowledgeable of unit aircraft operations and have sound decision- making ability. ODOs will possess either a Restricted Area Badge or an FAA pilot certificate. 49 FSS/AC ODO candidates must be submitted to the 49 MSG/CC by both the 49 FSS/AC Manager and 49 FSS/AC Chief Instructor Pilot. The 49 MSG/CC will appoint the 49 FSS/AC ODOs for 54 FG/CC approval. TG and AA must appoint ODOs and submit justification and appointment memorandum for 54 FG/CC approval. All ODO candidates must personally interview with 54 FG/CC prior to final approval. **Note**: All ODO appointments must be via official memorandum.

A10.6.2. ODO Training. The unit must develop a training program for ODOs. The training must include, but not be limited to:

A10.6.2.1. Airfield driver's training leading to an Airfield driver's license. This training will be conducted by the sponsoring unit's ADPM.

A10.6.2.2. AM will provide each unit with Airfield Inspection and Maintenance CBT and Airfield Criteria Course CBT. Prior to meeting with the AFM or designated representative the ODO must complete the CBT and provide the AFM with the certificates of completion.

A10.6.2.3. Briefing from the FD Chief or assistant.

A10.6.2.4. Briefing from the SFS Operations NCOIC.

A10.6.2.5. Airfield Lighting training conducted by 49 CES/CEOFE (if unit requires night ops).

A10.6.2.6. Review of service applicable regulations.

A10.6.2.7. Briefing from the AFM or designated representative.

A10.6.2.8. Airfield orientation with qualified ODO or AM personnel if no qualified ODO exists.

A10.6.2.9. ODOs will possess either a Restricted Area Badge or an FAA pilot certificate.

A10.6.2.10. Completed (see Figure A10.2.) ODO Training Checklist must be submitted to AM prior to assuming duties as ODO.

A10.7. UAO - ODO Responsibilities.

A10.7.1. Complete all necessary checklists.

A10.7.2. ODOs will possess either a Restricted Area Badge or an FAA pilot certificate and all items listed in paragraph A10.8.1.1. when acting as the ODO.

A10.7.3. Schedule training with 49 CES/CEOFE for lighting procedures and obtain access to the ALV control computer prior to scheduling night ops IAW (see Figure A10.2.) ODO Training Checklist.

A10.7.4. Ensure no unscheduled maintenance/PMIs are scheduled outside of what is published in the DoD FLIP *IFR Supplement US* when coordinating with CE and RAWS during setup checklist completion.

A10.7.5. Coordinate with WSRF CCTLR/ACCTLR and provide the following information: Date, time, and WSMR mission code. If there is no mission code, provide if operation will be on or off station (i.e., Fort Bliss airspace). Coordinate IAW UAO setup checklist to ensure R5107D IS NOT active for the proposed period of UAO. If R5107D is scheduled to be active, UAO must be postponed.

A10.7.6. Ensure flight plans and any amendments are on file.

A10.7.7. Obtain an in-person briefing with AFM or designated representative prior to commencing operations during setup checklist completion.

A10.7.8. Ensure setup checklist is accomplished (Figure A10.1., *UAO Setup Checklist*).

A10.7.9. Be familiar with each pilot's intentions and flight plans.

A10.7.10. Obtain a weather briefing for HMN and the surrounding area IAW Figure A10.3., *ODO Ops Checklist*.

A10.7.11. Be on duty at respective duty location 1 hour prior to any UAO. See paragraph A10.3.2.1.

A10.7.12. Determine runway in use based on current and forecasted wind direction and

coordinate with other ODOs IAW Figure A10.3., *ODO Ops Checklist*.

A10.7.13. Accomplish an in-person step brief with each aircrew prior to launch IAW Figure A10.3., *ODO Ops Checklist*. If the pilot is returning from a cross-country flight (or the pilot departed when the airfield was open), it is the responsibility of the ODO to ensure the pilot is contacted and given a step briefing prior to landing at HMN.

A10.7.14. Immediately prior to commencing ops, conduct an airfield check of all taxi routes and runways, and/or North Ramp to be utilized by the aircraft IAW Figure A10.3., *ODO Ops Checklist.*

A10.7.15. Notify 49 WG/CP when commencing or terminating UAO IAW Figure A10.3., *ODO Ops Checklist* and make blanket broadcasts IAW A10.3.14.3.

A10.7.16. All communication between the ODO and the aircraft will be via the CTAF.

A10.7.17. If more than one ODO is active, ODOs must identify a lead ODO IAW paragraph A10.3.14.2., inform 49 WG/CP, and all coordination must go through lead ODO.

A10.7.18. Monitor the weather and direct pilots as necessary to ensure safe operations.

A10.7.19. When called by a vehicle operator, approve or disapprove runway, Taxiway Golf, and/or North Ramp access based on UAO arriving/departing traffic.

A10.7.20. The ODO must transmit a blanket broadcast via LMR Tower Net of landing or departing aircraft.

A10.7.20.1. Check-out an LMR (with Tower Net), if needed, from AM. LMR will be hand receipted out on a first come, first served basis, when the ODO comes to AM to complete the Figure A10.1., UAO Setup Checklist. On the next duty day, the ODO will return the LMR to AM. Failure to return the LMR could result in the loss of the unit's UAO privilege.

A10.7.21. Report unauthorized aircraft/vehicles in the airfield environment to SFS.

A10.7.22. In the event of an aircraft mishap or emergency, immediately notify FD then notify 49 WG/CP. After an emergency, ensure the runway, Taxiway Golf, and/or North Ramp is clear of all debris, damage, and fluid spills.

A10.7.23. Notify and direct emergency response to the site of an aircraft mishap.

A10.7.24. Act as the on-scene final authority for UAO until relieved by a higher authority. Higher authority may include but is not limited to SFS, FD, AM, 54 FG/CC, or 54 FG Brickholder.

A10.7.25. Be accessible via provided contact numbers or risk suspension from UAO.

A10.7.26. If aircraft are departing/arriving at night, the ODO must turn on airfield lighting at the ALV no more than 30 minutes prior to taxi and no less than 30 minutes prior to landing. Airfield lighting must be turned off by the ODO no earlier than 30 minutes after departure and prior to terminating operations. **Note**: If airfield lighting is not turned on prior to landing, aircraft must divert to KALM or other airports as required until lighting is turned on.

A10.8. UAO – Participating Unit Manager Responsibilities. (Responsibilities may be delegated to or shared with designated representatives.)

A10.8.1. Ensure respective equipment is available to the ODO and is in good working order. At minimum the following equipment must be provided:

A10.8.1.1. A Government Owned Vehicle for airfield driving.

A10.8.1.2. VHF radio to include spare batteries or automotive adapter.

A10.8.1.3. Portable LMR (Tower Net) to include spare batteries and/or charger.

A10.8.1.4. Cellular phone including a spare battery or automotive adapter. **Note**: Each unit must have a dedicated ODO cellular phone number. The unit will provide AM with the ODO cell number to be distributed to all units directed under this instruction.

A10.8.1.5. Publications/continuity binder which includes all applicable unit/base directives and ODO checklists and procedures.

A10.8.1.6. Provide updates for critical phone number listing.

A10.8.2. Manage the ODO schedule.

A10.8.3. Assign qualified unit personnel to be appointed as ODO. The ODOs will be appointed by unit commanders and approved by the 54 FG/CC. Minimize change to the maximum extent possible.

A10.8.4. Develop a training program for ODOs. Refer to paragraph A10.6.2. for the minimum training criteria.

A10.8.5. Oversee initial and periodic training for ODOs on HMN UAO.

A10.8.6. Ensure unit specific checklists are created and maintained as necessary.

A10.9. Pilot UAO Responsibilities.

A10.9.1. Must depart the aerodrome following initial takeoff unless experiencing a situation that requires an immediate landing. Pilots returning to HMN to land must make one pattern to a full stop landing. UAO are normally limited to a single departure/arrival for each aircraft however, mission may require multiple departures/arrivals. Pattern work is prohibited.

A10.9.2. Ensure an ODO has been scheduled if planning to fly when HMN airfield is closed.

A10.9.3. File, activate, and close flight plan with ABQ FSS or other applicable facilities. Pilots must file a flight plan before participating in UAO. In the remarks section of the DD Form 1801, the pilot will annotate "UAO".

A10.9.4. Taxi in the most direct route to and from the runway consistent with safety and taxi restrictions. Pilots must not taxi into restricted areas (to include the West Ramp) and all taxi operations must be overt. Pilots must give right of way to all emergency vehicles.

A10.9.5. Adhere to recommended pattern procedures for UAO as described in the AIM and other related FAA regulations. Observe traffic patterns identified in this regulation.

A10.9.5.1. To ensure deconfliction with KALM, which is within 6 NM of HMN, the southeast sector of HMN Class D airspace must not be used. The southeast sector consists of an imaginary line extending off the departure ends of Runway 16 and 07 to extend to 5.0 NM. **Exception**: Standard downwind to Runway 16/34 is allowed.

A10.9.6. Obtain an altimeter setting from the KALM AWOS (127.825) or the HMN weather observer, if on duty.

A10.9.7. Adhere to the recommended radio procedures for UAO as described in the AIM. Departing pilots must establish contact with the ODO before taxiing to the runway for departure. Arriving pilots must establish contact with the ODO at least 10 NM prior to landing. All communication between the ODO and the aircraft will be via the CTAF.

A10.9.8. The PIC or the ODO have the authority to declare an emergency at any time.

A10.9.9. Ensure the ODO is familiar with each pilot's intentions and flight plans.

A10.9.10. If returning from a cross-country flight (or if pilot departed when the airfield was open) ensure the ODO is contacted and that pilot has been given a step briefing prior to returning to HMN to land.

A10.10. 54 FG/CC UAO Responsibilities.

A10.10.1. Approval authority for all unit ODOs and all UAO daily scheduled missions.

A10.10.1.1. Interview each ODO candidate and sign appointment letter prior to ODO assuming duties.

A10.11. AM UAO Responsibilities.

A10.11.1. Provide unit ODOs, 49 WG/CP, 49 WG/SE, CE, SFS, WSRF, Tower, FD/Crash Recovery, and Ambulance Response with the updated ODO contact list.

A10.11.2. Ensure critical information/procedures for HMN UAO are published in the

appropriate DoD FLIP.

A10.11.3. Confirm presence of the arrival/departure times received from 49 WG/CP on the schedule and maintain records, see Figure A10.2., *ODO Training Checklists,* and setup checklists.

A10.11.4. Provide in person briefing to ODO on airfield status prior to commencing operations and sign the completed setup checklist.

A10.11.5. Notify 49 WG/CP of upcoming UAO schedule via email.

A10.11.6. UAO Procedures for Airfield Driving have been added to DAFI 13-213_HAFBSUP, *Airfield Driving*.

A10.11.7. Provide each unit with Airfield Inspection and Maintenance CBT and the Airfield Criteria Course CBT.

A10.11.8. Brief ODO candidates prior to them assuming ODO responsibilities.

A10.11.9. Conduct airfield orientation for ODO nominees if a qualified ODO is not available.

A10.11.10. Solicit arrival/departure information from 49 WG/CP.

A10.11.11. Deconflict UAOs with live load operations that require runway access that conflicts with UAO proposed ops. Live load ops must have priority if a conflict exists.

A10.12. 49 WG/CP Responsibilities.

A10.12.1. Maintain ODO contact list and UAO schedule, as received from AM.

A10.12.2. When the airfield is closed, CP will abide by all requirements in paragraph A10.5. of this attachment to allow deconfliction of vehicle operations and UAOs. CP will monitor LMR Tower Net and respond to any vehicle requesting access to the runway, Taxiway Golf, and/or North Ramp. Refer to paragraph A10.5.1. for a list of agencies that are authorized access.

A10.12.3. Upon receiving notification of UAO commencement and termination, notify FD, SFS, Ambulance Response, and Medical Field Response Team.

A10.12.4. Notify FD, SFS, Ambulance Response, Medical Field Response Team, 49 WG/SE, 54 OSS/OSA/CC, or OSA/DO upon receiving notification of an emergency or mishap. If neither 54 OSS/OSA/CC nor OSA/DO can be contacted, 49 WG/CP will call 54 OSS/CC/DO. If they are unavailable, 49 WG/CP will call the 54 FG/CC or 54 FG Brickholder.

A10.12.5. Record arrival and departure times and pass to AM the next duty day.

A10.12.6. When contacted by any agency regarding airfield access/UAO during periods of

airfield closure, provide a schedule of UAO, ODO contact information, and positive information handoff between the vehicle operators and the ODO(s) if necessary. Track lead ODO if multiple ODOs exist.

A10.13. WSRF UAO Responsibilities.

A10.13.1. When ODO calls, advise whether R5107D is scheduled during proposed UAO window. If R5107D unexpectedly goes active, WSRF will call the 49 WG/CP to verify status of UAO. If UAO are in progress, WSRF will immediately advise 49 WG/CP and ODOs that ops are terminated.

A10.14. 49 CES/CEOFE UAO Responsibilities.

A10.14.1. 49 CES/CEOFE will train ODOs on airfield lighting procedures and ensure ODOs have access to the ALV lighting control computer for airfield lighting activation.

A10.14.2. BM, Airfield Maintenance, and 49 CES/CEOFE will deconflict any airfield work (scheduled or unscheduled) when ODO calls to coordinate. For maintenance during hours of airfield closure, contact 49 WG/CP and then ODO to deconflict any unexpected airfield work.

A10.14.3. FD will advise lead ODO and 49 WG/CP when emergency is terminated, and all emergency vehicles are off the airfield.

A10.15. RAWS UAO Responsibilities.

A10.15.1. RAWS will deconflict any potential airfield work when ODOs call to coordinate. Contact 49 WG/CP and then ODO via provided ODO contact information to coordinate any unexpected airfield work.

A10.16. 49 SFS UAO Responsibilities.

A10.16.1. Maintain situational awareness of UAO through 49 WG/CP notification of commencement/termination and related emergencies/mishaps.

A10.16.2. Investigate reports of suspicious activities to include unauthorized vehicle/aircraft movement.

A10.17. Approved Airfield Vehicle Operator UAO Responsibilities.

A10.17.1. To minimize the potential for incidents, only a select few agencies must have access to the runways, Taxiway Golf, and/or North Ramp when the airfield is closed. These agencies are FD, SFS, Ambulance Response, Medical Field Response Team, Crash Recovery, ODOs, Tower, AM, BM, Airfield Maintenance, 49 WG/SE, 49 CES/CEOFE, RAWS, 49 LRS/LGRF, 49 MXG, 82 ATRS, and 586 FLTS. **Note**: The 54 FG/CC or 54 FG Brickholder may authorize additional base agencies to come within 100 ft of any runway, Taxiway Golf, and or the North Ramp when the airfield is closed as the mission dictates.

A10.17.1.1. If any other unit needs access to a runway, Taxiway Golf, and/or North Ramp for mission essential reasons, prior coordination and training must be accomplished with the AFM.

A10.17.2. Call 49 WG/CP on LMR Tower Net (primary) or commercial (575) 572-4900 (secondary) prior to accessing the runway, Taxiway Golf, and/or North Ramp during periods when HMN airfield is closed. Abide by all requirements in paragraph A10.5. of this attachment to allow deconfliction of vehicle operations and UAOs.

A10.17.3. If there is any doubt as to whether UAO is in effect, call the 49 WG/CP.

A10.18. Legal Implications.

A10.18.1. Each pilot/unit will be responsible for their aircraft operating under these provisions if involved in an accident/incident. Under no circumstances will aircraft be scheduled during uncontrolled airfield conditions without proper notification to the appropriate agencies listed in this instruction.

Figure A10.1. UAO Setup Checklist.

Date:	**Unit:**	**Responsible ODO:**	**ODO Cell:**
A/C Call Sign:	**Number / Type A/C:**	**Date / Time of requested UAO:**	

Unit POC Phone List		
Unit	**DSN**	**Cell (add as required):**
49 WG / CP	572-4900	
704 TG	572-1229	
AA	679-1315	
49 FSS/AC	572-3752	
TDY Unit		
AM	572-5411	

Required Coordination (may be accomplished via telephone)			
Agency	**DSN**	**Commercial**	**Rank / Name of person contacted:**
WSRF	258-8001	(575) 678-8001	
CE / Airfield Maintenance	572-5883	(575) 572-5883	
CE / Barrier Maintenance	572-7268	(575) 572-7268	
CE / Exterior Electric	572-3781	(575) 572-3781	
RAWS	572-3132	(575) 572-3132	
CE / FD	572-7228	(575) 572-7228	
Ambulance	572-1893	(575) 572-1893 or (575) 491-6788	
SFS	572-7171	(575) 572-7171	
NOTE: UAO must be coordinated / de-conflicted with the above agencies prior to receiving AM brief.			

AM / ODO Brief (must be in person)	**AM Initials:**	**ODO Initials:**
Confirm flight plan is on file.		
Airfield Status Briefing.		
Live load operations (if live load ops will cross Runway 16/34 or interfere with proposed UAO, UAO WILL NOT occur).		
ODO received latest NOTAM sheet.		
Advise ODO of other scheduled UAO ops and provide contact information.		

AM – Print Rank / Name:	**AM Signature:**	**Date:**
ODO – Print Rank / Name:	**ODO Signature:**	**Date:**

Figure A10.2. ODO Training Checklist.

ODO Training Checklist. The unit will develop a training plan for ODOs that must include, but is not limited to:		
ODO Name:	Unit:	Training Start Date:
1. Airfield driver's training leading to an Airfield Driver's License.		AF Form 483 #:
2. Airfield Inspection and Maintenance CBT (see AM). Certificate will be turned in to AM.		AM Initials:
3. Airfield Criteria Course CBT (see AM). Certificate will be turned in to AM.		AM Initials:
4. Airfield Lighting Training by CE Exterior Electric. Exterior Electric coordinated ODO access to the ALV ALCMS. (Required for night ops).		Date Complete: 49 CES/CEOFE Initials:
5. Training from the AFM or designated representative: Includes parking plan (i.e. weight restrictions).		Date Complete: AFM Initials:
6. Training from the ODO and AFM, includes: Airfield Checks. Airfield orientation with AM personnel and qualified ODO.		Date Complete: AM / ODO Initials:
7. Briefing from FD Chief or assistant.		Date Complete: FD Initials:
8. Briefing from SFS Operations NCOIC.		Date Complete: SFS Initials:
9. Review of service applicable regulations.		ODO Initials:
10. 54 FG/CC Interview (**ODO rank/name**) _____ is approved to perform ODO duties for (**unit name**) _____. _____ **Commander, 54th Fighter Group** **Date**		
11. When complete, submit this checklist to AM.		AM Initials: ODO Initials:

Figure A10.3. ODO OPS Checklist.

Initials	Action / Responsibility
	UAO Preparation / Planning
	1. Be on duty at respective duty location one hour prior to any UAO. See HAFBI 13-250 paragraph A10.3.2.1.
	2. If more than one ODO is active, identify a lead ODO and inform 49 WG/CP. All coordination must go through lead ODO.
	3. Obtain a weather brief and review NOTAMs / FCIF / ORFs.
	4. Determine runway in use based on current and forecasted wind direction. If necessary, coordinate with other ODOs.
	5. Be familiar with each pilot's intentions / flight plans. Accomplish an in-person briefing with departing aircrew prior to launch. Accomplish an over-the-phone briefing with off-station arriving aircraft prior to arrival.
	6. Contact 49 WG/CP (572-4900) to coordinate commencement of UAO prior to your unit's first aircraft engine start.
	7. Perform an airfield check of intended taxi routes prior to aircraft movement.
	8. Activate airfield lighting as necessary. See HAFBI 13-250 paragraph A10.7.26.
	9. Be accessible via LMR (Tower Net) and provided contact number.
	UAO Departure / Arrivals
	1. Broadcast on LMR UAO commencement and departing / arriving aircraft intentions.
	2. Inform vehicle operators of arriving or departing aircraft. Inform arriving or departing aircraft of any known ground traffic.
	3. Visually observe all takeoffs and landings.
	4. Monitor the weather and direct pilot as necessary to ensure safe operations.
	5. Aircraft mishap or emergency: ☐ Immediately notify FD. ☐ Notify 49 WG/CP. After an emergency, ensure the runway, Taxiway Golf, and/or North Ramp is clear of debris, damage, and fluid spills. Coordinate with On-scene Commander.
	6. Report any unauthorized aircraft, vehicles, or equipment in the airfield environment to SFS.
	UAO Termination
	1. Broadcast UAO termination on LMR Tower Net.
	2. Report UAO termination to 49 WG/CP and FD.

Attachment 11

AIRFIELD OPERATIONS BOARD (AOB)

A11.1. The AOB provides a forum for discussing, updating, and tracking various activities associated with support of the HMN flying missions. The AOB will convene quarterly. The 54 OSS/OSA is the focal point for scheduling AOB meetings, preparing the agenda, and recording the minutes. The agenda will include mandatory items and any other pertinent issues the AOB Chairperson deems appropriate. See AFMAN 13-204V1 for additional AOB guidelines.

A11.1.1. The 49 WG/CC or designated representative (not delegated lower than the 54 FG/CC) is the Chairman of the HMN AOB.

A11.1.2. The members (or designated representative) and units/offices listed in Table A11.1. are members of the AOB and are therefore required to attend. Attendance by TDY units, contractors, or others will be coordinated through 54 OSS/OSA/CC.

A11.1.3. The following items will be reviewed annually.

A11.1.3.1. Local airfield and flight environment LOPs.

A11.1.3.2. TERPS. All instrument procedures to validate the continuing need for each.

A11.1.3.3. Air Installation Compatible Use Zone (AICUZ) program as required by AFI 32-1015 Integrated Installation Planning, Chapter 3.

A11.1.3.4. Results of annual self-inspections.

A11.1.3.5. Special interest items results.

A11.1.3.6. Results of the Annual Airfield Certification/Safety Inspection.

A11.1.3.7. Master Aircraft Parking Plan.

A11.1.3.8. Status of airfield waivers.

A11.1.3.9. Airfield projects.

Table A11.1. HMN AOB Members.

49 WG	
49 WG/SEF 49 WG/XP 49 WG/CP 54 FG/CC 54 FG/FGV 54 OSS/CC 8 FS/CC 311 FS/CC 314 FS/CC 54 OSS/OSA/CC 54 OSS/OSAA AFM 54 OSS/OSAM RAWS NCOIC 54 OSS/OSAR WSRF CCTLR 54 OSS/OSAT Tower CCTLR 54 OSS/OSA TERPS 54 OSS/OSOA HAM 54 OSS/OSOS Scheduling 54 OSS/OSW Leadership	49 OG/CC 49 OG/OGV 6 ATKS/CC 9 ATKS/CC 29 ATKS/CC 49 MSG/CC 49 CES/CC 49 CES/CEO Operations Flight 49 CES/CEN Engineering Flight 49 CS/CC 49 CS/SCO (Operations Flight) 49 CS/SCOT (Frequency Manager) 49 FSS/AC Manager 49 MXG/MOC
Tenant Units	**Off Station Organizations**
704 TG/CC 586 FLTS 82 ATRS AA	WSMR Airspace Manager ABQ ARTCC AETC/A3OF TERPS KALM Airport Manager

Attachment 12

F-16 OPERATIONS

A12.1. Introduction. This attachment outlines specific procedures for locally assigned AETC F-16 operations IAW AFMAN 11-2F-16V3_AETCSUP 1, *F-16 Operations Procedures,* AFMAN 13-204V3, and FAA JO 7610.14, *Non-Sensitive Procedures and Requirements for Special Operations*. 82 ATRS QF-16s will comply as applicable, IAW ACCMAN 11-2QF- 16V3, *QF-16 Operations Procedures*.

A12.2. General Policy. 54 FG/FGV will produce and manage local F-16 in flight publications (i.e., 49 WG IFG, Viper Annex, Smart Card) and is the primary POC for questions regarding, and changes to, these publications.

A12.3. Ground Operations.

A12.3.1. F-16 Taxi Operations.

A12.3.1.1. F-16s will call for taxi from their parking spots and be cleared to the designated arming area (see Table A12.1.). Clearance for the flight lead is clearance for the flight. If required, once armed, flight leads will request to continue taxiing to the active runway.

A12.3.1.2. F-16s unable to taxi with their flights from parking will call for taxi individually until rejoined with the flight, presuming the remainder of the flight is still on the ground. F-16s unable to rejoin with their flights will taxi single ship and comply with paragraph 4.1.5.

A12.3.1.3. F-16s will use full runway length for takeoffs. Ground Control will clear taxiing F-16s to the end of the active runway unless otherwise requested by the flight lead. Intersection departures at Runway 34 A/H intersection are approved by the 54 FG/CC. All other intersection departures require approval by the 54 FG/CC coordinated through the SOF.

A12.3.1.4. After landing, F-16s will request taxi to the applicable de-arm and/or parking area. Aircraft not ready to taxi with the flight lead will make their own radio calls separately. A taxi call for aircraft going from EOR Foxtrot to the West Ramp is not required.

A12.3.2. F-16 Aircraft Arresting System Configurations. F-16s must have 54 FG/CC or 54 FG Brickholder approval to depart or arrive without a departure end cable (overrun BAK-12s excluded). SOF (or Tower Watch Sup if there is no SOF on duty) will coordinate with F-16 Ops Sups for any expected or unexpected absence of departure end cables on the active runway.

A12.3.3. F-16 Radar Warning Receiver (RWR) Checks. 54 FG may conduct RWR checks south of Taxiway Juliet without any outside coordination. RWR checks in other locations require AM coordination and approval.

A12.3.4. F-16 Arm/De-arm. Recommended F-16 arm/de-arm locations are listed in Table A12.1.

Table A12.1. F-16 Arm/De-arm Locations.

Runway In Use	Arm EOR Options	De-arm EOR Options
16	Delta Echo Echo LOLA Foxtrot	Alpha Hotel Foxtrot
34	Alpha Hotel	Echo Echo LOLA Delta Foxtrot
22	Delta Echo Echo LOLA Foxtrot	Alpha Foxtrot
04	Foxtrot	Delta Echo Echo LOLA
25	Bravo	Alpha Foxtrot
07	Alpha Foxtrot	Bravo Hotel

A12.3.4.1. Squadron Ops Sups are responsible for determining their arm/de-arm locations. Setting an arm/de-arm location does not override guidance in paragraphs 2.2. and 5.3. Carefully consider weather, winds, restrictions, syllabus requirements, and airfield status when making the decision to change. The squadron Ops Sup will confer with the SOF and contact the appropriate Aircraft Maintenance Unit (AMU) (8 AMU DSN 572-2153/-2143, 311 AMU DSN 572-6704/-0167, or 314 AMU DSN 572-8315/-8316) with direction to change arm and/or de-arm locations. Expect changes to take up to 30 minutes. F-16 EOR operations are managed by 849 Aircraft Maintenance Squadron (849 AMXS/MX DSN 572-3294, -6924, or -3296).

A12.3.4.2. Table A12.1. assumes all taxiways and runways are available. Based on airfield construction, or other factors, if some taxiways or runways are unavailable the SOF may be required to coordinate with individual Ops Sups to create a new plan not included in Table A12.1.

A12.4. Departure Procedures. F-16s must confirm the desired departure type with clearance delivery upon initial contact.

A12.5. Range Operations/Weapons Employment. Local F-16 range operations will be IAW AFI 13-212V1 HAFB ADDENDA A and when inside WSMR airspace, WSMR FSOP governing the specific flight mission.

A12.6. Arrival Procedures.

A12.6.1. Chased F-16s will add *"WITH CHASE"* to the initial call to Tower when entering the pattern. Tower will assume "with chase" until informed otherwise by the chased aircraft.

A12.6.1.1. Chase aircraft will not overfly aircraft holding short of the runway awaiting departure.

A12.6.2. For formations arriving in the pattern together (i.e., high key, initial, straight-in SFO, or visual straight-in/instrument approach in trail), landing clearance for the flight lead constitutes clearance for the entire flight. Tower will only acknowledge subsequent flight members' gear down call with "*ROGER*." Upon low approach, the formation is considered terminated, and each aircraft will be handled individually.

A12.6.3. Locally assigned F-16s may conduct nighttime overhead patterns (Figures A12.1. – A12.3.) IAW paragraph 6.13.1.4.

Figure A12.1. F-16 Night Overhead Pattern – Runway 16/34.

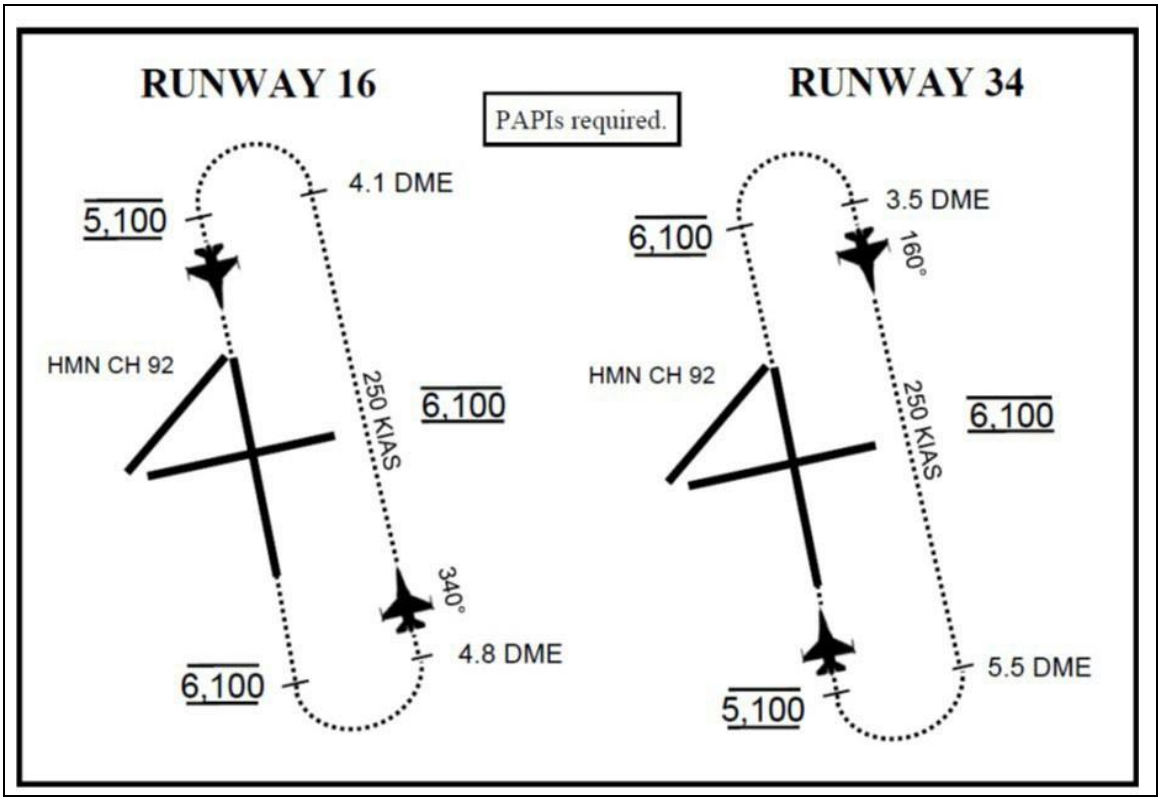


Figure A12.2. F-16 Night Overhead Pattern – Runway 22.

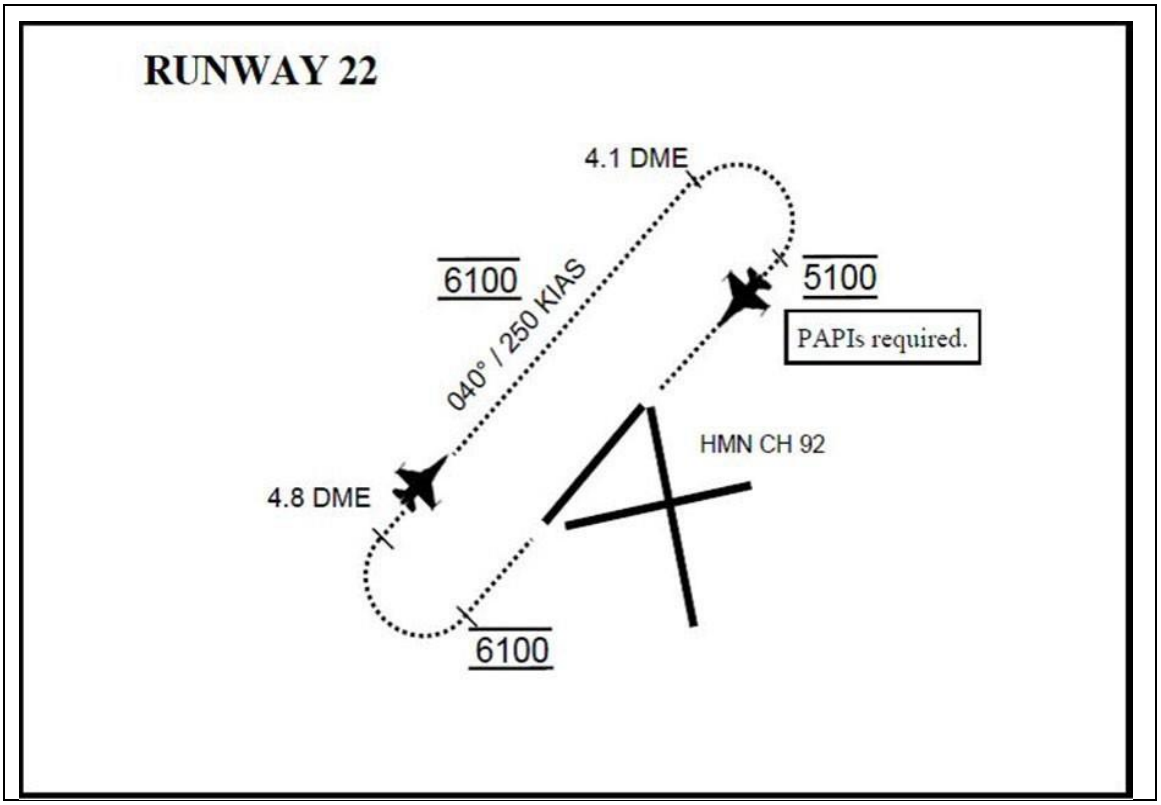
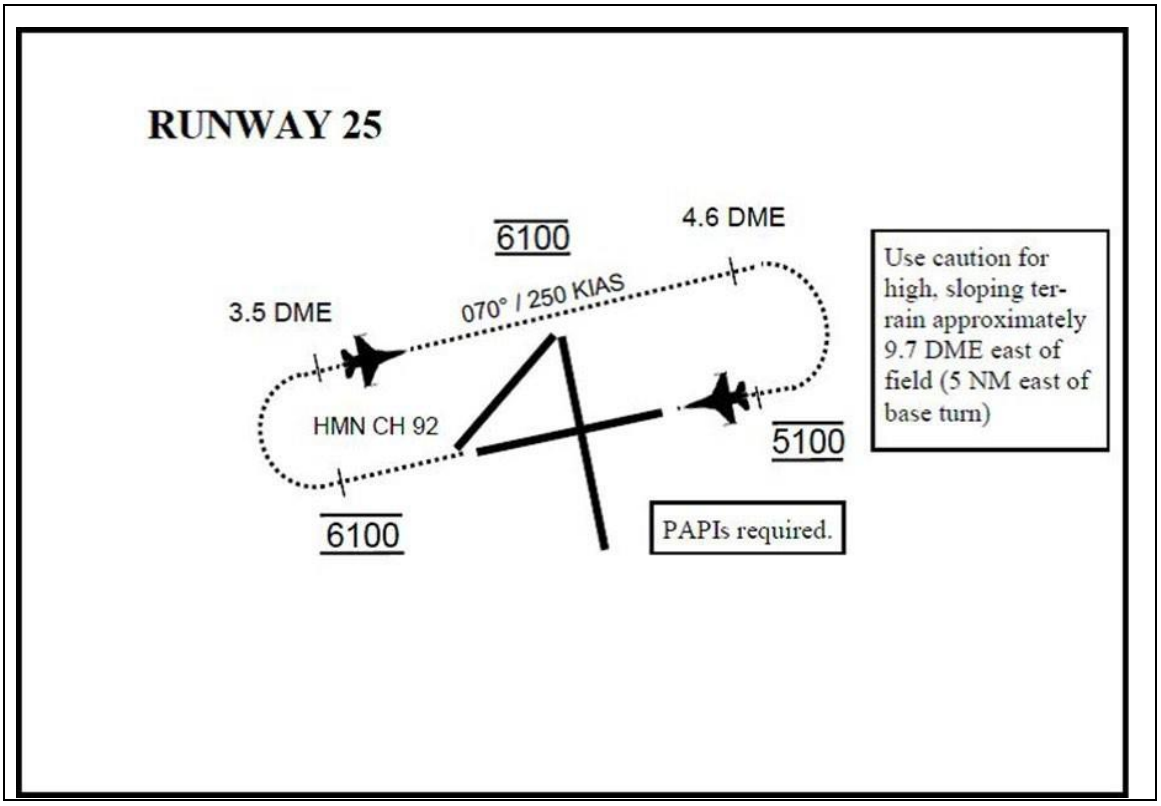


Figure A12.3. F-16 Night Overhead Pattern – Runway 25.



A12.6.4. F-16 Radar Trail Recovery Procedures.

A12.6.4.1. Radar-in-trail recovery must be requested and approved by WSRF prior to beginning the approach and spacing maneuver. Phraseology, *"(call sign) REQUEST RADAR TRAIL (type) APPROACH"*. Flight leads will execute IAW AFMAN 11-2F-16V3 and the 54 FG standards. **Note**: Pilots will not request radar-trail recoveries with ABQ ARTCC.

A12.6.4.2. Radar-in-Trail Spacing.

A12.6.4.2.1. Radar-in-Trail recovery is limited to a maximum of four aircraft.

A12.6.4.2.2. Aircrews conducting radar-in-trail recoveries are responsible for their own separation between elements of their flight while on final for full-stop landings. To ensure appropriate departure separation, multiple practice radar in-trail approaches that do not terminate with a full- stop landing must be conducted only in VMC. During practice approaches in VMC, after an executed low approach/landing, the flight is responsible for their own separation until ATC initiates flight split-ups for individual control.

A12.6.4.2.3. Spacing will be 2 NM between each aircraft in the formation or as defined by unit guidance. Flight leads will report the total spacing of the flight to ATC. The trail aircraft squawks 4000 on mode 3C. Attempt to establish spacing in VMC. If IMC, accomplish spacing maneuver in wings level flight. Requests other than this will be made with ATC.

A12.6.4.3. Missed Approach/Breakout/Go-Around procedures.

A12.6.4.3.1. Missed approach during radar-in-trail will be IAW published procedures. Unless a separate clearance is requested, aircraft will remain in radar trail during missed approach.

A12.6.4.3.2. Breakout/Go-around during radar-in-trail will be IAW ATC instructions. Pilot initiated breakout or Go-Around must advise ATC.

A12.6.4.4. In case of lost communications, execute lost comm procedures IAW paragraph 8.10.

A12.7. F-16 SFO PROCEDURES.

A12.7.1. F-16 Overhead SFO/Straight-In SFO (SI-SFO) Restrictions/Cut-offs.

A12.7.1.1. Tower WS has the authority to limit F-16 multiple approaches for safety reasons.

A12.7.1.2. Only 54 FG F-16s and 82 ATRS QF-16s may conduct overhead SFO/SI-SFO approaches at HMN.

A12.7.1.3. 54 FG F-16 and 82 ATRS QF-16 overhead SFO/SI-SFO patterns may terminate with a low approach or touch- and-go only.

A12.7.1.4. F-16 overhead SFO/SI-SFO patterns are not authorized at night. Remain VMC. Ceiling must be 1,000 ft above the F-16 overhead SFO high key/10-mile SI-SFO final and visibility 5 SM.

A12.7.1.5. F-16 SI-SFO patterns are not authorized to Runways 25 or 07.

A12.7.1.6. F-16 SFO Pattern Volume Restrictions.

A12.7.1.6.1. Typically, only one type of SFO will be accomplished at one time. Overhead, SI- SFO, or alternate entry SI-SFO may run concurrently at the discretion of the Tower WS. Additionally, only one overhead SFO pattern will be active at a time. All simultaneous SFO patterns must be accomplished to the same runway.

A12.7.1.6.2. No more than three flights (single aircraft or formation flights) of F-16s will be allowed to execute overhead SFO/SI-SFO/alternate entry SI-SFO procedures simultaneously. Additionally, no more than 20 aircraft operations will be conducted during an F-16 overhead SFO/SI-SFO approach/alternate entry SI-SFO. Aircraft operations include aircraft requesting departure, inbound aircraft within 15-miles (regardless of frequency), and overflights within the Class D airspace (regardless of frequency). Aircraft climbing to high key or holding at high key are not included in the three flights in the SFO pattern.

A12.7.1.6.3. Regardless of the maximum number allowed, the Tower WS will make the final determination on the number of aircraft that can safely operate in the VFR patterns. Tower must consider other aircraft characteristics, climb-out, missed approach, availability of the tower radar display, active restricted airspace, and FAA mandated intersecting/same runway separation, as well as Tower controller experience and capabilities before authorizing other operations during F- 16 overhead SFO/SI-SFO/alternate entry SI-SFO patterns. Tower will not issue instructions that require the division of the pilot's attention between the SFO and sequencing with traffic.

A12.7.1.7. F-16 SFO versus IFR/VFR Straight-In Arrivals. Other IFR/VFR aircraft inbound to the same or intersecting runway must be more than 10-flying miles out or inside 4-flying miles prior to SFO aircraft leaving high key. Aircraft inside the 10- to 4-mile cutoff may be sequenced behind the F-16 SFO or held outside the Class D airspace. F-16s may be instructed to hold at high key, 10- mile SI-SFO final, or at specific alternate entry SI-SFO final (10-14 DME) for the intended runway whenever a conflict exists with arriving or departing traffic.

A12.7.1.8. F-16 SFO Cut-Off Points.

A12.7.1.8.1. F-16 SFO versus IFR/VFR Departures: Departing aircraft must be rolling before the SFO reaches low key.

A12.7.1.8.2. SI-SFO versus IFR/VFR Departures: Departing aircraft must be rolling before the SI-SFO reports 5-mile SI-SFO final.

A12.7.1.9. For easterly turnouts on Runways 16 or 34 or northerly turnouts Runways 04, 22, or 25; if Tower has advised that other traffic is ahead in the SFO pattern, pilots will delay the turnout until 1-mile past the departure end unless the traffic is in sight and reported to Tower. If the traffic is still not in sight by 1-mile past the departure end, pilots will fly a wide pattern remaining below 7,100 ft MSL until 3 DME. Pilots may climb above 7,100 ft MSL once at 3 DME. Pilots may not turn inside 3 DME until the traffic is in sight and reported to Tower or instructed to do so by Tower issuing "*REPORT HIGH KEY*". When pilots report the preceding traffic in sight, they will maintain visual separation from the traffic in sight IAW FAA JO Order 7110.65.

A12.7.1.9.1. Aircraft established in the SFO pattern will remain within 3 DME to deconflict from aircraft climbing to high key.

A12.7.2. F-16 Overhead SFO Pattern.

A12.7.2.1. F-16s have three options for entering the overhead SFO pattern at high key: radar vectors from WSRF, outside downwind re-entry (see Figure A12.4. – A12.8.), or VFR climb after completion of an approach.

A12.7.2.2. All overhead SFO patterns begin with the F-16 aligned with the runway centerline, at the approach end of the landing runway, defined as the first one-third of the runway in use. Overhead SFO patterns will be flown toward the inside downwind (Figures A12.4. – A12.8.). Overhead SFO patterns to Runways 16 or 34 will remain to the east of the runway. Overhead SFO patterns to Runways 04, 22, 07. or 25 will remain to the north of the runway.

A12.7.2.3. Overhead SFO High Key: Over the runway approach end normally between 11,000 ft MSL and 14,000 ft MSL, airspeed approximately 200 to 300 knots.

A12.7.2.4. Overhead SFO Low Key: A point abeam the point of rollout on final, normally between 7,000 ft MSL and 9,500 ft MSL.

A12.7.2.5. Overhead SFO Base Key: A point halfway through the final turn at a minimum of 6,100 ft MSL.

A12.7.2.6. Overhead SFO Breakouts.

A12.7.2.6.1. Runways 16/34: Easterly climbing turn back to high key altitude. Runways 04/22/07/25: Northerly climbing turn back to high key altitude.

A12.7.2.6.2. Different breakout instructions may be directed by Tower for deconfliction purposes or approved upon pilot request. "*BREAKOUT*" will normally not be used once an F-16 has departed low key, instead Tower will use "*GO- AROUND (additional instructions as necessary).*"

A12.7.2.7. Overhead SFO Phraseology.

A12.7.2.7.1. Request overhead SFO approach with Tower by stating *"(call sign), REQUEST HIGH KEY.*"

A12.7.2.7.2. Traffic permitting, Tower will approve the request by stating *"(call sign), REPORT HIGH KEY.*"

A12.7.2.7.3. Report high key, *"(call sign), HIGH KEY."* If more than one aircraft is required to hold at high key, pilots will orbit east or north of the runway and advise Tower of holding altitude. When Tower instructs the aircraft "*REPORT LOW KEY*" the SFO is approved.

A12.7.2.7.3.1. For flights arriving from the airspace directly to high key, only the flight lead will report high key. SFO clearance for the flight lead is clearance for the entire flight.

A12.7.2.7.4. Tower will approve the SFO and request the F-16 to report low key *"(call sign), REPORT LOW KEY."*

A12.7.2.7.4.1. For flights that arrived from the airspace directly to high key, all flight members will report low key individually.

A12.7.2.7.5. Report low key, *"(call sign), LOW KEY."*

A12.7.2.7.6. At low key, Tower will NOT acknowledge unless issuing "*BREAKOUT*" instructions, or alternate instructions.

A12.7.2.7.7. Report base key, *"(call sign), BASE KEY, GEAR, LOW APPROACH."*

A12.7.2.7.7.1. For flights that arrived from the airspace directly to high key, all flight members will report base key individually.

A12.7.2.7.8. At base key, Tower will issue appropriate landing clearance or *"(call sign) GO- AROUND.*" During formation SFOs, for flight members other than the flight lead, Tower will respond IAW paragraph A12.6.2.

A12.7.2.8. Standard overhead outside downwind/re-entry (see paragraph 6.13.) may be used to enter high key. When requested, Tower will direct the F-16 to *"(call sign) RE-ENTER, SFO ON REQUEST*." Once approved, Tower will direct the F-16 to *"(call sign) REPORT HIGH KEY*." If overhead SFO is not available (because of traffic, weather, active airspace, etc.) by the 90-to-initial point, Tower will direct the F-16 "*UNABLE SFO, REPORT INITIAL*" or issue additional instructions.

A12.7.2.8.1. VFR "climb to" may be used to enter high key. When requested after completing an approach, Tower will direct the F-16 to *"(call sign) REPORT HIGH KEY."*

A12.7.2.8.2. F-16 Overhead SFO Holding. Aircraft may be instructed to hold at high key for the intended runway whenever a conflict exists with arriving or departing traffic. Holding must normally be left turns for Runways 16, 07, and 04, right turns for Runways 34, 22, and 25. Different turns may be used when necessary or requested. F-16s must remain within 3 NM of high key. Tower will issue *"(call sign) HOLD AT HIGH KEY, (time permitting, reason).*"

A12.7.2.9. F-16 Overhead SFO Conflicts.

A12.7.2.9.1. IFR departures from Runways 16, 22, or 25 may conflict with overhead SFO patterns (and holding) to Runways 16, 34, 04, 22, 07, and 25. Tower will not allow departing aircraft to execute IFR departure procedures that are a conflict for SFO traffic while F-16s are holding at high key or leaving high key until positive deconfliction is ensured.

A12.7.2.9.2. VFR aircraft in the overhead pattern (e.g., departure end break) may conflict with an F-16 VFR climb to high key. When necessary, Tower must issue traffic before allowing F-16s to climb to high key and issue sequencing instructions to ensure separation.

A12.7.2.9.3. An F-16 overhead SFO on the outside downwind climbing to high key may conflict with straight-in traffic inside 10-mile final. Tower must issue traffic before allowing F-16s to climb to high key and issue sequencing instructions to ensure separation.

A12.7.2.9.4. An F-16 overhead SFO on the outside downwind climbing to high key may conflict with an IFR aircraft on the radar downwind. Tower must issue traffic before allowing F-16s to climb to high key and issue sequencing instructions to ensure separation.

A12.7.2.9.5. An F-16 overhead SFO on the outside downwind climbing to high key may conflict with a VFR aircraft breaking out of the overhead pattern to the 90-to-initial point. Tower must issue traffic whenever an aircraft executes the overhead breakout maneuver.

A12.7.2.9.6. Base key with aircraft at 3-mile initial. This conflict can be prevented by:

A12.7.2.9.6.1. Traffic call issued to aircraft inside 10-miles for the overhead and aircraft near high key.

A12.7.2.9.6.2. Traffic call issued to aircraft inside 5-mile initial and aircraft near low key. **Note**: SFO conflicts frequently occur when aircraft are at initial and SFO aircraft do not have each other in sight.

A12.7.2.9.6.3. Aircraft inside of 5-mile initial will be directed to offset either south (Runway 22) or west (Runway 16) to deconflict with SFO traffic.

A12.7.2.9.7. If the SFO aircraft does not have initial traffic in sight or the pilot cannot complete the approach, do not descend below 6,600 ft MSL. If the SFO traffic has NOT started the base turn, the pilot will advise Tower of intentions to either breakout to the short re-entry point or climb back to high key. Pilots who initiate the base turn will maintain at or above 6,600 ft MSL, fly runway heading, and advise Tower of intentions. Do not breakout or climb back to high key after initiating the turn to base key.

A12.7.2.9.8. An F-16 overhead SFO leaving high key will conflict with aircraft on the inside downwind. Tower may allow F-16 overhead SFOs to leave high key and follow traffic that has passed midfield downwind to final. Tower will issue traffic information and the F-16 at high key must call traffic in sight and maintain visual separation.

A12.7.2.10. Standard Low Key SFO Entry. For use when ceiling is below 8,000 ft AGL, but greater than 4,000 ft AGL; or if required due to airspace restrictions.

A12.7.2.10.1. For use when training dictates low key SFO procedures. Random or alternate entry low key SFOs are not authorized at HMN.

A12.7.2.10.2. Low key procedures are identical to standard SFO procedures with the following additional restrictions:

A12.7.2.10.2.1. Weather Requirements. Ceiling 1,000 ft above the low-key altitude and 5 SM visibility. VFR re-entry pattern must be open during low key operations.

A12.7.2.10.2.2. Only one aircraft/flight at a time may use the SFO during low key operations.

A12.7.2.10.2.3. On a case-by-case basis, Tower may authorize aircraft to hold at low key. Aircraft must re-enter or climb to high key if the low key SFO cannot be approved. Breakout procedures will be directed by Tower.

A12.7.2.10.2.4. Overhead Direct. Pilots will request low key and state intentions to carry straight through initial for a right/left turn out to the low key SFO. Prior to initial, traffic permitting and with prior coordination, pilots may request direct low key from any direction.

A12.7.2.10.3. Standard Radio Calls.

A12.7.2.10.3.1. To request low key SFO on the go: *(call sign), REQUEST LOW KEY."*

A12.7.3. F-16 SI-SFO Patterns.

A12.7.3.1. F-16s have two options for entering the SI-SFO pattern at 10-mile final: radar vectors from WSRF or outside downwind re-entry (see Figures A12.9. – A12.12.).

A12.7.3.2. All SI-SFO patterns begin with the F-16 aligned with the runway centerline at 10 NM final (see Figures A12.9. – A12.12.).

A12.7.3.3. 10-mile SI-SFO final: located 10 NM final to Runway 04, 16, 22, or 34, altitude will normally be 14,500 ft MSL, or as requested, airspeed approximately 200 to 300 KIAS.

A12.7.3.4. 5-mile SI-SFO final: altitude approximately 8,000 ft MSL to 10,000 ft MSL, wheels should be down.

A12.7.3.5. SI-SFO Breakouts.

A12.7.3.5.1. As directed by Tower, an east or west climbing turn back to 10-mile SI-SFO final. Tower controllers must have visual contact, or see the aircraft on the tower radar display, by 3 NM or the SI-SFO/alternate entry SI-SFO will be broken out or directed to go-around.

A12.7.3.5.2. Different breakout instructions may be directed by Tower for deconfliction purposes or approved upon pilot request. "*BREAKOUT*" will not normally be used once an F-16 has departed 5-mile SI-SFO final, instead Tower will use *"(call sign) GO-AROUND (additional instructions as necessary)."*

A12.7.3.6. SI-SFO Phraseology.

A12.7.3.6.1. Request SI-SFO approach with WSRF by stating *"(call sign), REQUEST STRAIGHT-IN SFO*." WSRF will issue approval to fly the SI-SFO by stating *"(call sign), STRAIGHT-IN SFO APPROVED, (radar vectors, when/where to contact Tower, as necessary)"* or direct aircraft to hold, if required.

A12.7.3.6.2. On initial contact, Tower will direct the F-16 to *"(call sign), REPORT 10-MILE SFO FINAL"* or issue holding instructions, as necessary.

A12.7.3.6.3. Report 10-mile SI-SFO "*(call sign), 10-MILE SFO FINAL."* Tower will respond with, *"(call sign), REPORT 5-MILE SFO FINAL."*

A12.7.3.6.4. For flights arriving from the airspace directly to 10-mile SI-SFO final, only the flight lead will report 10-mile SI-SFO. SFO clearance for the flight lead is clearance for the entire flight.

A12.7.3.6.5. Report 5-mile SI-SFO final "*(call sign), 5-MILE SFO FINAL, GEAR, LOW APPROACH."*

A12.7.3.6.6. For flights that arrived from the airspace directly to 10-mile SI- SFO final, all flight members will report 5-mile SI-SFO final individually.

A12.7.3.6.7. At 5-mile SI-SFO final, Tower will issue appropriate landing clearance, issue *"(call sign) GO-AROUND*," or alternate instructions. During formation SFOs, for flight members other than the flight lead, Tower will respond IAW paragraph A12.6.2.

A12.7.3.7. Outside downwind/re-entry (see paragraph 6.13.) may be used to enter SI- SFO final. When requested, Tower will direct the F-16 to *"(call sign) RE-ENTER, STRAIGHT-IN SFO ON REQUEST*." Once approved, Tower will direct the F-16 to "*REPORT 10-MILE SFO FINAL."* If SI-SFO is not available (because of traffic, weather, active airspace, etc.) by the 90-to-initial point, Tower will direct the F-16 "*UNABLE SFO, REPORT INITIAL*" or issue additional instructions.

A12.7.3.8. F-16 SI-SFO Holding. Aircraft may be instructed to hold at 10-mile SI- SFO for the intended runway whenever a conflict exists with arriving or departing traffic. Tower or WSRF will issue *"(call sign) HOLD OVER 10-MILE FINAL."* Holding must normally be left turns for Runways 16 and 04, right turns for Runways 34 and 22. Different turns may be used when necessary or requested. F-16s must remain within 3 NM of 10-mile SI-SFO final. When holding is no longer necessary, Tower will issue *"(call sign), REPORT 10-MILE SFO FINAL."*

A12.7.3.9. F-16 SI-SFO Conflicts.

A12.7.3.9.1. An F-16 SI-SFO holding over the 10-mile SI-SFO final may conflict with IFR/VFR straight-in arrivals to the same runway. WSRF/Tower must issue traffic and issue sequencing instructions to ensure separation.

A12.7.3.9.2. An F-16 SI-SFO on the outside downwind climbing to the 10-mile SI-SFO final may conflict with aircraft in the radar pattern or IFR/VFR straight-in arrivals. WSRF/Tower must issue traffic and issue sequencing instructions to ensure separation.

A12.7.3.9.3. An F-16 SI-SFO may conflict with a VFR aircraft on the outside downwind turning 90-to-initial. Tower must issue traffic and ensure aircraft do not turn 90-to-initial in front of an F- 16 SI-SFO.

A12.7.3.10. F-16 Alternate Entry SI-SFO. An alternate entry SI-SFO is defined as an SFO approach that commences at a point other than aligned within +/- 5 degrees of the runway but intercepts the SI-SFO pattern prior to short final (no closer than 3-miles).

A12.7.3.10.1. F-16s have two options for entering an alternate entry SI-SFO pattern at 12-mile final: radar vectors from WSRF or outside downwind re-entry (see Figures A12.13. – A12.17.).

A12.7.3.10.2. Alternate entry SI-SFOs are only authorized to Runways 22, 16, and 34.

A12.7.3.10.3. North runway 22 alternate entry SI-SFO patterns begin with the F-16 at 12 DME anywhere between the HMN R-347 through R-029. Altitude will normally be between 12,500 ft MSL to 14,500 ft MSL, or as requested, airspeed approximately 200 to 300 KIAS (see Figure A12.13.).

A12.7.3.10.4. South Runway 22 alternate entry SI-SFO patterns begin with the F-16 at 12 DME anywhere between the R-039 through R-070. Altitude will normally be between 12,500 ft MSL to 14,500 ft MSL, or as requested, airspeed approximately 200 to 300 KIAS (see Figure A12.14.).

A12.7.3.10.5. East Runway 16 alternate entry SI-SFO patterns begin with the F-16 at 12 DME anywhere between the R-347 through R-034. Altitude will normally be between 12,500 ft MSL to 14,500 ft MSL, or as requested, airspeed approximately 200 to 300 KIAS (see Figure A12.15.).

A12.7.3.10.6. West runway 16 alternate entry SI-SFO patterns begin with the F-16 at 12 DME anywhere between the R-290 through R-337. Altitude will normally be between 12,500 – 14,500 ft MSL, or as requested, airspeed approximately 200 to 300 KIAS (see Figure A12.16.).

A12.7.3.10.7. West Runway 34 alternate entry SI-SFO patterns begin with the F-16 at 14 DME anywhere between the R-204 through R-159. Altitude will normally be between 12,500 – 14,500 ft MSL, or as requested, airspeed approximately 200 to 300 KIAS (see Figure A12.17.).

A12.7.3.10.8. 5-mile SI-SFO final: altitude approximately 8,000 ft MSL to 10,000 ft MSL, wheels should be down.

A12.7.3.10.9. All alternate entry SI-SFOs will intercept the runway extended centerline NLT 3 NM final (Runways 16 and 22 approximately 3 DME, Runway 34 approximately 5 DME).

A12.7.3.10.10. Alternate Entry SI-SFO Breakout. As directed by Tower or requested by the pilot. Runway 22 is either a northwest or southeast climbing turn back to alternate entry SI-SFO final. Runway 16 and 34 is either an easterly or westerly turn back to alternate entry SI-SFO final.

A12.7.3.10.11. Alternate Entry SI-SFO Holding. Left or right turns as necessary, holding must remain within 3 NM of the 12-mile alternate entry SI-SFO final. When possible, avoid crossing the runway extended centerline during holding.

A12.7.3.10.12. Alternate entry SI-SFO phraseology, outside downwind re-entry, holding, and conflicts are the same as for standard SI-SFOs (see paragraphs A12.7.3.6. – A12.7.3.9.) except the term *"ALTERNATE ENTRY"* will be included and a *"12-MILES"* call will be made at 12-miles instead of 10 (14 DME for Runway 34).

Figure A12.4. Runway 16 Overhead SFO (Drawing not to scale).

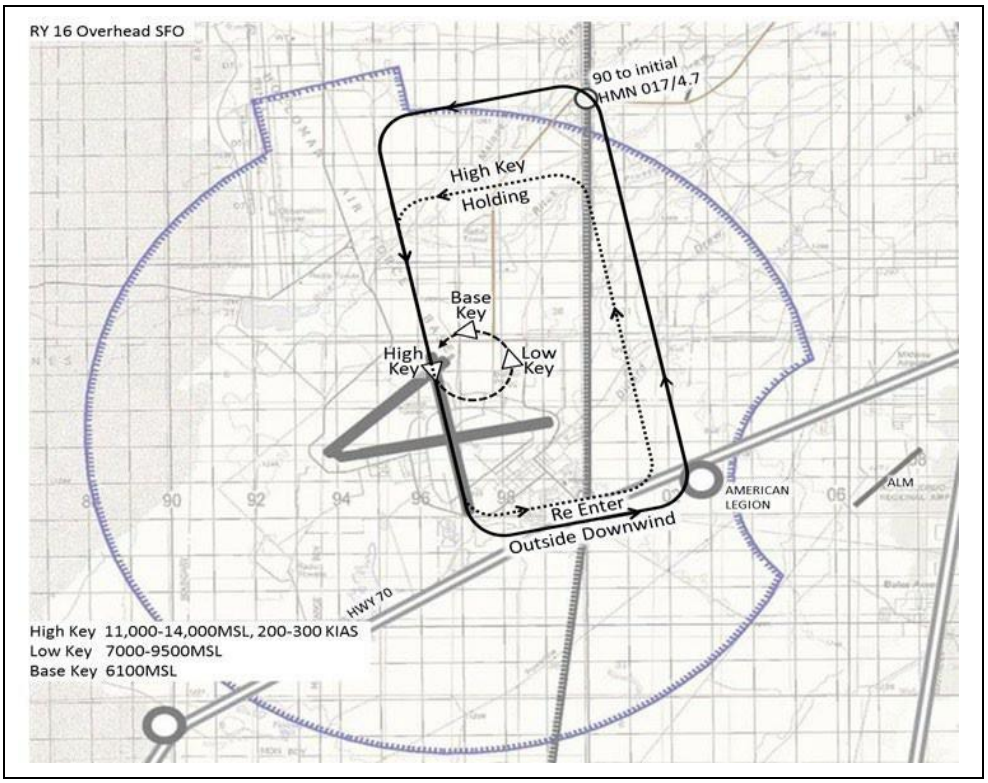


Figure A12.5. Runway 34 Overhead SFO (Drawing not to scale).

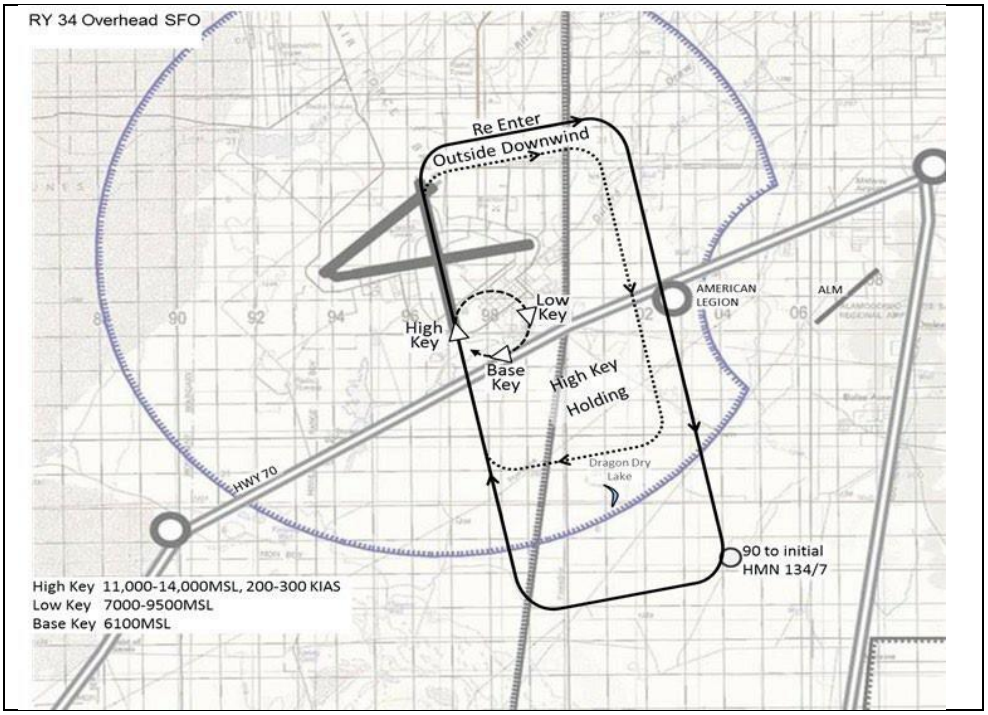


Figure A12.6. Runway 22 Overhead SFO (Drawing not to scale).

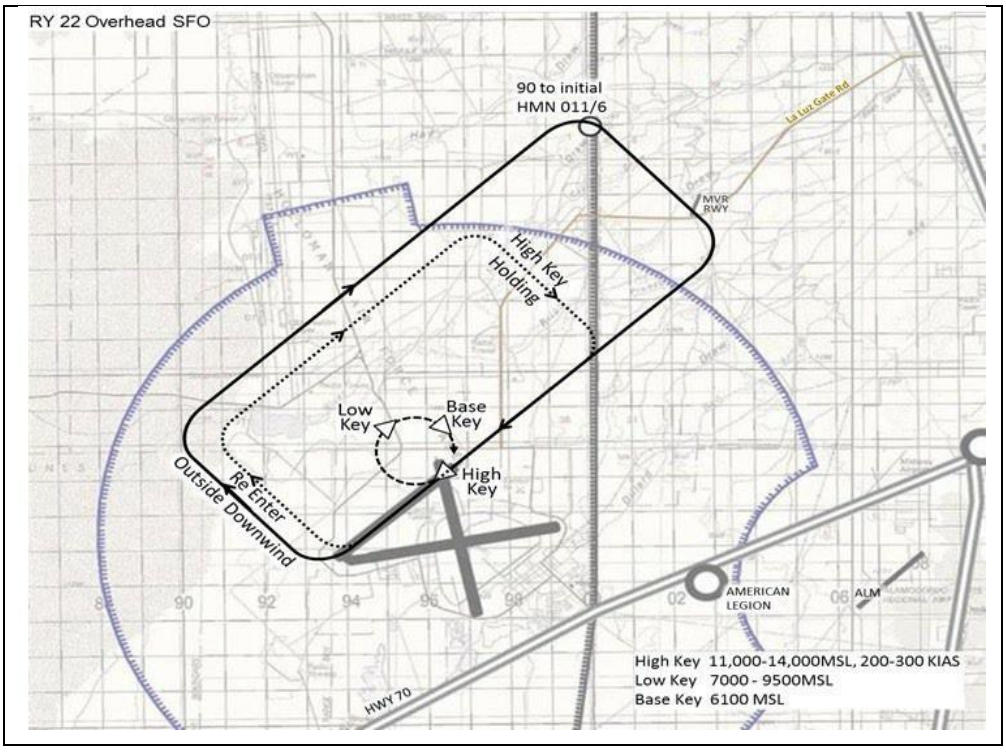


Figure A12.7. Runway 04 Overhead SFO (Drawing not to scale).

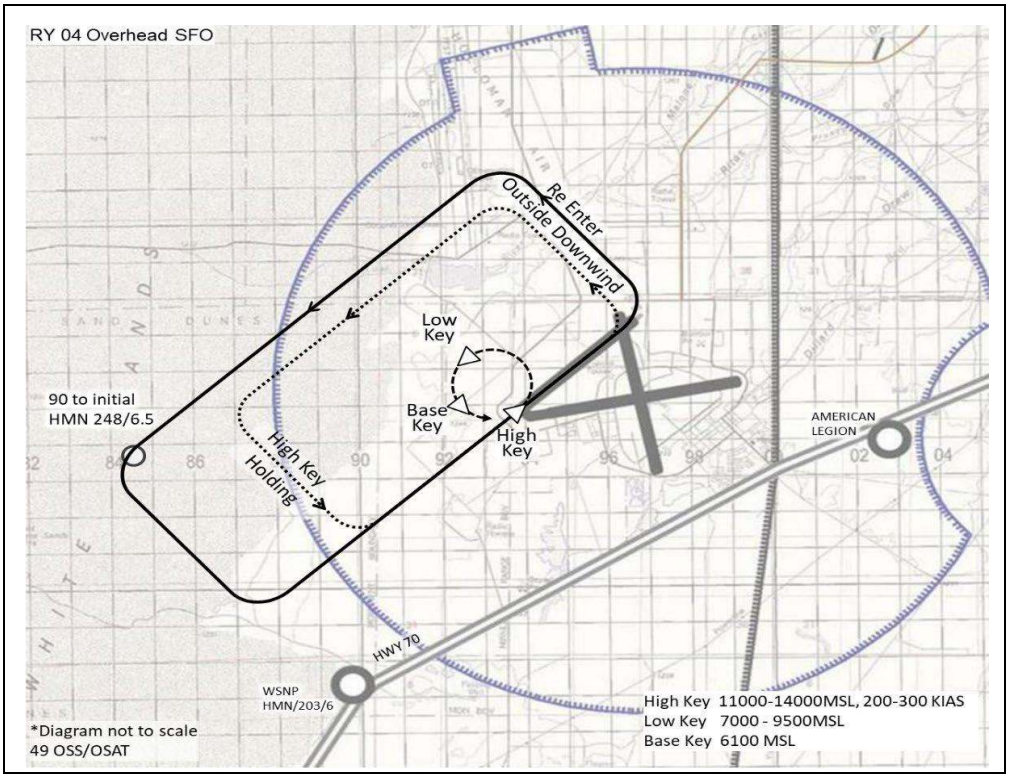


Figure A12.8. Runway 25 Overhead SFO (Drawing not to scale).

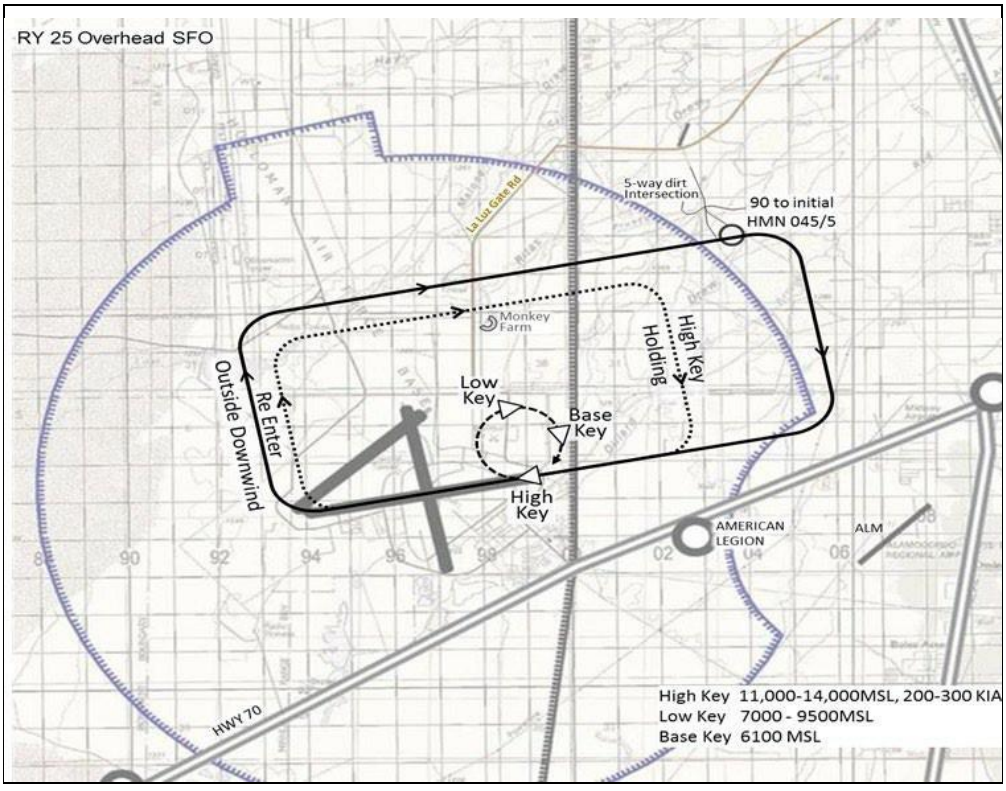


Figure A12.9. Runway 07 Overhead SFO (Drawing not to scale).

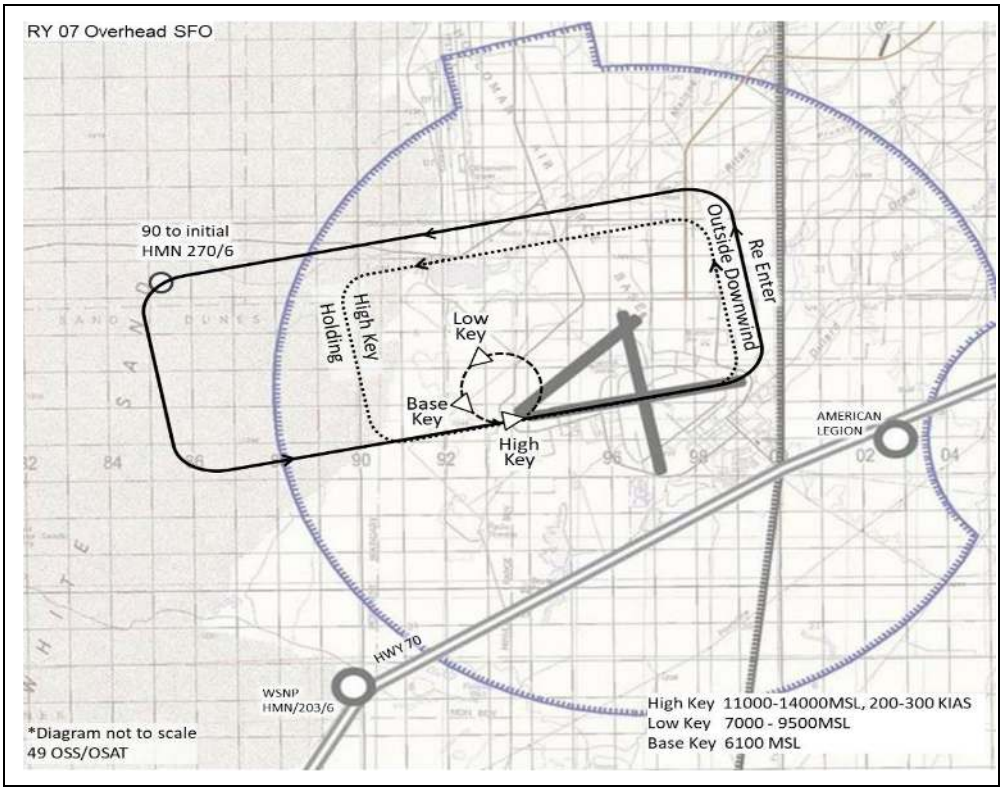


Figure A12.9. Runway 16 SI-SFO (Drawing not to scale).

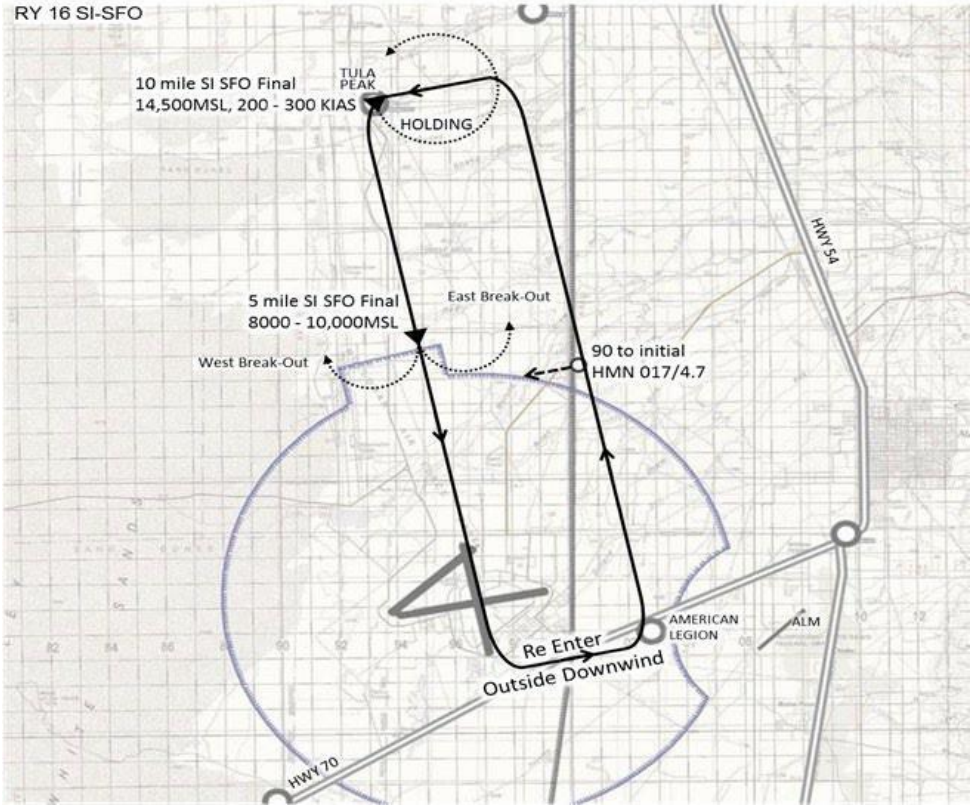


Figure A12.10. Runway 34 SI-SFO (Drawing not to scale).

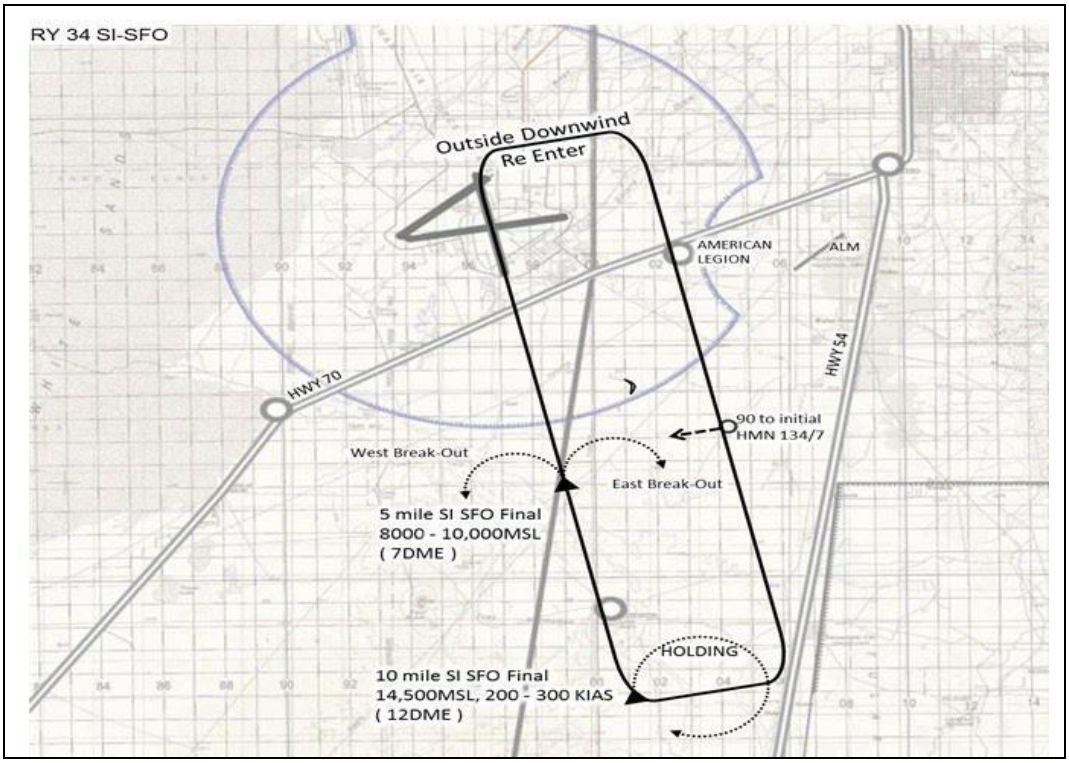


Figure A12.11. Runway 04 SI-SFO (Drawing not to scale).

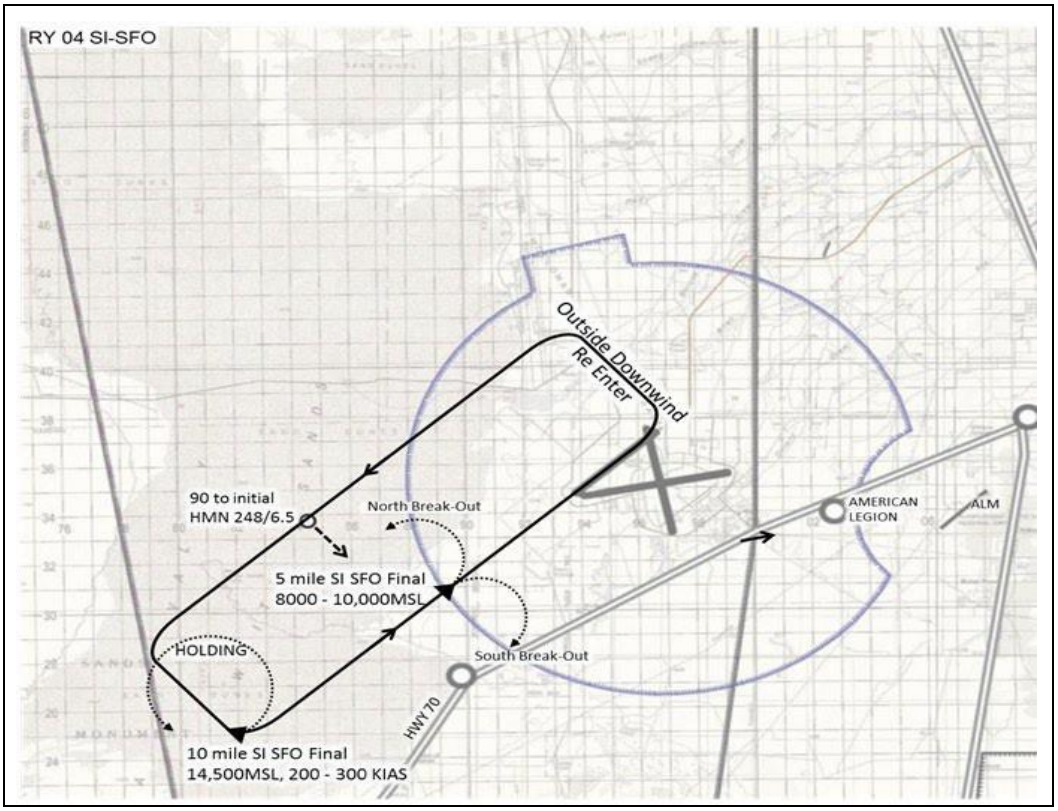


Figure A12.12. Runway 22 SI-SFO (Drawing not to scale).

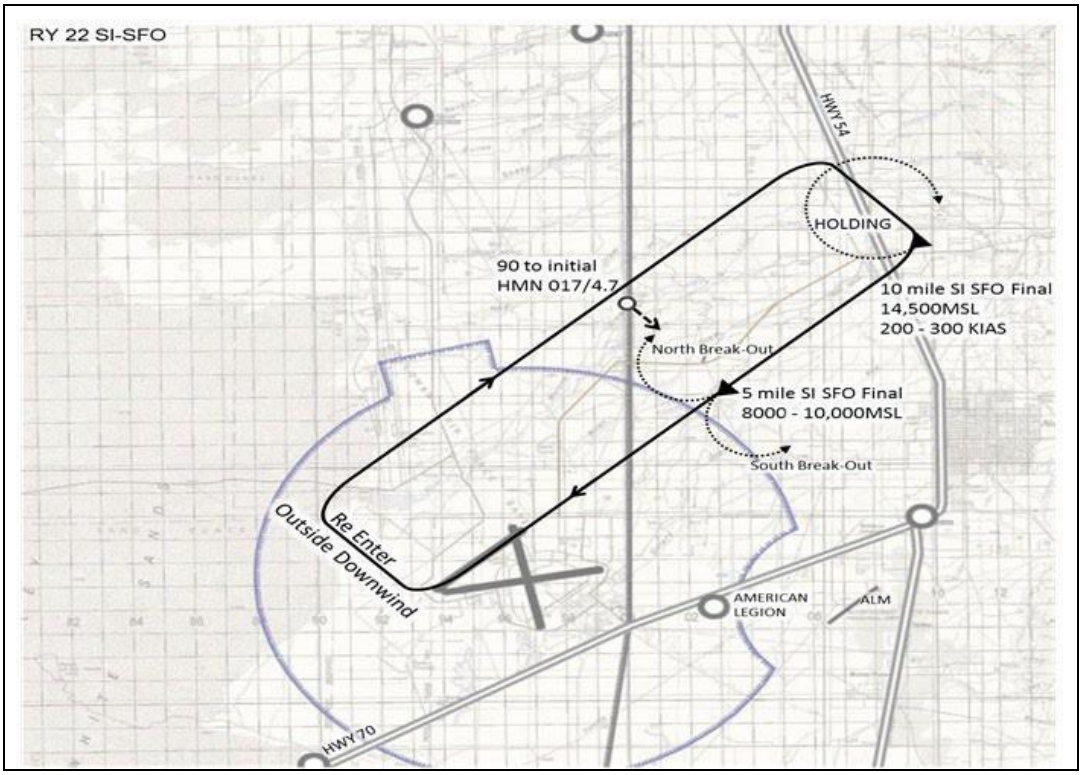


Figure A12.13. North Runway 22 Alternate Entry SI-SFO (Drawing not to scale).

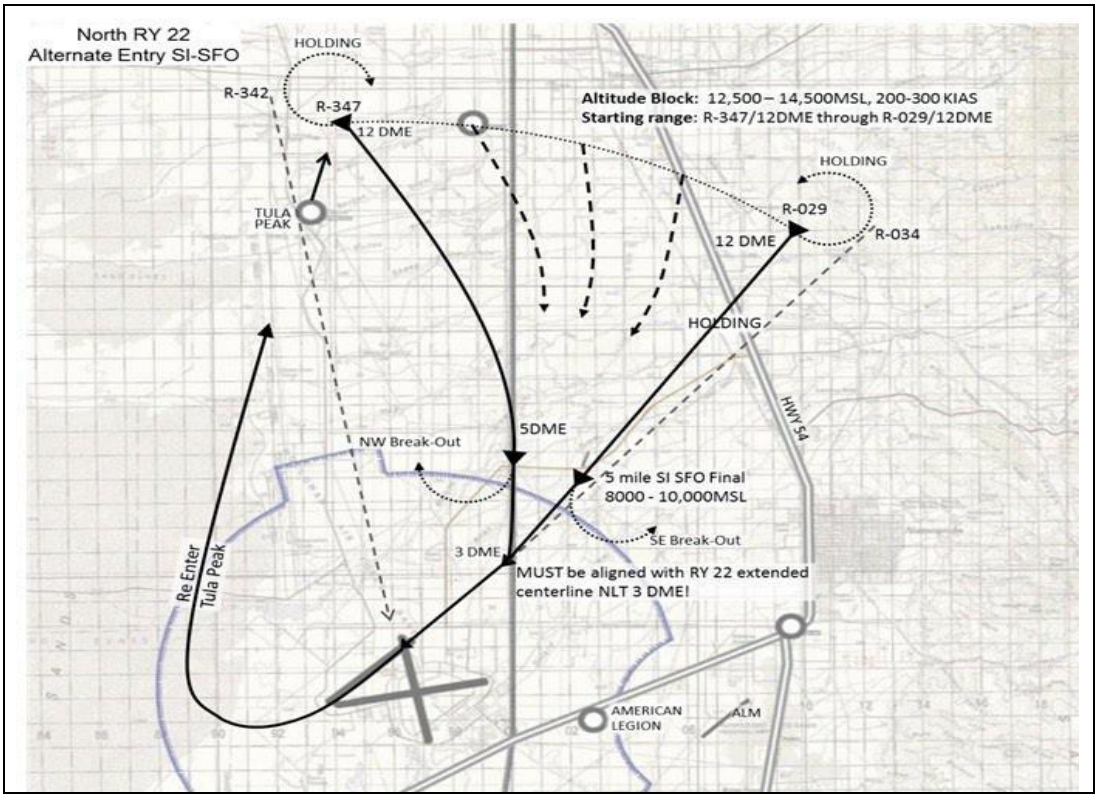


Figure A12.14. South Runway 22 Alternate Entry SI-SFO (Drawing not to scale).

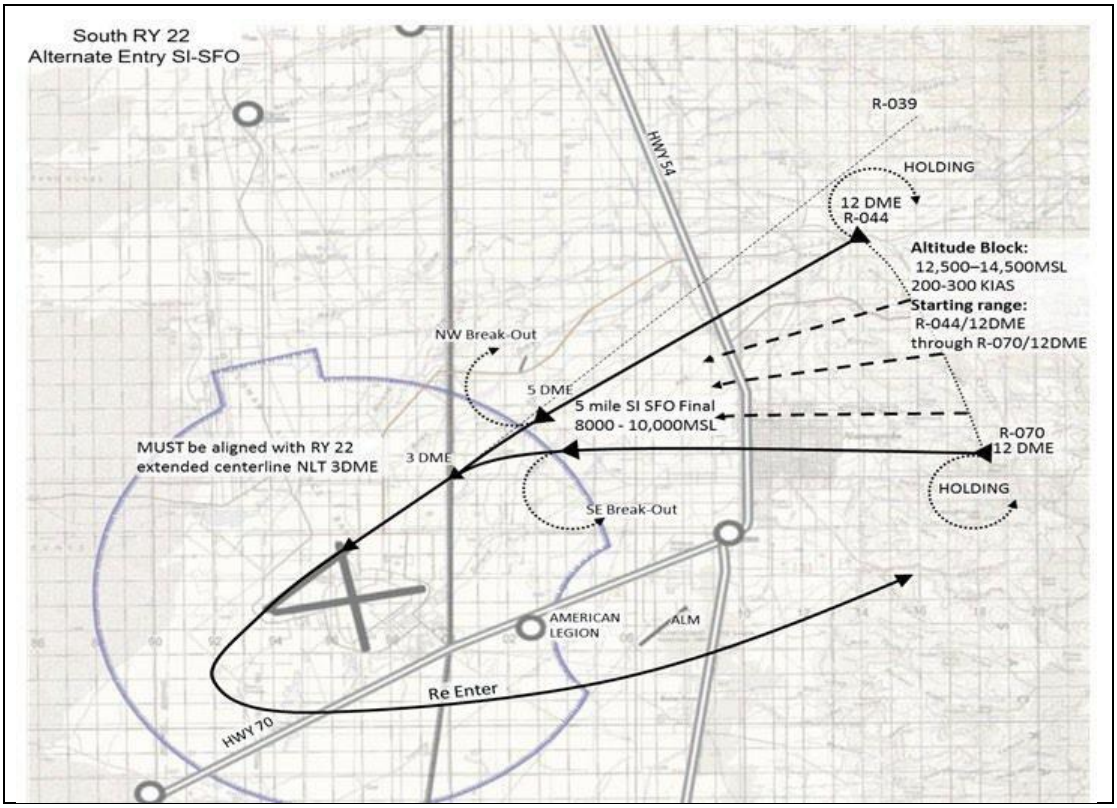


Figure A12.15. East Runway 16 Alternate Entry SI-SFO (Drawing not to scale).

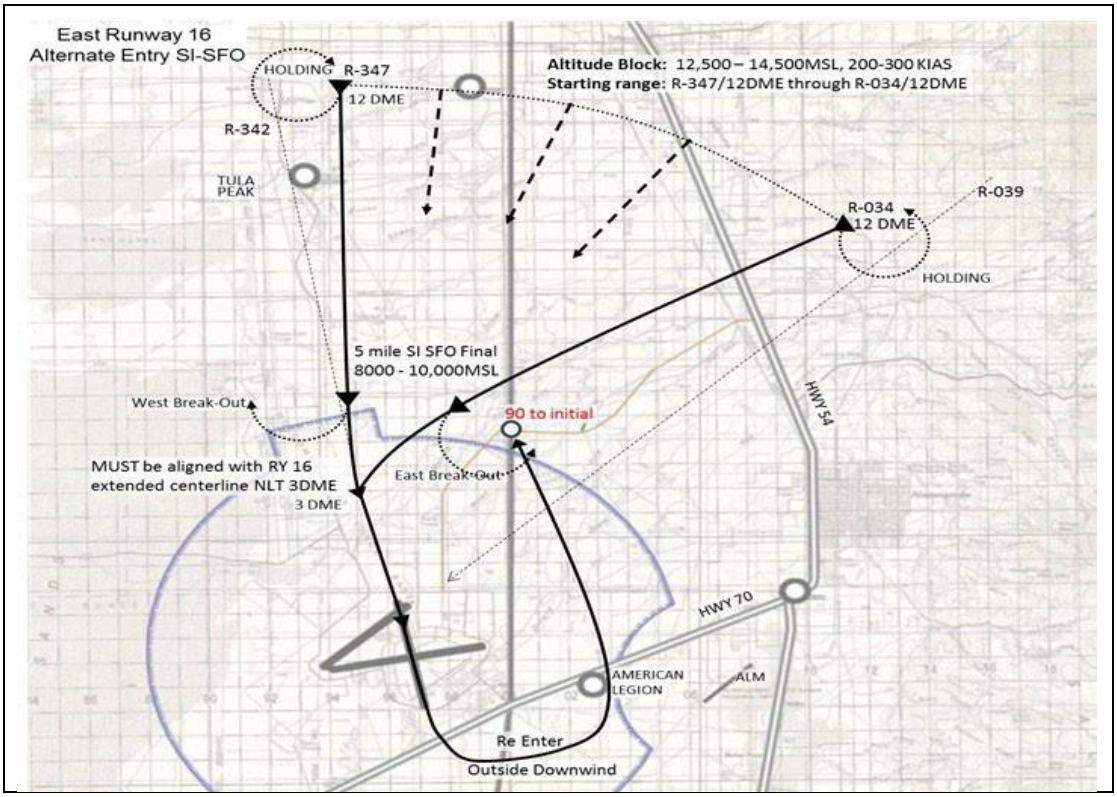


Figure A12.16. West Runway 16 Alternate Entry SI-SFO (Drawing not to scale).

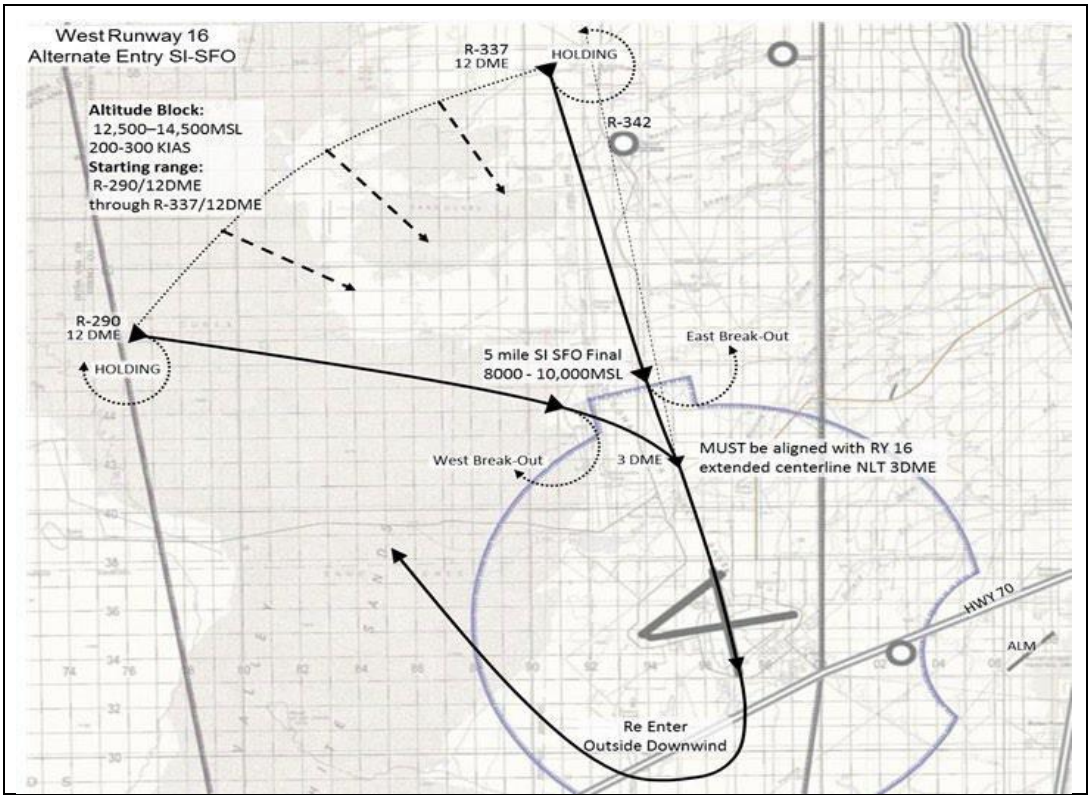
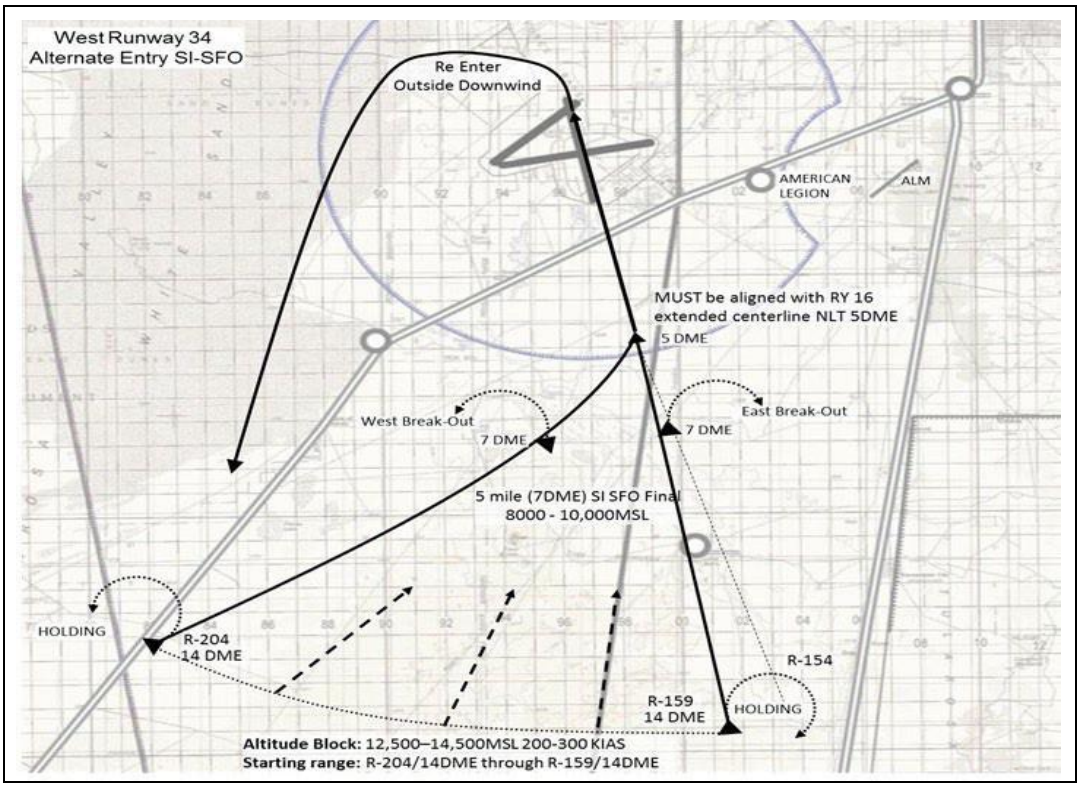


Figure A12.17. West Runway 34 Alternate Entry SI-SFO (Drawing not to scale).



Attachment 13

AIRCRAFT CARRYING HAZARDOUS CARGO

A13.1. Procedures. When notified of any aircraft arriving or departing HMN with hazardous/explosive materials (Class 1.1, 1.2.1, 1.2.2, 1.2.3, 1.3, or 1.4) on board, AM will notify all agencies in paragraph A13.2 of the following information:

A13.1.1. Type aircraft.

A13.1.2. Call sign and aircraft tail number.

A13.1.3. Estimated arrival or departure time.

A13.1.4. Parking location.

A13.1.5. Hazardous/explosive cargo information.

A13.1.6. Any special requests or remarks.

A13.2. Notification. The following agencies will be notified by AM.

A13.2.1. Transient Alert.

A13.2.2. 49 LRS/LGRF.

A13.2.3. Tower.

A13.2.4. 49 LRS Deployment and Distribution Flight (49 LRS/LGRD).

A13.2.5. FD.

A13.2.6. 49 SFS.

A13.2.7. 49 WG/SEW.

A13.2.8. Vehicle Operations, Dispatch Section.

A13.2.9. Deployment Control Center/Installation Deployment Officer.

A13.2.10. 49 WG/CP.

A13.3. Support Agencies.

A13.3.1. When notified of hazardous/explosive cargo movement, each support agency will implement actions to provide priority to aircraft landing, parking, servicing, loading and/or off-loading, and departing.

A13.3.2. Cargo security, safety, and custody will be strictly enforced during ground times.

A13.3.3. For nuclear cargo, the owning MAJCOM will provide an advisory message to support units/agencies.

A13.4. Responsibilities.

A13.4.1. Installation Commander: Prohibit the receipt or transfer of radioactive material (to include radioactive material used in classified operations) without prior coordination and/or approval by the installation Radiation Safety Officer (RSO).

A13.4.2. Permittees/Commanders: Coordinate with the installation RSO prior to receiving, possessing, using, distributing, storing, transporting, transferring, or disposing of any radioactive material, or commodity containing radioactive material.

A13.4.3. 49 LRS/Operations Officer: Establish procedures, with the installation RSO, for the safe movement of radioactive material within or on an AF installation.

A13.4.4. 49 LRS/LGRDAS will send notification to 49 WG/CP, 49 CES/CEF, 49 CES/CED, 49 SFS, and 49 WG/SEW regarding net explosive weight and division of ordnance originating/terminating at HMN IAW 49 LRS OI 91-2, *Explosives Safety*. Review and check emergency procedures prior to movement, and ensure all personnel are explosives handling qualified, to include any augmentees if used. Provide appropriate Material Handling Equipment (MHE) and ensure back-up MHE is readily available if required. Finalize cargo manifests, passenger manifest and air load plans for outbound cargo. Assist with on-load/off-load of cargo. Notify concerned agencies when explosive ordnance has departed HMN, whether by truck or by air. 49 LRS will provide all stanchions, signs, and fire extinguishers needed for hot cargo operations.

A13.4.4.1. Ordnance that is part of a Time-Phased Force Deployment Data movement originating from within the 49 WG must be palletized within the 49 EMS Ammo Compound by the owning unit and then trucked to the Hot Cargo Pad. In the event of out-sized cargo it may be first trucked to the Hot Cargo Pad and built on site by the owning unit. Cargo will be inspected by a qualified Joint Inspector for airworthiness.

A13.4.4.2. Ordnance originating from outside of the 49 WG will be met at the commercial cargo (west) gate by 49 LRS/LGRDAS personnel and escorted to the Hot Cargo Pad and built on site by the owning unit, with assistance from Air Transportation personnel. Cargo will be inspected by a qualified Joint Inspector for airworthiness.

A13.4.5. All support units/agencies involved will develop appropriate checklists to be used in support of nuclear cargo movements so that priority support can be rendered to the maximum extent possible. Support agencies will provide support as determined by the 49 MSG/CC or designated representative. An advisory message will be sent by the originating station to all enroute and destination stations that are affected by each mission carrying hazardous cargo.

A13.4.6. 49 SFS will coordinate movement of nuclear materials with 49 WG/CP, Munitions Control, AM, Tower, and the on-scene commander. All 49 SFS actions will be conducted IAW security directives. Materials exposure time will be always kept to an absolute minimum. The convoy haul route is depicted in HAFB *SAFE HAVEN PLAN*. During ground transport and storage of hazardous materials, use procedures outlined in HAFB *SAFE HAVEN PLAN* (Annex C).

A13.4.6.1. Receipt, storage, or transport of radioactive material, including nuclear devices, must be coordinated with the installation RSO at DSN 572-7938.

A13.4.7. Movement of support personnel and/or vehicles in the immediate area of aircraft with nuclear cargo will be approved by the on-scene commander or the designated representative using the advisory message provided by MAJCOM to support agencies. The advisory message will be authenticated by the Escort/Aircrew and Security Supervisor. The escort will be the approving authority for entry into the secure area and to board the aircraft during all nuclear/hazardous cargo operations. The aircraft commander is the final authority of flight safety and operations.

A13.4.8. Local flight and ground operations will be curtailed only for the period necessary to preclude direct conflict with vehicles carrying hazardous/explosive or nuclear materials.

A13.5. Parking and Storage.

A13.5.1. Primary Parking Areas. The authorized parking locations and amounts for aircraft carrying hazardous/explosive or nuclear cargo is outlined below.

A13.5.1.1. MMG Ramp is sited for the following:

A13.5.1.1.1. No hazard class/division (HC/D) 1.1.

A13.5.1.1.2. No HC/D 1.2.

A13.5.1.1.3. No HC/D 1.2.2.

A13.5.1.1.4. No HC/D 1.2.3.

A13.5.1.1.5. 1,500 lbs. of HC/D 1.3.

A13.5.1.1.6. Mission Essential Quantities (MEQ) of HC/D 1.4.

A13.5.1.2. At no time will Class 1.1 or Class 1.2 be stored, loaded/unloaded, or transported

to or on the MMG Ramp or CDF yard.

A13.5.1.3. HC Ramp at the end of Taxiway Tango is sited for the following:

A13.5.1.3.1. 29,000 lbs. of HC/D 1.1.

A13.5.1.3.2. 28,000 lbs. of HC/D 1.2.1 $\leq$ 450 Maximum Credible Event (MCE).

A13.5.1.3.3. 28,000 lbs. of HC/D 1.2.2.

A13.5.1.3.4. (13) 28,000 lbs. of HC/D 1.2.3. $\leq$ 450 lbs. MCE.

A13.5.1.3.5. 100,000 lbs. of HC/D 1.3.

A13.5.1.3.6. MEQ of HC/D 1.4.

A13.5.2. HMN does not have the capability to store or handle nuclear weapons/cargo. Aircraft with emergencies that land at HMN with nuclear weapons/cargo will only be provided support outlined in paragraphs A13.4.2. thru A13.4.6.

A13.5.3. Primary and alternate storage areas for hazardous/explosive cargo is the munitions storage area and the flightline munitions holding area or Hazardous Aircraft Munitions Storage. The munitions storage area will provide for the storage or parking of vehicles and cargo as required.